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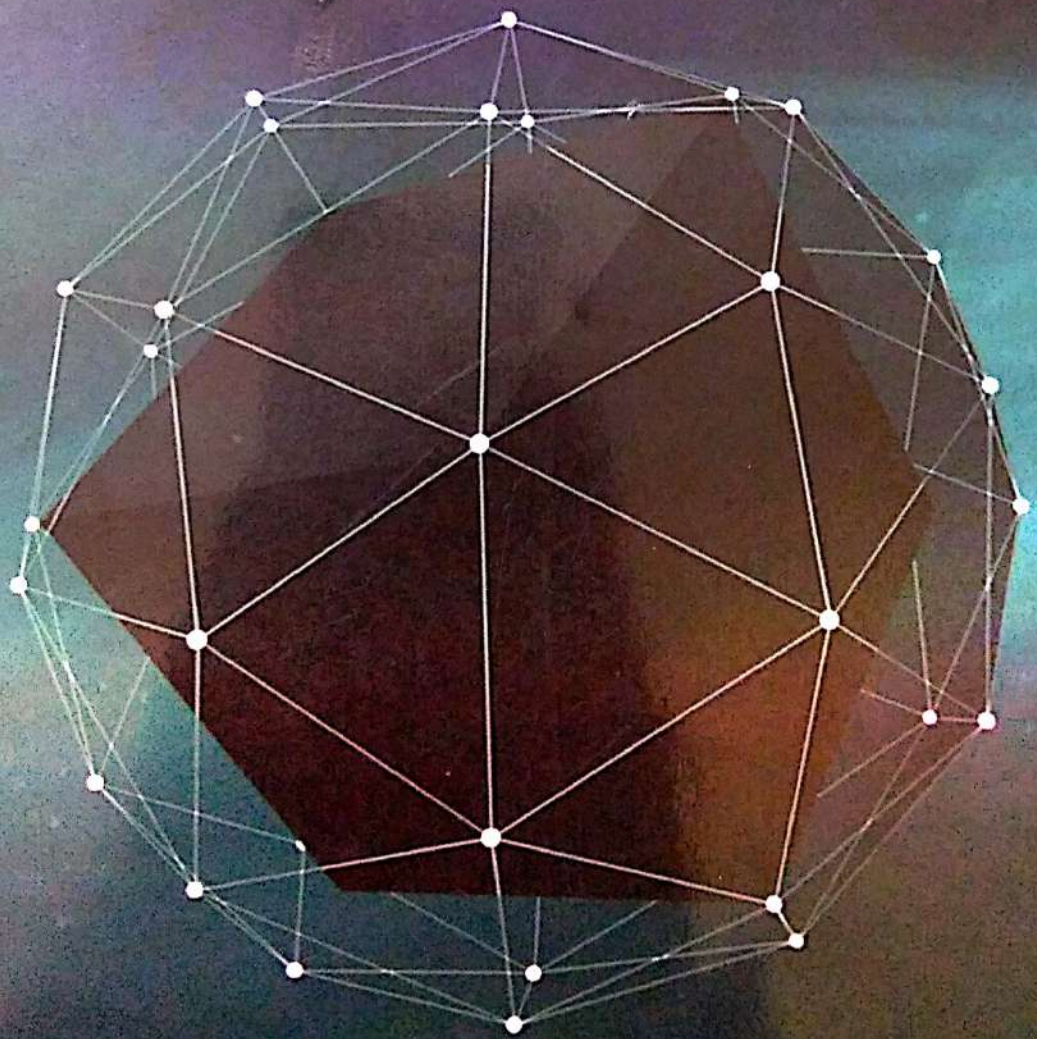
CSIR-NET

GENERAL APTITUDE - A New Outlook

For Mathematical Sciences, Life Sciences, Earth Sciences, Physical Sciences and Chemical Sciences



Christy Varghese

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CSIR–NET
GENERAL APTITUDE – A NEW OUTLOOK

Christy Varghese

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ABOUT THE AUTHOR

Christy Varghese

Christy Varghese has proved himself as an expert in General Aptitude and he is currently working as a faculty for General Aptitude in UV Physics Academy, Hyderabad. He is also giving General Aptitude classes in various NET coaching institutions across South India. He is also serving as a resource person for general aptitude training in Jamal Mohamed College, Trichy and in many other colleges across South India. He is the co-developer of "CSIR-NET General Aptitude" mobile application, the first ever interactive android app for the general aptitude section. He did his Post Graduation in Mathematics at Pondicherry Central University and his graduation at SB College, Changanacherry. His Post-graduate Diploma in Computer Application from Pondicherry University adds more to his achievements.

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Dedicated to

My Appa and Amma...

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PREFACE TO THE THIRD EDITION

I am delighted to introduce the latest edition of one among the best-selling preparatory guides "CSIR-NET General Aptitude – A New Outlook". The first two editions of this book served thousands of students across the country to better prepare for the general aptitude questions of CSIR-NET examination. Market statistics and public opinion do bear testimony for the success of this book. In my opinion, this book fulfils its purpose and is a worthy companion for all CSIR-NET aspirants. The overwhelming positive reviews for the book in the online shopping sites is worth-mentioning. I also came to learn that my students who used this book felt it to be a great boon and I am happy to extend a word of thanks to the author Mr. Christy Varghese for his assiduous endeavour.

Furthermore, the author and his team took this venture to a new level by developing an official Android mobile application titled "CSIR-NET General Aptitude" which I believe is absolutely one of a kind for CSIR-NET. This comes handy for all the busy ones, tech savvies and for those who want to take up real time mock tests and wish to have quick feedback for their queries. Learning process is made simple using this app. As I am informed, this application can cater to the needs of NET aspirants belonging to the science stream viz. Mathematical Sciences, Physical Sciences, Chemical Sciences, Life Sciences and Earth Sciences. I believe that this book as well as the mobile app will turn out to be really useful for all who ardently seek and meticulously pursue CSIR-NET examination.

R. Nageswara Rao
Director
UV Physics Academy
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Hyderabad

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FOREWORD

FR.DR. GIGI THOMAS

Development, as we all know, depends primarily and heavily on advancement in science and technology. One of the main objectives of all scientific research is to improve the life and living conditions of all living beings on earth. Educationists always underline the symbiotic relation between higher education and development. No human development is possible without progress in scientific research. It is in this context that our Honourable Prime Minister emphasized in the 102nd Indian Science Congress that "*Science may be the product of human brain, but it is also driven by the compassion of human heart- the desire to make human life better*". Science and technology has always been promoted in the post-Independence India, especially through the great vision of Pandit Jawaharlal Nehru who championed the cause of a scientific temper to be inculcated in all citizens and 'placed science and technology at the heart of national development'. It gives a ray of hope that, though importance given to study and research in basic sciences by students in higher education suffered a setback in the past, trends have changed in the last decade and science has once again begun to occupy the centre stage.

The pivotal role played by the National Eligibility Test in Science subjects jointly organized by UGC-CSIR is known to all. This highly prestigious competitive examination serves the most important gateway for all who wish to enter the world of teaching and research in sciences in the higher education sector. Anything highly crucial and pivotal will be hard to crack and achieve, and the same is true about the NET as well. Students try hard to pass through the test and obtain eligibility for teaching and/or research. Any type of guidance and training given in this field is therefore a great service to the student community and scientific fraternity of the country. The general aptitude guidance book written by Mr. Christy Varghese is highly useful and relevant in this background. This book opens wide horizons for those who wish to enter science teaching and research which will in turn lead the nation to more development and better days. This book contains fully solved question papers of General aptitude section of all previous CSIR-NET examinations. I realize this is beneficial for all science streams namely: Mathematical Sciences, Physical Sciences, Chemical Sciences, Life

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Sciences and Earth Sciences . Moreover the book contains topic-wise detailed theory of all topics mentioned in the latest syllabus issued by CSIR, strongly supported by topic wise illustrative examples for clear understanding and topic wise practice questions. Students will surely benefit from the sample practice question papers and the tips, tricks and shortcuts for solving quickly.

My joy of introducing this book to the student community is doubled because of the fact that Mr. Christy Varghese had been a student of mine at UG level Mathematics programme. I have known him to be a very sincere, committed and good-hearted person and a determined, hard-working student. It is not an easy task to write a book of this academic standard, and Mr. Christy has achieved this excellence through his passion for knowledge and hard work. I am sure this book will be an invaluable ally for all students preparing to crack UGC/CSIR National Eligibility Test as well as for all their trainers and teachers. Praying God's abundant blessings upon the author and all who make use of this book, I am very happy to introduce and dedicate this book to the student and teaching community.

Fr. Dr. Gigi Thomas
Principal
Mar Ivanios College (Autonomous)
Thiruvananthapuram
Kerala

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ACKNOWLEDGEMENT

Foremost of all, I humble myself before almighty God, for having bestowed upon me the strength and courage that was quintessential in undertaking and accomplishing an endeavour like this.

My heart abounds with gratitude to Dr Manil T Mohan, my teacher, who offered his guidance and support throughout the course of this work. I am thankful to Rev. Dr Gigi Thomas, for the 'foreword' and for being a source of inspiration through his positive strokes. The support and inspiration provided by R.Nageswara Rao, the Director of UV Physics Academy, Hyderabad is overwhelming and I thank him for the "Preface to the third edition". I am indebted to my review team comprising Aleena, Anjli, Cheriyachen, Dona, Lakshmi, Ligin, Lucy, Mahesh, Murali, Nikhil, Nitha, Renjini, Silja, Sobin, Sreejesh, Sreeram, Toby and Uma. A word of thanks to Mobin John Thomas who helped me tackle the problematics pertaining to language. I would also like to acknowledge my indebtedness to Jesse P Francis for his valuable support and crucial suggestions for the completion of the work. Thanks to Melvin Chacko Varkey and Akhil Sundar – the former crafted my ideas into a wonderful design for the cover page while the latter helped a lot with the layouts and formatting of this book. I extend my sincere gratitude to Dr Baiju KV and my students in Centre for Mathematics and Statistics for their prayers and support. I also thank Lilly Publishing House for their prompt publishing of the book.

Finally, I express my sincere thanks to my family, especially to my Appa, Dr Varghese Mathew, who always had a great deal of faith in me, who stood by my side and who provided the needful for this endeavour to materialize. Love and regards to my mother, brother, sister-in-law and my lovely niece, Eva – all, people who motivated me and without whom this book won't have been a reality.

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A WORD FROM THE AUTHOR

General aptitude is a section in the CSIR-NET examination question format, which most of the candidates are keen to avoid! They tend to avoid general aptitude questions as the questions are seemingly tricky or to quote what most of them complain - "We can't understand what the question means", "We don't know the basic formula to solve this question", "We want more time to solve this question" and it go like a long chain of troubles. The only way to gain command over general aptitude questions is through proper practice. However, even when you are determined to start practicing these questions you are still haunted by the same set of problems that is quoted above.

This book is designed with a vision to eliminate all these troubles and a proper use of this book will equip one to solve the questions without hassles and within the stipulated time. The common syllabus for general aptitude for all science subjects issued by CSIR is given below.

Syllabus: logical reasoning, graphical analysis, analytical and numerical ability, quantitative comparisons, series formation, puzzles, etc.

For general aptitude section, the questions are common for all the science streams and CSIR prepares only two sets of questions – one for the examinations held in the forenoon session of the examination day, and the other for the afternoon session. For the sake of practicing, one should better not avoid the set in which your subject doesn't belong to. All sets of question papers are equally useful for practice and preparation.

This book has three main sections – "Previous Question papers", "Topic wise theory and Solution of CSIR-NET questions (15 chapters)", and "7 sets of Model question papers" respectively. The set of previous question papers comprises all the previous question papers of all NET exams of all science subjects conducted after the release of the latest syllabus by CSIR. The second section has 15 chapters which cover the above mentioned syllabus. Each chapter clearly provides detailed descriptions about theories of all the concepts and formulas to solve the questions with illustrations. The next section comprises detailed solutions of CSIR-NET exam questions. All the questions asked in the different NET exams pertaining to various areas are discussed in the corresponding chapter which deals with that particular area.

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This will enable you to identify the question pattern very easily at the time of examination. Moreover, in each chapter, a bunch of practice questions are included to make you acquainted with more problems. The third section houses the Model Question papers. They are designed for self-assessment. Solving these questions serves the purpose of a mock test and will provide a room for you to assess yourself.

The "difficulty rating" and "average time" for each question is provided to help you select the right question at the time of exam. In the CSIR-NET exam, there are 20 questions in general aptitude section, in which you are supposed to attend 15 questions. Each correct answer earns you two marks and there is negative (0.5) mark for each of your wrong answer. Obviously, out of these twenty questions, you can avoid five questions. You should avoid the questions, which will take much time to solve. The average time given for solving each question will give you an idea of the question in hand. The questions with an average time (for finding a solution) more than 120 seconds are better to avoid. Again rarely some questions in the examination are real tough questions, which you may find difficult to solve even if you understand the concepts. The difficulty rating is given out of five and questions with difficulty rating 4 or above are tough questions and in the examination attempt these questions only if you are unable to get fifteen questions from the remaining set.

As part of my efforts to further serve the student community, I have took an initiative to develop an exclusive interactive mobile app for CSIR-NET General Aptitude. I believe that it will serve as an ally for all the readers of this book to take real-time mock test papers and to get updated with more model question papers and discussions on general aptitude.
To free download it, search "CSIR-NET General Aptitude" in play store or scan the QR code in the back cover of this book.

This book can be most efficiently used by following any of the two approaches listed below.

Approach 1:

- Read the theory section of each chapter and understand every concepts.
- Then solve the illustrative examples to get familiar with the questions.
- Try to solve the topic wise CSIR-NET exam questions. Only when doubt arises, look for the detailed solution given.
- For more practice, solve the practice questions.

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- For the final preparation of exam, do the mock test using the set of model question papers.
- Download the official mobile app and take real-time mock tests.

Approach 2: (Advisable for students who are familiar with general aptitude questions)

- Try to solve each of the CSIR NET exam question papers in the first section.
- Check with the answer key in the end and look for the detailed solution whose chapter name and page number is given.
- For more understanding of the theory, learn the theory section of each chapter.
- For further practice, solve the practice questions.
- For the final preparation of exam, do the mock test using the set of model question papers.
- Download the official mobile app and take real-time mock tests.

For further queries or clarifications:

Post your queries in the official Facebook page

"CSIR-NET General Aptitude – A New Outlook"

Or

Post your doubts clicking the POST A DOUBT tab in the official android app

"CSIR-NET General Aptitude"

Your doubts will be answered soon.

Hope this book along with the mobile app provide you a new window through which you can see the world of general aptitude with a newer perspective…. A New Outlook.

Best wishes.

Christy Varghese.

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HOW TO APPROACH A PROBLEM?

There are different methods (strategies) to solve a problem. Here I would like to quote some of them which are most relevant for objective type questions.

The challenge posed to every candidate in the CSIR-NET exam is to find the correct answer for the given question from the given four options. This can be done in two ways:

1. **(Selection)**: To arrive at an answer by deriving conclusions from the conditions/statements given in the question.
2. **(Rejection)**: To weed out the wrong options to arrive at the correct one.

Nearly all the questions in this book are solved using one of the following methods which involves either "selection" or "rejection" or both:

Direct method: In direct method, the conclusion is established by logically combining the axioms, definitions, and earlier known results or statements. Reader may be familiar with these, since it is the most commonly used methods to solve problems. In fact, all questions can be proved using this method, but it has its own disadvantages - it is often tiresome and time consuming in comparison to other methods.

Visual method: Drawing a picture will sometimes help solve the problems or have a better understanding of the problem. Especially problems involving group of objects (sets), can be better understood with Venn Diagrams. Sometimes data can be tabulated for more clear understanding if required.

Trial and error: By this method, we test possible solutions until the right one is found. Here, we pick each option given and see if it satisfies the given conditions (or equations) in the question. In many problems, this is the most effective alternative method as direct method is time consuming.

Eliminating options: When it is hard to say one option is right, you can try to see if any of the options are wrong, by any one of the above techniques. Sometimes you can easily eliminate all three wrong options, sometimes it eliminates one or two, simplifying the question.

Identifying "related" options:

1) Identify mutually contradicting options. One of them is definitely wrong. If you find a pair of mutually contradicting options, see if 2) can be applied to one of the options.
2) See if any one of the available options leads you to another option under the given conditions in the question – since only one can be right, both will be wrong.

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1. Suppose three meetings of a group of professors were arranged in Mumbai, Delhi and Chennai. Each professor of the group attended exactly two meetings. 21 professors attended Mumbai meeting, 27 attended Delhi meeting and 30 attended Chennai meeting. How many of them attended both the Chennai and Delhi meetings?

(a) 18 **(b) 24**
(c) 26 **(d) Cannot be found from the above information**

Answer (a) *Difficulty rating: 3.5/5* *Average time for solving: 90 seconds*

Let us picture the given data as a Venn-diagram.

Given: "Each professor of the group attended exactly two meetings."

That is, no professor attended single number of meetings or triple number of meetings.

Therefore,

Number of professors attending Mumbai meeting only = 0.

Number of professors attending Delhi meeting only = 0.

Number of professors attending Chennai meeting only = 0.

Number of professors attending Mumbai, Delhi and Chennai meetings = 0.

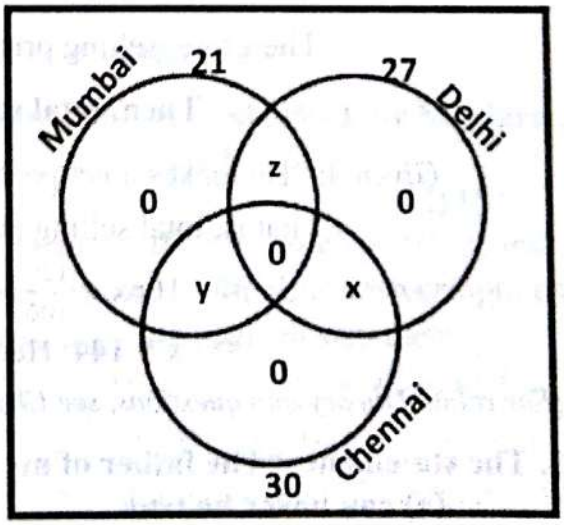
(Observe the regions having 0 values in the Venn-diagram.)

Let x be the number of professors attending both Chennai and Delhi meetings.
Let y be the number of professors attending both Chennai and Mumbai meetings.
Let z be the number of professors attending both Mumbai and Delhi meetings.

Note that in the Venn-diagram, circles corresponding to each cities is partitioned into four non-intersecting regions.

For example, in the circle corresponding to Mumbai, there are 4 non-intersecting regions which together sums up to the total of 21 professors who attended the Mumbai meeting.

That is, $21 = 0 + z + 0 + y$



Similar is the case for Delhi and Chennai meetings.

That is, $27 = z + 0 + x + 0$

$30 = y + 0 + x + 0.$

Hence, we obtain the three equations as,

$y + z = 21$ --------- (1)
$x + z = 27$ --------- (2)
$x + y = 30$ --------- (3)

We have to find the value of x by solving these three equations.

Subtracting equation (1) from equation (3), we obtain

$x - z = 9$ ---------- (4)

Adding equations (2) and (4), we obtain

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$2x = 36$

⇨ **$x = 18$.**

(For related theory and questions, see Chapter 14: Observational Ability)

2. A vendor sells articles having a cost price of Rs.100 each. He sells these articles at a premium price during first eight months, and at a sale price, which is half of the premium price, during next four months. He makes a net profit of 20% at the end of the year. Assuming that equal numbers of articles are sold each month, what is the premium price of the article?

(a) 122 **(b) 144** **(c) 150** **(d) 160**

Answer (b) *Difficulty rating: 3/5* *Average time for solving: 80 seconds*

Let x be the premium price of the article.

Let n be the number of articles sold in each month.

Total number of articles sold in one year = 12n

Given 1: Cost price of one article is Rs.100.

Therefore, total cost price for the articles sold in 1 year = $12n \times 100 =$ **$1200n$.**

Given 2: "He sells these articles at a premium price during first eight months"

Number of articles sold in the first 8 months = 8n.

That is, the vendor sells a total of 8n articles at Rs.x each.

Therefore, selling price for the first 8 months = $8n \times x =$ **$8nx$.**

Given 3: "sale price, which is half of the premium price, during next four months."

Number of articles sold in the last 4 months = 4n.

Half of the premium price = $\frac{x}{2}$

That is, the vendor sells a total of 4n articles at Rs.$\frac{x}{2}$ each.

Therefore, selling price for the last 4 months = $4n \times \frac{x}{2} =$ **$2nx$.**

Then, total selling price = $8nx + 2nx = 10nx$.

Given 4: "He makes a net profit of 20% at the end of the year."

That is, total selling price = 120 % of total cost price.

⇨ $10nx = \frac{120}{100} \times 1200n$

⇨ $x =$ **144.** Hence, **option (b).**

(For related theory and questions, see Chapter 10: Monetary Problems)

3. The statement: "The father of my son is the only child of your parents"

(a) can never be true

(b) is true in only one type of relation

(c) can be true for more than one type of relations

(d) can be true only in a polygamous family

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Answer (b) *Difficulty rating: 3/5* *Average time for solving: 60 seconds*

Given: **"The father of my son is the only child of your parents"**

Here there is a speaker and a hearer of this statement.

The statement says that the father of the speaker's son and the only child of hearer's parents are the same person.

Note that, "the only child of hearer's parents" **can only be the hearer himself.**

Therefore the given statement can be rephrased as

"The father of the speaker's son is the hearer."

If you consider the speaker to be a male, "father of the speaker's son" is the speaker himself.

That implies that the speaker and the hearer are the same person which is absurd!

Therefore the **speaker cannot be a male.**

If the speaker is a female, "father of the speaker's son" is speaker's husband.

That is, the speaker is a lady and the hearer is her husband.

Clearly, there is no other case when this statement is valid as the speaker can only be a female and the hearer can only be her husband.

Thus the statement is valid only in one type of relation - **"Wife to her Husband"**.

Hence **option (b).**

Those of you wondering why the options (c) and (d) are incorrect may read the following.

It is true that in a polygamous family, if a lady has two husbands, this statement is valid when she tells this statement to either of her husbands who have a child. But then also the type of relation "Wife to Husband" is the same in either cases. Hence option (c) is incorrect.

The statement is valid in a polygamous family but not **only** in a polygamous family, but in a monogamous family too. Hence option (d) is incorrect.

(For related theory and questions, see Chapter 14: Observational Ability)

4. "The clue is hidden in this statement", read the note handed to Sherlock by Moriarty, who hid the stolen treasure in one of the ten pillars. Which pillar is it?

(a) X **(b) II** **(c) III** **(d) IX**

Answer (d) *Difficulty rating: 2.5/5* *Average time for solving: 50 seconds*

The number of the pillar in which the treasure is hidden is one of the four answer options. Hence we just have to search the words - **ten, two, three, nine** in the given statement.

Given: "The clue is hidd**en in** this statement".

Clearly, the word **nine** is present in the reverse order.

Alternate solution

Given: "The clue is hidd**en in thi**s statement".

Clearly, the word **ninth** is hidden in the statement.

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In either case, **option (d)** is our obvious choice.

(For related theory and questions, see Chapter 14: Observational Ability)

5. Decode the below code

(a) GENT STUDENTS CAUSE LITTLE HEART BURNS
(b) STUDENTS ARE INTELLIGENT BUT PROBLEM IS NOT SOLVABLE
(c) THIS PROBLEM IS UNSOLVABLE BY ANY STUDENT
(d) THIS PROBLEM IS SOLVABLE BY INTELLIGENT STUDENTS

G	E	N	T	S	T	U
I	S	S	O	L	V	D
L	I	I	S	P	A	E
L	M	H	T	R	B	N
E	E	L	B	O	L	T
T	N	I	Y	B	E	S

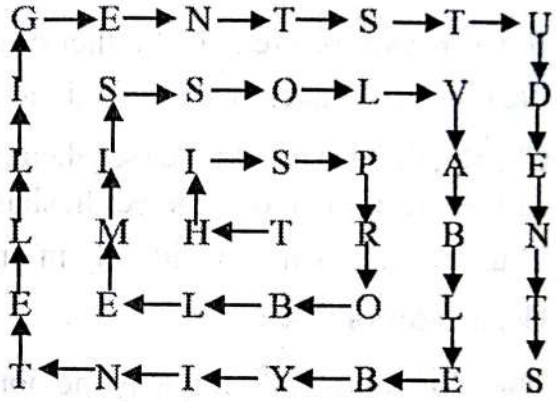
Answer (d) *Difficulty rating: 2.5/5* *Average time for solving: 50 seconds*

The quicker way to solve this problem is to look for the first word (or any word) of each of the answer options, so that you can pick up the sentence easily from the given code.

That is, search for, GENT, STUDENTS, THIS.

Hence, we can decode the code as shown.

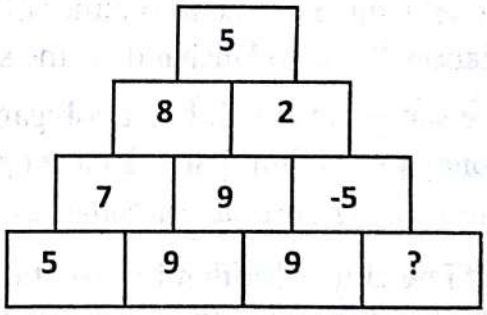
(Read in the direction shown through the arrows starting from T in the middle.)



Thus, **"THIS PROBLEM IS SOLVABLE BY INTELLIGENT STUDENTS"** is the decoded script. Hence **option (d).**

6. The missing number is

(a) -19
(b) -5
(c) 9
(d) -9



Answer (a) *Difficulty rating: 3/5*

Average time for solving: 60 seconds

Here the number in each box is the arithmetic mean (Average) of the corresponding numbers in the adjacent boxes below it.

That is, in the first row, 5 is the average of 8 and 2.

In the second row, 8 is the average of 7 and 9.

In the second row, 2 is the average of 9 and -5. etc.

Let x be the missing number.

Then -5 is the average of 9 and x.

That is, $-5 = \frac{9+x}{2}$

$\Rightarrow \quad -10 = 9 + x$

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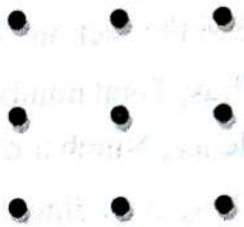
⇨ x = **-19.** Hence **option (a).**

(For related theory and questions, see Chapter 4: Series Formation)

7. The minimum number of straight lines required to connect the nine points without lifting the pen or retracing is

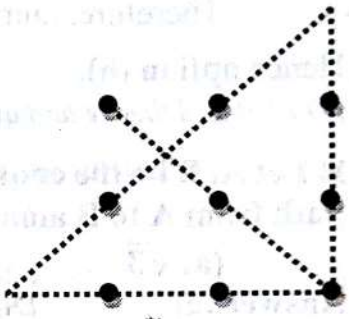


(a) 3
(b) 4
(c) 5
(d) 6

Answer (b) *Difficulty rating: 3.5/5* *Average time for solving: 50 seconds*

Here the only condition is to connect the nine points **without lifting the pen or retracing.**

Thus, it is legitimate even if we move the pen outside the area covered by the nine points. Hence, the way of connecting the points shown in the Figure is possible and it requires less number of straight lines than any other way of connecting.



Figure

Clearly, there are four straight lines in this way of connecting.

Hence, **option (b).**

(For related theory and questions, see Chapter 14: Observational Ability)

8. The number of diagonals of a convex dodecagon (12-gon) is

(a) 66 **(b) 54** **(c) 55** **(d) 60**

Answer (b) *Difficulty rating: 3.5/5* *Average time for solving: 50 seconds*

We know that a dodecagon is a polygon with 12 sides and a convex dodecagon is a dodecagon in which no line segment between two points on the boundary ever goes outside the dodecagon.

Remember: **Number of diagonals of a convex polygon (n-gon)** $= \frac{n(n-3)}{2}$

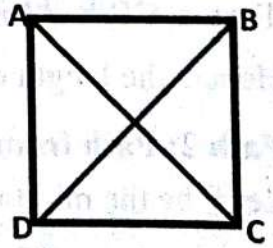
For a dodecagon (12-gon), n = 12.

Thus, Number of diagonals of a convex dodecagon $= \frac{12(12-3)}{2} = \frac{12 \times 9}{2} =$ **54.**

Hence **option (b).**

Alternate Solution

Let us take a 4-gon, say a square and logically determine the number of diagonals in it and hence try to extend the same logic to a 12-gon.



Consider the figure and count the number of straight lines in the figure.

Note that,

Total number of straight lines = Number of sides + Number of diagonals

That is, **Number of diagonals = Total number of straight lines - Number of sides.**

For a square, Number of sides = 4.

Here a straight line is formed by joining any two vertices.

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Thus, total number of straight lines is the number of distinct pair of vertices in the square.

For a square with 4 vertices, number of pairs = ${}^4C_2 = 6$.

(See the section "Permutations and Combinations" in "Chapter 6")

Thus, Total number of straight lines = ${}^4C_2 = 6$.

Hence, Number of diagonals = $6 - 4 = 2$.

Thus, extending this logic to n-gon, we obtain, **Number of diagonals of an n-gon = ${}^nC_2 - n$**

For a 12-gon, Number of sides = 12.

Therefore, number of diagonals = ${}^{12}C_2 - 12 = \frac{12\times11}{2} - 12 = 66 - 12 = \mathbf{54}$.

Hence **option (b).**

(For related theory and questions, see Chapter 11: Geometry)

9. Let A, B be the ends of the longest diagonal of the unit cube. The length of the shortest path from A to B along the surface is

(a) $\sqrt{3}$ **(b) $1+\sqrt{2}$** **(c) $\sqrt{5}$** **(d) 3**

Answer (c) *Difficulty rating: 4/5* *Average time for solving: 110 seconds*

Given a cube with side 1 unit.

Join AB.

Mark points C, D, X, Y, Z on the cube as shown in the Figure 1.

Given that AB is one of the longest diagonals of the unit cube.

Let us try to draw every distinct paths along the surface from A to B and check whether we can arrive at a single answer by eliminating the other options.

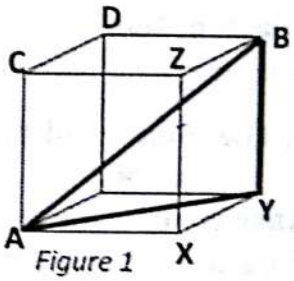

Figure 1

Path 1: Path from A to B via C (Figure 2)

Length of this path = Length of AC + Length of CB.

Obviously, Length of AC = **1 unit.**

We are required to find the length of CB.

Applying Pythagoras theorem on triangle CDB, we obtain

$$CB^2 = CD^2 + DB^2 = 1^2 + 1^2 = 2$$

That is, CB = **$\sqrt{2}$ units**

Hence, the length of this path = **$1+\sqrt{2}$.**

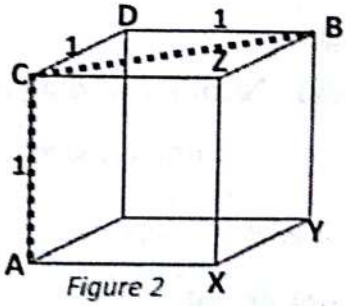

Figure 2

Path 2: Path from A to B via P (Figure 3)

Let P be the midpoint of the side ZX.

Consequently, PZ = $\frac{1}{2}$, PX = $\frac{1}{2}$

Length of this path = Length of AP + Length of PB.

Thus we have to find the lengths of AP and PB.

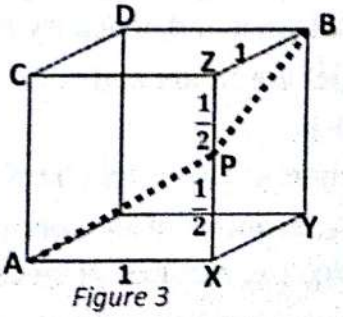

Figure 3

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Applying Pythagoras theorem on right triangle AXP, we obtain

$AP^2 = AX^2 + PX^2 = 1^2 + (\frac{1}{2})^2 = 1 + \frac{1}{4} = \frac{5}{4}$

That is, $AP = \sqrt{\frac{5}{4}} = \frac{\sqrt{5}}{2}$ **units.**

Applying Pythagoras theorem on right triangle PZB, we obtain

$PB^2 = PZ^2 + ZB^2 = (\frac{1}{2})^2 + 1^2 = \frac{1}{4} + 1 = \frac{5}{4}$

That is, $PB = \sqrt{\frac{5}{4}} = \frac{\sqrt{5}}{2}$ **units**

Hence the length of this path $= \frac{\sqrt{5}}{2} + \frac{\sqrt{5}}{2} = \sqrt{5}$.

Now, we have an obvious doubt about the shortest path among the two paths.

It is not easy to find the value of $\sqrt{5}$ and $1+\sqrt{2}$ without a calculator as we are unaware of the value of $\sqrt{5}$ and $\sqrt{2}$.

Therefore let us try to compare the square values of both the expressions.

$(\sqrt{5})^2 = 5$

$(1 + \sqrt{2})^2 = 1 + 2 + 2\sqrt{2} = 3 + 2\sqrt{2}$.

Now the value of $\sqrt{2}$ is causing us trouble. So let us analyse it logically whether $3 + 2\sqrt{2}$ is greater than 5 or not.

Consider $3 + 2\sqrt{2}$

Value of $\sqrt{2}$ is definitely greater than 1.

Any number greater than 1 when multiplied with 2 gives a product greater than 2.

3 added to a number greater than 2 gives a sum **greater than 5**.

Thus, $3 + 2\sqrt{2} > 5$.

$\Rightarrow (1 + \sqrt{2})^2 > (\sqrt{5})^2$

$\Rightarrow 1 + \sqrt{2} > \sqrt{5}$

Thus, we can eliminate the options (b) and (d).

Now, among the answer options, the value of option (a), that is $\sqrt{3}$ is less than that of $\sqrt{5}$.

Hence we have to eliminate one of them for finding the length of the shortest path.

For that, let us find the length of the diagonal AB.

Applying Pythagoras theorem on the right triangle AYB (See Figure 1)

$\Rightarrow AB^2 = AY^2 + BY^2 = (\sqrt{2})^2 + 1^2 = 2 + 1 = 3$.

[$AY = \sqrt{2}$ can be obtained be applying Pythagoras theorem on ΔAXY.]

$\Rightarrow AB = \sqrt{3}$ units.

From Figure 1, it is clear that the length of any path along the surface is greater than the longest diagonal AB.

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That is, any path along the surface is greater than $\sqrt{3}$ units.

Hence option (a) can be eliminated.

Thus, $\sqrt{5}$ is the length of the shortest path. Hence **option (c).**

(For related theory and questions, see Chapter 11: Geometry)

10. How many digits are there in 3^{16} when it is expressed in the decimal form?

(a) Three **(b) Six** **(c) Seven** **(d) Eight**

Answer (d) *Difficulty rating: 3.5/5* *Average time for solving: 100 seconds*

To find the number of digits of 3^{16}, we are not supposed to find the value of 3^{16} as it is time-consuming and tiresome. Instead, let us try to find an estimate of the size of the value of 3^{16}.

Let us use the "Laws of exponents" learnt in "Chapter 5: Numbers", for the same.

$3^{16} = (3^2)^8 = 9^8 = (9^2)^4 = 81^4$.

Finding the fourth power of 81 is not that simple.

Therefore let us find the fourth power of a number near to 81, say 80.

$80^4 = (8\times10)^4 = 8^4 \times 10^4 = 64^2 \times 10^4 = 4096 \times 10^4 = 40960000$.

Clearly, 80^4 has 8 digits.

Obviously, $81^4 > 80^4$.

⇨ 81^4 has at least 8 digits.

⇨ 3^{16} have 8 digits or more than 8 digits.

Observe that the answer options does not provide a value greater than 8.

Hence **option (d)** is our obvious choice.

Alternate Solution (Read only for learning a new concept.)

The following solution may appear tiresome for this particular question. But this description gives you some insights and ideas to approach similar and more complicated problems.

We know that, $3^{16} = 9^8$ [Since $3^{16} = (3^2)^8 = 9^8$]

9^8 means 9 multiplying 8 times with itself.

Let us try to estimate the value of the higher powers, by substituting a number which is **less than the** corresponding number to be multiplied with 9.

$9^2 = 9 \times 9 = 81$, (2 digits)

$9^3 \approx 81 \times 9 \approx 80 \times 9 \approx 720$, (3 digits)

$9^4 \approx 720 \times 9 \approx 700 \times 9 \approx 6300$, (4 digits)

$9^5 \approx 6300 \times 9 \approx 6000 \times 9 \approx 54000$, (5 digits)

$9^6 \approx 54000 \times 9 \approx 50000 \times 9 \approx 450000$, (6 digits)

$9^7 \approx 450000 \times 9 \approx 400000 \times 9 \approx 3600000$, (7 digits)

$9^8 \approx 3600000 \times 9 \approx 3000000 \times 9 \approx 27000000$, (8 digits)

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Since in each of the above steps, to multiply with 9, we took a number less than the original number, 9^8 is greater than 27000000.

That is, 27000000 is a lower bound of 9^8.

which means that an 8 digit number is a lower bound of 9^8.

Thus, 9^8 has at least 8 digits.

Since there are no option greater than 8, **option (d)** is our obvious choice.

Now, if there was an option which was greater than 8, we should have to find an upper bound of 9^8 to confirm that 9^8 has 8 digits only.

We can estimate the value of the higher powers, by substituting a number which is **greater than the** corresponding number to be multiplied with 9. Here we can take 10 in place of 9 in each step.

$9^2 = 9 \times 9 \approx 9 \times 10 \approx 90$, (2 digits)
$9^3 \approx 90 \times 9 \approx 90 \times 10 \approx 900$, (3 digits)
$9^4 \approx 900 \times 9 \approx 900 \times 10 \approx 9000$, (4 digits)
$9^5 \approx 9000 \times 9 \approx 9000 \times 10 \approx 90000$, (5 digits)
$9^6 \approx 90000 \times 9 \approx 90000 \times 10 \approx 900000$, (6 digits)
$9^7 \approx 900000 \times 9 \approx 900000 \times 10 \approx 9000000$, (7 digits)
$9^8 \approx 9000000 \times 9 \approx 9000000 \times 10 \approx 90000000$, (8 digits)

In each of the above steps, to multiply with 9, we took a number greater than 9.

Therefore, 9^8 is lesser than 90000000.

That is, 90000000 is an upper bound of 9^8.

which means that an 8-digit number is an upper bound of 9^8.

Thus, 9^8 has at most 8 digits.

Hence using the above two sets of operations, we can infer that:

$27000000 < 9^8 < 90000000$.

⇨ 8-digit number $< 9^8 <$ 8-digit number.

Hence, 9^8 is an 8 digit number.

(For related theory and questions, see Chapter 5: Numbers)

11. The probability that a ticketless traveller is caught during a trip is 0.1. If the traveller makes 4 trips, the probability that he/she will be caught during at least one of the trip is:

(a) $1-(0.9)^4$ **(b) $(1-0.9)^4$** **(c) $1-(1-0.9)^4$** **(d) $(0.9)^4$**

Answer (a) *Difficulty rating: 3/5* *Average time for solving: 90 seconds*

Probability to be caught in at least one of the trip

= Total probability – Probability to be caught in none of the trips.

We know that **the total probability is always 1.**

Now, we have to find the probability of getting caught in none of the trips.

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Probability of getting caught in none of the trips = Probability of not getting caught in the first trip AND not in the second trip AND not in the third trip AND not in the fourth trip.

As we have learnt in Chapter 6, AND can be replaced by the multiplication symbol.
(See the section "How to approach questions on Probability" in Chapter 6.)

Thus, probability of getting caught in none of the trips
= Probability of not getting caught in 1st trip × Probability of not getting caught in 2nd trip × Probability of not getting caught in 3rd trip × Probability of not getting caught in 4th trip.

Probability of not getting caught in the 1st trip
= Total Probability - Probability of getting caught in the 1st trip = 1 – 0.1 = 0.9.

Similarly with the 2nd, the 3rd and the 4th trip.

Thus, probability of getting caught in none of the trips = $0.9 \times 0.9 \times 0.9 \times 0.9 = (0.9)^4$

Hence, Probability of getting caught in at least one of the trip = $\mathbf{1 - (0.9)^4}$.

(For related theory and questions, see Chapter 6: Probability and Arrangements)

12. There is an inner circle and an outer circle around a square. What is the ratio of the area of the outer circle to that of the inner circle?

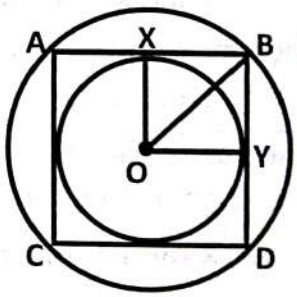
(a) $\sqrt{2}$ **(b) 2** **(c) $2\sqrt{2}$** **(d) $\sqrt{3/2}$**

Answer (b) *Difficulty rating: 3/5* *Average time for solving: 80 seconds*

We can picture the given data as shown in the Figure.

Let ABCD be the required square and O be the centre of both the circles.

Let X and Y be the midpoints of the sides AB and BD respectively. Construct OX and OY to form the radii of the inner circle and OB to form the radius of the outer circle.



Figure

We are required to find the value of $\frac{\text{Area of Outer Circle}}{\text{Area of Inner Circle}}$

Remember: Area of circle = π(radius)^2

Thus to find the ratio, we have to find the radii of the inner and the outer circle.

Let 2a be the length of each side of the square. That is, AB = 2a.

Then, **a is the radius of the inner circle.** That is, OX = OY = a.

Note that, BY = OX = a

Now we have to find the radius of the outer circle, that is OB.

Applying Pythagoras theorem on the right angled triangle OYB, we obtain

$OB^2 = OY^2 + BY^2 = a^2 + a^2 = 2a^2$

$\Rightarrow$ $OB = \sqrt{2}a$

Thus radius of outer circle is $\sqrt{2}$a.

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Now, Area of inner circle = πa^2

Area of outer circle = $\pi(\sqrt{2}a)^2 = 2\pi a^2$.

Hence, $\frac{\text{Area of Outer Circle}}{\text{Area of Inner Circle}} = \frac{2\pi a^2}{\pi a^2} = \mathbf{2.}$

(For related theory and questions, see Chapter 12: Mensuration and Quantitative comparison)

13. The base diameter of a glass is 20% smaller than the diameter at the rim. The glass is filled to half the height. The ratio of empty to filled volume of the glass is

(a) $\frac{\sqrt{10}-\sqrt{9}}{\sqrt{9}-\sqrt{8}}$ **(b)** $\frac{10-9}{9-8}$ **(c)** $\frac{10^2-9^2}{9-8}$ **(d)** $\frac{10^3-9^3}{9^3-8^3}$

Answer (d) *Difficulty rating: 3.5/5* *Average time for solving: 120 seconds*

The glass mentioned in the question is of the shape of a frustum of a cone and can be pictured as shown in the Figure.

Let d be the rim diameter of the glass.

Given: "base diameter of a glass is 20% smaller than the diameter at the rim."

Thus, Base diameter = (100-20) % of rim diameter = 80% of rim diameter = $\frac{80}{100}d = 0.8d$

Consequently, if r is the rim radius, then 0.8r is the base radius of the glass.

Let h be the height of the glass.

Given: "The glass is filled to half the height."

Remember: For a frustum, height is directly proportional to radius.

That is at half the height, length of the radius should be the average of r and 0.8r.

Hence, Rim radius of the filled region = $\frac{r + 0.8r}{2} = 0.9r$.

We are required to find the value of $\frac{\text{Empty Volume}}{\text{Filled Volume}}$

That is, we have to find the volume of two frustums, the empty one and the full-volume one.

Remember: Volume of frustum = $\frac{1}{3}\pi H\,(R_1^2 + R_2^2 + R_1R_2)$

where R_1 and R_2 are the two radii and H is the height of the frustum.

Thus, Empty Volume = $\frac{1}{3}\pi\frac{h}{2}(r^2 + (0.9r)^2 + 0.9r\times r)$

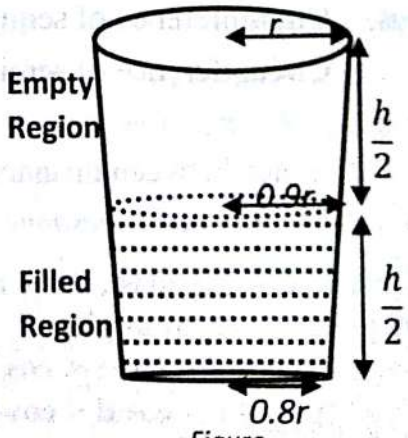
$= \frac{1}{3}\pi\frac{h}{2}(r^2 + 0.81r^2 + 0.9r^2)$

$= \frac{1}{3}\pi\frac{h}{2}r^2(1 + 0.81 + 0.9)$

$= \mathbf{\frac{1}{3}\pi\frac{h}{2}r^2 \times 2.71}$

Full Volume $= \frac{1}{3}\pi\frac{h}{2}((0.9r)^2 + (0.8r)^2 + 0.9r\times 0.8r)$

$= \frac{1}{3}\pi\frac{h}{2}(0.81r^2 + 0.64r^2 + 0.72r^2)$



Figure

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$$= \frac{1}{3}\pi\frac{h}{2} r^2(0.81 + 0.64 + 0.72)$$

$$= \frac{1}{3}\pi\frac{h}{2} r^2 \times \mathbf{2.17}$$

Now, $\frac{\text{Empty Volume}}{\text{Filled Volume}} = \frac{2.71}{2.17} = \frac{271}{217}$

But as the answer options are in a different form, we should check to which option does our obtained expression match.

$\frac{10^3-9^3}{9^3-8^3} = \frac{1000-729}{729-512} = \frac{271}{217}$. Hence **option (d).**

(For related theory and questions, see Chapter 12: Mensuration and Quantitative comparison)

14. A wheel barrow with unit spacing between its wheels is pushed along a semi-circular path of mean radius 10. The difference between distances covered by the inner and outer wheels is

(a) 0 **(b) 10** **(c) π** **(d) 2π**

Answer (c) *Difficulty rating: 3.5/5* *Average time for solving: 100 seconds*

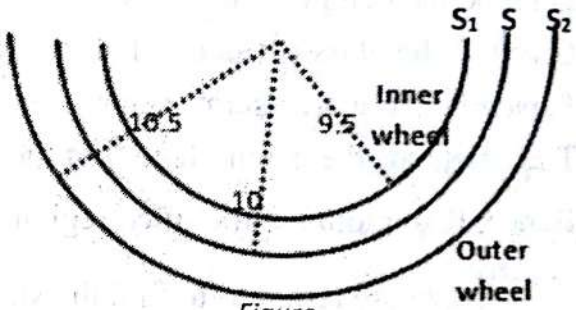
Given a wheel barrow with its two wheels separated by a distance of 1 unit.

As the wheel barrow moves in a semi-circular path, the two wheels trace two different semi-circles. [Denoted by S_1 and S_2 in the Figure]

Let S denote the semi-circular path of mean radius 10, along which the wheel barrow is pushed.

We are required to find the difference between distances covered by the inner and outer wheels.

In other words, we have to find the difference between circumferences of the two semicircles S_1 and S_2.



Figure

Since the mean radius is 10 units, the radius of inner semicircle is 9.5 units and that of the outer semicircle is 10.5 units as the two wheels are separated by a distance of 1 unit.

Remember: Circumference of semicircle = π × radius

Thus, Circumference of semicircle traced by outer wheel = 10.5π

Circumference of semicircle traced by inner wheel = 9.5π

Hence,

Difference between distances covered by the inner and outer wheels = 10.5π – 9.5π = **π.**

(For related theory and questions, see Chapter 12: Mensuration and Quantitative comparison)

15. Write d = 1 degree, r = 1 radian and g = 1 grad. Then which of the following is true? (100 grad = a right angle)

(a) cos d < cos r < cos g **(b) cos r < cos g < cos d**

(c) cos r < cos d < cos g **(d) cos g < cos d < cos r**

Answer (c) *Difficulty rating: 3/5* *Average time for solving: 70 seconds*

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We know that 180 degree = π radian.

That is, 1 degree = $\frac{\pi}{180}$ radian ≈ $\frac{3.14}{180}$ radian.

Obviously, the value of $\frac{3.14}{180}$ is less than 1.

Therefore, 1 degree is less than 1 radian.

Hence d < r ------ (1)

Given: "100 grad = a right angle".

⇨ 100 grad = 90 degree

⇨ 1 grad = $\frac{90}{100}$ degree

That is, 1 grad is only 90% of 1 degree.

Hence g < d ------ (2)

From inequalities (1) and (2), we obtain, **g < d < r.**

We know that for any angle between 0 degrees and 90 degrees, the value of cos θ decreases as the value of θ increases.

Clearly, the values of g, d and r are between 0 degrees and 90 degrees.

Therefore, **cos r < cos d < cos g.**

Hence **option (c).**

(For related theory and questions, see Chapter 7: Distances and Directions)

16. Three boxes are coloured red, blue and green and so are three balls. In how many ways can one put the balls one in each box such that no ball goes into the box of its own colour?

(a) 1 **(b) 2** **(c) 3** **(d) 4**

Answer (b) *Difficulty rating: 3/5* *Average time for solving: 70 seconds*

Let RED, BLUE and GREEN denote the colour of the box and Red, Blue and Green denote the colour of the ball.

We are required to place the balls in the boxes without violating the given condition that

"No ball goes into the box of its own colour."

Consider the RED box.

By the given condition, it can have either the Blue ball or the Green ball in it which are the only two possible cases.

Let us analyse them separately.

Case 1: The Blue ball is in the RED box.

Blue	?	?
RED	BLUE	GREEN

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Now, the BLUE box can have the Red ball or the Green ball in it. But if we put the Red ball in the BLUE box, then the GREEN box can have only the remaining Green ball placed in it, which will violate the given condition.

Thus in this case, the BLUE box can have only the Green ball in it.

That is,

Blue	Green	Red
RED	BLUE	GREEN

is the first way of arranging

Case 2: The Green ball is in the RED box.

Green	?	?
RED	BLUE	GREEN

Now, the BLUE box can only have the Red ball in it and hence the GREEN box can only have the remaining Blue ball in it.

That is,

Green	Red	Blue
RED	BLUE	GREEN

is the second way of arranging.

Thus, there are only two ways of arranging. Hence **option (b).**

(For related theory and questions, see Chapter 6: Probability and Arrangements)

17. A circle drawn in the x-y co-ordinate plane passes through the origin and has chords of lengths 8 units and 7 units on the x and y axes, respectively. The coordinates of its centre are

(a) (8, 7) **(b) (-8, 7)** **(c) (-4, 3.5)** **(d) (4, 3.5)**

Answer (c),(d) *Difficulty rating: 2.5/5* *Average time for solving: 70 seconds*

Unlike other questions, this question has two different answers, both of which are correct.

We are required to draw a circle that passes through the origin and has chords of lengths 8 units and 7 units on the x and y axes, respectively.

This can be done in 4 different ways according to the position of the centre of the circle in the 4 different quadrants.

Let us discuss only two of those cases which yields answer options given in the question. To check the invalidity of the other two cases are left as an exercise for the readers to do.

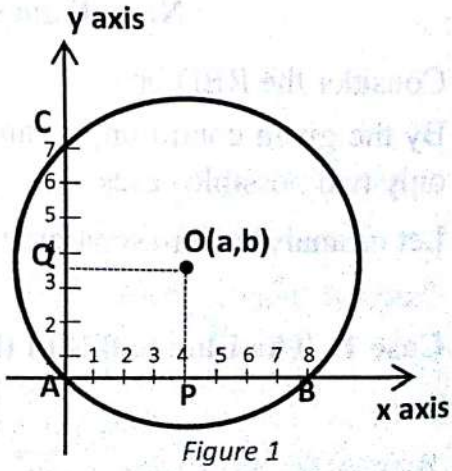

Figure 1

Case 1:

Consider Figure 1.

Let AB be the chord of length 8 units and AC be the chord of length 7 units.

Let O be the centre of the circle and let (a,b) be the coordinates of O.

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We have to find the values of a and b.

Remember: The perpendicular bisector of any chord of a circle must pass through the centre of that circle.

Thus we can construct perpendicular bisectors OP and OQ to the chords AB and AC.

From the figure, it can be observed that a = Length of OQ, b = Length of OP.

Thus we are required to find lengths of OP and OQ.

As OP and OQ are bisectors, P and Q are mid points of AB and AC respectively.

Hence, AP $= \frac{8}{2} = 4$ units, AQ $= \frac{7}{2} = 3.5$ units.

Hence a = OQ = AP = 4 units, b = OP = AQ = 3.5 units.

Thus the coordinates of the centre of the circle = **(4, 3.5)**

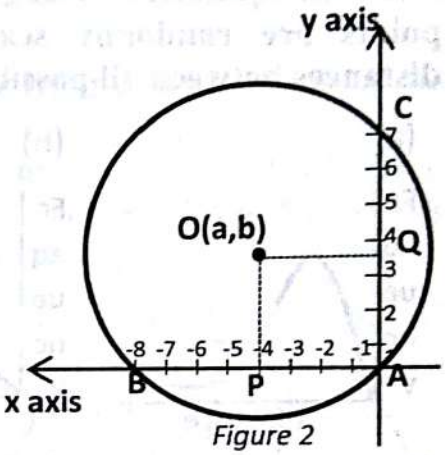

Figure 2

Case 2:

Consider Figure 2.

Proceeding as above we can obtain the coordinates of this circle as **(-4, 3.5).**

Hence **option (c)** and **option (d)** are correct.

(For related theory and questions, see Chapter 11: Geometry)

18. Most Indian tropical fruit trees produce fruits in April-May. The best possible explanation for this is

(a) optimum water availability for fruit production.

(b) the heat allows quicker ripening of fruit.

(c) animals have no other source of food in summer.

(d) the impending monsoon provides optimum conditions for propagation.

Answer (d) *Difficulty rating: 3/5* *Average time for solving: 60 seconds*

This is one such question that checks your common sense.

Let us consider each of the answers to weed out the wrong options and hence find the right one.

Consider (a)

This is wrong because it is summer in India during April-May and hence there is no optimum water availability.

Consider (b)

Wrong. Summer heat acts as a catalyst for increased production of hormones which causes quicker ripening of fruits but that doesn't contribute to the production of the fruits within a period of two months.

Consider (c)

Wrong. The process of reproduction does involves production of ripe fruits which attract birds and animals. This eventually helps in scattering the seeds to produce more trees.

The statement that animals have no other source of food in summer is true.

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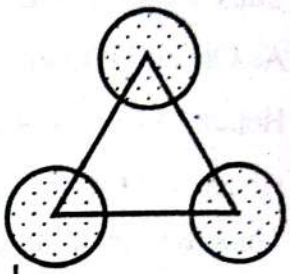
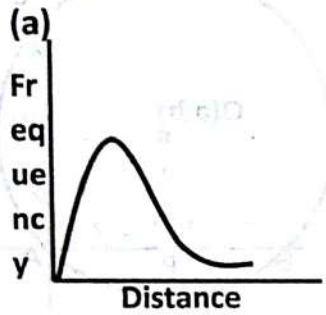
But the explanation do not have much relation with the question as the consumption of more fruits by animals doesn't contribute to the production of fruits but only to the production of trees.(Even if the production of trees eventually increases the production of fruits, it doesn't happen within a period of 2 months.)

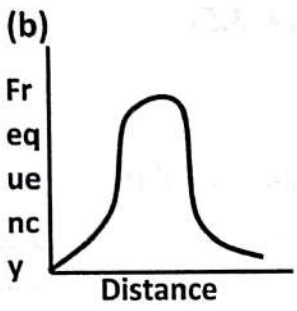
Thus, we weeded out the three wrong options.

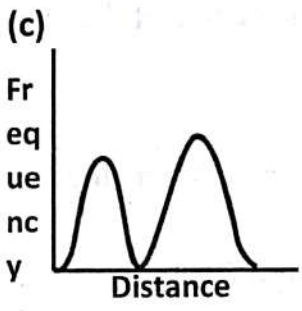
Hence our obvious choice is the remaining **option (d).**

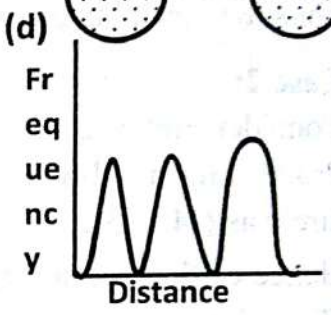
19. Three circles of equal diameters are placed such that their centres make an equilateral triangle as in the figure. Within each circle, 50 points are randomly scattered. The frequency distribution of distances between all possible pairs of points will look as











Answer (c) *Difficulty rating: 4.5/5* *Average time for solving: 110 seconds*

Note that "possible pairs of points" implies distinct pair of points.

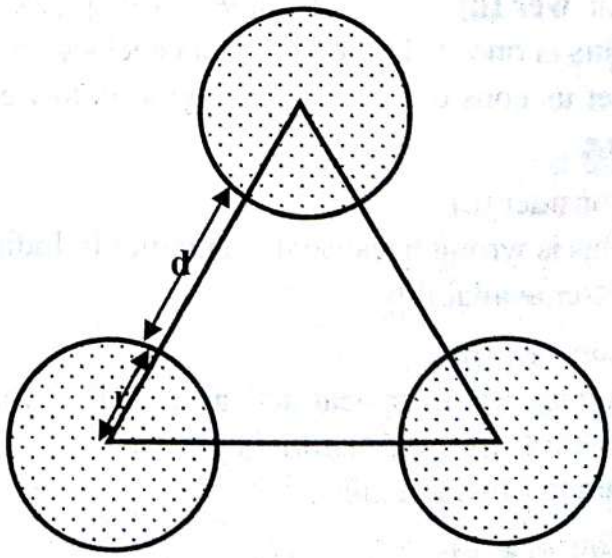
Label the distances as follows.

Here d is the distance between circles and r is the radius of each circle.

Note that, as the figure is only a representation, we cannot infer whether d > 2r or not.

Hence we have to consider both situations: that is d > 2r and d < = 2r.

Let us try to interpret the zero values of the frequencies in the answer options to find the solution.



For a distance "a", frequency is zero means that there are no two pair with such a distance "a".

Note that the frequency can be zero for a distance "a" only at three cases.

Case 1: a = 0

Case 2: a > d+2r+2r.

Case 3: For the situation d > 2r, a = d.

(Note that if d < = 2r, we can easily place two points at a distance d inside one circle, hence this case is not valid for that situation.)

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The fact that no other cases may arise is assured by the symmetry of the graph (due to equilateral triangle).

Thus, considering both the situations, we have exactly two zeros (if d <= 2r) or exactly three zeros (if d > 2r).

That is the answer graph can have a maximum of 3 zeros.

Now option (a) and (b) are ruled out as they have only one zero.

Option (d) can be ruled out as it have 4 zeros.

Hence we are left out with **option (c).**

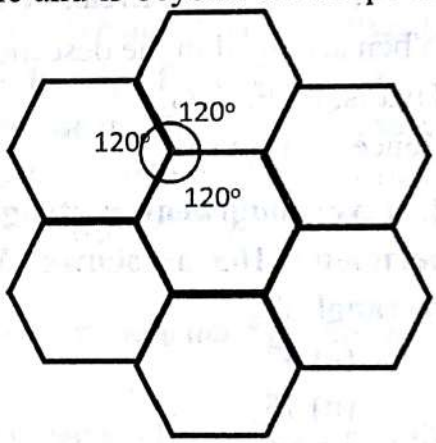
20. One is required to tile a plane with congruent regular polygons. With which of the following polygon is this possible?

(a) 6-gon **(b) 8-gon** **(c) 10-gon** **(d) 12-gon**

Answer (a) *Difficulty rating: 3/5* *Average time for solving: 60 seconds*

Here we are required to tile a plane using only one type of regular polygon.

The explanation of the underlying theory is a vast research topic and is beyond the scope of this discussion. Therefore, the theory is vaguely stated below without the explanation. Without knowing this theory also this question can be reasoned by those of you who have observed the tiling pattern of different floors. (See figure)



Figure

Theory: **There are only three types of regular polygons which can tile a plane using only one type of polygon. They are equilateral triangle, square and regular 6-gon (hexagon).**

Out of these three, only 6-gon is present in the answer options and hence the answer.

(For related theory and questions, see Chapter 11: Geometry)

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December 2015 – Life Sciences, Physical Sciences

1. In each of the following groups of words is a hidden number, based on which you should arrange them in descending order. Pick the correct answer:

E. Papers I Xeroxed
F. Wi-Fi veteran
G. Yourself ourselves
H. Breaks even

(a) E, F, G, H **(b) E, G, F, H** **(c) H, F, G, E** **(d) H, E, F, G**

Answer (d) *Difficulty rating: 2.5/5* *Average time for solving: 60 seconds*

Through observation, we can find the hidden number in each of the following groups of words.

That is,

E. Paper**s I X**eroxed -------- 6

F. Wi-**Fi ve**teran ---------- 5

G. Yoursel**f our**selves ----- 4

H. Break**s even** ---------- 7

When arranged in the descending order, the numbers can be listed as 7, 6, 5, 4.

That is, H, E, F, G.

Hence, **option (d).**

2. Five congruent rectangles are drawn inside a big rectangle of perimeter 165 as shown. What is the perimeter of one of the five rectangles?

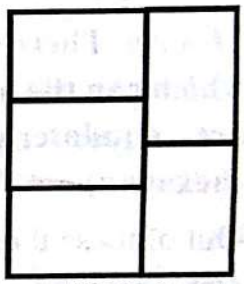

Figure 1

(a) 37
(b) 75
(c) 15
(d) 165

Answer (b) *Difficulty rating: 3/5* *Average time for solving: 100 seconds*

Remember: Perimeter of a rectangle = (2 × Length) + (2 × Breadth)

Let L and B respectively be the length and breadth of each of the small rectangles.

Then the given figure can be labelled as shown in the Figure 2.

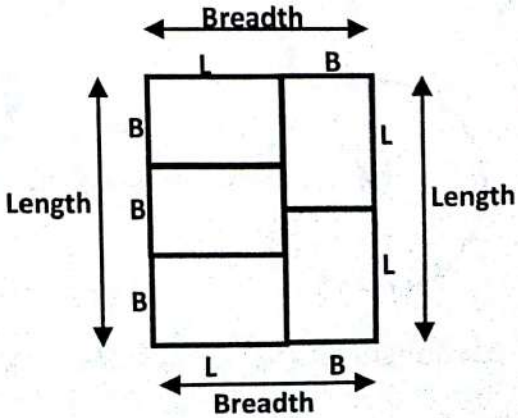

Figure 2

From Figure 2, it is evident that

Perimeter of the big rectangle = 4L + 5B

[Summing up all the Ls and Bs in the figure together.]

Given that the perimeter of the big rectangle = 165

Thus, 4L + 5B = 165 ----- (1)

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Now observe the lengths of the big rectangle.

As the two lengths of a rectangle are equal, we obtain

$$3B = 2L \text{ ---- (2)}$$

Substituting equation (2) in equation (1), we obtain

$$6B + 5B = 165$$

$$\Rightarrow 11B = 165$$

$$\Rightarrow B = \frac{165}{11} = 15$$

Substituting B = 15 in equation (2), we get $L = \frac{45}{2}$

We are required to find the perimeter of any of the smaller rectangle.

Perimeter of the smaller rectangle = $2L + 2B = 2\times\frac{45}{2} + 2\times15 = 45 + 30 = \mathbf{75}$.

(For related theory and questions, see Chapter 11: Geometry)

3. At one instant, the hour hand and the minute hand of a clock are one over the other in between the markings for 5 and 6 on the dial. At this instant, the tip of the minute hand

(a) is closer to the marking for 6 (b) is equidistant from the markings for 5 and 6

(c) is closer to marking for 5 (d) is equidistant from the markings for 11 and 12

Answer (c) *Difficulty rating: 3.5/5* *Average time for solving: 80 seconds*

Logical Solution

We are given the instant of the clock at which the hour hand and the minute hand are one over the other in between the markings for 5 and 6.

Analysing the answer options, we can understand that we are required to check if the tip of the minute hand is closer to 5 or closer to 6, at this particular instant.

As the minute hand and hour hand are one over the other at this particular instant, it is enough to **check if the tip of the hour hand is closer to 5 or it is closer to 6.**

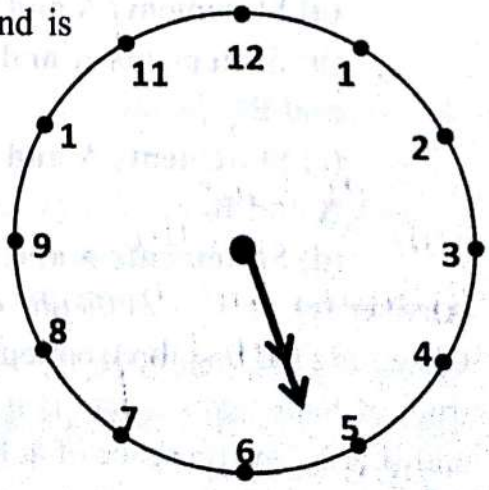
Therefore, let us try to estimate the position of the **hour hand** logically.

As this particular instant is in between 5 and 6, the hour hand is definitely in between the markings of 5 and 6.

Since the minute hand is over the hour hand, minute hand is also in between the markings of 5 and 6.

The time on the clock at which minute hand is between the markings of 5 and 6 is some time between 5:25 and 5:30.

For any time before 5:30, the hour hand can never go past half the way between the markings of 5 and 6.



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That is, the tip of the hour hand is closer to marking of 5 at this particular instant.

Consequently, the tip of the minute hand is also closer to marking of 5 at this particular instant of the clock. Hence **option (c).**

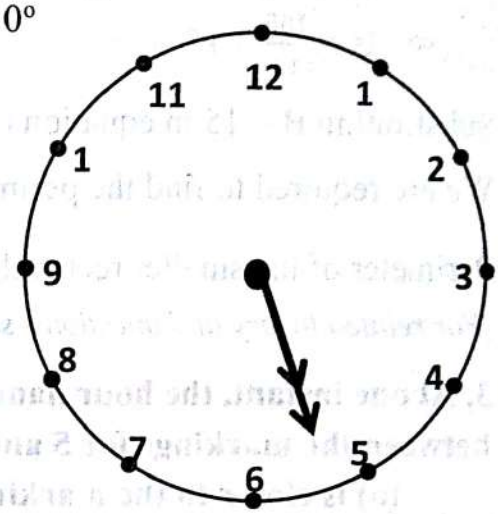
Formula-based solution

For the hour and minute hands to be one over the other, the angle between them should be 0^o.

We have the formula, **Angle between H hours and M minutes** $= \frac{1}{2} \times |60H - 11M|$

Here H = 5 Hours, Angle between H hours and M minutes = 0^o

We are required to find the time in minutes, i.e., M.



Thus, $0 = \frac{1}{2} | (60 \times 5) - 11M |$

$\Rightarrow 0 = |300 - 11M|$

$\Rightarrow 300 - 11M = 0$

$\Rightarrow M = \frac{300}{11} = 27\frac{3}{11}$ minutes.

Now convert $\frac{3}{11}$ minutes to seconds.

Remember: 1 minute = 60 seconds.

Therefore $\frac{3}{11}$ minutes $= \frac{3}{11} \times 60 = \frac{180}{11} \approx 16$ seconds.

Thus, **5 Hours 27 minutes 16 seconds** (05:27:16) is the instant at which the hour hand and the minute hand are one over the other in between the markings for 5 and 6.

It should be noted that at any time between 05:00:00 to 05:27:30, the tip of the minute hand is closer to the marking for 5.

The time 05:27:16 is obviously in between 05:00:00 and 05:27:30. Hence **option (c).**

(For formula based approach of similar problems, see Chapter 2: Clock Problem)

4. Statement A: The following statement is true.

Statement B: The preceding statement is false.

Choose the correct inference from the following.

(a) Statements A and B are always true.

(b) Statements A and B can be true if there is at least one statement between A and B.

(c) Statements A and B can be true if there are at least two statements between A and B.

(d) Statements A and B can never be true, independently.

Answer (c) *Difficulty rating: 3.5/5* *Average time for solving: 80 seconds*

Observing the first three options, we can understand that we are required to check whether the truth of Statements A and B is ensured by the presence of at least one statement between A and B or by the presence of at least two statements between A and B.

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Let us check each of the options separately.

Consider option (a)

Claim: Statements A and B are true even if there is no statement between A and B.

Statement A: The following statement is true

Statement B: The preceding statement is false.

Here Statement A says B is true.

Statement B says A is false.

Combining these two, we get

Statement A says that B saying "A is false" is true.

But that is a contradiction to our assumption and hence not possible.

Consider option (b)

Claim: Statements A and B are true if there is at least one statement between A and B.

Let X be one statement between A and B.

Statement A: The following statement is true

Statement X:

Statement B: The preceding statement is false.

Now, A says X is true and B says X is false which is a contradiction as no statement can be both true and false simultaneously.

Consider option (c)

Claim: Statements A and B are true if there is at least two statements between A and B.

Let X and Y be two statements between A and B.

Statement A: The following statement is true

Statement X:

Statement Y:

Statement B: The preceding statement is false.

Now A says X is true and B says Y is false which doesn't create any contradiction and hence possible.

Hence **option (c)** is true and the truth of any of the first three options ensures the falsity of option (d).

(For related theory and questions, see Chapter 15: Logical Puzzles)

5. A person walks downhill at 10 km/h, uphill at 6 km/h and on the plane at 7.5 km/h. If the person takes 3 hours to go from a place A to another place B, and 1 hour on the way

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back, the distance between A and B is

(a) 15 km. **(b) 23.5 km.**

(c) 16 km. **(d) Given data is insufficient to calculate the distance**

Answer (a) *Difficulty rating: 4/5* *Average time for solving: 100 seconds*

To understand the question well, let us picture such a situation described in the question.

Let us denote the distances travelled uphill by the letter "u", downhill by the letter "d" and the plane by the letter "p".

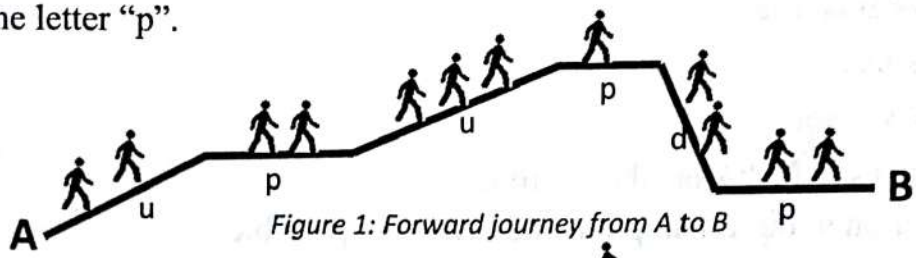

Figure 1: Forward journey from A to B

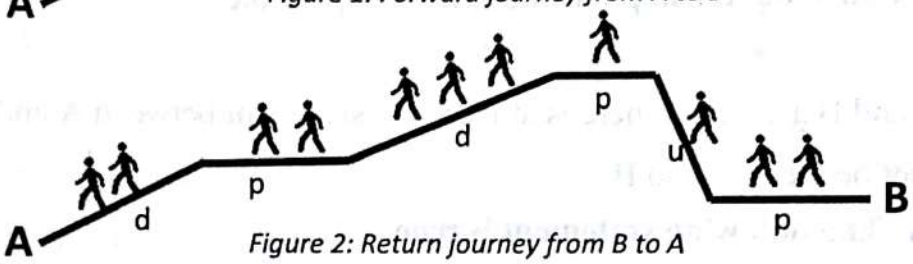

Figure 2: Return journey from B to A

Let us first observe the changes occurred in the return journey comparing the forward journey. The distance travelled on the plane is the same as that of the forward journey. (Note that the p-regions are the same in Figure 1 and Figure 2.)

But the regions of uphill in the return journey is the regions of downhill in the forward journey, and vice versa. (Note the interchange of u-regions and d-regions in Figure 1 and Figure 2.)

Let U, D and P be the total distance travelled in uphill, downhill and plane respectively.

That is U is the sum of distances of all u-regions and similar is the case for P and D.

Then, distance between A and B = U + D + P, which is what we are required to find.

Given: "A person walks downhill at 10 km/h, uphill at 6 km/h and on the plane at 7.5 km/h."

Given: "the person takes 3 hours to go from a place A to another place B"

Remember: $\text{Time} = \frac{\text{Distance}}{\text{Speed}}$

The total time taken is the sum of the time taken to walk uphill, downhill and on the plane.

$$\text{Hence, } \frac{U}{10} + \frac{D}{6} + \frac{P}{7.5} = 3 \text{ ---------- (1)}$$

Given: "and 1 hour on the way back"

As U and D interchanges, the equation for the return journey from B to A is given by

$$\frac{D}{10} + \frac{U}{6} + \frac{P}{7.5} = 1$$

which can be rearranged as

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$$\frac{U}{6}+\frac{D}{10}+\frac{P}{7.5}=1 \text{ ---------- (2)}$$

Note that here we are not required to find the values of U, D, P but the value of U+D+P.

Taking LCM on equation (1), we obtain: $\frac{3U+5D+4P}{30}=3$.

$\Rightarrow$ $3U + 5D + 4P = 90$ ------(3)

Similarly, taking LCM on equation (2), we obtain: $\frac{5U+3D+4P}{30}=1$.

$\Rightarrow$ $5U + 3D + 4P = 30$ ------(4)

Adding equations (3) and (4), we get,

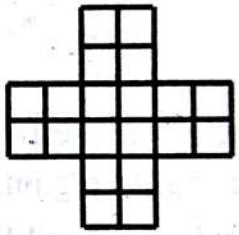
$8U + 8D + 8P = 120$

$\Rightarrow$ $8(U + D + P) = 120$

$\Rightarrow$ $U + D + P = \frac{120}{8} = 15$ km. Hence **option (a).**

(For related theory and questions, see Chapter 5: Numbers)

6. The number of squares in the figure is



(a) 30
(b) 29
(c) 25
(d) 20

Answer (b) *Difficulty rating: 2.5/5* *Average time for solving: 60 seconds*

The question demands to count the number of squares.

Number of 1×1 squares = 20

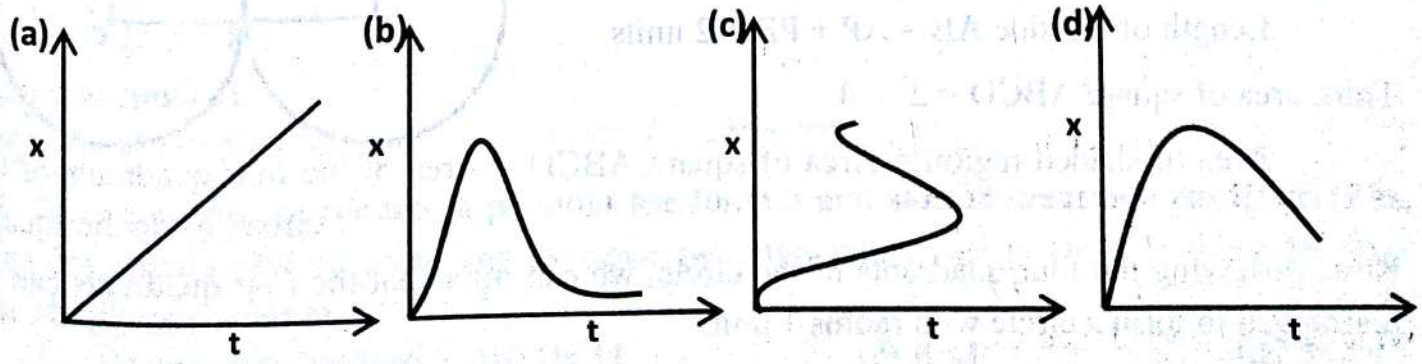
Number of 2×2 squares = 9

(When you count the number of 2×2 squares, ensure that the 2×2 square in the centre is not counted twice.)

Hence, total number of squares = 20 + 9 = 29.

(For related theory and questions, see Chapter 14: Observational Ability)

7. A bird leaves its nest and flies away. Its distance x from the nest is plotted as a function of time t. Which of the following plots cannot be right?



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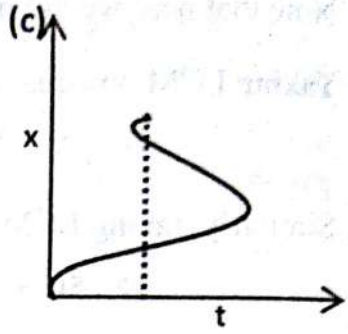
Answer (c) *Difficulty rating: 2.5/5* *Average time for solving: 60 seconds*

Consider graph (c).

Draw a vertical line on the graph as shown in the figure.



The vertical line cuts the graph at three distinct points.

That means, at the same instant of time, three distinct values of distance exists.

In simple words, the bird is positioned at three different places at the same instant of time!

Obviously impossible.

Thus, **graph (c)** cannot be right.

(For related theory and questions, see Chapter 13: Data Interpretation)

8. Four circles of unit radius each are drawn such that each one touches two others and their centres lie on the vertices of a square. The area of the region enclosed between the circles is

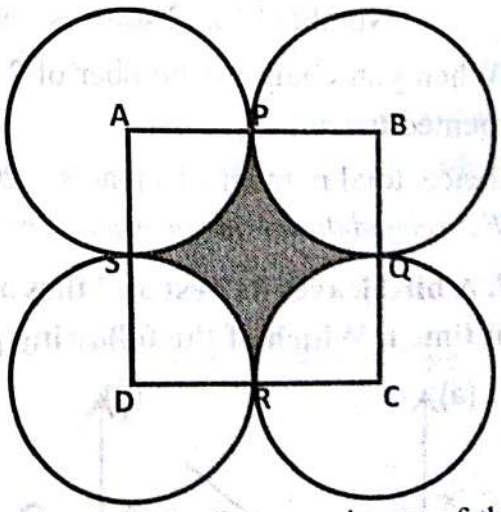
(a) π-1 **(b) π-2** **(c) 3-π** **(d) 4-π**

Answer (d) *Difficulty rating: 3.5/5* *Average time for solving: 100 seconds*

The given question can be pictured as shown in the figure. We are required to find the area of the shaded region.

Let A, B, C and D respectively be the centres of each circles and consequently ABCD is the square on whose vertices the centre of the four circles lie.

Let P, Q, R and S be the points where the circles meet each other. Evidently, P, Q, R and S are the mid points of the sides of the square ABCD.



Given that all the four circles have unit radius.

Therefore, AP = AS = 1 unit

BP = BQ = 1 unit

CQ = CR = 1 unit

DR = DS = 1 unit

Clearly, for the square ABCD,

Length of the side AB = AP + PB = 2 units.

Thus, area of square ABCD = $2^2 = 4$

Area of shaded region = Area of square ABCD – Area of the four quadrants of the circle inside the square.

Now, observing the four quadrants of the circle, we can infer that the four quadrants can be rearranged to form a circle with radius 1 unit.

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Area of the circle with radius 1 unit = $\pi (1)^2 = \pi$.

Thus, Area of shaded region = Area of square ABCD – Area of the circle with radius 1 unit

$= 4 - \pi$

(For related theory and questions, see Chapter 12: Mensuration and Quantitative comparison)

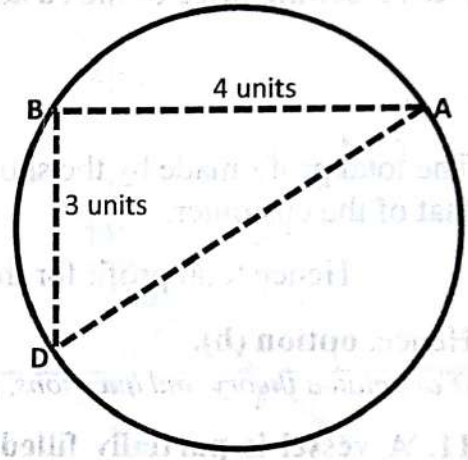
9. A turtle starts swimming from a point A located on the circumference of a circular pond. After swimming for 4 meters in a straight line it hits point B on the circumference of the pond. From there it changes direction and swims for 3 meters in a straight line and arrives at point D diametrically opposite to point A. How far is point D from A?

(a) 3m **(b) 4m** **(c) 7m** **(d) 5m**

Answer (d) *Difficulty rating: 3/5* *Average time for solving: 90 seconds*

Given: "After swimming for 4 meters in a straight line it hits point B on the circumference of the pond."

That is AB = 4 units.



Given: "From there it changes direction and swims for 3 meters in a straight line and arrives at point D"

So the length of BD = 3 units.

But the direction in which the turtle moves is dependent on the position of D from B.

Given: "D is diametrically opposite to point A."

Therefore to know the direction in which the turtle swims from B to reach D, first we have to fix the point D diametrically opposite to A. Or in simple words, place D on the circle such that AD is the diameter of the circle.

Thus we can join BD to complete the figure.

As AD is the diameter of the circle, $\Delta ABD = 90^\circ$

[Since the angle subtended in a semicircle is a right angle.]

We are required to find the length of AD.

Applying Pythagoras theorem on the right triangle ABD, we obtain

$AD^2 = AB^2 + BD^2 = 4^2 + 3^2 = 25$

$\Rightarrow$ AD = 5.

Hence **option (d).**

(For related theory and questions, see Chapter 11: Geometry)

10. A shopkeeper purchases a product for Rs.100 and sells it making a profit of 10%. The customer resells it to the same shopkeeper incurring a loss of 10%. In these dealings the shopkeeper makes

(a) no profit no loss **(b) Rs.11** **(c) Re.1** **(d) Rs.20**

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Answer (b) *Difficulty rating: 3/5* *Average time for solving: 70 seconds*

Given: "A shopkeeper purchases a product for Rs.100 and sells it making a profit of 10%."

That is the selling price of the shopkeeper is 10% more than his cost price, Rs.100.

Hence, selling price of the shop keeper $= 100 + 10\%$ of 100

$= 100 + 10$

$= 110$

Given: "The customer resells it to the same shopkeeper incurring a loss of 10%."

That is, the selling price of the customer is 10% less than the cost price of the customer.

Here the cost price of the customer is Rs.110 and we are required to find the selling price.

Hence, selling price of the customer $= 110 - 10\%$ of 110

$= 110 - 11$

$= 99$

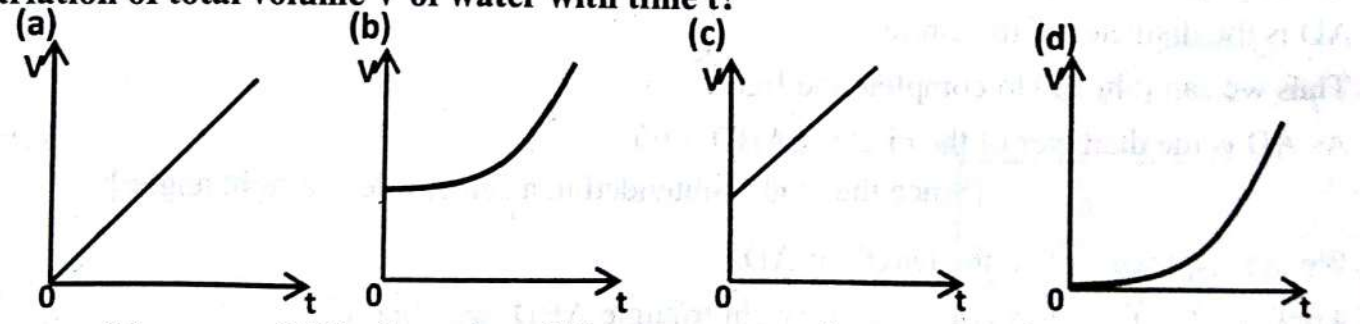
The total profit made by the shop keeper is the difference between the selling price of him and that of the customer.

Hence total profit for the shop keeper $= 110 - 99 = 11$.

Hence, **option (b).**

(For related theory and questions, see Chapter 10: Monetary Problems)

11. A vessel is partially filled with water. More water is added to it at a rate directly proportional to time [$\frac{dV}{dt} \alpha$ t]. Which of the following graphs depicts correctly the variation of total volume V of water with time t?



Answer (b) *Difficulty rating: 3/5* *Average time for solving: 80 seconds*

At the initial point, the vessel is partially filled with water.

That is, the volume of water in the vessel is not 0 at the initial point.

Hence we can rule out option (a) and option (d).

Given: "More water is added to it at a rate directly proportional to time [$\frac{dV}{dt} \alpha$ t]."

That is, the rate of change of volume ($\frac{dV}{dt}$) is directly proportional to time.

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Graph (3) is the graph in which volume (V) is directly proportional to time and hence can be ruled out.

Thus we are left out with a single option. Hence **option (b).**

(For related theory and questions, see Chapter 13: Data Interpretation)

12. The triangle formed by the lines y = x, y = 1 - x and x = 0 in a two dimensional plane is (x and y axes have the same scale)

(a) isosceles and right-angled **(b) isosceles but not right-angled**

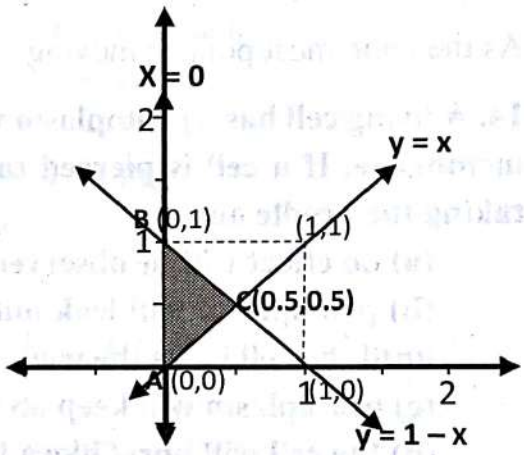
(c) right-angled but not isosceles **(d) neither isosceles nor right-angled**

Answer (a) *Difficulty rating: 3/5* *Average time for solving: 100 seconds*

For this question, drawing a rough graph helps to solve quickly.

Let ΔACB be the required triangle.

Join the points (1, 0) and (0, 1) to (1, 1) on the graph to form a unit square.



Remember: Diagonals of a square bisect each other.

Hence, ∠ACB = 90° [Since the sum of angles around a point is 360°]

That is ΔACB is a right-angled triangle.

Again from the property of the unit square along with the application of the Pythagoras theorem on the right triangle ΔACB we can derive that AB = 1 unit, BC = AC = $\sqrt{2}$ unit making ΔACB an isosceles right-angled triangle.

Hence **option (a).**

(For related theory and questions, see Chapter 11: Geometry)

13. A car is moving at 60 km/h. The instantaneous velocity of the upper most points of its wheels is

(a) 60 km/h forward **(b) 120 km/h forward**

(c) 60 km/h backward **(d) 120 km/h backward**

Answer (b) *Difficulty rating: 4/5* *Average time for solving: 90 seconds*

This question demands the knowledge of some basic physics concepts you learnt in High school classes.

Consider any of the upper most point of any one wheel of the car.

As the car is moving, the upper most point is exerted both by the translatory and the rotatory force of the car.

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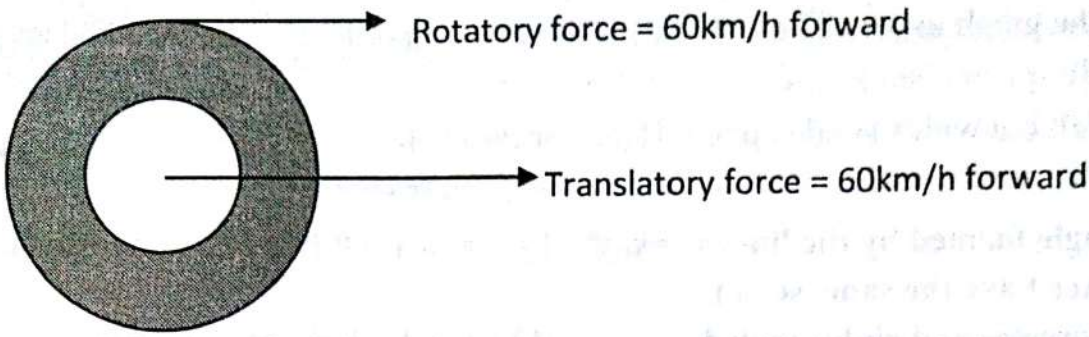

Therefore,

Instantaneous velocity of the upper most point = Translatory force + Rotatory force
= 60 Km/Hr. + 60 Km/Hr.
= 120 Km/Hr.

As the uppermost point is moving forward, **option (b)** is our obvious choice.

14. A living cell has a protoplasm which is water based and demarcated by a lipid bilayer membrane. If a cell is pierced to 1/5th of its diameter with a very sharp needle, after taking the needle out

(a) no effect will be observed.

(b) protoplasm will leak out from the hole made by the needle for a few minutes until the cell heals the wound.

(c) protoplasm will keep on leaking out till the cell is dead

(d) the cell will burst like a balloon.

Answer (a) *Difficulty rating: 3.5/5* *Average time for solving: 70 seconds*

The lipid bilayer is a thin polar membrane made of two layers of lipid molecules.

The lipid bilayer membrane will reseal immediately after getting pierced.

Therefore no effect will be observed when a living cell is pierced to 1/5th of its diameter.

Hence **option (a).**

15. If D + I + M = 1501

C + I + V + I + L = 157

L + I + V + I + D = 557

C + I + V + I + C = 207

What is V + I + M = ?

(a) Cannot be found **(b) 1009** **(c) 1006** **(d) 509**

Answer (c) *Difficulty rating: 3/5* *Average time for solving: 80 seconds*

In these set of equations, there are 6 unknown variables – I, V, L, C, D, M.

Remember: **In Roman Number System, I = 1, V = 5, L = 50, C = 100, D = 500, M = 1000.**

Clearly, these numbers satisfies each of the above 4 equations.

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That is, D + I + M = 500 + 1 + 1000 = 1501

C + I + V + I + L = 100 + 1 + 5 + 1 + 50 = 157

L + I + V + I + D = 50 + 1 + 5 + 1 + 500 = 557

C + I + V + I + C = 100 + 1 + 5 + 1 + 100 = 207

Thus, V + I + M = 5 + 1 + 1000 = **1006.** Hence **option (c).**

Alternate solution [If you do not know all the above Roman letters]

Even if you don't know all of these Roman letters, you might surely know that I = 1, V = 5.

We are required to find V + I + M.

That is, we just have to find the value of M from the above set of equations.

Substituting the values of I and V in the above set of equations, we obtain

D + 1 + M = 1501 => **D + M = 1500 ----- (1)**

C + 1 + 5 + 1 + L = 157 => **C + L = 150 ------ (2)**

L + 1 + 5 + 1 + D = 557 => **L + D = 550 ------ (3)**

C + 1 + 5 + 1 + C = 207 => **2C = 200 ------- (4)**

From equation (4), C = 100

Substituting C = 100 in equation (2), we obtain L = 50

Substituting L = 50 in equation (3), we obtain D = 500

Substituting D = 500 in equation (1), we obtain M = 1000

Thus, V + I + M = 5 + 1 + 1000 = **1006.**

(For related theory and questions, see Chapter 14: Observational Ability)

16. Density of a rice grain is 1.5 g/cc and bulk density of rice heap is 0.80 g/cc. If a 1 litre container is completely filled with rice, what will be the approximate volume of pore space in the container?

(a) 350 cc **(b) 465 cc** **(c) 550 cc** **(d) 665 cc**

Answer (b) *Difficulty rating: 3.5/5* *Average time for solving: 100 seconds*

Remember

1: cc means cubic centimetre or cm^3

2: Mass = Density × Volume

3: One litre is the volume of the cube with side 10 cm.

That is, 1 Litre = 10 × 10 × 10 cm^3 = 1000 cm^3 = 1000 cc

Given: "1 litre container is completely filled with rice"

Mass of the 1 litre rice-filled container = Density of rice heap × Volume of 1 litre container

= 0.80 × 1000 = 800 g.

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It should be noted that

Mass of the 1 litre rice-filled container = Mass of the rice in the container.

We are required to find the volume of the pore space in the container.

Volume of the rice in the container $= \frac{\text{Mass of the rice in the container}}{\text{Density of the rice grain}} = \frac{800}{1.5} \approx 533.3$ cc

Volume of pore space = Volume of 1 litre container – Volume of the rice in the container.

$\approx 1000 - 533.3$

$\approx 465.$

(For related theory and questions, see Chapter 12: Mensuration and Quantitative comparison)

17. A cubical cardboard box made of 1 cm thick card board has outer side of 29 cm. A tight-fitting cubical box of the same thickness is placed inside it, then another one inside it and so on. How many cubical boxes will be there in the entire set?

(a) 29 **(b) 28** **(c) 15** **(d) 14**

Answer (d) *Difficulty rating: 3/5* *Average time for solving: 100 seconds*

Given a cardboard box whose length of the outer side is **29 cm.**

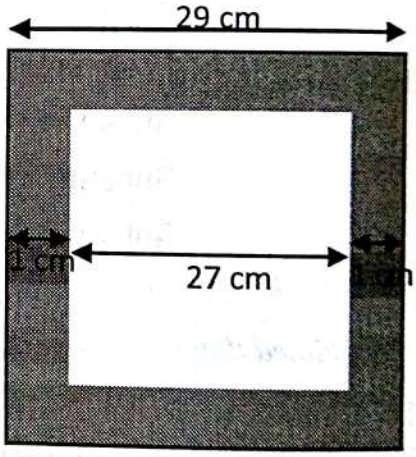
Given that its thickness is 1 cm.

Therefore, the length of the outer side of the tight-fitting box inside it is **27 cm.**

Hence, with each box, the length of the outer side reduces by 2 cm.

Thus, totally we can reduce 2 cm each 14 times.

Hence **option (d).**



(For related theory and questions, see Chapter 14: Observational Ability)

18. A film projector and microscope give equal magnification. But a film projector is not used to see living cells because

(a) a living cell cannot be placed in a film projector.

(b) the viewer's eye is close to a microscope whereas it is far away from the projector screen.

(c) a microscope produces a virtual image whereas a projector produces a real image.

(d) a microscope has greater resolving power than a projector.

Answer (d) *Difficulty rating: 3.5/5* *Average time for solving: 70 seconds*

This question also demands some basic physics knowledge.

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In the case of living cells, the cells can be so close together that there exists a need for identifying them distinctly which can only be done using an optical instrument of higher resolving power.

Fact: Resolving power is the power of an optical instrument to separate far away objects that are close together, into individual images.

Hence **option (d).**

19. Secondary colours are made by a mixture of three primary colours: Red, Green and Blue in different proportions; each of the primary colours comes in 8 possible levels. Grey corresponds to equal proportions of Red, Green and Blue. How many shades of grey exist in this scheme?

(a) 8^3 **(b) 8** **(c) 3^8** **(d) 8×3**

Answer (b) *Difficulty rating: 4/5* *Average time for solving: 100 seconds*

Given: "each of the primary colours comes in 8 possible levels."

Let us denote the 8 levels of each of the colours in the increasing order of the proportion of the colour as follows.

Let $R_1, R_2, R_3, R_4, R_5, R_6, R_7, R_8$ be the 8 levels of Red.

Let $G_1, G_2, G_3, G_4, G_5, G_6, G_7, G_8$ be the 8 levels of Green.

Let $B_1, B_2, B_3, B_4, B_5, B_6, B_7, B_8$ be the 8 levels of Blue.

That is R_1 has less proportion of red than R_2, R_2 has less proportion of red than R_3 etc… Similarly for the colours green and blue.

Also, R_1, G_1, B_1 have the same proportion of their respective colours and similar is the case for the other indices.

Given: "Grey corresponds to equal proportions of Red, Green and Blue."

R_1, G_1, B_1 are of the same proportion and hence can be mixed to obtain a shade of grey, say GREY-1.

R_2, G_2, B_2 are of the same proportion and hence can be mixed to obtain another shade of grey, say GREY-2.

Similarly, we can obtain GREY-3, GREY-4, GREY-5, GREY-6, GREY-7 and GREY-8 and they are the only shades of grey we can obtain.

Hence **option (b).**

(For related theory and questions, see Chapter 6: Probability and Arrangements)

20. There are two buckets A and B. Initially A has 2 litres of water and B is empty. At every hour 1 litre of water is transferred from A to B followed by returning ½ litre back

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to A from B half an hour later. The earliest A will get empty is in:

(a) 1.5h **(b) 4h** **(c) 3h** **(d) 2h**

Answer (c) *Difficulty rating: 3.5/5* *Average time for solving: 100 seconds*

For reference, let us call the transfer of water from A to B as the first process and transfer of water from B to A as the second process.

It should be noted that the first process happens at every hour and does not depend on the second process. But the second process depends on the first process.

For convenience, let us fix an initial time and analyse the question with respect to that time.

Let us fix the initial time as 12 pm.

Then by the given condition, the first process will occur after at every hour, that is, 1 pm, 2 pm etc…

But the second process occurs exactly half an hour after the occurrence of the first process.

That is, the second process happens at 1:30 pm, 2:30 pm etc…

Let us tabulate our observation as follows

Process	Time	Water in A (In Litres)	Water in B (In Litres)
Initial condition	12 pm	2	0
1st	1 pm	1	1
2nd	1:30 pm	$1\frac{1}{2}$	$\frac{1}{2}$
1st	2 pm	$\frac{1}{2}$	$1\frac{1}{2}$
2nd	2:30 pm	1	1
1st	3 pm	0	2

We started our procedure at 12 pm and obtained our required condition at 3 pm, that is a difference of 3 hours.

Hence, earliest A will get empty in 3 hours. Hence **option (c).**

(For related theory and questions, see Chapter 14: Observational Ability)

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June 2015 – Mathematical Sciences, Life Sciences, Earth Sciences

1. In each of the following groups of words is a hidden number, based on which you should arrange them in ascending order. Pick the right answer:

A. Tinsel event **B. Man in England** **C. Good height** **D. Last encounter**

(a) A,B,C,D **(b) C,B,D,A** **(c) A,C,D,B** **(d) C,D,B,A**

Answer (b) *Difficulty rating: 2.5/5* *Average time for solving: 50 seconds*

Through observation, we can find the hidden number in each of the following groups of words.

That is

A. Tins**el even**t ------ 11

B. Ma**n in E**ngland------ 9

C. Good h**eight** ------ 8

D. Las**t en**counter ------ 10

When arranged in the ascending order, the numbers can be listed as 8, 9, 10, 11.

That is, C, B, D, A. Hence, **option (b).**

2. Let m and n be two positive integers such, that m + n + mn = 118.
Then the value of m + n is

(a) not uniquely determined **(b) 18**

(c) 20 **(d) 22**

Answer (d) *Difficulty rating: 4/5* *Average time for solving: 120 seconds*

Given: mn + m + n = 118

Add 1 to both sides of the equation.

⇨ mn + m + n + 1 = 119

⇨ m(n + 1) + n + 1 = 119

⇨ (m + 1) (n + 1) = 119

That is, product of the numbers (m + 1) and (n + 1) gives 119.

Prime factorization of 119 is given by

$119 = 17 \times 7$.

We know that prime factorization of a number is unique.

Hence, m + 1 = 17

⇨ m = 16.

And n + 1 = 7

⇨ n = 6.

Further, if we took (n + 1) = 17 and (m + 1) = 7, we obtain m = 6 and n = 16.

In either case, m + n = 22.

Thus, value of m + n is uniquely determined as 22.

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Hence, **option (d).**

(For related theory and questions, see Chapter 5: Numbers)

3. AB is the diameter of a circle. The chord CD is perpendicular to AB intersecting it at P. If CP = 2 and PB = 1, the radius of the circle is

(a) 1 (b) 2.5 (c) 2 (d) 5

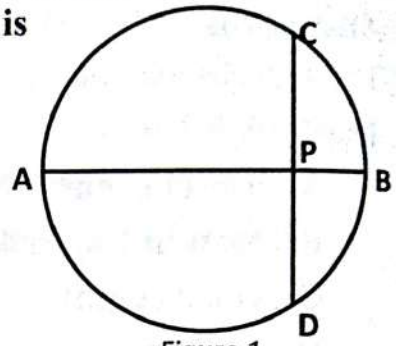

Figure 1

Answer (b)

Difficulty rating: 3/5 *Average time for solving: 80 seconds*

Let x be the length of the radius of the given circle.

Let O be the centre of the given circle.

Join OC.

Now, OC and OB are the radius of the circle. (See Figure 2)

That is, OC = OB = x

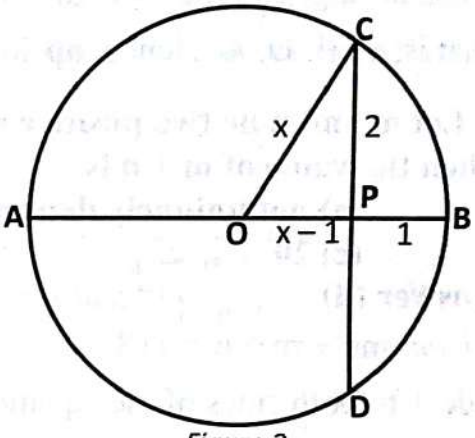

Figure 2

Given: PB = 1

⇨ OP = x – 1.

Applying Pythagoras theorem on ΔOPC

$(x-1)^2 + 2^2 = x^2$

⇨ $x^2 - 2x + 1 + 4 = x^2$

⇨ $2x = 5$

⇨ $x = \mathbf{2.5}$

(For related theory and questions, see Chapter 11: Geometry)

4. Jar W contains 40 white marbles and jar B contains 40 black marbles. Ten black marbles from B are transferred to W and mixed thoroughly. Now, ten randomly selected marbles from W are put back in Jar B to make 40 marbles in each jar. The number of black marbles in W

(a) would be equal to the number of white marbles in B

(b) would be more than the number of white marbles in B

(c) would be less than the number of white marbles in B

(d) cannot be determined from the information given

Answer (a) *Difficulty rating: 3.5/5* *Average time for solving: 90 seconds*

Let us tabulate our observations from each of the given statements.

Given 1:"Jar W contains 40 white marbles and jar B contains 40 black marbles."

	Jar W	**Jar B**
White	40	0
Black	0	40

Given 2: "Ten black marbles from B are transferred to W and mixed thoroughly."

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	Jar W	Jar B
White	40	0
Black	10	30

Given 3: "Now, ten randomly selected marbles from W are put back in Jar B to make 40 marbles in each jar."

Let k be the number of black marbles and 10 – k be the number of white marbles in the batch of 10 marbles moved from Jar W to Jar B.

Thus now,

	Jar W	Jar B
White	40 – (10 – k)	0 + (10 – k)
Black	10 – k	30 + k

which can be written as:

	Jar W	Jar B
White	30 + k	**10 – k**
Black	**10 – k**	30 + k

Clearly, the number of black marbles in Jar W is equal to the number of white marbles in B.

Hence, **option (a).**

(For related theory and questions, see Chapter 6: Probability and Arrangements.)

5. Of three persons A, B and C, one always lies while the others always speak the truth. C asked A. "Do you always speak the truth, yes or no?" He said something that C could not hear. So C asked B. "what did A say?" B replied. "A said No" So, who is the liar?

(a) A (b) B (c) C (d) cannot be determined

Answer (b) *Difficulty rating: 3.5/5* *Average time for solving: 90 seconds*

Here there are only two replies: Reply of A and Reply of B.

Consider Reply of A to the question "Do you always speak the truth. yes or no?"

If A lies, A will reply "**Yes",** because for this question, the lie reply is Yes for a liar.

If A speaks truth, A will reply **"Yes",** because for this question, the truth reply is Yes for a person speaking truth.

Hence, whatever the case, the reply of A can only be "Yes"

B replied "No" to the question "What did A say?"

That is, B said that, what A said is "No".

We are sure that, what A said is "Yes", and hence B lied.

Since, given there is only one liar, only B lied.

Hence, **option (b).**

(For related theory and questions, see Chapter 15: Logical Puzzles)

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6. If aN ⇨ S
eF ⇨ I
gH ⇨ M
then nS ⇨ ?
(a) T **(b) A** **(c) L** **(d) K**

Answer (a) *Difficulty rating: 3/5* *Average time for solving: 50 seconds*

aN when reversed gives Na, that is Sodium. Hence, the first letter, **S**.

Similarly eF ≡ Fe = Iron ⇨ **I.**

gH ≡ Hg = Mercury ⇨ **M.**

Therefore nS ≡ Sn = Tin ⇨ **T.**

Hence, **option (a).**

7. From a group of 40 players, a cricket team of 11 players is chosen. Then, one of the eleven is chosen as the captain of the team. The total number of ways this can be done is

[$\binom{m}{n}$ below means the number of ways *n* objects can be chosen from *m* objects]

(a) $\binom{40}{11}$ **(b)** $11\binom{40}{11}$ **(c)** $29\binom{40}{11}$ **(d)** $\binom{39}{10}$

Answer (b) *Difficulty rating: 2.5/5* *Average time for solving: 60 seconds*

Given $\binom{m}{n}$ means the number of ways n objects can be chosen from m objects.

First, 11 players should be chosen from 40 players.

That is, m = 40 and n = 11.

Therefore, number of ways 11 players can be chosen from 40 players is $\binom{40}{11} = {}^{40}C_{11}$.

Now 1 captain should be chosen from the selected 11 players.

Thus, now m = 11 and n = 1.

The number of ways 1 captain can be chosen from 11 players = $\binom{11}{1} = {}^{11}C_1 = 11$.

[Remember ${}^{n}C_1 = n$]

Therefore, the total number of ways = ${}^{11}C_1 \times {}^{40}C_{11} = 11 \times \binom{40}{11}$. Hence, **option (b).**

(For related theory and questions, see Chapter 6: Probability and Arrangements)

8. What is angle x in the schematic diagram given?

(a) 60 **(b) 50** **(c) 40** **(d) 30**

Answer (d)

Difficulty rating: 2.5/5 Average time for solving: 90 seconds

Extend the line as shown in the Figure 2 and mark the vertices accordingly.

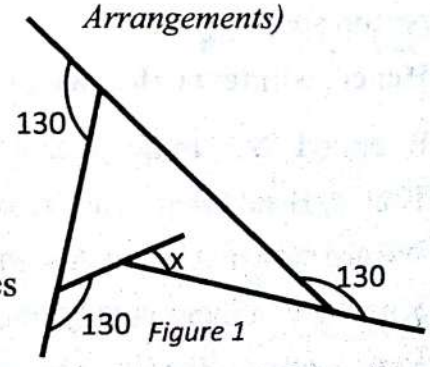

Figure 1

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Remember: Angles on one side of a straight line will always add to 180 degrees [Linear Pair].

Hence, $\angle BAC = \angle CED = \angle ABC = 180^\circ - 130^\circ = 50^\circ$.

Consider ΔABC

$\angle ACB = 180^\circ - (\angle ABC + \angle BAC)$ [Angle Sum Property]

$= 180^\circ - (50+50) = 80^\circ$

Consider ΔDCE

$\angle DCE = 180 - \angle ACB$ [Linear Pair]

$= 180 - 80 = 100^\circ$

Now, $x = 180 - (50 + 100)$ [Angle Sum Property]

⇨ **$x = 30^\circ$.**

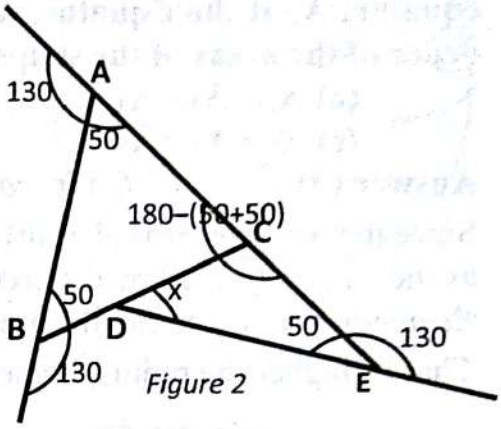

Figure 2

(For related theory and questions, see Chapter 11: Geometry)

9. I bought a shirt at 10% discount and sold it to a friend at a loss of 10%. If the friend paid me Rs. 729.00 for the shirt, what was the undiscounted price of the shirt?

(a) Rs. 900 **(b) Rs. 800** **(c) Rs 1000** **(d) Rs.911.25**

Answer (a) *Difficulty rating: 3/5* *Average time for solving: 90 seconds*

In monetary problems, even if we have the formula for profit or loss percentage, try to do the problem logically, which saves whole lot of time.

Let us discuss the same for the solution of this problem.

Let x be the undiscounted price of the shirt.

Given 1: "I bought a shirt at 10% discount."

I bought the shirt at 10% discount ≡ I bought the shirt at 90% price $= \frac{90}{100}x$.

Given 2: "sold it to a friend at a loss of 10%."

That is, Price I sold it to the friend $= (100 - 10)\%$ of the price I bought

$= 90\%$ of the price I bought

$= 90\%$ of $\frac{90}{100}x$

$= \frac{90}{100} \times \frac{90}{100}x$

$= \frac{81}{100}x$.

Given 3: "friend paid me Rs. 729.00"

That is, $\frac{81}{100}x = 729$

⇨ $x = \frac{729 \times 100}{81}$

⇨ $x =$ **Rs.900.**

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(For the formula based approach of similar problems, see Chapter 10: Monetary Problems)

10. Consider 3 parallel strips of 10 m width running around the Earth, parallel to the equator; A_1 at the Equator, A_2 at the Tropic of Cancer and A_3at the Arctic Circle. The order of the areas of the strips is

(a) $A_1 < A_2 < A_3$ **(b) $A_1 = A_2 > A_3$**

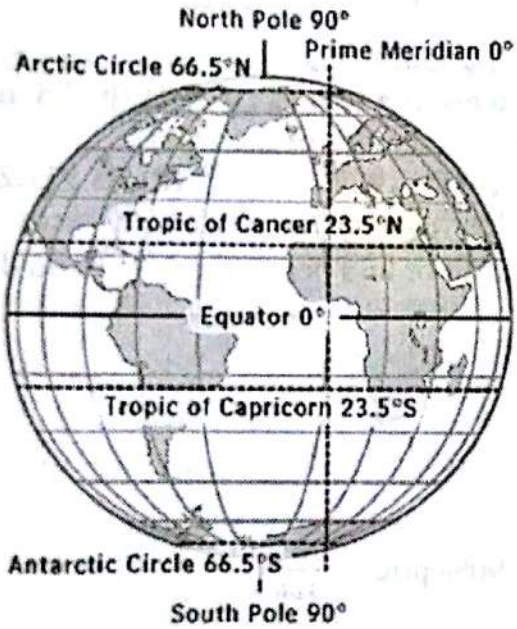
(c) $A_1 > A_2 = A_3$ **(d) $A_1 > A_2 > A_3$**

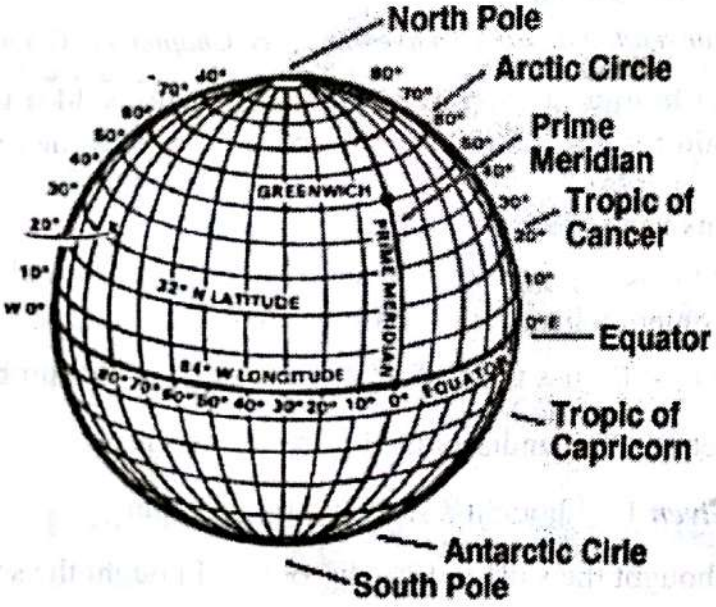
Answer (d) *Difficulty rating: 3/5* *Average time for solving: 60 seconds*

Since the width is same (10 cm) for the three strips, the only parameter that differs is the radius, as the strip is formed by different circles.

Remember: For a circle, the area is directly proportional to the square of the radius.

That is, higher the radius, higher will be the area.





From the two figures, it is evident that the equator will have the highest radius followed by the radius of the Tropic of cancer and then by the radius of the Arctic Circle.

Hence $A_1 > A_2 > A_3$.

(For related theory and questions, see Chapter 7: Distances and Directions)

11. Starting from a point A you fly one mile south, then one mile east, then one mile north which brings you back to point A. Point A is NOT the north pole. Which of the following MUST be true?

(a) You are in the Northern Hemisphere
(b) You are in the Eastern Hemisphere
(c) You are in the Western Hemisphere
(d) You are in the Southern Hemisphere

Answer (d) *Difficulty rating: 4.5/5* *Average time for solving: 120 seconds*

Here, first we can find a point where the given conditions are satisfied.

Consider a ring in the southern hemisphere having a circumference of one mile.

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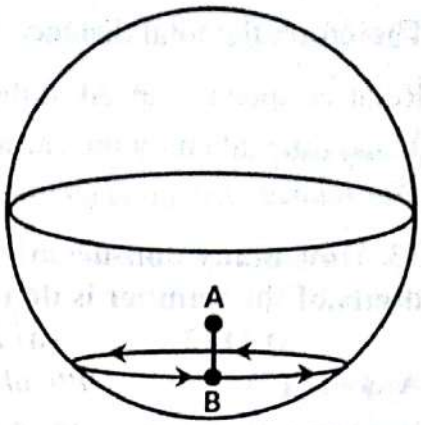
Assume that you are standing at a point one mile north of the ring and mark the point as A. (See figure)



Now read the question again starting with point A.

Given: "Starting from a point A you fly one mile south."

That is you reach a point on the ring with the circumference 1 mile. Denote the point as B.

Given: "then one mile east"

This will bring you to the same point B, as the circumference of the ring is 1 mile.

Given: "then one mile north which brings you back to point A."

One mile north will bring you back to point A.

Hence A, a point in the southern hemisphere satisfies the conditions in the question.

When you analyse other answer options, you can see that the only other point which satisfies the conditions is North pole. However, it is given that point A is not the North Pole.

(Proceeding with the above procedure, you can infer that no point in the eastern or western hemisphere satisfies the conditions, which is left as an exercise.)

Hence, **option (d).**

(For related theory and questions, see Chapter 7: Distances and Directions)

12. A 3m long car goes past a 4 m long truck at rest on the road. The speed of the car is 7 m/s. The time taken to go past is

(a) 4/7 s **(b) 1 s** **(c) 7/4 s** **(d) 10/7 s**

Answer (b) *Difficulty rating: 2.5/5* *Average time for solving: 60 seconds*

Remember: $\text{Time} = \frac{\text{Distance}}{\text{Speed}}$

Here, the distance to go past starts when the front end of the car meets the back end of the truck and ends when the back end of the car meets the front end of the truck.

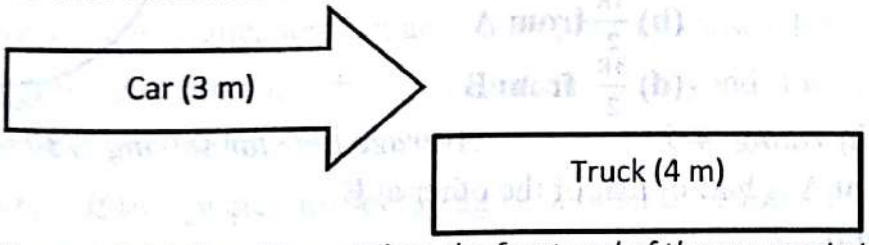


Figure 1: Initial position – When the front end of the car meets the back end of the truck

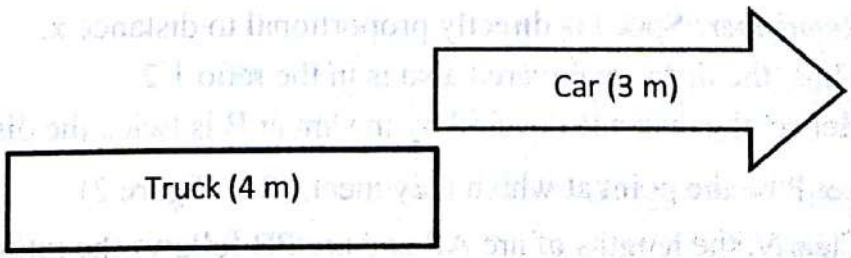


Figure 2: Final position – When the back end of the car meets the front end of the truck

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Therefore, the total distance = Length of the car + Length of the truck = 4m + 3m = 7 m.

Relative speed = Speed of the car – Speed of the truck = 7 m/s – 0 m/s = 7 m/s.

Thus, time taken by the car to go past the truck = 7/7 = 1 s.

(For related theory and questions, see Chapter 3: Moving Locomotive Problem)

13. How many non-negative integers less than 10,000 are there such that the sum of the digits of the number is divisible by three?

(a) 1112 **(b) 2213** **(c) 2223** **(d) 3334**

Answer (d) *Difficulty rating: 3/5* *Average time for solving: 70 seconds*

We know that if the sum of the digits of the number is divisible by three, then the number is divisible by three.

Hence, the number is a multiple of 3.

9999 is the highest such number less than 10000.

Starting from 3, such a number occur in every set of three consecutive numbers.

Thus 9999/3 = 3333 such numbers are there.

However, in the question, the number of such **non-negative integers** are asked.

0 is such a nonnegative integer which is divisible by 3.

Hence, there are 3333 + 1 = 3334 non-negative integers less than 10000 such that the sum of digits of the number is divisible by 3.

Hence, **option (d).**

(For related theory and questions, see Chapter 5: Numbers)

14. Two ants, initially at diametrically opposite points A and B on a circular ring of radius R, start crawling towards each other. The speed of the one at A is half that of the one at B. The point at which they meet is at a straight line distance of

(a) R from A **(b) $\frac{3R}{2}$ from A**

(c) R from B **(d) $\frac{3R}{2}$ from B**

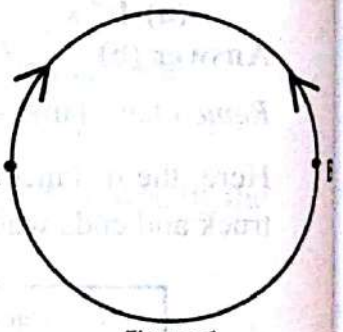

Figure 1

Answer (a) *Difficulty rating: 4/5* *Average time for solving: 130 seconds*

Given: The speed of the one at A is half of that of the other at B.

That is, the speed is in the ratio 1:2.

Remember: Speed is directly proportional to distance x.

Thus, the distance covered also is in the ratio 1:2.

Hence, the distance covered by the ant at B is twice the distance covered by the one at A.

Let P be the point at which they meet. (See Figure 2)

Clearly, the lengths of arc AP and arc PB follows the ratio 1:2.

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Therefore, their central angle also follows the same ratio.

⇨ ∠AOP = 60°, ∠POB = 120°.

If we analyse the answer options, we can infer that we are required to find the straight-line distance AP or the straight line distance PB.

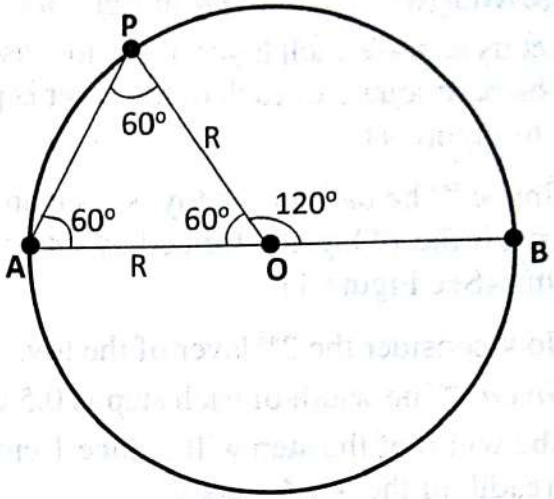

Figure 2

Consider ΔAOP

Here ∠AOP = 60° and AO = OP = R.

Hence, ΔAOP is an isosceles triangle.

Remember: For an isosceles triangle, opposite angles of the equal sides are equal.

That is, ∠APO = ∠PAO.

We know that, ∠AOP = 60°.

Then, by angle sum property,

∠APO + ∠PAO = 120°.

⇨ ∠APO = ∠PAO = ∠AOP = 60°.

That is, ΔAOP is an equilateral triangle.

Thus, length of the line segment, AP = OA = OP = **R.**

Therefore, the point at which they meet is at a straight line distance of R from A.

Hence **option (a).**

(For related theory and questions, see Chapter 11: Geometry)

15. Two plane mirrors facing each other are kept at 60° to each other. A point is located on the angle bisector. The number of images of the point is

(a) 6 **(b) 3** **(c) 5** **(d) Infinite**

Answer (c) *Difficulty rating: 4/5* *Average time for solving: 50 seconds*

Unfortunately, this question demands a bit physics knowledge.

The following formula, whose derivation is beyond the scope of this discussion, is quintessential for doing this question.

Remember: If two plane mirrors facing each other are kept at θ°, to each other and if a point is located at the angle bisector, then the number of images (n) of the point is given by

$$n = \left(\frac{360}{\theta}\right) - 1.$$

Here θ = 60°.

$$n = \left(\frac{360}{60}\right) - 1 = 6 - 1 = \mathbf{5}.$$

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16. A pyramid shaped toy is made by tightly placing cubic blocks of 1 × 1 × 1 cm³. The base of the toy is a square 4×4 cm². The width of each step is 0.5 cm. How many blocks are required to make the toy?

(a) 30 **(b) 34** **(c) 36** **(d) 40**

Answer (a) *Difficulty rating: 3/5* *Average time for solving: 110 seconds*

Let us analyse each layer of the toy, assuming the base layer as the 1st layer. The base square of each of the layer is pictured in the figures shown. (Figure 1 to Figure 4)

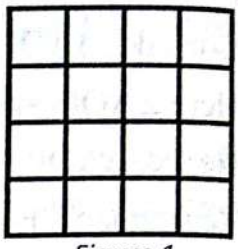

Figure 1

Given: "The base of the toy is a square 4 × 4 cm²."
That is the 1st layer is formed by 4×4 = **16 cubic boxes** of dimension 1×1×1 cm³. (See Figure 1)

Now consider the 2nd layer of the toy.

Given: "The width of each step is 0.5 cm."

The width of the step will reduce 1 cm (0.5 + 0.5) from both the length and breadth of the 4 × 4 square.

Thus, now the base of the second layer would be a square of 3 × 3 cm².

Hence, the 2nd layer is formed by 3×3 = **9 cubic boxes** of dimension 1×1×1 cm³. (See Figure 2)

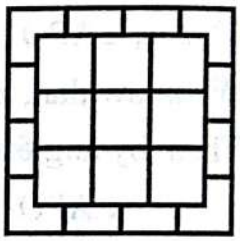

Figure 2

Consider the 3rd layer of the toy.

The width of the step will reduce 1 cm (0.5 + 0.5) from both the length and breadth of the 3 × 3 square.

Thus, now the base of the 3rd layer would be a square of 2 × 2 cm².

Hence, the 3rd layer is formed by 2×2 = **4 cubic boxes** of dimension 1×1×1 cm³. (See Figure 3)

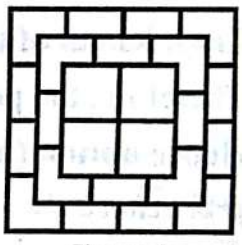

Figure 3

Consider the 4th layer of the toy.

The width of the step will reduce 1 cm (0.5 + 0.5) from both the length and breadth of the 2 × 2 square.

Thus, now the base of the 4th layer would be a square of 1 × 1 cm².

Hence, the 4th layer is formed by **1 cubic box** of dimension 1×1×1 cm³. (See Figure 4)

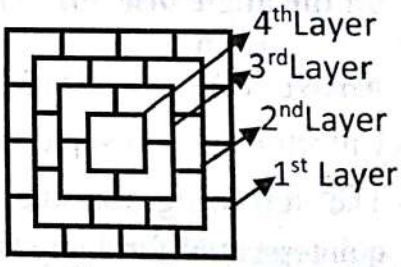

Figure 4

Thus, total number of cubic blocks of 1 × 1 × 1 cm³ = 16 + 9 + 4 + 1 = **30.**

(For related theory and questions, see Chapter 11: Geometry)

17. Suppose

1) **x = 4**
2) **Then x - 4 = x² - 4² (as both sides are zero)**
3) **Therefore (x - 4) = (x - 4) (x + 4)**
   Cancelling (x - 4) from both sides
4) **1 = (x + 4)**

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5) Then x = -3

Which is the wrong step?

(a) 1 to 2 **(b) 2 to 3** **(c) 3 to 4** **(d) 4 to 5**

Answer (c) *Difficulty rating: 2.5/5* *Average time for solving: 60 seconds*

Here you should be careful about analysing each step keeping in mind the validity of the previous steps.

Consider each of the steps.

Step 1

Given: "x = 4"

The variable x is assigned the value 4.

Step 2

Given: "Then $x - 4 = x^2 - 4^2$(as both sides are zero)"

From step 1: $x = 4$

⇨ $x - 4 = 4 - 4 = 0$

⇨ $x^2 - 4^2 = 4^2 - 4^2 = 0$

Hence, step 2 is correct.

Step 3

Given: "Therefore $(x - 4) = (x - 4)(x + 4)$"

Remember: $a^2 - b^2 = (a-b)(a+b)$

Thus, $x^2 - 4^2 = (x-4)(x+4)$ and $x - 4 = (x-4)(x+4)$ [From step 2]

Given: "Cancelling (x - 4) from both sides."

From step 1: $x - 4 = 0$.

That is, cancelling by (x–4) means dividing both sides by 0 which is a nonsense.

Hence, step 4 derived from step 3 is wrong.

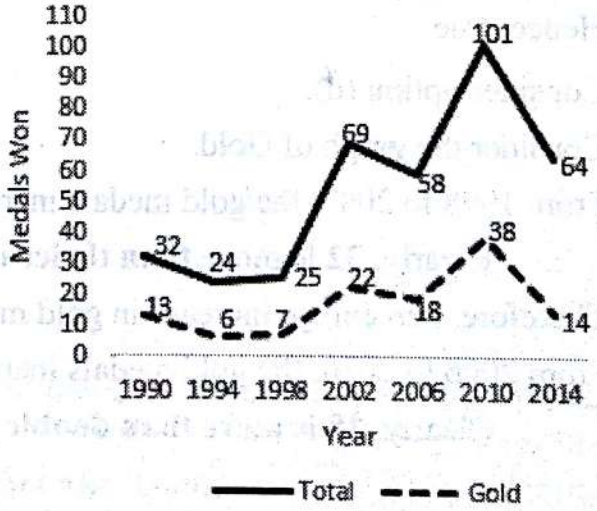
Hence, **option (c).**

18. Based on the graph, which of the following statements is NOT true?

(a) Number of gold medals increased whenever total number of medals increased

(b) Percentage increase in gold medals in 2010 over 2006 is more than the corresponding increase in total medals

(c) Every time non-gold medals together account for more than 50% of the total medals



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(d) Percentage increase in gold medals in 2010 over 2006 is more than the corresponding increase in 2002 over 1998

Answer (d) *Difficulty rating: 2.5/5* *Average time for solving: 120 seconds*

In data interpretation questions, when you are expected to find percentage, do not try to find the exact percentage using formula, which consumes much of your valuable time in examination. Instead, try to estimate logically the percentage, which is illustrated in the following solution.

Consider each of the answer options.

Consider option (a).

It is evident that, whenever the slope of gold went uphill, the slope of the total also went uphill.

Hence, true.

Consider option (b).

Consider the graph of 2006 and 2010.

The gold medals increased from 18 to 38.

Clearly, **38 is more than double of 18.**

Therefore, percentage increase in gold medals in 2010 over 2006 is **more than 100%.**

The total medals increased from 58 to 101.

Clearly, **101 is less than double of 58.**

Therefore, percentage increase in total medals in 2010 over 2006 is **less than 100%.**

Thus, Percentage increase in gold medals in 2010 over 2006 is more that the percentage increase in total medals in 2010 over 2006.

Hence, true.

Consider option (c).

Non gold medals = Total medals – Gold medals

Clearly, in each year, the number of non-gold medals is more than half of the total number of medals in that year.

Hence, true.

Consider option (d).

Consider the graph of Gold.

From 1998 to 2002, the gold medals increased from 7 to 22.

Clearly, **22 is more than thrice of 7.**

Therefore, percentage increase in gold medals in 2002 over 1998**is more than 200%.**

From 2006 to 2010, the gold medals increased from 18 to 38.

Clearly, **38 is more than double but less than thrice of 18.**

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Therefore, percentage increase in gold medals in 2010 over 2006 **is more than 100% but less than 200%.**

Thus, Percentage increase in gold medals in 2010 over 2006 is not more than the corresponding increase in 2002 over 1998.

Hence, **option (d)** is not true.

(For related theory and questions, see Chapter 13: Data Interpretation)

19. Information in DNA is in the form of sequence of 4 bases namely A, T, G and C. The proportion of G is the same as that of C, and that of A is the same as that of T. Which of the following strands of DNA will potentially have maximum diversity (i.e., maximum information content per base)?

(a) length 1000 bases with 10% G **(b) length 2000 bases with 10% A**

(c) length 2000 bases with 40% T **(d) length 1000 bases with 25% C**

Answer (d) *Difficulty rating: 3.5/5* *Average time for solving: 90 seconds*

Given 1: "Proportion of G is same as that of C"

That is Percentage of G = Percentage of C.

Given 2: "Proportion of A is same as that of T"

That is Percentage of A = Percentage of T.

Given: Maximum diversity is defined as the maximum information content per base.

The term "per base" eliminates the significance of length of the base.

That is "length 1000 bases" or "length 2000 bases" in the answer options does not affect the maximum diversity.

Now consider each of the answer options.

Consider (a)

By Given 1 and Given 2:

G = 10%, C = 10%, A = 40%, T = 40%.

Consider (b)

By Given 1 and Given 2:

G = 40%, C = 40%, A = 10%, T = 10%.

Consider (c)

By Given 1 and Given 2:

G = 10%, C = 10%, A = 40%, T = 40%.

Consider (d)

By Given 1 and Given 2:

G = 25%, C = 25%, A = 25%, T = 25%.

Clearly **option (d)** has the maximum diversity as the bases occur in same proportion and hence more combinations of G, C, A, T bases are possible.

(For related theory and questions, see Chapter 14: Observational Ability)

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20. A single celled spherical organism contains 70% water by volume. If it loses 10% of its water content, how much would its surface area change by approximately?

(a) 3% **(b) 5%** **(c) 6%** **(d) 7%**

Answer (b) *Difficulty rating: 4.9/5* *Average time for solving: 180 seconds*

Remember 1: Volume of a sphere $= \frac{4}{3}\pi$ (radius)^3.

Remember 2: Surface area of a sphere $= 4\pi$ (radius)^2

Let V_1 be the initial volume and V_2 be the volume after it loses 10% water content.
Let R_1 be the initial radius and R_2 be the radius after it loses 10% water content.

Then $V_1 = \frac{4}{3}\pi {R_1}^3$.

Its water content is 70% of V_1, that is $\frac{70}{100} V_1$.

Given: It loses 10% of its water content.

Hence it loses $\frac{10}{100}(\frac{70}{100} V_1) = \frac{7}{100} V_1$

Thus $V_2 = V_1 - \frac{7}{100} V_1 = \frac{93}{100} V_1$.

Therefore, $\frac{4}{3}\pi {R_1}^3$ becomes $\frac{93}{100}\frac{4}{3}\pi {R_1}^3$

Or $\frac{93}{100}\frac{4}{3}\pi {R_1}^3 = \frac{4}{3}\pi {R_2}^3$

$\Rightarrow \frac{93}{100}{R_1}^3 = {R_2}^3$

$\Rightarrow .93\ {R_1}^3 = {R_2}^3$

$\Rightarrow (\sqrt[3]{.93}R_1)^3 = {R_2}^3$

$\Rightarrow {R_2}^3 = (0.976\ R_1)^3$.

(Here taking cube root within the stipulated time is almost impossible without a calculator.)

Now consider the change in the surface area.

Let A_1 and A_2 be initial and final surface areas.

$A_1 = 4\pi {R_1}^2$ and $A_2 = 4\pi {R_2}^2$

$= 4\pi (0.976R_1)^2 = 4\pi {R_1}^2(0.976)^2 = 0.95 \times 4\pi {R_1}^2 = 0.95\ A_1$

Thus, A_1 become $0.95\ A_1$.

Hence the change in surface area $= A_1 - 0.95\ A_1 = 0.05\ A_1 = \frac{5}{100}\ A_1 = 5\%$ of A.

Hence, **option (b).**

(For related theory and questions, see Chapter 12: Mensuration and Quantitative comparison)

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June 2015 – Physical Sciences, Chemical Sciences

1. Let r be a positive number satisfying

$r^{(1/1234)} + r^{(-1/1234)} = 2$

Then $r^{4321} + r^{-4321} = ?$

(a) 2 **(b)** $2^{(4321/1234)}$ **(c)** 2^{3087} **(d)** 2^{1234}

Answer (a) *Difficulty rating: 2.5/5* *Average time for solving: 60 seconds*

Given: r is a positive number satisfying

$r^{(1/1234)} + r^{(-1/1234)} = 2.$

The quickest way to solve this problem is to find such an r satisfying the above equation.

Clearly r = 1 is a positive number which satisfies the above equation.

That is, $1^{(1/1234)} + 1^{(-1/1234)} = 1 + 1 = 2.$

Substituting r = 1 in the second equation gives us the solution.

$1^{4321} + 1^{-4321} = 1 + 1 = \mathbf{2.}$

(For related theory and questions, see Chapter 5: Numbers)

2. Find the height of a box of base area 24 cm × 48 cm, in which the longest stick that can be kept is 56 cm long.

(a) 8 cm **(b) 32 cm** **(c) 37.5 cm** **(d) 16 cm**

Answer (b) *Difficulty rating: 4/5* *Average time for solving: 110 seconds*

Consider a rectangular box and mark the vertices as shown in the figure.

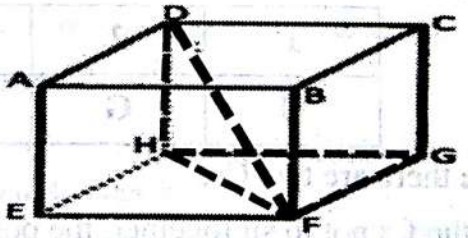
Given: GF = 24 cm, HG = 48 cm.

We are required to find DH.

Let DH = x.

The length of the longest stick that can be kept in this box is equal to the length of DF.

Therefore DF = 56 cm.



Consider ΔDHF.

By Pythagoras theorem,

$DH^2 + HF^2 = DF^2$

⇨ $x^2 + HF^2 = 56^2$ -------- (1)

Now to find x, we have to first find the length of HF.

Let us consider a triangle involving HF for the same.

Hence, consider ΔHGF.

By Pythagoras theorem,

$HF^2 = HG^2 + GF^2$

⇨ $HF^2 = 48^2 + 24^2$ ------(2)

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Substitute equation (2) in equation (1).

⇨ $x^2 + 48^2 + 24^2 = 56^2$

⇨ $x^2 = 56^2 - 48^2 - 24^2$

⇨ $x^2 = (8\times7)^2 - (8\times6)^2 - (8\times3)^2$

⇨ $x^2 = 8^2\times7^2 - 8^2\times6^2 - 8^2\times3^2$ (Since $(a\times b)^2 = a^2 \times b^2$)

⇨ $x^2 = 8^2 \times (7^2 - 6^2 - 3^2)$

⇨ $x^2 = 64 \times (49 - 36 - 9)$

⇨ $x^2 = 64 \times 4$

⇨ $x^2 = 256$

⇨ $x = 16$

(The above procedure using the "Laws of exponents" makes our task easy, as a calculator is not allowed for the examination. See the theory section of Chapter 5 (Numbers) to learn about the laws of exponents.)

Thus, height of the box = **16 cm.**

(For related theory and questions, see Chapter 11: Geometry)

3. Consider a series of letters placed in the following way:

U_G_C_C_S_I_R

Each letter moves one step to its right and the extreme right letter takes the first position, completing one operation. After which of the following numbers of operations do the Cs not sit side by side?

(a) 3 **(b) 10** **(c) 19** **(d) 25**

Answer (d) *Difficulty rating: 2.5/5* *Average time for solving: 110 seconds*

Mark the positions of letters from 1 to 7 as shown below.

1	2	3	4	5	6	7
U	G	C	C	S	I	R

Here there are two Cs.

For the Cs not to sit together, the position of the 2nd C (from left) should be **1** and the position of 1st C should be **7**.

That is,

1	2	3	4	5	6	7
C	S	I	R	U	G	C

We are required to find among the given options, the number of operations after which this situation will occur.

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Let us observe the first few operations.

Operation #1

1	2	3	4	5	6	7
R	U	G	C	C	S	I

Operation #2

1	2	3	4	5	6	7
I	R	U	G	C	C	S

Operation #3

1	2	3	4	5	6	7
S	I	R	U	G	C	C

Operation #4

1	2	3	4	5	6	7
C	S	I	R	U	G	C

Thus, operation #4 is the first operation after which we achieve our required condition.

We should continue the process until we reach a number present in the answer options.

As there are 7 positions, 7 more operations are required to achieve the required condition again.

That is, after operations #11, #18, #25, #32, …, the required condition is achieved again.

Among them, #25 is the number present in the answer options.

Hence, **option (d).**

(For related theory and questions, see Chapter 6: Probability and Arrangements)

4. An infinite row of boxes is arranged. Each box has half the volume of the previous box. If the largest box has a volume of 20 cc, what is the total volume of all the boxes?

(a) Infinite **(b) 400 cc** **(c) 40 cc** **(d) 80 cc**

Answer (c) *Difficulty rating: 3.5/5* *Average time for solving: 80 seconds*

Remember: Sum of terms of an infinite geometric progression $= \frac{a}{1-r}$, $r < 1$.

where a is the first term and r is the common ratio.

Given 1: "Each box has half the volume of the previous box."

Given 2: "The largest box has a volume of 20 cc."

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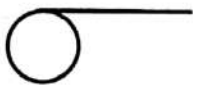
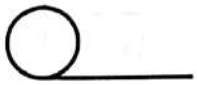
Hence the total volume of boxes $= 20 + \frac{20}{2} + \frac{20}{4} + \frac{20}{8} + \frac{20}{16} + \cdots$

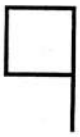
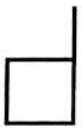
This is an infinite geometric progression with $a = 20$ and $r = \frac{1}{2}$

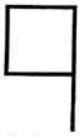
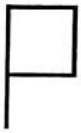
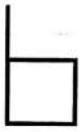
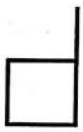
Therefore, the total volume of boxes $= \frac{20}{1-\frac{1}{2}} = \frac{20}{\frac{1}{2}} =$ **40 cc.**

(For related theory and questions, see Chapter 5: Numbers)

5. Find the missing element based on the given pattern

A.  **B.**  **C.**

A.  **B.**  **C.** **?**

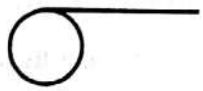
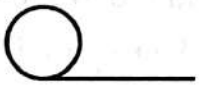
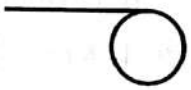
(a)  **(b)**  **(c)**  **(d)** 

Answer (b) *Difficulty rating: 4/5* *Average time for solving: 90 seconds*

Here the first and second set of figures should follow common predefined rules.

As the first set of figures is the completed one, we can find the rule from it and apply the same to the second pattern to obtain the missing figure.

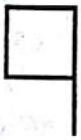
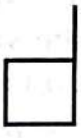
Consider the first set of figures.

A.  **B.**  **C.** 

B is the mirror image of A when the mirror is kept below A horizontally.

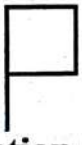
C is the mirror image of A when the mirror is kept to the left of A vertically.

Therefore, the second set of figures should also follow the same rules.

A.  B.  C. **?**

Clearly, B is the mirror image of A when the mirror is kept below A horizontally.

Hence, C should be the mirror image of A when the mirror is kept to the left of A vertically.

That is, 

Hence, **option (b).**

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(For related theory and questions, see Chapter 4: Series Formation)

6. Each of the following pairs of words hides a number, based on which you can arrange them in ascending order. Pick the correct answer:

I. Cloth reel J. Silent wonder K. Good tone L. Bronze rod

(a) L, K, J, I (b) I, J, K, L (c) K, L, J, I (d) K, J, I, L

Answer (a) *Difficulty rating: 2.5/5* *Average time for solving: 60 seconds*

Through observation, we can find the hidden number in each of the following groups of words.

I. Clo**th ree**l -------- 3

J. Silen**t wo**nder ---- 2

K. Good t**one**--------- 1

L. Bron**ze ro**d-------- 0

When arranged in the ascending order, the numbers can be listed as 0, 1, 2, 3.

That is, L, K, J, I.

Hence, **option (a).**

7. A man start his journey at 0100 Hrs local time to reach another country at 0900 Hrs local time on the same date. He starts his return journey on the same night at 2100 Hrs local time to his original place, taking the same to travel back. If the time zone of his country of visit lags by 10 hours, the duration for which the man was away from his place is

(a) 48 hours (b) 20 hours (c) 25 hours (d) 36 hours

Answer (a) *Difficulty rating: 3/5* *Average time for solving: 90 seconds*

Given 1:"The man reach another country at 0900 Hrs local time.

Given 2:"Time zone of his country of visit lags by 10 hours.

That is, the time in his country when the man reached the other country = 0900 Hrs + 10 Hrs.

= 1900 Hours.

The time taken for the forward journey = 1900 Hrs – 0100 Hrs = **18 Hours.**

Given 3: He starts his return journey to his original place on the same night at 2100 Hrs local time.

Hence, the time he spent in the other country = 2100 Hrs – 0900 Hrs = **12 Hours.**

Given 4: He takes the same time to travel back.

Thus, time taken for the return journey = The time taken for the forward journey = **18 Hours.**

Duration for which the man was away from his place

= Time for forward journey + Time he spent in the other country + Time for return journey

= 18 + 12 + 18 = **48 Hours.**

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(For related theory and questions, see Chapter 14: Observational Ability)

8. Which of the following values is same as $2^{2^{2^2}}$?

(a) 2^6 **(b) 2^8** **(c) 2^{16}** **(d) 2^{222}**

Answer (c) *Difficulty rating: 2.5/5* *Average time for solving: 50 seconds*

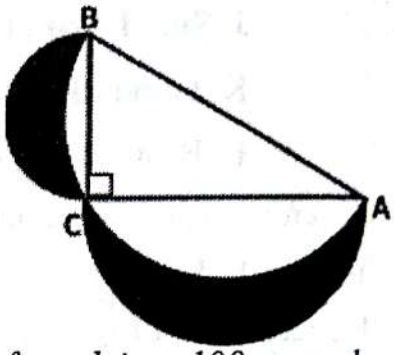
By applying the laws of exponents, we get

$2^{2^{2^2}} = 2^{[2^{(2^2)}]} = 2^{[2^4]} = 2^{16}$. (Since $a^{x^y} = a^{[x^y]}$)

Hence, **option (c).**

(For related theory and questions, see Chapter 5: Numbers)

9. ABC is a right angled triangle inscribed in a semicircle. Smaller semicircles are drawn on sides BC and AC. If the area of the triangle is a, what is the total area of the shaded lunes?



(a) a **(b) πa**

(c) a/π **(d) a/2π**

Answer (a)

Difficulty rating: 3.5/5 *Average time for solving: 100 seconds*

Here there are three semicircles on three sides of the triangle ABC.

That is, the semicircle on AB (semicircle made by the arc ACB), the semicircle on BC and the semicircle on AC.

Let us denote them by S_{AB}, S_{BC} and S_{AC} respectively.

Area of the shaded lunes = (Area of S_{BC} + Area of S_{AC} + Area of ΔABC) - Area of S_{AB}.

Given: Area of $\Delta ABC = a$.

Remember 1: Area of semicircle $= \frac{\pi}{2}$ radius^2.

Here the radius of the semicircles are half the length of the corresponding sides of the triangle.

Let BC = b and AC = h.

Then, $AB = \sqrt{b^2 + h^2}$. [By Pythagoras theorem]

Therefore, radius of $S_{BC} = \frac{b}{2}$, radius of $S_{AC} = \frac{h}{2}$, radius of $S_{AB} = \frac{\sqrt{b^2+h^2}}{2}$.

Thus,

$$\text{Area of } S_{BC} = \frac{\pi}{2}\left(\frac{b}{2}\right)^2 = \frac{\pi b^2}{8}$$

$$\text{Area of } S_{AC} = \frac{\pi}{2}\left(\frac{h}{2}\right)^2 = \frac{\pi h^2}{8}$$

$$\text{Area of } S_{AB} = \frac{\pi}{2}\left(\frac{\sqrt{b^2+h^2}}{2}\right)^2 = \frac{\pi(b^2+h^2)}{8}$$

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Therefore,

$$\text{Area of the shaded lunes} = \frac{\pi b^2}{8} + \frac{\pi h^2}{8} + a - \frac{\pi(b^2+h^2)}{8}$$

$$= \frac{\pi(b^2+h^2)}{8} + a - \frac{\pi(b^2+h^2)}{8}$$

$$= a.$$

Hence **option (a).**

(For related theory and questions, see Chapter 12: Mensuration and Quantitative Comparison)

10. The product of the perimeter of a triangle, the radius of its in-circle, and a number gives the area of the triangle. The number is

(a) ¼ **(b) 1/3** **(c) ½** **(d) 1**

Answer (c) *Difficulty rating: 4/5* *Average time for solving: 120 seconds*

Consider ΔABC with length of sides AB = a, BC = b and CA = c. (See Figure)

Let r be the radius and O be the centre of the incircle of the triangle.

Join O to each of the vertices of the triangle to obtain OA, OB and OC.

Join O to meet the points of intersection of the incircle and the triangle. (See the dotted lines)

These three lines are in fact the radius of the incircle.

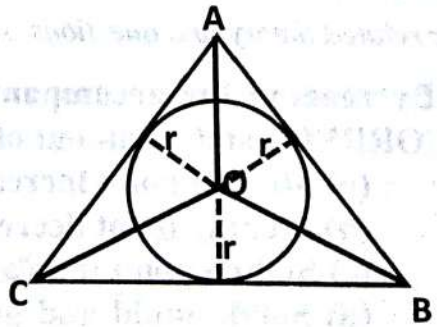
Observe that the sides AB, BC and AC are tangent to the incircle and we know that **tangent and radius meet at right angles.**

Therefore, the dotted lines are the corresponding heights (or altitudes) of the three newly formed triangles - ΔAOC, ΔBOC and ΔBOA.

From the figure, it is clear that,

Area of ΔABC = Area of ΔAOC + Area of ΔBOC + Area of ΔBOA

Remember: Area of triangle $= \frac{1}{2}$ Base × Height.



Thus,

$$\text{Area of } \Delta ABC = \frac{1}{2}ar + \frac{1}{2}br + \frac{1}{2}cr.$$

$$= \frac{1}{2} \times (a+b+c) \times r$$

Here a + b + c is the perimeter of the ΔABC.

Therefore,

Area of $\Delta ABC = \frac{1}{2} \times$ Perimeter × Radius

Thus the required number is $\frac{1}{2}$.

(For related theory and questions, see Chapter 11: Geometry)

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11. If

$$\begin{array}{r} 2\ a \\ \times\ b\ 2 \\ \hline c\ 6 \\ \hline 8\ 4 \\ \hline 8\ d\ 6 \end{array}$$

Here a, b, c and d are digits.
Then a + b =

(a) 4 **(b) 9** **(c) 11** **(d) 16**

Answer (c) *Difficulty rating: 2.5/5* *Average time for solving: 70 seconds*

Given:

$$\begin{array}{r} 2\ a \\ \times\ b\ 2 \\ \hline c\ 6 \end{array}$$

That is, 2 times a gives the last digit 6.

Clearly, a is either 3 or 8.

Given:

$$\begin{array}{r} 2\ a \\ \times\ b\ 2 \\ \hline c\ 6 \\ 8\ 4 \end{array}$$

If a = 3

23b = 84.which is impossible as 84 is not a multiple of 23.

If a = 8

28b = 84

⇨ b = 3.

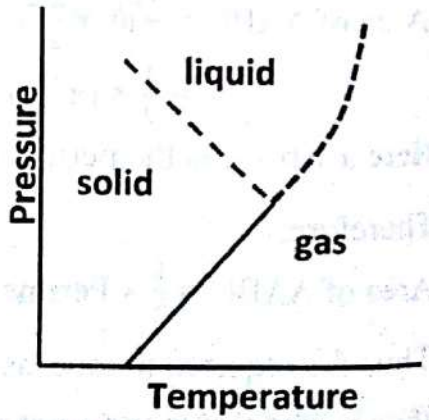
Hence a = 8, b = 3.

Therefore, **a + b = 11.**

(For related theory and questions, see Chapter 5: Numbers)

12. By reading the accompanying graph, determine the INCORRECT statement out of the following.

(a) Melting point increases with pressure
(b) Melting point decreases with pressure
(c) Boiling point increases with pressure
(d) Solid, liquid and gas can co-exist at the same pressure and temperature



Answer (a)

Difficulty rating: 3.5/5

Average time for solving: 90 seconds

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Remember: **The melting point of a solid is the temperature at which it changes from solid to liquid. The boiling point of a liquid is the temperature at which a liquid boils and turns to vapour (gas).**

Thus, for a particular value of the pressure in the y axis, the value of melting point or boiling point is the corresponding value of temperature in the x axis.

Here the melting point is the dotted line in between the solid and liquid region and boiling point is the dotted curve in between the liquid and gas region.

Consider options (a) and (b).

Both the options analyses the variation of melting point with respect to the increase in pressure.

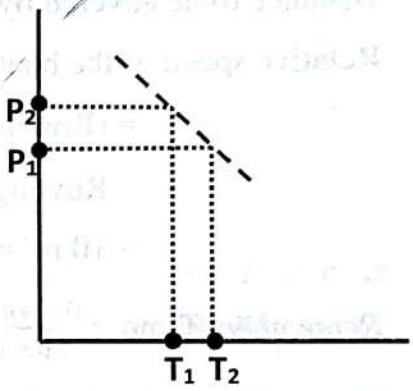
Consider the dotted line that of the melting point.

Let P_1 and P_2 be two points in the pressure axis such that $P_1 < P_2$.

Pressure changing from P_1 to P_2can be considered as an increase in pressure.

From the figure, it is evident that when P_1 changes to P_2, the temperature changes from T_2 to T_1 and $T_1 < T_2$.

That is, the temperature (melting point) decreases.



Figure

Hence, **Melting point decreases with pressure.**

Therefore, the statement that melting point increases with pressure is wrong.

It is left as an exercise for the readers to check the correctness of option (c), proceeding the same way as above. (Note: Boiling point is the dotted curve in between the liquid and gas region.)

Consider option (d).

In the graph, the meeting point of the three curves (lines) is the point at which solid, liquid and gas can co-exist at the same pressure and temperature. Hence, correct.

Therefore, **option (a)** is the only incorrect option.

(For related theory and questions, see Chapter 13: Data Interpretation)

13. A float is drifting in a river, 10 m downstream of a boat that can be rowed at a speed of 10 m/minute in still water. If the boat is rowed downstream, the time taken to catch up with the float

(a) will be 1 minute
(b) will be more than 1 min
(c) will be less than 1 min

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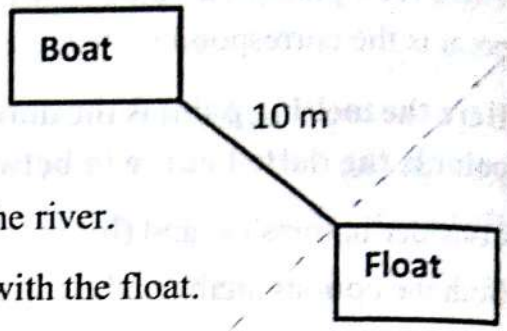
(d) can be determined only if the speed of the river is known

Answer (a) *Difficulty rating: 2.5/5* *Average time for solving: 50 seconds*

Here 'float' means 'a moving object in water'.

Given: The float is 10 m downstream of the boat.



Since there is no external force on the float,

Speed of the float = Speed of the river.

However,

Speed of the boat = Rowing speed of the boat + Speed of the river.

We are required to find the time taken by boat to catch up with the float.

Distance to be covered by the boat = 10 m.

Relative speed of the boat = Speed of the boat – Speed of the float.

= (Rowing speed of the boat + Speed of the river) –Speed of the river

= Rowing Speed of the boat.

= 10 m/ minute.

Remember: $\text{Time} = \frac{\text{Distance}}{\text{Speed}}$

Hence, Time taken by the boat to catch up with the float $= \frac{\text{Distance to be covered by the boat}}{\text{Relative speed of the boat}}$

$= \frac{10}{10}$

= 1 minute.

Hence, **option (a).**

(For related theory and questions, see Chapter 3: Moving Locomotive Problem)

14. If you change only one observation from a set of 10 observations, which of the following will definitely change?

(a) Mean **(b) Median** **(c) Mode** **(d) Standard deviation**

Answer (a) *Difficulty rating: 4/5* *Average time for solving: 90 seconds*

Here "change only one observation" means removing one observation and adding a different observation in place of the removed observation.

We are required to find the answer option that will definitely change even if we take any set of 10 observations and change any one observation in that set.

Consider (a)

Remember: Mean is defined as

Mean = (Sum of Observations)/(Number of observations)

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Here number of observations = 10

Thus, Mean = (Sum of Observations)/10.

If we change only one observation, the sum of observations will definitely change.

Hence, the numerator will definitely change.

As the numerator only changes, the fraction will definitely change.

Therefore, the mean will definitely change.

Now those of you who are inquisitive enough to understand why the other options fail, can read further.

To show that the other options does not change in every case, we just have to provide one counter example in which the change of one observation does not affect the median or mode or standard deviation of that corresponding set of 10 observations.

Constructing such sets of observations is the tricky task before us.

Consider (b)

Consider the set of observations {1, 2, 3, 4, 10, 20, 31, 32, 33, 34}

Remember: To find the median, we arrange the observations in increasing order. If there is an odd number of observations, the median is the middle value. If there is an even number of observations, the median is the average of the two middle values.

Here there are 10 observations.

Therefore, median is the average of 5^{th} and 6^{th} observations.

Hence, Median $= \frac{10+20}{2} = 15$.

Now, if we change the value 4 by the value 6, the median is still 15, unchanged.

Consider (c)

Consider the set of observations {1, 2, 3, 4, 5, 6, 6, 7, 7, 7}

Remember: The mode is the most frequently occurring value.

Here 7 is the most frequently occurring observation and hence the mode.

Now, if we change the value 1 by the value 10, the mode is still 7, unchanged.

Consider (d)

Consider the set of observations {1, 3, 5, 7, 9, 11, 13, 15, 17, 19}

Remember: Variance for N observations is defined as:

$$\textbf{Variance} = \sum \frac{1}{N}(x_i - \textbf{Mean})^2$$

Here the Mean $= \frac{1+3+5+7+9+11+13+15+17+19}{10} = \frac{100}{10} = 10$

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Then, Variance =

$$\frac{(1-10)^2 + (3-10)^2 + (5-10)^2 + (7-10)^2 + (9-10)^2 + (11-10)^2 + (13-10)^2 + (15-10)^2 + (17-10)^2 + (19-10)^2}{10} \text{---(1)}$$

$$= \frac{(-9)^2 + (-7)^2 + (-5)^2 + (-3)^2 + (-1)^2 + (1)^2 + (3)^2 + (5)^2 + (7)^2 + (9)^2}{10} \quad \text{--- (2)}$$

$$= \frac{81 + 49 + 25 + 9 + 1 + 1 + 9 + 25 + 49 + 81}{10}$$

$$= \frac{330}{10}$$

$$= 33.$$

Standard Deviation = $\sqrt{\text{Variance}} = \sqrt{33}$.

Now, let us change the value 1 by the value 19 in the set of observations.

Observe the change in equation (1).

$(1 - 10)^2$ is changed as $(19 - 10)^2$.

In equation (2), $(-9)^2$ is changed as 9^2.

However $(-9)^2 = 9^2 = 81$.

Hence, the Standard deviation is $\sqrt{33}$ itself.

Thus, this is a case when standard deviation doesn't change even if we change one observation.

Hence **option (a).**

(For related theory and questions, see Chapter 9: Average)

15. In a fast moving car with open windows, the driver feels a continuous incoming breeze. The pressure inside the car, however, does not keep increasing because,

(a) air coming in from the front window goes out from the rear.

(b) air comes in as well as goes out through every window but the driver only feels the incoming one.

(c) no air actually comes in and the feeling of breeze is an illusion.

(d) cool air reduces the temperature therefore the pressure does not increase.

Answer (b) *Difficulty rating: 2.5/5* *Average time for solving: 80 seconds*

This is one such question that checks your common sense.

Let us consider each of the answers to weed out the wrong options and hence find the right one.

Consider (a)

This is wrong as air coming in from the front window goes out not only through the rear window.

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Consider (c)

Wrong. Air does come in.

Consider (d)

Wrong. In the question, it is nowhere mentioned that the breeze is cool. So it can be cool air or hot air depending on the outside temperature.

Thus, the three options are wrong and the remaining one should be the correct option.

Hence, **option (b).**

16. The maximum number of points formed by intersection of all pairs of diagonals of convex octagon is

(a) 70 **(b) 400** **(c) 120** **(d) 190**

Answer (a) *Difficulty rating: 4.5/5* *Average time for solving: 100 seconds*

We know that an octagon is a polygon with 8 sides and a convex octagon is an octagon in which no line segment between two points on the boundary ever goes outside the octagon.

Remember: **The maximum number of points formed by intersection of all pairs of diagonals of an n-sided polygon = ${}^{n}C_4$.**

Here n = 8.

Therefore, maximum number of points formed by intersection of all pairs of diagonals of an n-sided polygon = ${}^{8}C_4$ = **70.**

(For related theory and questions, see Chapter 11: Geometry)

17. A 12 m × 4 m rectangular roof resting on four 4 m tall thin poles. Sunlight falls on the roof at an angle of 45° from the east, creating a shadow on the ground. What will be the area of the shadow?

(a) 24 m² **(b) 36 m²** **(c) 48 m²** **(d) 60 m²**

Answer (c) *Difficulty rating: 4/5* *Average time for solving: 90 seconds*

When the sunlight falls at an angle of 45°, the object and its shadow has the same shape and dimension.

That is, the shadow will be a rectangle of dimension 12 m × 4m.

Therefore, Area of the shadow = Area of the rectangular roof = $12 \times 4 = 48$ m².

Hence, **option (c).**

(For related theory and questions, see Chapter 7: Distances and Directions)

18. An ant can lift another ant of its size whereas an elephant cannot lift another elephant of its size, because

(a) ant muscle fibers are stronger than elephant muscle fibers.

(b) ant has proportionately thicker legs than elephant

(c) strength scales as the square of the size while weight scales as cube of the size

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(d) ants work cooperatively, whereas elephants work as individuals

Answer (c) *Difficulty rating: 3/5* *Average time for solving: 90 seconds*

As this question is asked in the general aptitude section, we are supposed to answer the question logically.

Let us try to eliminate the wrong options.

Consider (a)

Clearly not.

Consider (b)

Wrong. It is evident that the leg of an ant is much thinner than its overall size compared to that of the elephant.

Consider (d)

Wrong. The question is asked about a single ant. In this option, it is stated about a group of ants.

Thus, we could eliminate three wrong options.

Hence, we can induce that **option (c)** is the correct option even without understanding the exact physical theory behind it.

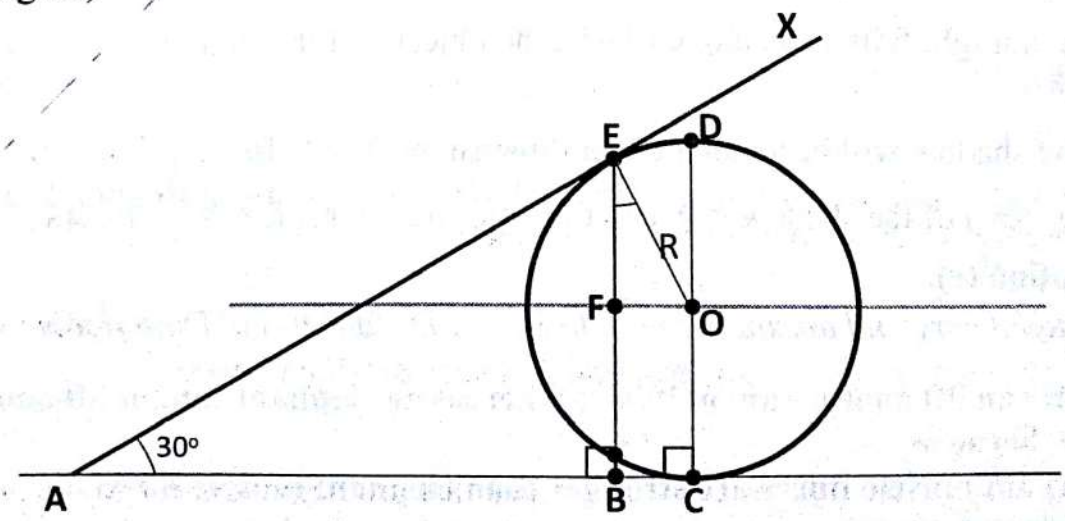
19. An inclined plane rests against a horizontal cylinder of radius R. If the plane makes an angle of 30° with the ground, the point of contact of the plane with the cylinder is at a height of

(a) 1.500 R **(b) 1.866 R** **(c) 1.414 R** **(d) R**

Answer (b) *Difficulty rating: 4.5/5* *Average time for solving: 150 seconds*

For simplicity, let us deal this question in two dimension.

So the horizontal cylinder can be considered as a circle and the inclined plane as a line segment say AX. (See figure)



Construct BE and CD perpendicular to the ground as shown in the figure.

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Note that OE = DO = OC = Radius = R.

*G*iven: "The plane makes an angle of 30° with the ground."

That is, ∠EAB = 30°.

⇨ ∠BEA = 180 – (30 + 90) = 60° [Angle sum property]

We are required to find the height at which the plane touches with the cylinder.

That is, we are required to find BE.

Remember: A tangent and radius meet at right angles.

AX is the tangent and OE is the radius of the circle.

Therefore, ∠OEA = 90°.

⇨ ∠OEF = 30° [Since ∠BEA = 60°]

Consider the right triangle OFE

$$\cos 30^\circ = \frac{\text{Adjacent side}}{\text{Hypotenuse}} = \frac{EF}{R}$$

$$\Rightarrow \frac{\sqrt{3}}{2} = \frac{EF}{R}$$

$$\Rightarrow EF = \frac{\sqrt{3}R}{2}$$

Clearly BF = OC = R

$$BE = EF + BF$$

$$= \frac{\sqrt{3}R}{2} + R$$

$$= \left(\frac{\sqrt{3}+2}{2}\right)R$$

$$\approx \left(\frac{1.73+2}{2}\right)R$$

$$\approx \mathbf{1.866R.}$$

(For related theory and questions, see Chapter 11: Geometry)

20. What is the maximum number of parallel, non-overlapping cricket pitches (length 24 m, width 3 m) that can be laid in a field of diameter 140 m, if the boundary is required to be at least 60 m from the centre of any pitch?

(a) 6 **(b) 7** **(c) 12** **(d) 4**

Answer (b) *Difficulty rating: 3.5/5* *Average time for solving: 100 seconds*

Consider a circular field and consider any diameter of length 140 m.

We are asked to place parallel pitches.

So assume that we are placing parallel pitches that are perpendicular to the above diameter. (See figure)

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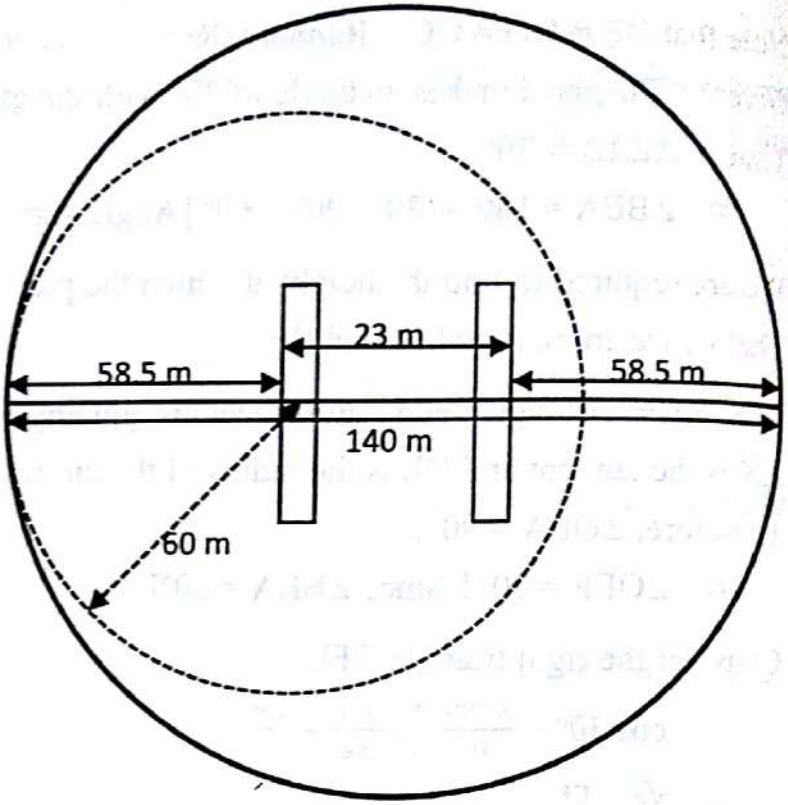
Given: The boundary is required to be at least 60 m from the centre of any pitch.



As the width of each pitch is 3 cm, distance to the centre of the pitch will be half of the width that is 1.5 cm.

Therefore, the first and last pitch should be $60 - 1.5 = 58.5$ m away from the corresponding boundaries.

Hence, the total width available for placing the pitches is

$140 - (58.5 + 58.5) = 23$ m.

In 23 m, a maximum of 7 non-overlapping pitches of width 3 m can be placed perpendicular to the above-mentioned diameter.

[Some of you may have the doubt that if we arrange in this way, whether the rule that the boundary is required to be at least 60 m from the centre of any pitch is satisfied in every direction??
The possibility of constructing a circle of diameter 60 m from the centre of the left most pitch clarifies your doubt. (See the dotted circle in the figure)]

(For related theory and questions, see Chapter 11: Geometry)

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PREVIOUS YEARS' QUESTION PAPERS

(Topic-wise detailed solution in the next section)

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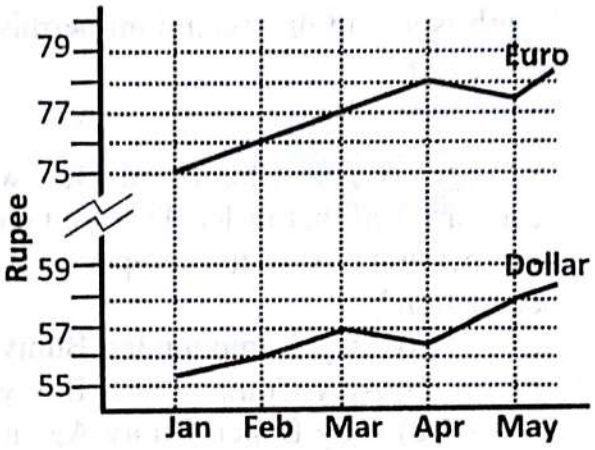
1. DECEMBER 2014
MATHEMATICAL SCIENCES, LIFE SCIENCES, EARTH SCIENCES

1. What is the 94^{th} term of the following sequence?
1, 1, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3, 4, 4, 4, 4, 4, 4, 4, 4, ...
(a) 8 (b) 9 (c) 10 (d) 11

2. Which of the following numbers is a perfect square?
(a) 1022121 (b) 2042122 (c) 3063126 (d) 4083128

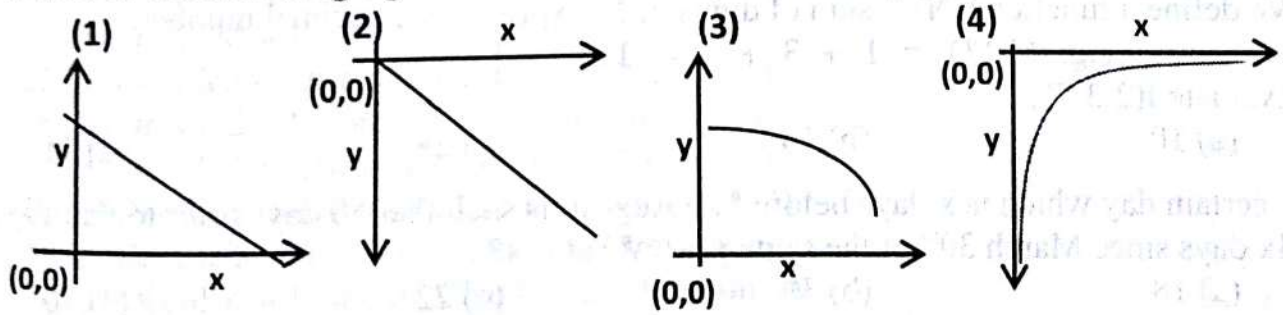
3. The equation $m^2 - 33n + 1 = 0$, where m & n are integers has
(a) no solution
(b) exactly one solution
(c) exactly two solutions
(d) infinitely many solutions

4. The following graphs depict variation in the value of Dollar and Euro in terms of Rupee over six months. Which of the following statements is true?
(a) Values of Dollar and Euro rose steadily from January to June.
(b) Values of Dollar and Euro rose by equal rate between January to March.
(c) The rise in the value of Dollar from April to May is three times the fall in Euro during the same period.
(d) Values of Dollar and Euro rose equally between May and June.



5. What is the maximum number of whole laddoos having diameter of 6 cm that can be packed in a box whose inner dimensions are $24 \times 18 \times 17$ cm^3?
(a) 24 (b) 30 (c) 33 (d) 36

6. Which of the following figures best shows that y is inversely proportional to x?



(a) 1 (b) 2 (c) 3 (d) 4

7. What is the next term in the following sequence?
7, 11, 13, 17, 19, 23, 29, ...
(a) 37 (b) 35 (c) 31 (d) 33

8. What is the area of the triangle bounded by the lines
$y = 2x$, $y = -2x$ and $y = 6$?
(a) 38 (b) 18 (c) 12 (d) 24

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9. Three volumes of a Hindi book, identical in shape and size, are next to each other in a shelf, all up-right, so that their spines are visible, left to right: I, II, III. A worm starts eating from the outside front cover of volume I, and eats its way horizontally to the outside back cover of volume III. What is the distance travelled by the worm, if each volume is 6 cm thick?

(a) 6 cm (b) 12 cm (c) 18 cm (d) a little more than 18 cm.

10. A cubical piece of wood was filed to make it into the largest possible sphere. What fraction of the original volume was removed?

(a) more than ¾ (b) ½ (c) slightly less than ½ (d) slightly more than 1/2

11. Two platforms are separated horizontally by a distance A and vertically by a distance B. They are to be connected by a staircase having identical steps. If the minimum permissible length is a, and the maximum permissible step height is b, the number of steps the staircase can have is?

(a) $\geq B/b$ (b) $\leq A/a$ (c) $\geq B/b$ and $\leq A/a$ (d) $\leq B/b$ and $\geq A/a$

12. Ajay, Bunty, Chinu and Deb were agent, baker, compounder and designer, but not necessarily in that order. Deb told baker that Chinu is on his way. Ajay is sitting across the designer and next to the compounder. The designer didn't say anything. What is each person's occupation?

(a) Ajay-Compounder, Bunty-Designer, Chinu-Baker, Deb-Agent
(b) Ajay-Compounder, Bunty-Baker, Chinu-Agent, Deb-Designer
(c) Ajay-Baker, Bunty-Agent, Chinu-Designer, Deb-Compounder
(d) Ajay-Baker, Bunty-Designer, Chinu-Agent, Deb-Compounder

13. Every month, the price of a particular commodity falls in this order:

1024, 640, 400, 250 ...

What is the next value?

(a) 156.25 (b) Approximately 39 (c) 64 (d) 40

14. We define a function f(N) = sum of digits of N, expressed as decimal number.

e.g. f(137) = 1 + 3 + 7 = 11:

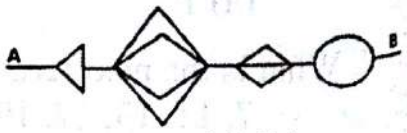
Evaluate $f(2^7 3^5 5^6)$

(a) 10 (b) 18 (c) 28 (d) 11

15. A certain day which is x days before 17th August, is such that 50 days prior to that day, it was 4x days since March 30th of the same year. What is x?

(a) 18 (b) 30 (c) 22 (d) 16

16. A mouse has to go from a point A to B without retracing any part of the path, and never moving backwards. What is the total number of distinct paths that the mouse may take to go from A to B?



(a) 11 (b) 48 (c) 72 (d) 24

17. The sum of first n natural numbers with one of them missed is 42. What is the number that was missed?

(a) 1 (b) 2 (c) 3 (d) 4

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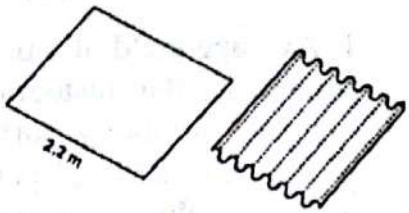
18. A 2.2 m wide rectangular steel plate is corrugated as shown in the diagram. Each corrugation is a semi-circle in cross section having a diameter of 7 cm. What will be the width of the steel after it is corrugated?

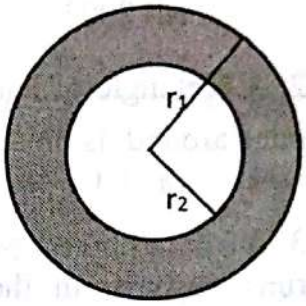


(a) 1.4 m (b) 1.6 m (c) 0.7 m (d) 1.1 m

19. If N, E and T are distinct positive integers such that $N \times E \times T = 2013$, then which of the following is the maximum possible sum of N, E and T?

(a) 39 (b) 2015 (c) 675 (d) 671

20. The areas of inner circle and the shaded ring are equal. The radii r_1 and r_2 are related by



(a) $r_1 = r_2$ (b) $r_1 = r_2\sqrt{2}$ (c) $r_1 = r_2\sqrt{3}$ (d) $r_1 = 2r_2$

Answer key

1. c (Detailed solution on Chapter 5, Question 1).
2. a (Detailed solution on Chapter 5, Question 2).
3. a (Detailed solution on Chapter 5, Question 3).
4. c (Detailed solution on Chapter 13, Question 1).
5. c (Detailed solution on Chapter 11, Question 1).
6. d (Detailed solution on Chapter 13, Question 2).
7. c (Detailed solution on Chapter 4, Question 1).
8. b (Detailed solution on Chapter 12, Question 1).
9. a (Detailed solution on Chapter 14, Question 1).
10. c (Detailed solution on Chapter 12, Question 2).
11. c (Detailed solution on Chapter 14, Question 2).
12. d (Detailed solution on Chapter 15, Question 1).
13. a (Detailed solution on Chapter 4, Question 2).
14. b (Detailed solution on Chapter 5, Question 4).
15. a (Detailed solution on Chapter 1, Question 1).
16. b (Detailed solution on Chapter 5, Question 5).
17. c (Detailed solution on Chapter 5, Question 6).
18. a (Detailed solution on Chapter 12, Question 3).
19. c (Detailed solution on Chapter 5, Question 7).
20. b (Detailed solution on Chapter 12, Question 4).

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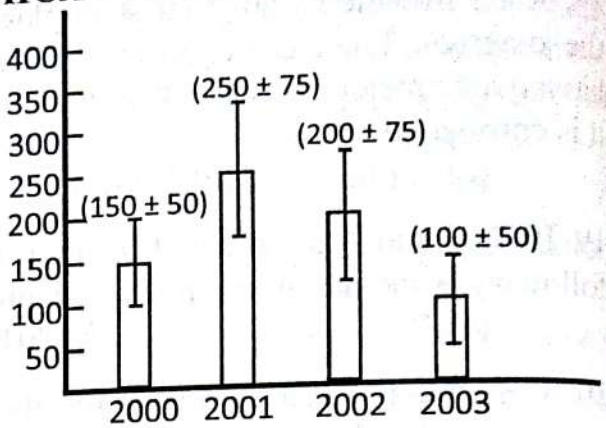
2 . DECEMBER 2014

PHYSICAL SCIENCES, CHEMICAL SCIENCES

1. Average yield of a product in different years is shown in the histogram. If the vertical bars indicate variability during the year, then during which year was the percent variability over the average of the year is least?

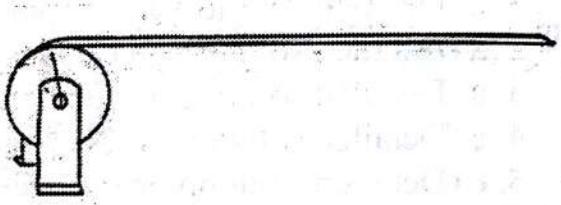


(a) 2000
(b) 2001
(c) 2002
(d) 2003

2. A rectangle of length d and breadth $\frac{d}{2}$ is revolved once completely around its length and once around its breadth. The ratio of volumes swept in the two cases is

(a) 1:1 (b) 1:2 (c) 1:3 (d) 1:4

3. A long ribbon is wound around a spool up to a radius R. Holding the tip of the ribbon, a boy runs away from the spool with a constant speed maintaining the unwound portion of the ribbon horizontal. In 4 minutes, the radius of the wound portion becomes $\frac{R}{\sqrt{2}}$. In what further time, it will become $\frac{R}{2}$



(a) $\sqrt{2}$ min (b) 2 min (c) $2\sqrt{2}$ min (d) 4 min

4. If n is a positive integer, then

$n(n+1)(n+2)(n+3)(n+4)(n+5)(n+6)$ is divisible by

(a) 3 but not 7 (b) 3 and 7 (c) 7 but not 3 (d) neither 3 nor 7

5. The area (in m^2) of a triangular park of dimension 50 m, 120 m and 130 m is

(a) 3000 (b) 3250 (c) 5550 (d) 7800

6. Lunch-dinner pattern of a person of m days is given below. He has a choice of a VEG or a NON-VEG meal for his lunch/dinner

1. If he takes a NON-VEG lunch, he will have only VEG for dinner
2. He takes NON-VEG dinner for exactly 9 days
3. He takes VEG lunch for exactly 15 days
4. He takes a total of 14 NON-VEG meals.

What is m?

(a) 18 (b) 24 (c) 20 (d) 38

7. A bank offers a scheme wherein deposits made for 1600 days are doubled in value, the interest being compounded daily. The interest accrued on the deposit of Rs.1000/- over the first 400 days would be Rs.

(a) 250 (b) 183 (c) 148 (d) 190

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8. What is the next number in the following sequence?

2, 3, 4, 7, 6, 11, 8, 15, 10, ...

(a) 12 (b) 13 (c) 17 (d) 19

9. Two locomotives are running towards each other with speeds of 60 and 40 km/hr. An object keeps on flying to and fro from the front tip of one locomotive to the front tip of the other with a speed of 70 km/hr. After 30 minutes, the two locomotives collide and the object is crushed. What distance did the object cover before being crushed?

(a) 50 km (b) 45 km (c) 35 km (d) 10 km

10. Weights (in kg) of 13 persons are given below:

70, 72, 74, 76, 78, 80, 82, 84, 86, 88, 90, 92, 94

Two new persons having weights 100 kg and 79 kg join the group. The average weight of the group increases by

(a) 0 kg (b) 1 kg (c) 1.6 kg (d) 1.8 kg

11. A code consists of at most two identical letters followed by at most four identical digits. The code must have at least one letter and one digit. How many distinct codes can be generated using letters A to Z and digits 1 to 9?

(a) 936 (b) 1148 (c) 1872 (d) 2574

12. Two solid iron spheres are heated to 100°C and then allowed to cool. One has the size of a football, the other has the size of a pea. Which sphere will attain the room temperature (constant) first?

(a) The bigger sphere
(b) The smaller sphere
(c) Both spheres will take same time
(d) It will depend on the room temperature

13. Find the missing letter.

(a) L
(b) Q
(c) N
(d) O

A	?	Q	E
C	M	S	C
E	K	U	A
G	I	W	Y

14. The least significant bit of an 8-bit binary number is zero. A binary number whose value is 8 times the previous number has

(a) 12 bits ending with 3 zeros (b) 11 bits ending with 4 zeros
(c) 11 bits ending with 3 zeros (d) 12 bits ending with 4 zeros

15. A person sells two objects at Rs.1035/- each. On the first object he suffers a loss of 10% while on the second he gains 15%. What is his net loss/gain percentage?

(a) 5 % gain (b) < 1% gain (c) < 1% loss (d) no loss, no gain

16. Continue the sequence 2, 5, 10, 17, 28, 41, _, _, _

(a) 58, 77, 100 (b) 64, 81, 100 (c) 43, 47, 53 (d) 55, 89, 113

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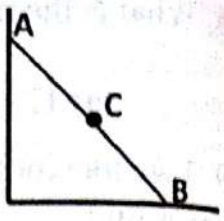
17. A ladder rests against a wall as shown. The top and bottom ends of the ladder are marked A and B. The base B slips. The central point C of the ladder falls along



(a) a parabola (b) the arc of a circle
(c) a straight line (d) a hyperbola

18. 20% of students of a particular course get jobs within one year of passing. 20% of the remaining students gets jobs by the end of second year of passing. If 16 students are still jobless, how many students had passed the course?

(a) 32 (b) 64 (c) 25 (d) 10

19. Binomial theorem in algebra gives $(1 + x)^n = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$, where $a_0, a_1, \ldots, a_n$ are constants depending on n.

What is the sum of $a_0 + a_1 + a_2 + \cdots + a_n$?

(a) 2^n (b) n (c) n^2 (d) $n^2 + n$

20. A sphere is made up of very thin concentric shells of increasing radii (leaving no gaps). The mass of an arbitrary chosen shell is

(a) equal to the mass of the preceding shell (b) proportional to its volume
(c) proportional to its radius (d) proportional to its surface area

Answer key

1. b (Detailed solution on Chapter 13, Question 3).
2. b (Detailed solution on Chapter 12, Question 5).
3. b (Detailed solution on Chapter 12, Question 6).
4. b (Detailed solution on Chapter 5, Question 8).
5. a (Detailed solution on Chapter 12, Question 7).
6. c (Detailed solution on Chapter 15, Question 2).
7. d (Detailed solution on Chapter 10, Question 1).
8. d (Detailed solution on Chapter 4, Question 3).
9. c (Detailed solution on Chapter 3, Question 1).
10. b (Detailed solution on Chapter 9, Question 1).
11. c (Detailed solution on Chapter 6, Question 1).
12. a (Detailed solution on Chapter 14, Question 3).
13. d (Detailed solution on Chapter 4, Question 4).
14. b (Detailed solution on Chapter 5, Question 9).
15. b (Detailed solution on Chapter 10, Question 2).
16. a (Detailed solution on Chapter 4, Question 5).
17. b (Detailed solution on Chapter 11, Question 2).
18. c (Detailed solution on Chapter 8, Question 1).
19. a (Detailed solution on Chapter 5, Question 10).
20. b (Detailed solution on Chapter 14, Question 4).

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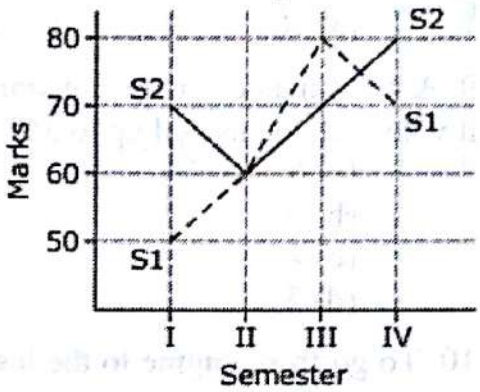
3. JUNE 2014

MATHEMATICAL SCIENCES, LIFE SCIENCES, EARTH SCIENCES

1. Marks obtained by two students S1 and S2 in a four semester course are plotted in the following graph. Which of the following statements is true?



(a) S2 got higher marks than S1 in all four semesters
(b) Over four semesters, S1 improved by a higher percentage compared to S2
(c) Total marks of S1 and S2 are equal
(d) S1 and S2 did not get the same marks in any semester

2. The following table shows the price of diamond crystals of a particular quality:

Wt. of a diamond crystal in carat	Price per carat (in lakhs)
1	4
2	8
3	12
4	16

What will be the price (in lakh Rs.) of a 2.5 carat diamond crystal?

(a) 10 (b) 20 (c) 25 (d) 50

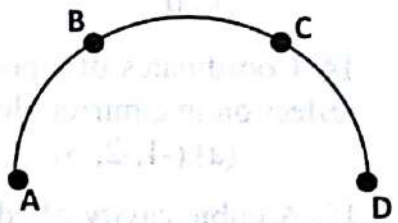
3. How many digits are there in $2^{17} \times 3^2 \times 5^{14} \times 7$?

(a) 14 (b) 15 (c) 16 (d) 17

4. A man on the equator moves along 0° longitude up to 45° N. He then turns east and moves up to 90° E, and turns to the equator along 90° E. The distance covered in multiples of Earth's radius R is

(a) $(\frac{\sqrt{3}}{4}\pi)$ R (b) $(\frac{\pi}{2}+\frac{\pi}{4\sqrt{2}})$ R (c) $(\frac{\pi}{2}+\frac{\pi}{2\sqrt{2}})$R (d) $(\frac{\pi}{4}+\frac{\pi}{\sqrt{2}})$ R

5. On a semi-circle of diameter 10m drawn on a horizontal ground are standing 4 boys A,B, C and D with distances AB = BC = CD. The length of the line segment joining A and B is



(a) 5m (b) 6m (c) 7m (d) $\frac{5\pi}{3}$

6. What is the next number in the following sequence?

2, 3, 5, 6, 3, 4, 7, 12, 4, 5, 9 ... ...

(a) 10 (b) 20 (c) 13 (d) 16

7. The following sum is: $1 + 1 - 2 + 3 - 4 + 5 - 6 \ldots - 20 = ?$

(a) 10 (b) -10 (c) -11 (d) -9

8. You are given 100 verbs using which you have to form sentences containing at least one verb, without repeating the verbs, under the condition that the number of verbs (from this set

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of 100) in any two sentences should not be equal. The maximum number of sentences you can form is

(a) 10 (b) 13 (c) 14 (d) 100

9. A 4 × 4 magic square is given: How many 2 × 2 squares are there in it whose elements add up to 34?

1	15	14	4
12	6	7	9
8	10	11	5
13	3	2	16

(a) 6
(b) 9
(c) 4
(d) 5

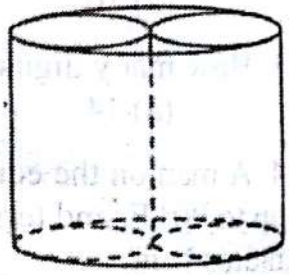
10. To go from engine to the last coach of his train of length 200m, a man jumped from his train to another train moving on a parallel track in the opposite direction, waited till the last coach of his original train appeared and then jumped back. In how much time did he reach the last coach if the speed of each train was 60km/hr.?

(a) 5 s (b) 6 s (c) 10 s (d) 12 s

11. November 9, 1994 was a Wednesday. Then which of the following is true?

(a) November 9, 1965 is a Wednesday and November 9, 1970 is a Wednesday.
(b) November 9, 1965 is not a Wednesday and November 9, 1970 is a Wednesday.
(c) November 9, 1965 is a Wednesday and November 9, 1970 is not a Wednesday.
(d) November 9, 1965 is not a Wednesday and November 9, 1970 is not a Wednesday.

12. Two identical cylinders of the same height as a bigger hollow cylinder were put vertically into the latter to fit exactly into it. The volume of the bigger cylinder is V. The volume of each of the smaller cylinder is



(a) $\frac{V}{8}$ (b) $\frac{V}{4}$ (c) $\frac{V}{2}$ (d) V

13. You get 20% returns on your investment annually, but also pay a 20% tax on the gain. At the end of 5 years, the net gain made by you (as percentage of the capital) is approximately

(a) 0 (b) 16 (c) 80 (d) 100

14. Coordinates of a point in x, y, z space is (1, 2, 3). What would be the coordinates of its reflection in a mirror along the x, z plane?

(a) (-1,-2,-3) (b) (1,-2,-3) (c) (1,-2,3) (d) (-1,2,3)

15. A cubic cavity of edge 20μm is filled with a fluid with a cubic solid of edge 2μm. What percent of the cavity volume is occupied by the fluid?

(a) 10.0 (b) 20.0 (c) 90.0 (d) 99.9

16. Find the missing number in the sequence? 61, 52, 63, 94, __, 18, 001, 121

(a) 46 (b) 70 (c) 66 (d) 44

17. What is the length of the longest rod that can be put in a hemispherical bowl of radius 10 cm such that no end of the rod is outside the bowl? (Assume that the rod has negligible thickness.)

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(a) $10\sqrt{2}$ (b) $10\sqrt{3}$ (c) $10\sqrt{4}$ (d) $10\sqrt{5}$

18. If a 4 digit year (e. g. 1927) is chosen randomly, what is the probability that it is NOT a leap year?

(a) ¾ (b) ¼ (c) < ¼ (d) > ¾

19. After giving 20% discount on the marked price to a customer, the seller's profit was 20%. Which of the following is true?

(a) Sale price $= \frac{(\text{Marked price} + \text{Cost price})}{2}$

(b) Sale price $< \frac{(\text{Marked price} + \text{Cost price})}{2}$

(c) $\frac{2(\text{Marked price} + \text{Cost price})}{3} > \text{Sales price} > \frac{(\text{Marked price} + \text{Cost price})}{2}$

(d) Sale price $> \frac{2(\text{Marked price} + \text{Cost price})}{3}$

20. Three years ago, the difference in the ages of two brothers was 2 years. The sum of their present ages will double in 10 years. What is the present age of the elder brother?

(a) 6 (b) 11 (c) 7 (d) 9

Answer key

1. b (Detailed solution on Chapter 13, Question 4).
2. c (Detailed solution on Chapter 13, Question 5).
3. d (Detailed solution on Chapter 5, Question 11).
4. c (Detailed solution on Chapter 7, Question 1).
5. a (Detailed solution on Chapter 11, Question 3).
6. b (Detailed solution on Chapter 4, Question 6).
7. d (Detailed solution on Chapter 5, Question 12).
8. b (Detailed solution on Chapter 5, Question 13).
9. d (Detailed solution on Chapter 14, Question 5).
10. b (Detailed solution on Chapter 3, Question 2).
11. d (Detailed solution on Chapter 1, Question 2).
12. b (Detailed solution on Chapter 12, Question 8).
13. c (Detailed solution on Chapter 10, Question 3).
14. c (Detailed solution on Chapter 11, Question 4).
15. d (Detailed solution on Chapter 12, Question 9).
16. a (Detailed solution on Chapter 4, Question 7).
17. c (Detailed solution on Chapter 11, Question 5).
18. d (Detailed solution on Chapter 1, Question 3).
19. b (Detailed solution on Chapter 10, Question 4).
20. b (Detailed solution on Chapter 8, Question 2).

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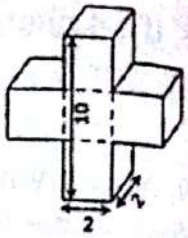
4. JUNE 2014

PHYSICAL SCIENCES, CHEMICAL SCIENCES

1. The following diagram shows 2 perpendicularly inter-grown prismatic crystals (twins) of identical shape and size. What is the volume of the object shown (units are arbitrary)?



(a) 60 (b) 65 (c) 72 (d) 80

2. Suppose in a box there are 20 red, 30 black, 40 blue and 50 white balls. What is the minimum number of balls to be drawn, without replacement, so that you are certain about getting 4 red, 5 black, 6 blue and 7 white balls?

(a) 140 (b) 97 (c) 104 (d) 124

3. In the growing years of a child, the height increases as the square root of the age while the weight increases in direct proportion to the age. The ratio of the weight to the square of the height in this phase growth

(a) is constant
(b) reduces with age
(c) increases with age
(d) is constant only if the weight and height at birth are both zero

4. Students in a group A obtained the following marks: 40, 80, 70, 50, 60, 90, 30. Students in group B obtained 40, 80, 35, 70, 85, 45, 50, 75, 60 marks.
Define Dispersion (D) = (maximum marks - minimum marks),

and Relative Dispersion (RD) $= \frac{Dispersion}{Mean}$

(a) RD of group A = RD of group B (b) RD of group A > RD of group B
(c) RD of group A < RD of group B (d) D of group A < D of group B

5. In 450 g of pure coffee powder, 50 g of chicory is added. A person buys 100 g of the mixture and adds 5 g of chicory to that. What would be the round off percentage of chicory in this final mixture?

(a) 10 (b) 5 (c) 14 (d) 15

6. The time gap between the two instants, one before and one after 12:00 noon, when the angle between the hour hand and the minute hand is 66°, is

(a) 12 min (b) 16 min (c) 18 min (d) 24 min

7. Suppose $x \Delta y = (x-y)^2$
$xoy = (x+y)^2$
$x * y = (x \times y)^{-1}$
$x.y = x \times y$
+, - and × have their usual meanings.
What is the value of $\{(197\ o\ 315) - (197\ \Delta\ 315)\}.(197 * 315)$?

(a) 1018 (b) 512 (c) 2 (d) 4

8. If $A \times B = 24, B \times C = 32, C \times D = 48$, then $A \times D$

(a) cannot be found (b) is a perfect square (c) is a perfect cube (d) is odd

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9. If all horses are donkeys, some donkeys are monkeys, and some monkeys are men, then which statement must be true?

(a) All donkeys are men
(b) Some donkeys may be men
(c) Some horses are men
(d) All horses are also monkeys

10. A rectangular area of sides 9 and 6 units is to be covered by square tiles of sides 1, 2 and 5 units. The minimum number of tiles needed for this is

(a) 3 (b) 11 (c) 12 (d) 15

11. Suppose n is a positive integer. Then $(n^2 + n)(2n + 1)$

(a) may not be divisible by 2
(b) is always divisible by 2 but may not be divisible by 3
(c) is always divisible by 3 but may not be divisible by 6
(d) is always divisible by 6

12. There is a train of length 500 m, in which a man is standing at the rear end. At the instant the rear end crosses a stationary observer on a platform, the man starts walking from the rear to the front and the front to the rear of the train at constant speed of 3 km/hr. The speed of the train is 80 km/hr. The distance of the man from the observer at the end of 30 minutes is

(a) 41.5 km (b) 40.5 km (c) 40.0 km (d) 41 km

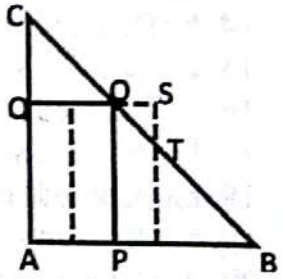
13. Three identical equilateral-triangular plates of side 5 cm each are placed together such that they form a trapezium. The length of the longer of the two parallel sides of this trapezium is

(a) $5\sqrt{\frac{3}{4}}$ cm (b) $5\sqrt{2}$ cm (c) 10 cm (d) $10\sqrt{3}$ cm

14. An archer climbs to the top of a 10 m high building and aims at a bird on top of a tree 17 m away. The line of sight from the archer to the bird makes an angle of 45° to the horizontal. What is the height of the tree?

(a) 17 m (b) 27 m (c) 37 m (d) 47 m

15. Consider a right-angled triangle ABC where AB = AC = 3. A rectangle APOQ is drawn inside it, as shown, such that the height of the rectangle is twice its width. The rectangle is moved horizontally by a distance 0.2 as shown schematically in the diagram (not to scale) What is the value of the ratio $\frac{\text{Area of } \Delta ABC}{\text{Area of } \Delta OST}$?



(a) 625 (b) 400 (c) 225 (d) 125

16. 80 gsm paper is cut into sheets of 200 mm × 300 mm size and assembled in packets of 500 sheets. What will be the weight of a packet? (gsm =g/m^2)

(a) 1.2 kg (b) 2.4 kg (c) 3.6 kg (d) 4.8 kg

17. Find the missing letter

(a) P
(b) K
(c) J
(d) L

A	B	C	D
F	I	L	O
K	P	U	Z
P	W	D	?

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18. A merchant buys equal numbers of shirts and trousers and pays Rs.38000. If the cost of 3 shirts is Rs.800 and that of a trouser is Rs.1000, then how many shirts were bought?

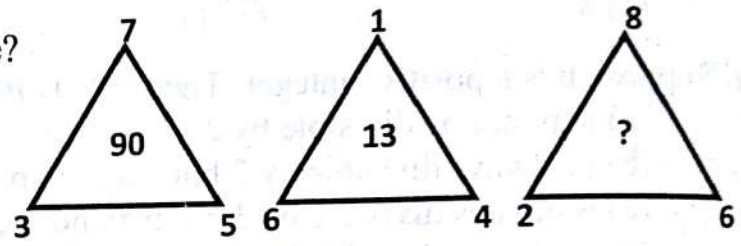
(a) 60 (b) 30 (c) 15 (d) 10

19. Consider the set of numbers $\{17^1, 17^2, \ldots, 17^{300}\}$. How many of these numbers end with the digit 3?

(a) 60 (b) 75 (c) 100 (d) 150

20. Find the missing number in the triangle?

(a) 16
(b) 96
(c) 50
(d) 80



Answer key

1. c (Detailed solution on Chapter 12, Question 10).
2. d (Detailed solution on Chapter 6, Question 2).
3. d (Detailed solution on Chapter 14, Question 6).
4. b (Detailed solution on Chapter 9, Question 2).
5. c (Detailed solution on Chapter 12, Question 11).
6. d (Detailed solution on Chapter 2, Question 1).
7. d (Detailed solution on Chapter 5, Question 14).
8. b (Detailed solution on Chapter 5, Question 15).
9. b (Detailed solution on Chapter 15, Question 3).
10. c (Detailed solution on Chapter 11, Question 6).
11. d (Detailed solution on Chapter 5, Question 16).
12. b (Detailed solution on Chapter 3, Question 3).
13. c (Detailed solution on Chapter 11, Question 7).
14. b (Detailed solution on Chapter 7, Question 2).
15. c (Detailed solution on Chapter 11, Question 8).
16. b (Detailed solution on Chapter 12, Question 12).
17. b (Detailed solution on Chapter 4, Question 8).
18. b (Detailed solution on Chapter 10, Question 5).
19. b (Detailed solution on Chapter 5, Question 17).
20. d (Detailed solution on Chapter 4, Question 9).

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5. DECEMBER 2013

MATHEMATICAL SCIENCES, LIFE SCIENCES, EARTH SCIENCES

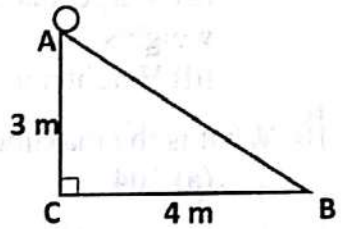
1. A cart wheel rolls along a straight line. If the distance covered is equal to the diameter of the wheel, what is the angle through which the wheel has turned?

(a) 90°
(b) between 90° and 120°
(c) between 120° and 150°
(d) between 150° and 180°

2. In a class of 10 students, 3 failed in History, 6 failed in Geography and 2 failed in both. How many passed in both the subjects?

(a) 1 (b) 2 (c) 3 (d) 0

3. As shown in the diagram, a sphere is placed on the top of an incline. It rolls down the incline without slipping in exactly 50 turns. The radius of the sphere is

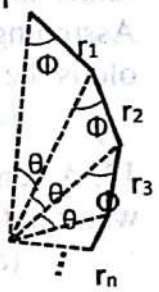


(a) $(\frac{5}{\pi})$ cm
(b) $(\frac{5}{\pi})$ m
(c) $(\frac{10}{\pi})$ cm
(d) 10 cm

4. A set of concentric circles of integer radii 1, 2,…,N is shown in the figure. An ant starts at point A_1, goes round the first circle, returns to A_1, moves to A_2, goes round the second circle, returns to A_2, moves to A_3 and repeats this until it reaches A_N. The distance covered by the ant is

(a) N(N + 1)
(b) 2Nπ+ N
(c) π(N+1)N + N-1
(d) π(N-1)N + N-1

5. The figure shows an infinite series of triangles, in which $r_1 > r_2 > r_3$. What is the total length of the solid line segments in the figure?

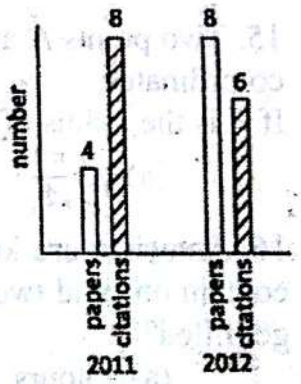


(a) $\frac{r_1}{r_2} + \frac{r_2}{r_3} + \ldots$
(b) $\frac{{r_1}^2}{r_1 - r_2}$
(c) $\frac{{r_2}^2}{r_1 + r_2}$
(d) $\frac{r_1 - r_2}{{r_1}^2}$

6. If a_i, b_i and c_i are distinct, how many terms will the expansion of the product $(a_1 + a_2 + a_3)(b_1 + b_2 + b_3 + b_4)(c_1 + c_2 + c_3 + c_4 + c_5)$ contain?

(a) 12 (b) 30 (c) 23 (d) 60

7. The plot depicts the number of research publications of a scientist along with the number of citations. Which of the following statements is not correct?



(a) In the year 2012, 50% more papers were published but citations decreased by 25%
(b) In the year 2012, 100% more papers were published but citations were 75% of the number of papers in that year.
(c) The papers published in the year 2011 are only 33.33% of the total number of papers in both years.
(d) The total number of citations for both years in 16.66% more than the total number of papers.

8. The next number of the sequence 1, 5, 14, 30, 55… is

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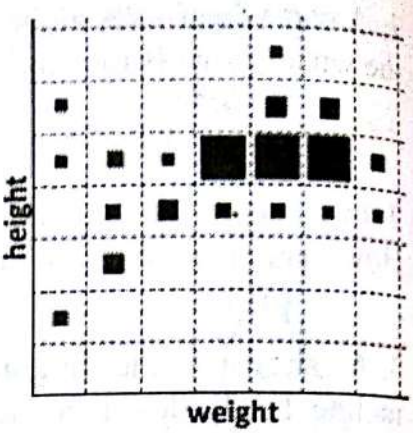
(a) 85 (b) 90 (c) 91 (d) 95

9. The distribution of heights and weights in a population is shown in a 2-parameter scatter plot. The size of the square is proportional to the number of persons having particular combination of weight and height. Which statement best describes the trend in the population?



(a) Height and weight are strongly correlated.
(b) Height and weight are anti-correlated.
(c) Large heights do not imply proportionately large weights.
(d) Height and weight are independent characteristics.

10. What is the maximum sum of the numbers of Saturdays and Sundays in a leap year?
(a) 104 (b) 105 (c) 106 (d) 107

11. Two trains of lengths 150 m and 250 m pass each other with constant speeds on parallel tracks in opposite directions. The drivers and guards are at extremities of the trains. The time gap between the drivers passing each other and first driver-guard pair passing each other is 30 s. How much later will other driver-guard pair pass by?
(a) 10 s (b) 20 s (c) 30 s (d) 50 s

12. In a room, we have one grandfather, two fathers, two sons, and a grandson. The age of one father is seven times the age of his son. The age of the other father is twice his son's age. Assuming that there are only 3 people in the room and the grandfather is 70 years old, how old is the grandson?
(a) 1 (b) 2 (c) 5 (d) cannot be determined

13. A hemispherical bowl is being filled with water at a constant volumetric rate. The level of water in the bowl increases
(a) in direct proportion of time (b) in inverse proportion of time
(c) faster than direct proportion of time (d) slower than direct proportion of time

14. Equal masses of two liquids of densities 6 kg/m^3 and 4 kg/m^3 are mixed thoroughly. The density of the mixture is
(a) 4.8 kg/m^3 (b) 5.0 kg/m^3 (c) 5.2 kg/m^3 (d) 5.4 kg/m^3

15. Two points A and B on the surface of the Earth have the following latitude and longitude co-ordinates. 30°N, 45°E and 30°N, 135°W
If R is the radius of the Earth, the length of the shortest path from A to B is
(a) $\sqrt{3}\frac{\pi R}{2}$ (b) $\frac{\pi R}{3}$ (c) $\frac{\pi R}{6}$ (d) $\frac{2\pi R}{3}$

16. Amoebae are known to double in 3 min. Two identical vessels A and B, respectively contain one and two amoebae to start with. The vessel B gets filled in 3 hours. When will A get filled?
(a) 3 hours (b) 2 hours and 57 minutes (c) 3 hours 3 minutes (d) 6 hours

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17. Students of a school are divided into 4 groups. What is the probability that three friends get into the same group?

(a) $\frac{3}{4}$ (b) $\frac{1}{64}$ (c) $\frac{1}{16}$ (d) $\frac{1}{3}$

18. A fruit vendor buys 120 Shimla apples at 4 for Rs.100, and 120 Golden apples at 6 for Rs.100. She decides to mix them and sell at 10 for Rs.200. She will make

(a) no profit, no loss (b) a loss of 4% (c) a gain of 4% (d) a loss of 10%

19. $4^0 + 4^2 + (\frac{1}{4})^2 + 4^{\frac{1}{2}} + \frac{1}{4^{\frac{1}{2}}}$

(a) 4^0 (b) $4^{2\frac{1}{2}} + 4^{-2\frac{1}{2}}$ (c) $19\frac{9}{16}$ (d) $22\frac{9}{16}$

20. In an enclosure there were both crows and cows. If there are 30 heads and 100 legs, what fraction of them are crows?

(a) 1:3 (b) 1:4 (c) 1:10 (d) 3:10

Answer key

1. b (Detailed solution on Chapter 11, Question 9).
2. c (Detailed solution on Chapter 14, Question 7).
3. a (Detailed solution on Chapter 11, Question 10).
4. d (Detailed solution on Chapter 11, Question 11).
5. b (Detailed solution on Chapter 11, Question 12).
6. d (Detailed solution on Chapter 5, Question 18).
7. a (Detailed solution on Chapter 13, Question 6).
8. c (Detailed solution on Chapter 4, Question 10).
9. c (Detailed solution on Chapter 13, Question 7).
10. c (Detailed solution on Chapter 1, Question 4).
11. b (Detailed solution on Chapter 3, Question 4).
12. c (Detailed solution on Chapter 8, Question 3).
13. d (Detailed solution on Chapter 12, Question 13).
14. a (Detailed solution on Chapter 12, Question 14).
15. a (Detailed solution on Chapter 7, Question 3).
16. c (Detailed solution on Chapter 14, Question 8).
17. c (Detailed solution on Chapter 6, Question 3).
18. b (Detailed solution on Chapter 10, Question 6).
19. c (Detailed solution on Chapter 5, Question 19).
20. a (Detailed solution on Chapter 8, Question 4).

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6. DECEMBER 2013

PHYSICAL SCIENCES, CHEMICAL SCIENCES

1. Three fishermen caught fishes and went to sleep. One of them woke up, took away one fish and $\frac{1}{3}^{rd}$ of the remainder as his share, without others' knowledge. Later, the three of them divided the remainder equally. How many fishes were caught?

(a) 58 (b) 19 (c) 76 (d) 88

2. What is the arithmetic mean of $\frac{1}{1\times2}, \frac{1}{2\times3}, \frac{1}{3\times4}, \frac{1}{4\times5}, \ldots, \frac{1}{100\times101}$?

(a) 0.01 (b) $\frac{1}{101}$ (c) 0.00111... (d) $\frac{\frac{1}{49\times50}+\frac{1}{50\times51}}{2}$

3. Every time a ball falls to ground, it bounces back to half the height it fell from. A ball is dropped from a height of 1024 cm. The maximum height from the ground to which it can rise after the tenth bounce is

(a) 102.4 cm (b) 1.24 cm (c) 1 cm (d) 2 cm

4. A circle of radius 7 units lying in the fourth quadrant touches the x-axis at (10,0). The center of the circle has coordinates

(a) (7,7) (b) (-10,7) (c) (10, -7) (d) (7, -7)

5. A cylinder of radius 1 cm and height 1 cm is broken into three pieces. Which of the following MUST be true?

(a) At least one piece has volume equal to 1 cm^3.
(b) At least two pieces have equal volumes.
(c) At least one piece has volume less than 1 cm^3.
(d) At least one piece has volume greater than 1 cm^3

6. One of the four - A, B, C and D - committed a crime. A said "I did it.", B said, "I didn't." C said, "B did it." D said, "A did it.". Who is lying?

(a) A (b) B (c) C (d) D

7. Consider the sequence of ordered sets of natural numbers: {1}, {2,3}, {4,5,6},... What is the last number in the 10^{th} set?

(a) 10 (b) 19 (c) 55 (d) 67

8. If a + b + c + d + e = 10 (all positive numbers), then the maximum value of a × b × c × d × e is

(a) 12 (b) 32 (c) 48 (d) 72

9. A farmer gives 7 full, 7 half-full and 7 empty bottles of honey to his three sons and asks them to share these among themselves such that each of them gets the same amount of honey and the same number of bottles. In how many ways can this be done? (bottles cannot be distinguished otherwise, they are sealed and cannot be broken).

(a) 0 (b) 1 (c) 2 (d) 3

10. A car is moving along a straight track. Its speed is changing with time as shown. Which of the following statements is correct?

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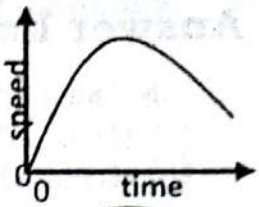
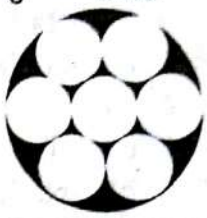
(a) The speed is never zero.
(b) The acceleration is zero once on the path.
(c) The distance covered initially increases and then decreases.
(d) The car comes back to its initial position once.

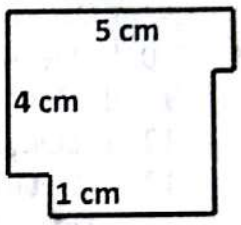


11. A circle circumstances identical close-packed circles of unit diameter as shown. What is the total area of the shaded portion?



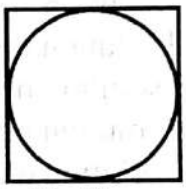
(a) 2 (b) 2π
(c) $\frac{1}{2}$ (d) $\frac{\pi}{2}$

12. What is the perimeter of the given figure, where adjacent sides are right angles to each other?



(a) 20 cm (b) 18 cm
(c) 21 cm (d) Cannot be determined.

13. What does the diagram establish?
Note: The diagram is a circle inside a square.



(a) $\pi > 3$ (b) $\pi \geq 2\sqrt{2}$
(c) $\pi < 4$ (d) π is closer to 3 than to 4.

14. What is the next number in the following sequence? 39, 42, 46, 50,…
(a) 52 (b) 53 (c) 54 (d) 55

15. $(25 \div 5 + 3 - 2 \times 4) + (16 \times 4 - 3) =$
(a) 61 (b) 22 (c) 41/24 (d) 16

16. A student buys a book from an online shop at 20% discount. His friend buys another copy of the same book in a book fair for Rs.192 paying 20% less than his friend. What is the full price of the book?
(a) Rs.275 (b) Rs.300 (c) Rs.320 (d) Rs.392

17. How many nine-digit positive integers are there, the sum of squares of whose digits is 2?
(a) 8 (b) 9 (c) 10 (d) 11

18. 366 players participate in a knock-out tournament. In each round all competing players pair together and play a match, the winner of each match moving to the next round. If at the end of a round there is an odd number of winners, the unpaired one moves to the next round without playing a match. What is the total number of matches played?
(a) 366 (b) 282 (c) 365 (d) 418

19. For real numbers x and y, $x^2 + (y - 4)^2 = 0$. Then the value of x + y is
(a) 0 (b) 2 (c) $\sqrt{2}$ (d) 4

20. There are 2 hills, A and B, in a region. If hill A is located N30°E of hill B, what will be the direction of hill B when observed from hill A? (N30° means 30° from north towards east).
(a) S30°W (b) S60°W (c) S30°E (d) S60°E

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Answer key

1. b (Detailed solution on Chapter 8, Question 5).
2. b (Detailed solution on Chapter 9, Question 3).
3. c (Detailed solution on Chapter 5, Question 20).
4. c (Detailed solution on Chapter 11, Question 13).
5. d (Detailed solution on Chapter 14, Question 9).
6. c (Detailed solution on Chapter 15, Question 4).
7. c (Detailed solution on Chapter 4, Question 11).
8. b (Detailed solution on Chapter 14, Question 10).
9. c (Detailed solution on Chapter 14, Question 11).
10. b (Detailed solution on Chapter 13, Question 8).
11. d (Detailed solution on Chapter 12, Question 15).
12. a (Detailed solution on Chapter 11, Question 14).
13. c (Detailed solution on Chapter 12, Question 16).
14. d (Detailed solution on Chapter 4, Question 12).
15. a (Detailed solution on Chapter 5, Question 21).
16. b (Detailed solution on Chapter 10, Question 7).
17. a (Detailed solution on Chapter 6, Question 4).
18. c (Detailed solution on Chapter 6, Question 5).
19. d (Detailed solution on Chapter 5, Question 22).
20. a (Detailed solution on Chapter 7, Question 4).

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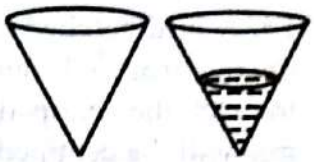
7. DECEMBER 2013

MATHEMATICAL SCIENCES, LIFE SCIENCES, EARTH SCIENCES

1. What is the angle between the minute and hour hand of a clock at 7: 35?
   (a) 0^o (b) $17{:}5^o$ (c) $19{:}5^o$ (d) 20^o

2. A stream of ants go from a point A to point B and return to A along the same path. All the ants move at a constant speed and from any given point 2 ants pass per second one way. It takes 1 minute for an ant to go from A to B. How many returning ants will an ant meet in its journey from A to B?
   (a) 120 (b) 60 (c) 240 (d) 180

3. The capacity of the conical vessel shown is V. It is filled with water up to half its height. The volume of water in the vessel is
   (a) $\frac{V}{2}$ (b) $\frac{V}{4}$ (c) $\frac{V}{8}$ (d) $\frac{V}{16}$



4. A large tank filled with water is to be emptied by removing half of the water present in it every day. After how many days will there be closest to 10% water left in the tank?
   (a) One (b) Two (c) Three (d) Four

5. n is a natural number. If n^5 is odd, which of the following is true?
   A. n is odd B. n^3 is odd C. n^4 is even
   (a) A only (b) B only (c) C only (d) A and B only

6. Suppose you expand the product $(x_1 + y_1)(x_2 + y_2)\ldots(x_{20} + y_{20})$. How many terms will have only one x and rest y's?
   (a) 1 (b) 5 (c) 10 (d) 20

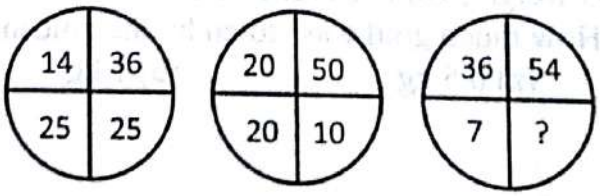
7. A 16.2 m long wooden log has a uniform diameter of 2 m. To what length the log should be cut to obtain a piece of 22 m^3 volume?
   (a) 3.5 m (b) 7.0 m (c) 14.0 m (d) 22.0 m

8. What is the last digit of 7^{73}?
   (a) 7 (b) 9 (c) 3 (d) 1

9. A lucky man finds 6 pots of gold coins. He counts the coins in the first four pots to be 60, 30, 20 and 15 respectively. If there is a definite progression, what would be the numbers of coins in the next two pots?
   (a) 10 and 5 (b) 4 and 2 (c) 15 and 15 (d) 12 and 10

10. A bee leaves its hive in the morning and after flying for 30 minutes due south reaches a garden and spends 5 minutes collecting honey. Then it flies for 40 minutes due west and collects honey in another garden for ten minutes. Then it returns to the hive taking the shortest route. How long was the bee away from its hive? (Assume that the bee flies at constant speed)
   (a) 85 min (b) 155 min (c) 135 min (d) less than 1 hour

11. Find the missing number:
   (a) 1 (b) 0
   (c) 2 (d) 3



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12. In solving a quadratic equation of the form $x^2 + ax + b = 0$, one student took the wrong value of a and got the roots as 6 and 2; while another student took the wrong value of b and got the roots as 6 and 1. What are the correct values of a and b, respectively?

(a) 7 and 12 (b) 3 and 4 (c) -7 and 12 (d) 8 and 12

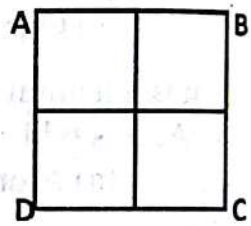
13. The distance between two oil rigs is 6 km. What will be the distance between these rigs in maps of 1:50000 and 1:5000 scales, respectively?

(a) 12 cm and 1.2 cm (b) 2 cm and 12 cm
(c) 120 cm and 12 cm (d) 12 cm and 120 cm

14. A bird perched at the top of 12 m high tree sees a centipede moving towards the base of the tree from a distance equal to twice the height of the tree. The bird flies along a straight line to catch the centipede. If both move at the same speed, at what distance from the base of the tree will the centipede be picked up by the bird?

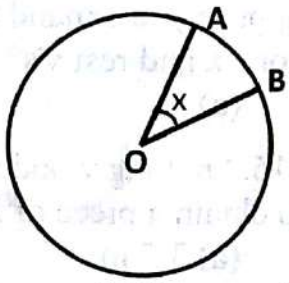
(a) 16 m (b) 9 m (c) 12 m (d) 14 m

15. An ant goes from A to C in the figure crawling only on the lines and taking the least length of path. The number of ways in which it can do so is



(a) 2
(b) 4
(c) 5
(d) 6

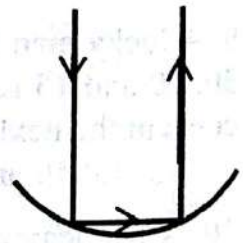
16. A point is chosen at random from a circular disc shown. What is the probability that the point lies in the sector OAB?



(where $\angle AOB = x$ radians)

(a) $\frac{2x}{\pi}$
(b) $\frac{x}{\pi}$
(c) $\frac{x}{2\pi}$
(d) $\frac{x}{4\pi}$

17. A ray of light, after getting reflected twice from a hemispherical mirror of radius R (see the figure), emerges parallel to the incident ray. The separation of the original incident ray and the final reflected ray is



(a) R (b) $R\sqrt{2}$
(c) 2R (d) $R\sqrt{3}$

18. A king ordered that a golden crown be made for him from 8 kg of gold and 2 kg of silver. The gold-smith took away some amount of gold and replaced it by an equal amount of silver and the crown when made, weighed 10 kg. Archimedes knew that under water gold lost $1/20^{th}$ of its weight, while silver lost $1/10^{th}$. When the crown was weighed under water, it was 9:25 kg. How much gold was stolen by the goldsmith?

(a) 0.5 kg (b) 1 kg (c) 2 kg (d) 3 kg

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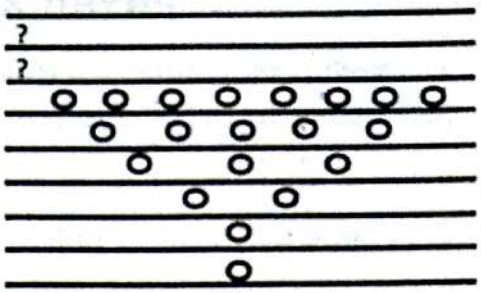
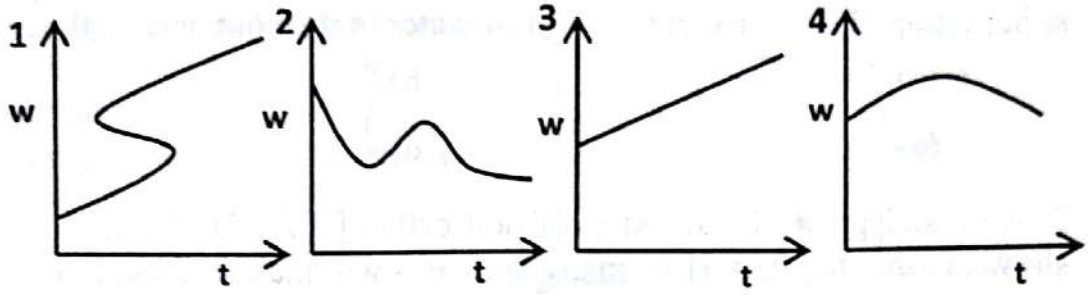
19. In the figure the numbers of circles in the blank rows must be



(a) 12 and 20
(b) 13 and 20
(c) 13 and 21
(d) 10 and 11

20. If we plot the weight (w) versus age (t) of a child in a graph, the one that will never be obtained from amongst the four graphs is



(a) 1 (b) 2 (c) 3 (d) 4

Answer key

1. b (Detailed solution on Chapter 2, Question 2).
2. c (Detailed solution on Chapter 14, Question 12).
3. c (Detailed solution on Chapter 12, Question 17).
4. c (Detailed solution on Chapter 14, Question 13).
5. d (Detailed solution on Chapter 5, Question 23).
6. d (Detailed solution on Chapter 5, Question 24).
7. b (Detailed solution on Chapter 12, Question 18).
8. a (Detailed solution on Chapter 5, Question 25).
9. d (Detailed solution on Chapter 4, Question 13).
10. c (Detailed solution on Chapter 7, Question 5).
11. d (Detailed solution on Chapter 4, Question 14).
12. c (Detailed solution on Chapter 5, Question 26).
13. d (Detailed solution on Chapter 12, Question 19).
14. b (Detailed solution on Chapter 7, Question 6).
15. d (Detailed solution on Chapter 14, Question 14).
16. c (Detailed solution on Chapter 6, Question 6).
17. b (Detailed solution on Chapter 7, Question 7).
18. d (Detailed solution on Chapter 8, Question 6).
19. c (Detailed solution on Chapter 4, Question 15).
20. a (Detailed solution on Chapter 13, Question 9).

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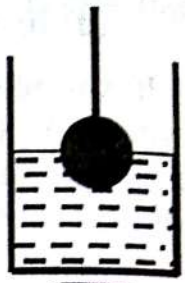
8. JUNE 2013

PHYSICAL SCIENCES, CHEMICAL SCIENCES

1. There is an equilateral triangle in the XY - plane with its center at the origin. The distance of its sides from the origin is 3.5 cm. The area of its circumcircle in cm^2 is

(a) 38.5 (b) 49 (c) 63.65 (d) 154

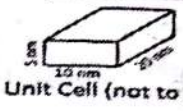
2. A sphere of iron of radius $\frac{R}{2}$ fixed to one end of a string was lowered into water in a cylindrical container of base radius R to keep exactly half the sphere dipped. The rise in the level of water in the container will be



(a) $\frac{R}{3}$ (b) $\frac{R}{4}$

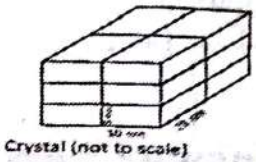
(c) $\frac{R}{8}$ (d) $\frac{R}{12}$

3. A crystal grows by stacking of unit cells of $10 \times 20 \times 5$ nm size as shown in the diagram. How many unit cells will make a crystal of 1 cm^3 volume?



(a) 10^6 (b) 10^9

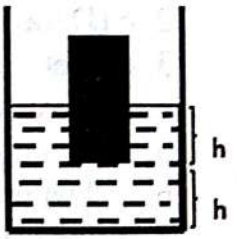
(c) 10^{12} (d) 10^{18}



4. What is the value of $\frac{1}{1\times2} + \frac{1}{2\times3} + \frac{1}{3\times4} + \ldots$ to ∞?

(a) 2/3 (b) 1 (c) 2 (d) ∞

5. A solid cylinder of basal area A was held dipped in water in a cylindrical vessel of basal area 2A vertically such that a length h of the cylinder is immersed. The lower tip of the cylinder is at a height h from the base of the vessel. What will be the height of the water in the vessel when the cylinder is taken out?



(a) 2h (b) $\frac{3}{2}$ h (c) $\frac{4}{3}$ h (d) $\frac{5}{4}$ h

6. Of all the triangles that can be inscribed in a semi-circle of radius R with the diameter as one side, the biggest one has the area

(a) R^2 (b) $R^2\sqrt{R}$ (c) $R^2\sqrt{3}$ (d) $2R^2$

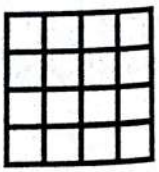
7. Choose the largest number:

(a) 2^{500} (b) 3^{400} (c) 4^{300} (d) 5^{200}

8. A daily sheet calendar of the year 2013 contains sheets of 10×10 cm size. All the sheets of the calendar are spread over the floor of a room of $5m \times 7.3m$ size. What percentage of the floor will be covered by these sheets?

(a) 0.1 (b) 1 (c) 10 (d) 100

9. How many rectangles (which are not squares) are there in the figure?



(a) 56 (b) 70

(c) 86 (d) 100

10. Define $a \otimes b = LCM(a,b) + GCD(a,b)$ and $a \oplus b = a^b + b^a$.
What is the value of $(1 \oplus 2) \otimes (3 \oplus 4)$?

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Here LCM = least common multiple and GCD = greatest common divisor.

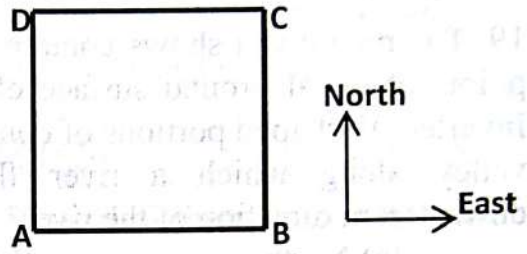
(a) 145 (b) 286 (c) 436 (d) 572

11. A square pyramid is to be made using a wire such that only one strand of wire is used for each edge. What is the minimum number of times that the wire has to be cut in order to make the pyramid?

(a) 3 (b) 7 (c) 2 (d) 1

12. A person has to cross a square field by going from A to C. The person is only allowed to move towards the east or towards the north or use a combination of these movements. The total distance travelled by the person

(a) depends on the length of each step
(b) depends on the total number of steps
(c) is different for different paths
(d) is the same for all paths



13. A crow is flying along a horizontal circle of radius R at a height R above the horizontal ground. Each of a number of men on the ground found that the angular height of the crow was fixed angle ($< 45^\circ$) when it was closest to him. Then all these men must be on a circle on the ground with a radius

(a) $R + R \sin\theta$ (b) $R + R \cos\theta$ (c) $R + R \tan\theta$ (d) $R + R \cot\theta$

14. How many pairs of positive integers have GCD 20 and LCM 600? (GCD = greatest common divisor, LCM = least common multiple)

(a) 4 (b) 0 (c) 1 (d) 7

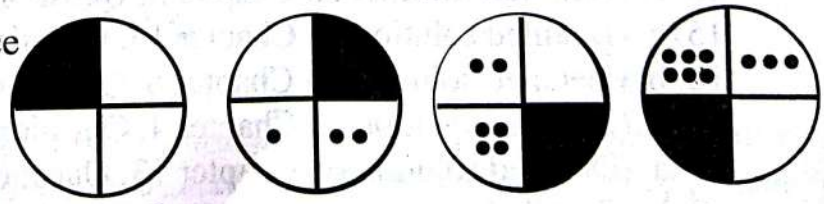
15. During an evening party, when Ms.Black, Ms.Brown and Ms.White met, Ms.Brown remarked, "It is interesting that our dresses are white, black or brown, but for each of us the name does not match the colour of the dress!". Ms.White replied, "But your white dress does not suit you!". Pick the correct answer.

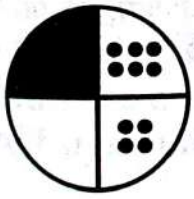
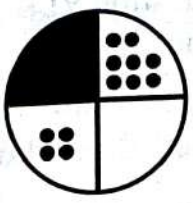
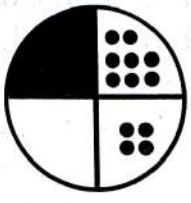
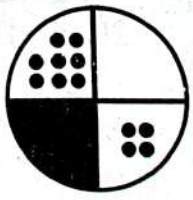
(a) Ms.White's dress was brown.
(b) Ms.Black's dress was white.
(c) Ms.White's dress was black.
(d) Ms.Black's dress was black.

16. Two integers are picked at random from the first 15 positive integers without replacement. What is the probability that the sum of the two numbers is 20?

(a) $\frac{3}{4}$ (b) $\frac{1}{21}$ (c) $\frac{1}{105}$ (d) $\frac{1}{20}$

17. Identify the next figure in the sequence



(a) 
(b) 
(c) 
(d) 

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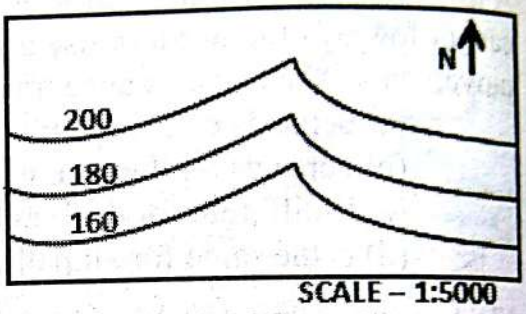
18. In a customer survey conducted during Monday to Friday, of the customers who asked for child care facilities in super markets, 23% were men and the rest, women. Among them, 19.9% of the women and 8.8% of the men were willing to pay for the facilities.
A. What is the ratio of the men to women customers who wanted child care facilities?
B. If the survey had been conducted during the weekend instead, how will the result change?
With the above data,

(a) Only A can be answered.
(b) Only B can be answered.
(c) Both A and B can be answered.
(d) Neither A nor B can be answered.

19. The map given shows contour lines which connect points of equal ground surface elevation in a region. Inverted 'V' shaped portions of contour lines represent a valley along which a river flows. What is the downstream direction of the river?



(a) North
(b) South
(c) East
(d) West

20. During a summer vacation, of 20 friends from a hostel, each wrote a letter to each of all others. The total number of letters written was

(a) 20
(b) 400
(c) 200
(d) 380

Answer key

1. d (Detailed solution on Chapter 11, Question 15).
2. d (Detailed solution on Chapter 12, Question 20).
3. d (Detailed solution on Chapter 12, Question 21).
4. b (Detailed solution on Chapter 4, Question 16).
5. b (Detailed solution on Chapter 12, Question 22).
6. a (Detailed solution on Chapter 12, Question 23).
7. b (Detailed solution on Chapter 5, Question 27).
8. c (Detailed solution on Chapter 12, Question 24).
9. b (Detailed solution on Chapter 11, Question 16).
10. c (Detailed solution on Chapter 5, Question 28).
11. d (Detailed solution on Chapter 14, Question 15).
12. d (Detailed solution on Chapter 14, Question 16).
13. d (Detailed solution on Chapter 7, Question 8).
14. a (Detailed solution on Chapter 5, Question 29).
15. c (Detailed solution on Chapter 15, Question 5).
16. b (Detailed solution on Chapter 6, Question 7).
17. c (Detailed solution on Chapter 4, Question 17).
18. a (Detailed solution on Chapter 13, Question 10).
19. b (Detailed solution on Chapter 7, Question 9).
20. d (Detailed solution on Chapter 6, Question 8).

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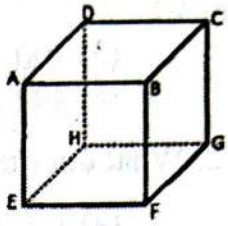
9. DECEMBER 2012

MATHEMATICAL SCIENCES, PHYSICAL SCIENCES, EARTH SCIENCES

1. Which of the following numbers is the largest? $2^{3^4}, 2^{4^3}, 3^{2^4}, 3^{4^2}, 4^{2^3}, 4^{3^2}$

(a) 2^{3^4} (b) 3^{4^2} (c) 4^{3^2} (d) 4^{2^3}

2. The cube ABCDEFGH in the figure has each edge equal to a. The area of the triangle with vertices A, C and F is



(a) $\frac{\sqrt{3}}{4}a^2$ (b) $\frac{\sqrt{3}}{2}a^2$

(c) $\sqrt{3}\,a^2$ (d) $2\sqrt{3}\,a^2$

3. What is the number of distinct arrangements of the letters of the word UGCCSIR so that U and I cannot come together?

(a) 2520 (b) 720 (c) 1520 (d) 1800

4. Suppose the sum of the seven positive numbers is 21. What is the minimum possible value of the average of the squares of these numbers?

(a) 63 (b) 21 (c) 9 (d) 7

5. Let

$A = \frac{1^{13}+2^{13}+3^{13}+\cdots 100^{13}}{100}$ $B = \frac{1^{13}+3^{13}+5^{13}+\cdots 99^{13}}{50}$ $C = \frac{2^{13}+4^{13}+6^{13}+\cdots 100^{13}}{50}$

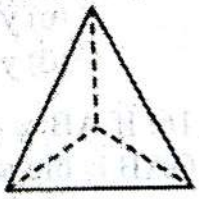
Which of the following is true?

(a) $B < C < A$ (b) $A < B < C$ (c) $B < A < C$ (d) $C < A < B$

6. A circle of radius 5 units in the XY plane has its center in the first quadrant touches the X-axis and has a chord of length 6 units on the Y-axis. The coordinates of its center are

(a) (4,6) (b) (3,5) (c) (5,4) (d) (4,5)

7. A wire of length 6 m is used to make a tetrahedron of each edge 1 m, using only one stand of wire for each edge. The minimum number of times the wire has to be cut is



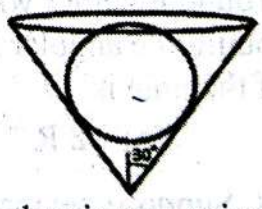
(a) 2 (b) 3 (c) 1 (d) 0

8. If the sum of the next two terms of the series below is x, what is the value of $\log_2 x$?

2, -4, 8, -16, 32, -64, 128,…

(a) 128 (b) 10 (c) 256 (d) 8

9. A conical vessel with semi-vertical angle 30° and height 10.5 cm has a thin lid. The radius of the sphere in cm is



(a) 3.5 (b) 5

(c) 6.5 (d) 7

10. Amar, Akbar and Antony are 3 friends, one of whom is a doctor, another is an engineer and the third is a professor. Amar is not an engineer. Akbar is the shortest. The tallest person is a doctor. The engineer's height is the geometric mean of the heights of the other two. Then which of the following is true?

(a) Amar is a doctor and he is the tallest

(b) Akbar is a professor and he is the tallest

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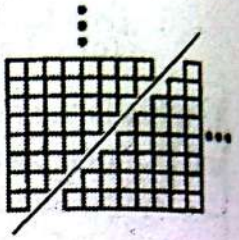
(c) Antony is a doctor and he is the shortest
(d) Antony is a doctor and he is the tallest

11. If 100 cats catch 100 mice in 100 minutes, then how long will take for 7 cats to catch 7 mice?

(a) 100 / 7 minutes
(b) 100 minutes
(c) 49 / 100 minutes
(d) 7 minutes

12. What does this diagram demonstrate?



(a) $1 + 2 + 3 + \ldots + n = \frac{n(n+1)}{2}$
(b) $1^2 + 2^2 + 3^2 + \ldots + n^2 = \frac{n(n+1)(2n+1)}{6}$
(c) $1 + 3 + \ldots (2n + = -1) = n^2$
(d) $2^2 + 4^2 + \ldots + (2n)^2 = \frac{2n(n+1)(2n+1)}{3}$

13. Suppose there are socks of N different colours in a box. If you take out one sock at a time, what is the maximum number of socks that you have to take out before a matching pair is found? Assume that N is an even number.

(a) N (b) N + 1 (c) N – 1 (d) N/2

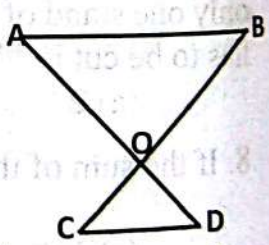
14. At what time after 4 O' clock, the hour and the minute hands will lie opposite to each other?

(a) 4-50'-31'' (b) 4-52'-51'' (c) 4-53'-23'' (d) 4-54'-33''

15. Which of the following curve just touches the x-axis?

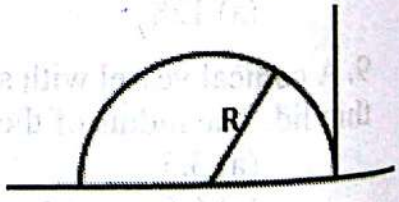
(a) $y = x^2 - x + 1$
(b) $y = x^2 - 2x + 2$
(c) $y = x^2 - 10x + 25$
(d) $y = x^2 - 7x + 12$

16. If AB is parallel to CD and AO = 2OD, then the area of the triangle OAB is bigger than the area of the triangle OCD by a factor of



(a) 2
(b) 3
(c) 4
(d) 8

17. A semi-circular arch of radius R has a vertical pole put on the ground together with one of its legs. An ant on the top of the arch founds the angular height of the tip of the pole to be 45°. The height of the pole is



(a) $\sqrt{2}$ R (b) $\sqrt{3}$ R (c) $\sqrt{4}$ R (d) $\sqrt{5}$ R

18. Suppose we make N identical smaller spheres from a big sphere. The total surface area of the smaller spheres is X times the total surface area of the big sphere, where X is

(a) $\sqrt{N}$ (b) 1 (c) $N^{1/3}$ (d) N^3

19. What is the next number of the sequence 24, 30, 33, 39, 51, ?

(a) 57 (b) 69 (c) 54 (d) 81

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20. Four lines are drawn on a plane with no two parallel and no three concurrent. Lines are drawn joining the points of intersection of the previous four lines. The number of new lines obtained in this way is

(a) 3 (b) 5 (c) 12 (d) 2

Answer key

1. a (Detailed solution on Chapter 5 , Question 30).
2. b (Detailed solution on Chapter 11 , Question 17).
3. d (Detailed solution on Chapter 6 , Question 9).
4. c (Detailed solution on Chapter 9, Question 4).
5. c (Detailed solution on Chapter 9 , Question 5).
6. d (Detailed solution on Chapter 11, Question 18).
7. c (Detailed solution on Chapter 11, Question 19).
8. d (Detailed solution on Chapter 4, Question 18).
9. a (Detailed solution on Chapter 11, Question 19).
10. a (Detailed solution on Chapter 15, Question 6).
11. b (Detailed solution on Chapter 14, Question 18).
12. a (Detailed solution on Chapter 11, Question 20).
13. a (Detailed solution on Chapter 6, Question 10).
14. d (Detailed solution on Chapter 2, Question 3).
15. c (Detailed solution on Chapter 5, Question 31).
16. c (Detailed solution on Chapter 11, Question 21).
17. c (Detailed solution on Chapter 7, Question 10).
18. c (Detailed solution on Chapter 12, Question 25).
19. a (Detailed solution on Chapter 4, Question 19).
20. a (Detailed solution on Chapter 11, Question 22).

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10. DECEMBER 2012

LIFE SCIENCES, CHEMICAL SCIENCES

1. A granite block of 2 m × 5 m × 3 m size is cut into 5 cm thick slabs of 2 m × 5 m size. These slabs are laid over a 2 m wide pavement. What is the length of the pavement that can be covered with these slabs?

(a) 100 m (b) 200 m (c) 300 m (d) 500 m

2. Which is the least among the following?

(a) $0.33^{0.33}$ (b) $0.44^{0.44}$ (c) $\pi^{\frac{-1}{\pi}}$ (d) $e^{\frac{-1}{e}}$

3. What is the next number in this "see and tell" sequence?

1, 11, 21, 1211, 111221,...

(a) 312211 (b) 1112221 (c) 1112222 (d) 1112131

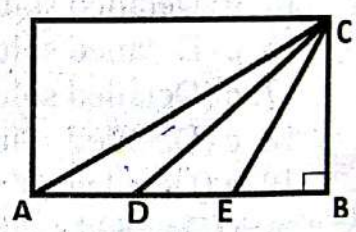
4. A vertical pole of length a stands at the center of a horizontal regular hexagonal ground of side a. A rope that is fixed taut in between a vertex on the ground and the tip of the pole has a length

(a) a (b) $\sqrt{2}a$ (c) $\sqrt{3}a$ (d) $\sqrt{6}a$

5. A peacock perched on the top of a 12 m high tree spots a snake moving towards its hole at the base of the tree from a distance equal to thrice of the height of the tree. The peacock flies towards the snake in a straight line and they both move at the same speed. At what distance from the base of the tree will the peacock catch the snake?

(a) 16 m (b) 18 m (c) 14 m (d) 12 m

6. In the figure $\angle ABC = \frac{\pi}{2}$, AD = DE = EB. What is the ratio of the area of triangle ADC to that of triangle CDB?

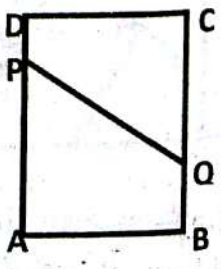


(a) 1 : 1 (b) 1 : 2
(c) 1 : 3 (d) 1 : 4

7. The cities of a country are connected by intercity roads. If a city is directly connected to an odd number of other cities, it is called an odd city. If a city is directly connected to an even number of other cities, it is called an even city. Then which of the following is impossible?

(a) There are even number of odd cities (b) There are odd number of odd cities
(c) There are even number of even cities (d) There are odd number of even cities

8. A rectangular sheet ABCD is folded in such a way that vertex A meets vertex C, thereby forming a line PQ. Assuming AB = 3 and BC = 4, find PQ. Note that AP = PC and AQ = QC.



(a) $\frac{13}{4}$ (b) $\frac{15}{4}$
(c) $\frac{17}{4}$ (d) $\frac{19}{4}$

9. A string of diameter 1 mm is kept on a table in the shape of a close flat spiral. i.e., a spiral with no gap between turns. The area of the table occupied by the spiral is 1 m^2. Then the length of the string is

(a) 10 m (b) 10^2 m (c) 10^3 m (d) 10^6 m

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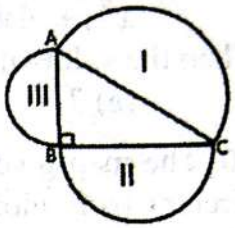
10. 25% of 25% of a quantity is x% of the quantity, where x is
(a) 6.25 % (b) 2.5 % (c) 25 % (d) 50 %

11. In a sequence $\{a_n\}$, every term is equal to the sum of all its previous terms. If $a_0 = 3$, then $\lim_{n\to\infty} \frac{a_{n+1}}{a_n}$ is
(a) 3 (b) 2 (c) 1 (d) e

12. In the figure, angle ABC = $\frac{\pi}{2}$. I, II, III are the areas of semicircles on the sides opposite to angles B, A and C respectively. Which of the following is always true?

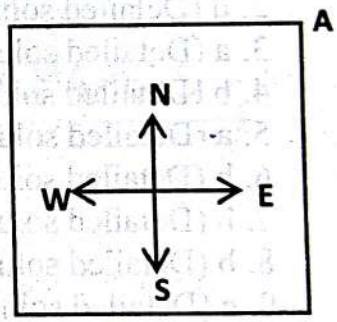


(a) $II^2 + III^2 = I^2$ (b) II + III = I
(c) $II^2 + III^2 > I^2$ (d) II + III < I

13. What is the minimum number of days between one Friday the 13th and the next Friday the 13th? (Assume that the year is a leap year)
(a) 28 (b) 56 (c) 91 (d) 84

14. Cucumber contains 99% water. Ramesh buys 100 kg of cucumbers. After 30 days of storing, the cucumbers lose some water. They now contain 98% water. What is the total weight of cucumbers now?
(a) 99 kg (b) 50 kg (c) 75 kg (d) 2 kg

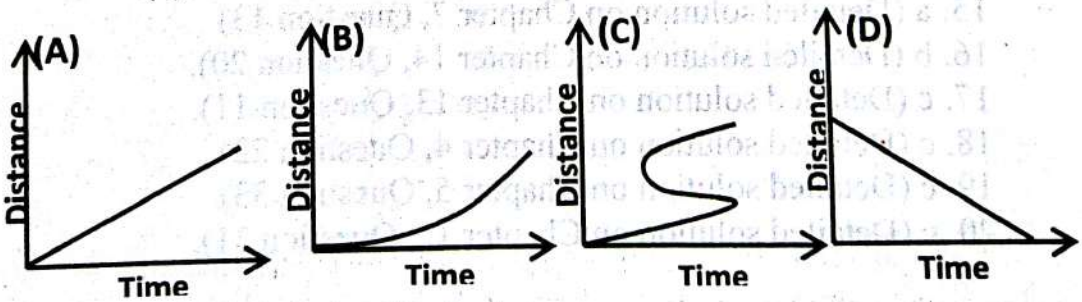
15. Suppose a person A is at the North-East corner of a square (see the figure). From that point he moves along the diagonal and after covering $\frac{1}{3}$rd portion of the diagonal, he goes to his left and after sometimes he stops, rotates 90° clockwise and moves straight. After a few minutes he stops, rotates 180° anticlockwise. Towards which direction he is facing now?



(a) North-East (b) North-West
(c) South-East (d) South-West

16. In a museum there were old coins with their respective years engraved on them, as follows.
A. 1837 AD B. 1907 AD C. 1947 AD D. 200 BC
Identify the fake coin(s)
(a) coin A (b) coin D (c) coin A & B (d) coin C

17. A student observes the movement of four snails and plots the graphs of distance moved as a function of time as given in figures (A), (B), (C) and (D). Which one of the following is not correct?



(a) Graph (A) (b) Graph (B) (c) Graph (C) (d) Graph (D)

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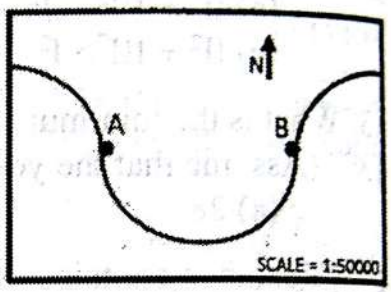
A	EGK	C
?		P
U		R
Q		V
B	OJF	D

18. Find the missing letter?
(a) H
(b) L
(c) Z
(d) V

19. Consider the following equation
$x^2 + 4y^2 + 9z^2 = 14x + 28y + 42z - 147$, where x, y and z are real numbers. Then the value of x + 2y + 3z is
(a) 7 (b) 14 (c) 21 (d) not unique

20. The map given shows a meandering river following a semi-circular path, along which two villages are located at A and B. The distance between A and B along the east-west direction in the map is 7 cm. What is the length of the river between A and B in the ground?



(a) 1.1 km (b) 3.5 km
(c) 5.5 km (d) 11 km

Answer key

1. c (Detailed solution on Chapter 12, Question 26).
2. d (Detailed solution on Chapter 5, Question 32).
3. a (Detailed solution on Chapter 4, Question 20).
4. b (Detailed solution on Chapter 7, Question 11).
5. a (Detailed solution on Chapter 7, Question 12).
6. b (Detailed solution on Chapter 12, Question 27).
7. b (Detailed solution on Chapter 14, Question 19).
8. b (Detailed solution on Chapter 11, Question 23).
9. c (Detailed solution on Chapter 12, Question 28).
10. a (Detailed solution on Chapter 8, Question 7).
11. b (Detailed solution on Chapter 4, Question 21).
12. b (Detailed solution on Chapter 12, Question 29).
13. c (Detailed solution on Chapter 1, Question 5).
14. b (Detailed solution on Chapter 12, Question 30).
15. a (Detailed solution on Chapter 7, Question 13).
16. b (Detailed solution on Chapter 14, Question 20).
17. c (Detailed solution on Chapter 13, Question 11).
18. c (Detailed solution on Chapter 4, Question 22).
19. c (Detailed solution on Chapter 5, Question 33).
20. c (Detailed solution on Chapter 12, Question 31).

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TOPIC WISE THEORY

AND

SOLUTION OF PREVIOUS YEARS' QUESTION PAPERS

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CHAPTER 1

CALENDAR PROBLEM

Average Difficulty rating: 2.7/5

Average time for each question: 80 seconds

Number of questions expected for each CSIR-NET exam: 1

1.1 THE HISTORY - WHY DO WE HAVE LEAP YEARS?

The total number of days of a year is the time taken by the earth to rotate once around the sun. It takes the earth about 365.2422 days to rotate around the sun, but a normal calendar year is only 365 days! Thus, the .2422 is the extra fraction of a day and it adds up over the years. Earth rotating around the sun four times takes 1460.9688 days, but four calendar years would only be 1460 days. That .9688 is nearly a whole day, so every four years we add an extra day to our calendar, February 29. We call that year a leap year. To make things easier, leap years are always divisible by four: 2008 and 2012 are both leap years.

For hundreds of years, people used a calendar called the Julian calendar that followed this rule, adding a leap year every four years. However, because .9688 is not *exactly* a whole day, the Julian calendar slowly began to disagree with the real seasons. In 1582, Pope Gregory fixed this problem by ordering everyone to use a new set of rules. These rules are named the Gregorian calendar, after him. They work like this:

The Gregorian calendar

Rule	Examples
Every fourth year is a leap year.	2004, 2008, and 2012 are leap years.
However, no hundredth year is a leap year.	1900 and 2100 are not leap years.
Every four hundred years, there is a leap year after all.	2000 and 2400 are leap years.

People in English-speaking countries did not start using the Gregorian calendar until 1752. Some countries, such as Iran, still use other systems.

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1.2 THEORY

1. **Odd Days:**

   In any year, there are 52 complete weeks.

   In a given period, the number of days more than the complete weeks are called odd days.

2. **Leap Year:**

   For an year to be a leap year,

   1. It should be divisible by 4.
   2. If it is divisible by 100, it should also be divisible by 400.

   Note: A leap year has 366 days.

   Examples:

   - The years 884, 1564, 1600, 2000, 2012 are leap years.
   - The years 431, 777, 1722, 2001, 2007, 2200 are not leap years.

3. **Ordinary Year:**

   The year, which is not a leap year, is called an *ordinary year* or a *non-leap year.*

   An ordinary year has 365 days.

   Examples:

   - The years 573, 998, 2009 are ordinary years.
   - The years 1700, 1800, 1900, 2100 are ordinary years.

4. **Counting of Odd Days:**

   First thing to remember is that the, 1st January of the year AD 1 was a Monday and therefore, we must set Sunday as the 0th day, Monday as the 1st day,..., Saturday as the 6th day.

Day of the week related to Odd days:

No. of odd days:	0	1	2	3	4	5	6
Day:	Sun.	Mon.	Tues.	Wed.	Thurs.	Fri.	Sat.

Month of a year related to Odd days:

Month	Jan	Feb	Mar	April	May	June	July	Aug	Sep	Oct	Nov	Dec
Days	31	28/29	31	30	31	30	31	31	30	31	30	31
Odd days	3	0/1	3	2	3	2	3	3	2	3	2	3

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That is, 3 odd days for every month with 31 days, 2 odd days for every month with 30 days and 0/1 odd days for February.

- $1\ ordinary\ year\ =\ 365\ days\ =\ (52\ weeks\ +\ 1\ day.)$

  Therefore, **1 ordinary year has 1 odd day.**

- $1\ leap\ year\ =\ 366\ days\ =\ (52\ weeks\ +\ 2\ days)$

  Therefore, **1 leap year has 2 odd days.**

- $100\ years\ =\ 76\ ordinary\ years\ +\ 24\ leap\ years$

  (Why not 25 leap years? Because the 100th year is a non-leap year.)

Thus $100\ years = (76 \times 1 + 24 \times 2)\ odd\ days$

$= 124\ odd\ days$

$= (17 \times 7) + 5$

$= (17\ weeks + 5\ days)$

Thus, there are 5 more days after the 17 complete weeks.

Therefore, Number of odd days in 100 years = 5 odd days.

- Number of odd days in $200\ years$

  $= (5 \times 2)(mod\ 7) = 10(mod\ 7) \equiv 3$ odd days.

- Number of odd days in $300\ years$

  $= (5 \times 3)\ (mod\ 7) = 15(mod\ 7) \equiv 1$ odd day.

- Number of odd days in $400\ years$

  $= (5 \times 4) + 1 = 21(mod\ 7) \equiv 0$ odd day.

(The +1 is due to the second condition of a leap year. That is, unlike the previous centuries, 400 is a leap year and hence one extra odd day.)

Similarly, each one of 800 years, 1200 years, 1600 years, 2000 years etc. has 0 odd days.

(Note: Taking mod 7 of a number means obtaining the remainder after dividing the number with 7.)

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1.2.1 Tricks and shortcuts for quick calculation

- For making calculations quicker, the number of days/odd days in an year is given by

Months	Ordinary year		Leap year	
	Days	Odd days	Days	Odd days
January 1 to March 31	90	6	91	0
April 1 to June 30	91	0	91	0
July 1 to September 30	92	1	92	1
October 1 to December 31	92	1	92	1

For example

To get the number of odd days of **November 17th** of a particular year:

Odd days from January 1 to September 30 = 6 + 0 + 1 = 7 = 0 odd days.

Odd days from October 1 to October 31 = 3 odd days.

Odd days from November 1 to November 17 = 17 = 3 odd days.

Thus, total odd days of November 17th = 0 + 3 + 3 = **6 odd days.**

- **Number of odd days of 1900 years = 1 odd day.** Memorizing this is useful, as in most of the questions the year will be some year after 1900.
- The last day of a century must be Sunday, Monday, Wednesday or Friday. (The reason is given in Illustration 4.)
- An ordinary year always begins and ends on the same day of the week.
- For an ordinary year, February has exactly 4 weeks, so February and March always consecutively start on the same day of the week.
- Any date in March is the same day of the week as the corresponding date in November of that year. Another pair showing similar corresponding feature is April-July.

1.2.2 Working rule for finding the day of a given date

[For proper understanding, read this section only after reading all the above theory sections]

Step 1: Find the odd days until the previous century of the year in the given date.

For example, if 1991 is the given year, find the number of odd days until 1900 years. The fact that every multiple of 400 years have 0 odd days, can be used to find the odd days.

That is, 1900 years = 1600 years + 300 years = 0 odd days + 1 odd day = 1 odd day.

Step 2: Find the odd days from the first year after that century **until the previous year of the given year.**

That is, find the number of odd days of the **90 years** from **1901 to 1990.**

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Step 3: Find the number of odd days of the completed months of the year in the given date.

Step 4: Find the number of odd days of the day in the given date.

Step 5: Add the results of step 1, step 2, step 3 and step 4 to obtain the number of odd days corresponding to the given date. If the number of odd days is greater than 7, then take modulo 7 to get the number of odd days corresponding to the given date.

Step 6: Match the day corresponding to the number of odd days.

No. of odd days:	0	1	2	3	4	5	6
Day:	Sun.	Mon.	Tues.	Wed.	Thurs.	Fri.	Sat.

If the reference day is given; then

Step 1: Find the net number of odd days for the period between the reference day and the given date.

Step 2: The day of the week on the particular date is equal to number of odd days ahead or behind the reference day depending on whether the reference day falls before or after the given date.

Note that while counting the number of odd days, always check

1. Whether the interval has leap years and if yes,
2. Whether February of the leap year falls in the interval or not.

ILLUSTRATION 1:

What was the day of the week on 17th October 1991?

(a) Monday (b) Tuesday (c) Wednesday (d) Thursday

Answer (d)

The given year is 1991.

Step 1: Find the odd days until the previous century of the year in the given date.

That is, find the number of odd days of 1900 years.

1900 years = 1600 years + 300 years = 0 odd days + 1 odd day = **1 odd day.**

Step 2: Find the number of odd days from the first year after that century until the previous year of the given year.

That is, find the number of odd days of the 90 years from 1901 to 1990.

First, we can find the number of leap years in 90 years.

As we know, generally a leap year occurs once in every 4 years. Therefore to obtain the number of leap years in 90 years, we may find the quotient when 90 is divided by 4.

Thus, Number of leap years in 90 years = Quotient when 90 is divided by 4 = 22

Number of non-leap years = 90 – 22 = 68.

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As we learnt, every leap year has 2 odd days and every non-leap year has 1 odd day.

Thus, Number of odd days in 90 years = (22 × 2) + (68 × 1) = 112 odd days
= 112 (mod 7) = **0 odd days.**

Step 3: Find the number of odd days of the completed months of the year in the given date.

That is find the number of odd days from January to September in the year 1991.

Since 1991 is a non-leap year, February would have 28 days, that is 0 odd days.

Number of odd days from January to September = $3 + 0 + 3 + 2 + 3 + 2 + 3 + 3 + 2$
$= 21(mod\ 7) =$ ***0 odd days.***

Step 4: Find the number of odd days of the day in the given date.

That is, find the number of odd days from 1st October to 17th October.

Number of odd days from 1st October to 17th October. $= 17(mod\ 7) =$ ***3 odd days.***

Step 5: Add the results of step 1, step 2, step 3 and step 4 to obtain the number of odd days corresponding to the given date.

Hence, Total number of odd days for 17 October 1991 $= 1 + 0 + 0 + 3 = 4$ *odd days.*

Step 6: Match the day corresponding to the number of odd days.

4 odd days corresponds to **Thursday**. Thus 17th October 1991 is a Thursday.

ILLUSTRATION 2:

Today is Wednesday. 76th day from today will be?

(a) Sunday (b) Tuesday (c) Wednesday (d) Saturday

Answer (b)

A day repeats itself in every 7 days. That is, a day repeats itself in every multiple of 7.

Thus if today is a Wednesday, 77th day is Wednesday.

Hence, 76th day will be a **Tuesday**.

ILLUSTRATION 3:

On 21st June 2016, Tuesday falls. What day of the week was 21st June 2015?

(a) Sunday (b) Tuesday (c) Monday (d) Saturday

Answer (a)

As 2016 is a leap year, between 21st June 2016 and 21st June 2015 there is one complete leap year. Note that the extra day, February 29th falls on this period.

In other words there are 366 days and hence two odd days between these two dates.

Therefore if 21st June 2016 falls on Tuesday, 21st June 2015 falls on two days before.

Hence, 21st June 2015 is a **Sunday**.

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ILLUSTRATION 4:

The last day of a century cannot be?

(a) Monday (b) Tuesday (c) Wednesday (d) Friday

Answer (b)

Prerequisite: For proper understanding of this question, fully read the theory section above.

First century means 100 years and number of odd days in 100 years is 5 which corresponds to Friday. Therefore, last day of the 1st century is **Friday.**

200 years has 3 odd days and 3 corresponds to Wednesday.

Therefore, last day of the 2nd century is **Wednesday.**

300 years has 1 odd day and 1 corresponds to Monday.

Therefore, last day of the 3rd century is **Monday.**

400 years has 0 odd day and 0 corresponds to Sunday.

Therefore, last day of the 4th century is **Sunday.**

This cycle gets repeated.

Thus, the last day of a century cannot be Tuesday or Thursday or Saturday. Hence, **option (b).**

1.2.3 Conditions for calendars of two different years to be same

1. Both the years must be of the same type. That is, either both must be non-leap years or both must be leap years.
2. The net number of odd days for the period between these two years should be zero. That is, 1st January of both the years must fall on the same day of the week.

ILLUSTRATION 5:

The calendar for the year 2015 will be same for the year?

(a) 2016 (b) 2020 (c) 2026 (d) 2027

Answer (c)

A calendar repeats itself if the number of days between those two years is a multiple of 7.

That is, count the number of odd days from the year 2015 onwards to get the sum equal to 0 odd day and the next year will be our required year.

2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
1	2	1	1	1	2	1	1	1	2	1

Sum = 14 odd days ≡ 0 odd days.

Thus, by completing the year 2025, there are zero odd days with reference to the year 2015.

The next year is 2026. 2015 and 2026 are both non-leap years.

Hence, Calendar for the year 2026 will be the same as for the year 2015.

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1.3 CSIR-NET Exam Questions – Detailed Solution

1. A certain day which is x days before 17th August, is such that 50 days prior to that day, it was 4x days since March 30th of the same year. What is x?

(a) 18 **(b) 30** **(c) 22** **(d) 16**

(CSIR-NET DEC 2014)

Answer (a)

Difficulty rating: 3/5 *Average time for solving: 100 seconds*

Note: Here "prior" means "before", and "since" means "after".

Here two dates, 30th March and 17th August are given and some dates in between these two dates are given in terms of an unknown variable x.

We are required to find the value of x.

First, we can find the number of days from 30th March to 17th August.

31st March = 1 day.

1st April – 30th April = 30 days.

1st May – 31st May = 31 days.

1st June – 30th June = 30 days.

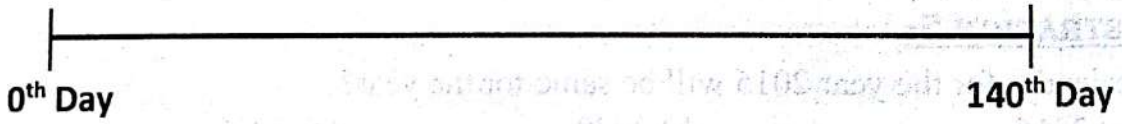
1st July – 31st July = 31 days.

1st August – 17th August = 17 days.

That is, $1 + 30 + 31 + 30 + 31 + 17 = \mathbf{140\ \boldsymbol{days}}$.

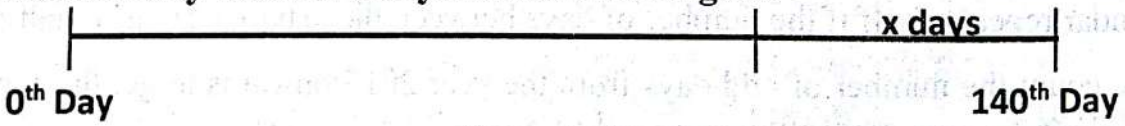
Therefore, 17th August is the 140th day from 30th March.

Thus, we can construct a timeline with 30th March as the 0th day and 17th August as the 140th day.

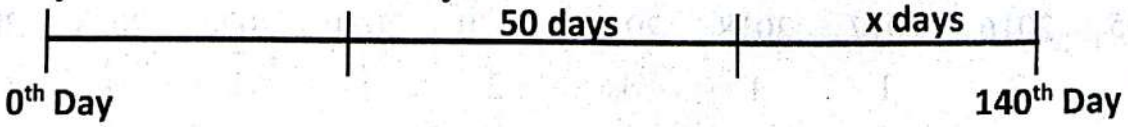


Now analyse each of the statements in the question and mark the obtained data in the timeline.

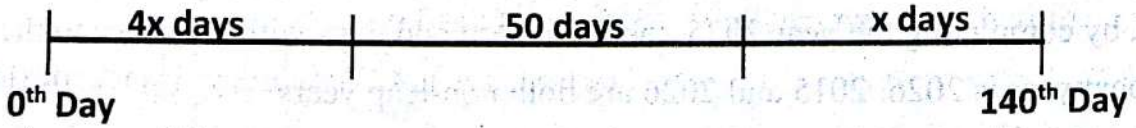
Given 1: **"A certain day which is x days before 17th August."**



Given 2: **"50 days before that certain day."**



Given 3: ***"it was*** **4x days since March 30th"**



Thus, by the given conditions,

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$4x + 50 + x = 140.$

$\Rightarrow 5x = 90$

$\Rightarrow x = 18.$

2. November 9, 1994 was a Wednesday. Then which of the following is true?

(a) November 9, 1965 is a Wednesday and November 9, 1970 is a Wednesday.

(b) November 9, 1965 is not a Wednesday and November 9, 1970 is a Wednesday.

(c) November 9, 1965 is a Wednesday and November 9, 1970 is not a Wednesday.

(d) November 9, 1965 is not a Wednesday and November 9, 1970 is not a Wednesday.

Answer (d) **(CSIR-NET JUNE 2014)**

Difficulty rating: 3/5 *Average time for solving: 100 seconds*

Given that, November 9, 1994 is a Wednesday, which is our reference day.

In the given four options, there are only two dates: November 9, 1965 and November 9, 1970.

We have to find the number of odd days between each of these days and the reference day.

November 9, 1970:

We have to find the number of odd days between November 9, 1994 and November 9, 1970.

Between these two dates, there are 24 complete years.

Number of leap years in these 24 years is the quotient when 24 is divided by 4.

Thus there are 6 leap years and 24 – 6 = 18 non-leap years.

Remember: Every *leap year* have 2 odd days and *non-leap year* has 1 odd day.

That is, Number of odd days in 24 years $= (6 \times 2) + (18 \times 1) = 30(mod\ 7) \equiv 2\ odd\ days$

The given reference day is a Wednesday.

So to find the day on which November 9, 1970 falls, we have to subtract 2 days from the reference day. Thus November 9, 1970 is a **Monday.**

November 9, 1965:

We have to find the number of odd days between November 9, 1994 and November 9, 1965.

Between these two dates, there are 29 complete years.

Number of leap years in these 29 years is the quotient when 29 is divided by 4.

Thus there are 7 leap years and 29 – 7 = 22 non-leap years.

That is, Number of odd days in 29 years $= (7 \times 2) + (22 \times 1) = 36(mod\ 7) \equiv 1\ odd\ day$

The given reference day is a Wednesday.

So to find the day on which November 9, 1965 falls, we have to subtract 1 day from the reference day. That is, November 9, 1965 is a **Tuesday.**

Thus both the dates given in the answer options are not Wednesday. Hence, **option (d).**

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3. If a 4-digit year (e.g. 1927) is chosen randomly, what is the probability that it is NOT a leap year?

(a) ¾ **(b) ¼** **(c) < ¼** **(d) > ¾**

(CSIR-NET JUNE 2014)

Answer (d)

Average time for solving: 60 seconds

Difficulty rating: 2.5/5

Remember: For a year to be leap year, there are two criteria:

1. It should be divisible by 4.
2. If it is divisible by 100, it should also be divisible by 400.

In every 4 consecutive numbers, there will be three numbers that are not divisible by 4.

Thus by the first criterion alone, chances for a randomly chosen year to be non-leap year is ¾ and by the second criterion, more years become non leap years and hence our **answer is > ¾.**

For further understanding, we can take a set of four consecutive years in which no leap years occurs. For example, 1899, 1900, 1901, 1902.

Already by the first criterion itself, 1899, 1901, 1902 are non-leap years.

Again, by the second criterion 1900 is also a non-leap year.

So in these 4 years, the probability of non-leap years is 1; which is greater than ¾.

4. What is the maximum sum of the number of Saturdays and Sundays in a leap year?

(a) 104 **(b) 105** **(c) 106** **(d) 107**

(CSIR-NET DEC 2013)

Answer (c)

Average time for solving: 50 seconds

Difficulty rating: 2/5

In a leap year, there are 366 days, that is 52 complete weeks and 2 odd days.

[Since $366 = 52 \times 7 + 2$]

Thus there are 52 Saturdays and 52 Sundays which sums to 104 days.

To maximize the sum of the number of Saturdays and Sundays, let the two odd days be Saturday and Sunday respectively.

Hence maximum sum of the number of Saturdays and Sundays = 104 + 2 = **106.**

5. What is the minimum number of days between one Friday the 13th and the next Friday the 13th? (Assume that the year is a leap year)

(a) 28 **(b) 56** **(c) 91** **(d) 84**

(CSIR-NET DEC 2012)

Answer (c)

Average time for solving: 100 seconds

Difficulty rating: 3/5

Note that the given year is a leap year. Hence February has 29 days.

For the sake of convenience, assume that January 13th of the given year falls on a Friday.

We know that a day gets repeated in every 7 days.

Consequently, the next Friday falls on 13th of another month if the number of days counted from January 13th till the 13th of that month is a multiple of 7,

Now, check in which month's 13th, the sum of days until then is a multiple of 7.

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$Jan\ 13th\ to\ Feb\ 13th = 31$, not a multiple of 7.

$Jan\ 13th\ to\ March\ 13th = 31 + 29 = 60$, not a multiple of 7.

$Jan\ 13th\ to\ April\ 13th = 31 + 29 + 31 = \mathbf{91}(mod\ 7) = 0\ days$, a multiple of 7.

Hence, **option (c).**

Note that, if it were a non-leap year, the answer would have been 28 as February 13th and March 13th are the same day of the week for a non-leap year.

1.4 PRACTICE QUESTIONS

1. What was the day of the week on 3 October 1992?
   (a) Monday (b) Wednesday (c) Saturday (d) Sunday

2. Today is Sunday. Before 63 days it was?
   (a) Sunday (b) Monday (c) Tuesday (d) Saturday

3. The year next to 2009 having the same calendar as that of 2009 is
   (a) 2010 (b) 2015 (c) 2016 (d) 2018

4. What was the day of the week on Independence Day of India (15th August 1947)?
   (a) Thursday (b) Friday (c) Saturday (d) Sunday

5. September 13, 1992 was a Sunday. Then which of the following is true?
   (a) September 13, 1985 is a Sunday and September 13, 1959 is a Sunday.
   (b) September 13, 1985 is not a Sunday and September 13, 1959 is a Sunday.
   (c) September 13, 1985 is a Sunday and September 13, 1959 is not a Sunday.
   (d) September 13, 1985 is not a Sunday and September 13, 1959 is not a Sunday.

6. What is the minimum sum of the number of Saturdays and Sundays in a leap year?
   (a) 104 (b) 105 (c) 106 (d) 107

7. For a non-leap year, find the minimum number of days between one Friday the 13th and the next Friday the 13th?
   (a) 28 (b) 56 (c) 91 (d) 84

8. If 3rd of the month is Friday what day will be on the 4th day after the 21st of the month?
   (a) Monday (b) Tuesday (c) Saturday (d) Sunday

9. Which of the following months will have the same day on the last day of the months?
   (a) January – October (b) April – June (c) February – October (d) Both (a) and (c)

10. A certain day which is x days before 12th December, is such that 30 days prior to that day, it was 3x days since September 13th of the same year. What is x?
   (a) 15 (b) 16 (c) 20 (d) 22

Answer Key

1. (c) 2. (a) 3. (b) 4. (b) 5. (b) 6. (a) 7. (a) 8. (c) 9. (c) 10. (a)

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CHAPTER 2

CLOCK PROBLEM

Average Difficulty rating: 3.3/5

Average time for each question: 90 seconds

Number of questions expected for each CSIR-NET exam: 1

2.1 THEORY

As you know, a clock is a device that measures and shows time. The clock consists of two arms called the minute hand and hour hand. (In clock problems, generally the existence of the second hand is neglected.) The dial of a clock is a circle (360^O) whose circumference is divided into 12 parts, called hour spaces. Each hour space is divided into 5 parts, called minute spaces. Thus, the whole circumference of the clock is divided into $12 \times 5 = 60$ minute spaces.

2.1.1 Important facts and shortcuts for quick calculation

- In 12 hours, the hour hand will cover 360°.
- So in 1 hour, hour hand will cover $\frac{360^o}{12} = 30^o$.
- In 60 minutes, the minute hand will cover 360°.
- So in 1 minute, minute hand will cover $\frac{360^o}{60} = 6^o$.
- For the minute hand and hour hand to coincide, the angle between them should be 0°.
- During every hour, minute hand and hour hand coincide once.
- Both the hands coincide after every $65\frac{5}{11}$ minutes.
- For the minute hand and hour hand to lie opposite to each other, the angle between them should be 180°.
- In 60 minutes, the minute hand gains 55 minutes on the hour hand.
- In $\frac{12}{11}$ minutes, the minute hand gains 1 minute over the hour hand.
- To convert minutes into hours, divide the corresponding minutes by 60.
- To convert seconds into hours, divide the corresponding seconds by 3600.

2.1.2 Important formulas

- Angle between H hours and M minutes $= \frac{1}{2} \times |60H - 11M|$

where, H is the time in hours.
M is the time in minutes.

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Note: |x| means absolute value or modulus value of x. It simply takes the positive value of the given number.

- To find the mirror image of any given time (except any time between 11 and 1), subtract the given value from 11:60.
- To find the mirror image of any time between 11 and 1, subtract the given value from 23:60.

2.1.3 Some important types of Clock problems

Type 1: What is the angle between minute hand and hour hand of a clock at a given time?

Type 2: What is the time when the angle between minute hand and hour hand is given?

Type 3: What is the time gap between two instants, when angle between minute hand and hour hand is given?

2.1.3.1 Type 1: What is the angle between minute hand and hour hand of a clock at a given time?

Here a time on the clock will be given. You are expected to find the smaller angle between the minute hand and hour hand at the given time.

Angle between H hours and M minutes $= \frac{1}{2} \times |60H - 11M|$

Step 1: Identify H and M from the given question.

Step 2: Substitute H and M in the above formula.

Note: For any time between 12 and 1, take H = 0.

ILLUSTRATION 1:

What is the angle between minute hand and hour hand of a clock at 4:50?

(a) 155° (b) 173° (c) 175° (d) 180°

Answer (a)

Step 1: Here H = 4, M = 50,

Step 2: Substituting in the formula.

Angle between H hours and M minutes $= \frac{1}{2} \times |60H - 11M|$

$$= \frac{1}{2}|(60 \times 4) - (11 \times 50)|$$

$$= \frac{1}{2}|240 - 550|$$

$$= \frac{1}{2}|-310|$$

$$= \frac{1}{2} \times 310$$

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$= 155$

Thus, angle between minute hand and hour hand at 4:50 is **155°.**

ILLUSTRATION 2:

What is the angle between minute hand and hour hand of a clock at 6:40?

(a) 40° (b) 42° (c) 50° (d) 90°

Answer (a)

Here H = 6, M = 40,

Substituting in the formula.

Angle between H hours and M minutes $= \frac{1}{2} \times |60H - 11M|$

$= \frac{1}{2}|(60\times6) - (11\times40)|$

$= \frac{1}{2}|360 - 440|$

$= \frac{1}{2}|-80|$

$= \frac{1}{2} \times 80$

$= 40$

Thus, angle between minute hand and hour hand at 6:40 is **40°.**

2.1.3.2 Type 2: What is the time on clock if the angle between minute hand and hour hand is given?

ILLUSTRATION 3:

At what time after 8 O' clock, the angle between hour hand and the minute hand will be 20°?

(a) 8 hours 20 minutes (b) 8 hours 33 minutes

(c) 8 hours 40 minutes (d) 8 hours 50 minutes

Answer (c)

We know the formula,

Angle between H hours and M minutes $= \frac{1}{2} \times |60H - 11M|$

Given: Angle between minute hand and hour hand = 20°, H = 8, M = ?

$\Rightarrow \quad 20 = \frac{1}{2}|(60\times8) - (11M)|$

$\Rightarrow \quad 20 = \frac{1}{2}|480 - 11M|$

$\Rightarrow \quad 40 = |480 - 11M|$

Now, since modulus sign is there, 480 – 11M can be positive and negative depending on the value of M.

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If 480 – 11M is positive,

$\Rightarrow 40 = 480 - 11M$

$\Rightarrow 11M = 480 - 40$

$\Rightarrow 11M = 440$

$\Rightarrow M = \frac{440}{11} =$ **40 minutes.**

If 480 – 11M is negative,

$\Rightarrow 40 = 11M - 480$

$\Rightarrow 11M = 480 + 40$

$\Rightarrow 11M = 520$

$\Rightarrow M = \frac{520}{11} = 47\frac{3}{11}$ **minutes.**

(Note: For learning how to convert improper fraction into mixed fraction, read the section "Fractions" in Chapter 5.)

Thus, at both 8 hours 40 minutes and 8 hours $47\frac{3}{11}$ minutes, the angle between the minute and hour hands is 20°.

Among both, 8 hours 40 minutes is the option present in the given options.

Hence, **option (c).**

ILLUSTRATION 4:

At what time after 6 O' clock, the hour and the minute hands will coincide?

(a) 6-31'-54'' (b) 6-31'-59'' (c) 6-32'-44'' (d) 6-32'-54''

Answer (c)

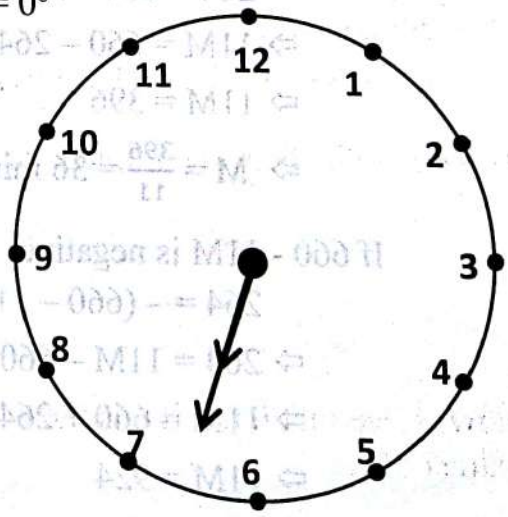
For the hour hand and minute hand to coincide, the angle between them should be 0°.

Thus, our question can be restructured as "At what time after 6 O' clock, the angle between the hour and the minute hands is 0°".

Angle between H hours and M minutes $= \frac{1}{2} \times |60H - 11M|$

Here H = 6 Hours, Angle between H hours and M minutes = 0°

We are required to find the time in minutes, i.e., M.



Thus, $0 = \frac{1}{2} |(60 \times 6) - 11M|$

$\Rightarrow 0 = \frac{1}{2} |360 - 11M|$

$\Rightarrow 0 = |360 - 11M|$

Here even though modulus sign exists, since the LHS = 0, we may consider only the positive value for 360 – 11M.

$\Rightarrow 360 - 11M = 0$

$\Rightarrow M = \frac{360}{11} = 32\frac{8}{11}$ minutes.

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Now convert $\frac{8}{11}$ minutes to seconds.

Remember: 1 minute = 60 seconds.

Therefore $\frac{8}{11}$ minutes $= \frac{8}{11} \times 60 = \frac{480}{11} = 43.6$ seconds ≈ 44 seconds.

Thus, **6 Hours 32 minutes 44 seconds** is the required time.

2.1.3.3 Type 3: What is the time gap between two instants, when angle between minute hand and hour hand is given?

Here two instants with the same angle will be given. That is the angle between minute hand and hour hand is same at two different time in the clock. We are expected to find the time gap between these two instants.

ILLUSTRATION 5:

The time gap between the two instants, one before and one after 12 am, when the angle between the hour hand and the minute hand is 132°, is

(a) 24 min (b) 36 min (c) 40 min (d) 48 min

Answer (d)

The two instants can be represented as shown in the figure.

Consider the instant before 12.

For this instant, the time is in between 11 and 12.

Hence H = 11, M = ?

By using the formula, **Angle between H hours and M minutes** $= \frac{1}{2} \times |60H - 11M|$

$$132^\circ = \frac{1}{2} \times |60 \times 11 - 11M|$$

$$\Rightarrow 264 = |660 - 11M|$$

Here 660 - 11M can be positive or negative depending on the value of M.

If 660 - 11M is positive,

$$264 = 660 - 11M$$

$$\Rightarrow 11M = 660 - 264$$

$$\Rightarrow 11M = 396$$

$$\Rightarrow M = \frac{396}{11} = 36 \text{ minutes}$$

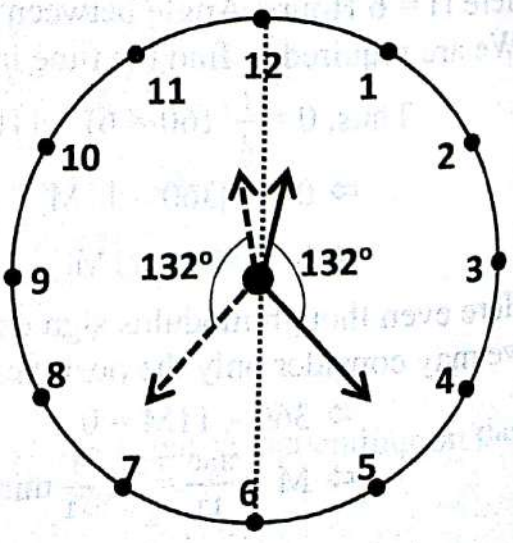
If 660 - 11M is negative,

$$264 = -(660 - 11M)$$

$$\Rightarrow 264 = 11M - 660$$

$$\Rightarrow 11M = 660 + 264$$

$$\Rightarrow 11M = 924$$



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$\Rightarrow M = \frac{924}{11} = 84$ minutes

Since the minute M cannot take any value greater than 60, 84 minutes is impossible.

Hence, **11:36 is the time corresponding to the instant before 12 am.**

Now consider the instant after 12.

For this instant, the time is in between 12 and 1.

Hence H = 0, M = ?

By using the formula

$$132^{o} = \frac{1}{2} \times |60 \times 0 - 11M|$$

$$\Rightarrow 264 = |-11M|$$

Here -11M can be positive or negative depending on the value of M.

If -11M is positive,

$$264 = -11M$$

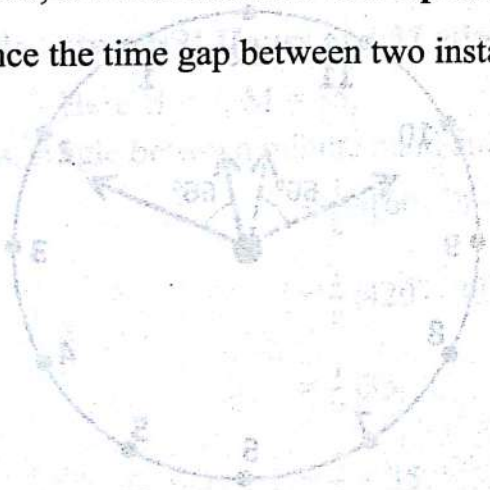
$$\Rightarrow M = -\frac{264}{11} = -24$$

If -11M is negative, M = 24.

Since time can take positive value only, we have M = 24 minutes.

Hence, **12:24 is the time corresponding to the instant after 12 am.**

Hence the time gap between two instants = 12:24 – 11:36 = **48 minutes.**



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2.2 CSIR-NET Exam Questions – Detailed Solution

1. The time gap between the two instants, one before and one after 12:00 noon, when the angle between the hour hand and the minute hand is 66°, is

(a) 12 min **(b) 16 min** **(c) 18 min** **(d) 24 min**

Answer (d) **(CSIR-NET JUNE 2014)**

Difficulty rating: 3.5/5 *Average time for solving: 100 seconds*

The formula we use here is our usual formula to find the angle between minute hand and hour hand.

$$\textbf{Angle between H hours and M minutes} = \frac{1}{2} \times |60H - 11M|$$

Here we are given two instants in a clock – one before 12 and one after 12. The two instants can be represented as in the figure. .

Consider the instant before 12 noon.

For this instant the time is in between 11 and 12.

So, H = 11, M = ?

By using the formula

$$66^\circ = \frac{1}{2} \times |60 \times 11 - 11M|$$

$$\Rightarrow 132 = |660 - 11M|$$

Here 660 - 11M can be positive or negative depending on the value of M.

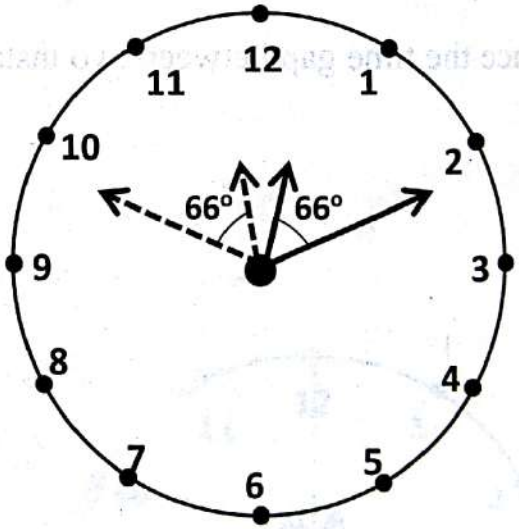
If 660 - 11M is positive,

$$132 = 660 - 11M$$

$$\Rightarrow 11M = 660 - 132$$

$$\Rightarrow 11M = 528$$

$$\Rightarrow M = \frac{528}{11} = 48 \text{ minutes}$$



If 660 - 11M is negative,

$$132 = -(660 - 11M)$$

$$\Rightarrow 132 = 11M - 660$$

$$\Rightarrow 11M = 660 + 132$$

$$\Rightarrow 11M = 792$$

$$\Rightarrow M = \frac{792}{11} = 72 \text{ minutes}$$

72 minutes is impossible, since the minute M cannot take any value greater than 60.

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Hence, **11:48 am is the time corresponding to the instant before 12 noon.**

Now consider the instant after 12 noon.
For this instant, the time is in between 12 and 1.
So, H = 0, M =?
By using the formula

$$66 = \frac{1}{2} \times |60\times0 - 11M|$$

$$\Rightarrow 132 = |-11M|$$

Here -11M can be positive or negative depending on the value of M.

If -11M is positive,

$$132 = -11M$$

$$\Rightarrow M = -\frac{132}{11} = -12 \text{ minutes.}$$

If -11M is negative, M = 12 minutes.

Since time can take positive value only, M = -12 minutes can be eliminated.

Hence, **12:12 pm is the time corresponding to the instant after 12 am.**

Hence the time gap between two instants = 12:12 – 11:48 = **24 minutes.**

2. What is the angle between the minute and hour hand of a clock at 7: 35?

(a) 0° **(b) 17.5°** **(c) 19.5°** **(d) 20°**

Answer (b) **(CSIR-NET JUNE 2013)**

Difficulty rating: 2.5/5 *Average time for solving: 60 seconds*

Angle between H Hours and M minutes $= \frac{1}{2} \times |60H - 11M|$

Here H = 7, M = 35,

Thus, Angle between minute hand and hour hand at 7:35

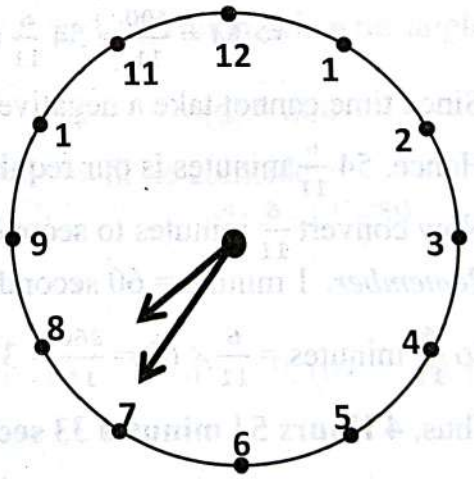
$$= \frac{1}{2} |(60\times7) - (11\times35)|$$

$$= \frac{1}{2} |420 - 385|$$

$$= \frac{1}{2} |35|$$

$$= \frac{1}{2} \times 35$$

$$= \mathbf{17.5^o.}$$



3. At what time after 4 O' clock, the hour and the minute hands will lie opposite to each other?

(a) 4 - 50'- 31'' **(b) 4 - 52'- 51''** **(c) 4 - 53'- 23''** **(d) 4 - 54'- 33''**

Answer (d) **(CSIR-NET DEC 2012)**

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Difficulty rating: 4/5 *Average time for solving: 120 seconds*

For the hour hand and minute hand to lie opposite to each other, the angle between them must be 180°.

Thus our question can be rephrased as "At what time after 4 O' clock will the angle between the hour and the minute hands be 180°".

Angle between minute hand and hour hand $= \frac{1}{2} \times |60H - 11M|$

Here, Angle between hour and minute hands = 180°, H = 6 Hours, M = ?

Thus, $180^\circ = \frac{1}{2}|(60\times4) - (11M)|$

$\Rightarrow 360 = |240 - 11M|$

240 – 11M can be positive or negative depending on the value of M.

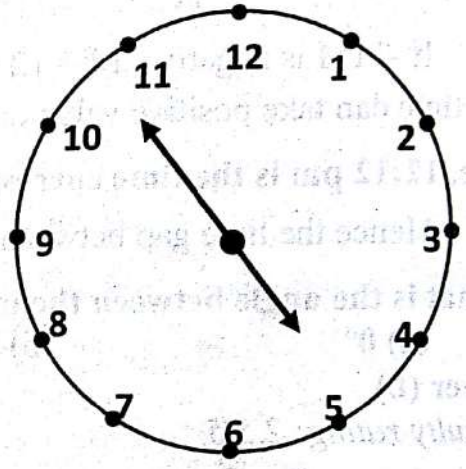
If 240 – 11M is positive,

$360 = 240 - 11M$

$\Rightarrow 11M = 240 - 360$

$\Rightarrow 11M = -120$

$\Rightarrow M = -\frac{120}{11} = -10\frac{10}{11}$ minutes.



If 240 – 11M is negative,

$360 = -(240 - 11M)$

$\Rightarrow 360 = 11M - 240$

$\Rightarrow 11M = 600$

$\Rightarrow M = \frac{600}{11} = 54\frac{6}{11}$ minutes .

Since time cannot take a negative value, we can avoid $-10\frac{10}{11}$ minutes

Hence, $54\frac{6}{11}$ minutes is our required time in minutes.

Now convert $\frac{6}{11}$ minutes to seconds.

Remember: 1 minute = 60 seconds.

So $\frac{6}{11}$ minutes $= \frac{6}{11} \times 60 = \frac{360}{11} = 32.7$ seconds ≈ 33 seconds.

Thus, **4 Hours 54 minutes 33 seconds** is the required time.

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2.3 Practice Questions

1. How many times the hands of a clock will be in vertically opposite directions from 10 am to 10 pm.

(a) 12 times (b) 25 times (c) 36 times (d) 48 times

2. The time between 7 and 8 o' clock, correct to the nearest minute, when the hands of a clock will form an angle of 84 degrees are:

(a) 7:23 and 7:53 (b) 7:20 and 7:52 (c) 7:22 and 7:53 (d) 7:23 and 7:52

3. Henry starts a trip when the hands of the clock are together between 8 a.m. and 9 a.m. He arrives at his destination between 2 p.m. and 3 p.m. when the hands of the clock are exactly 180° apart. The trip takes:

(a) 6 hr. (b) 6 hr. $43\frac{7}{11}$ min (c) 6 hr. 30 min (d) 5 hr.30 min

4. How many times do the hands of a clock coincide in a day?

(a) 22 (b) 24 (c) 44 (d) 48

5. The smaller angle between the hands of a clock at 12:25 p.m. is:

(a) 132° 30' (b) 137° 30' (c) 150° (d) 137°

6. What is the angle between minute hand and hour hand of a clock, when the time is 11:58?

(a) 9° (b) 10° (c) 11° (d) 12°

7. A clock is started at noon. At 10 minutes past 5, the hour hand has turned through:

(a) 145° (b) 150° (c) 155° (d) 160°

8. An accurate clock shows 8 o'clock in the morning. Through how many degrees will the hour hand rotate when the clock shows 2 o'clock in the afternoon?

(a) 160° (b) 172° (c) 173° (d) 180°

9. The time gap between the two instants, one before and one after 12:00 noon, when the angle between the hour hand and the minute hand is 11°, is

(a) 1 min (b) 2 min (c) 3 min (d) 4 min

10. At what time after 2 O' clock, the hour and the minute hands will lie coincide?

(a) 2 - 10'- 55'' (b) 2 - 11'- 32'' (c) 2 - 12'- 28'' (d) 2 - 13'- 59'

Answer Key

1. (a) 2. (a) 3. (a) 4. (a) 5. (b) 6. (c) 7. (c) 8. (d) 9. (d) 10. (a)

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CHAPTER 3

MOVING LOCOMOTIVE PROBLEM

Average Difficulty: 3.5/5

Average time for each question: 90 seconds

Number of questions expected for each CSIR-NET exam: 1

3.1 THEORY

A moving locomotive problem normally poses the challenge of finding out the time taken, the speed, or the distance covered while a particular object pass another. The moving locomotive can be a train, a car, a flight etc. and the objects can be a truck, a man, a platform, a pole, a bridge, a tunnel, a stationary train, a moving train etc.

A moving locomotive is said to have crossed or passed an object (stationary or moving) only when the total length of the moving locomotive crosses the total length of the object.

For instance, for a train to cross an object,

Distance covered by the train = length of train + length of object.

3.1.1 Important Facts, Tips and Shortcuts for Calculation

- km/hr. to m/s conversion:
  x km/hr. = $(x \times \frac{5}{18})$ m/s
- m/s to km/hr. conversion:
  x m/s = $(x \times \frac{18}{5})$ km/hr
- Speed = $\frac{\text{Distance}}{\text{Time}}$
- Suppose two objects are moving in the same direction at u m/s and v m/s, where u > v, then their relative speed = (u - v) m/s.
- Suppose two objects are moving in opposite directions at u m/s and v m/s, then their relative speed = (u + v) m/s.

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Based on the object (including its type, positioning and properties) that a moving locomotive has to cross, the formula to find the time taken by a moving locomotive to cross the object can be tabulated as follows:

Case	Type of Objects	Time to cross
1.	Object is stationary and is of negligible length. e.g. A locomotive crossing a lamp post, pole, standing man etc.	$t = \frac{\text{Length of locomotive}}{\text{Speed of locomotive}}$
2.	Object is stationary and has some length. e.g. A locomotive crossing a bridge, a tunnel, a platform or another moving locomotive at rest.	$t = \frac{\text{Length of locomotive+Length of object}}{\text{Speed of locomotive}}$
3.	Object is moving in the same direction and has negligible length. e.g. A locomotive crossing a running man, a running car, etc.	$t = \frac{\text{Length of locomotive}}{\text{Speed of locomotive}- \text{Speed of object}}$
4.	Object is moving in opposite direction and has negligible length. e.g. A locomotive crossing a running man, a running car, etc.	$t = \frac{\text{Length of locomotive}}{\text{Speed of locomotive+ Speed of object}}$
5.	Object is moving in same direction and has some length. e.g. A locomotive crossing another running moving locomotive (with lesser speed)	$t = \frac{\text{Length of locomotive+ Length of object}}{\text{Speed of locomotive}-\text{Speed of object}}$
6.	Object is moving in the opposite direction and has some length. e.g. A locomotive crossing another running locomotive	$t = \frac{\text{Length of locomotive+ Length of object}}{\text{Speed of locomotive+ Speed of object}}$

Even though we have all the above listed simple and easy to master formulas; for CSIR-NET exams the questions require in-depth analysis so that one can solve what is to be solved. What matters more is not just the application of formula, rather your ability to analyse the given situation and express it mathematically. It may also be noted that some questions, does not even require the application of any of the above formulas.

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However, to get familiar with the moving locomotive problems, we can start with simple questions, which do not demand much analysis. In this section, problems involving trains are frequently asked for examination.

ILLUSTRATION 1:

A train is running at a speed of 72 km/hr. It crosses a platform whose length is half that of the train in 30 seconds. What is the length of the platform?

(a) 100 m (b) 200 m (c) 300 m (d) 400 m

Answer (b)

Here the answer options are in metres.

So first, we shall convert the speed 72 Km / hr. into m/s $= 72 \times \frac{5}{18} = 20$ m /s.

Let x metre be the length of the platform.
Then, length of the train is 2x metre.

Remember: If the **object is stationary and is of some length**, the formula is

$$\text{Time} = \frac{\text{Length of the train+Length of the platform}}{\text{Speed of the train}}$$

Here, Speed = 20 m/s, Time = 30 s.

$$30 = \frac{2x+x}{20}$$

$\Rightarrow$ 3x = 600 m.

$\Rightarrow$ x = 200 m.

Therefore, Length of the platform = **200 m**.

ILLUSTRATION 2:

A 220 metres long train running at the speed of 110 km/hr crosses another train running in opposite direction at the speed of 70 km/hr in 7 seconds. What is the length of the other train?

(a) 100 m (b) 130 m (c) 150 m (d) 155 m

Answer (b)

Given: Speed of train 1 = 110 km/hr.
Speed of train 2 = 70 km/hr.
Time, t = 7 sec
Length of train 1 = 220 m.

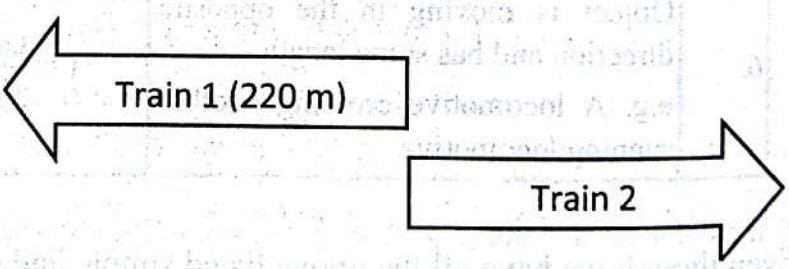

Figure: When the two trains cross.

Let x be the length of train 2.

Here, **two objects are moving in opposite direction and both have some length.**
Then, the formula is

$$t = \frac{\text{Length of train 1 + Length of train 2}}{\text{Speed of train 1 + Speed of train 2}}$$

Relative speed = Speed of train 1 + Speed of train 2 = 110 + 70 = 180 km/ hr.

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Converting km/hr into m/s, $180 \times \frac{5}{18} = 50$ m/s.

Thus, substituting in the formula, we get

$$7 = \frac{220+x}{50}$$

$$\Rightarrow 350 = 220 + x$$

$$\Rightarrow x = 130$$

Thus, length of the other train = **130 m.**

ILLUSTRATION 3:

A jogger running at 10 km/hr alongside a railway track in 150 metres ahead of the engine of a 300 metres long train running at 64 km/hr in the same direction. In how much time will the train pass the jogger?

(a) 10 s (b) 15 s (c) 20 s (d) 30 s

Answer (d)

Here the jogger is an object with negligible length and moving in the same direction as of the train.

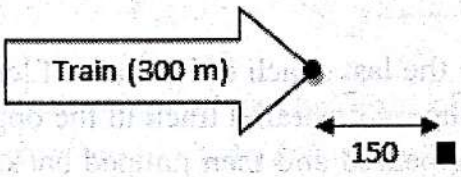

Figure 1: Initial position – When jogger is 150 m away from the train

Let us represent the given data pictorially as shown in the figure. Here the dark circle denotes the reference point on the train and the dark square denotes the position of the jogger.

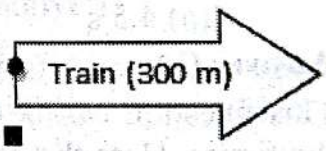

Figure 2: Final position – When the train passes the jogger

Speed of the train = 64 km/hr.

Speed of the jogger = 10 km/hr.

Length of the train = 300 m.

Then, speed of train relative to jogger = (64 – 10) km/hr = $54 \times \frac{5}{18} = 15$ m/s.

For the train to cover the jogger, the train should cover the initial gap (150 m) between the train and jogger plus the length of the train (300 m).

Thus, Distance = 150 + 300 = 450 m.

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}} = \frac{450}{15} = \mathbf{30\ s.}$$

ILLUSTRATION 4:

Two trains are moving in opposite directions at 72 km/hr and 90 km/hr. Their lengths are 500 m and 400 m respectively. The time taken by the slower train to cross the faster train is

(a) 15 s (b) 18 s (c) 20 s (d) 25 s

Answer (c)

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Length of train 1 = 500 m.,
Length of train 2 = 400 m.

Speed of train 1 = 72 km/hr

$$= 72 \times \frac{5}{18} = 20 \text{ m/s.}$$

Speed of train 2 = 90 km/hr

$$= 90 \times \frac{5}{18} = 25 \text{ m/s.}$$

Here train 1 is the slower train.

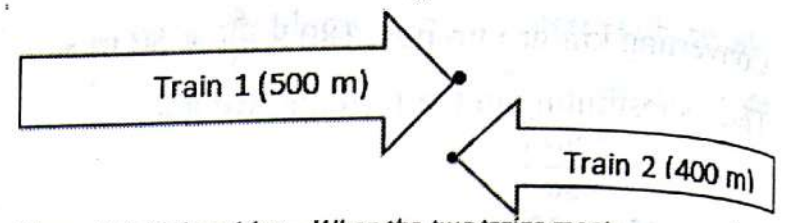

Figure 1: Initial position – When the two trains meet

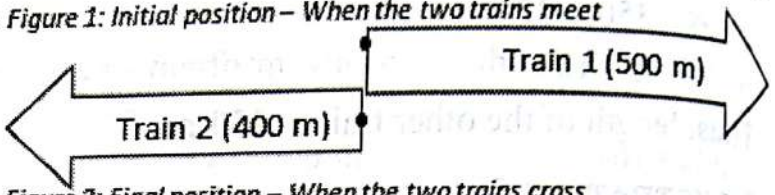

Figure 2: Final position – When the two trains cross

To cross train 2, train 1 has to cover the length of train 2 plus the length of itself.

We know that if two objects is moving in the opposite direction and both has some length,

$$\text{Then, } t = \frac{\text{Length of train 1+ Length of train 2}}{\text{Speed of train 1+ Speed of train 2}} = \frac{500+400}{20+25} = \frac{900}{45} = \mathbf{20\ s.}$$

ILLUSTRATION 5:

To go from engine to the last coach of his train of length 180 m, a man jumped from his train to another train moving on a parallel track in the opposite direction, waited till the last coach of his original train appeared and then jumped back. In how much time did he reach the last coach if the speed of each train was 72 km/hr?

(a) 4.5 s (b) 5 s (c) 5.5 s (d) 7 s

Answer (a)

This question can be pictured as shown in the figure. Here the dark square denotes the position of the man. E and L denote Engine and Last coach respectively of train 1.
Clearly, the length of train 2 does not matter, as we are only concerned about the movement of a particular point inside the train 2.

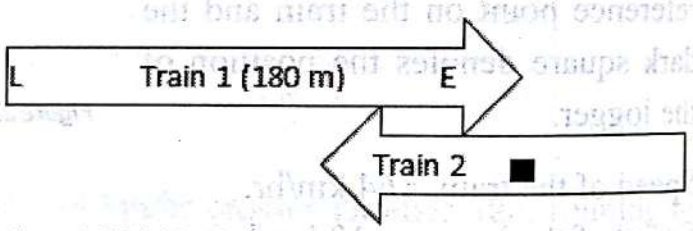

Figure 1: Initial Position - When the man jumps from train 1 to train 2

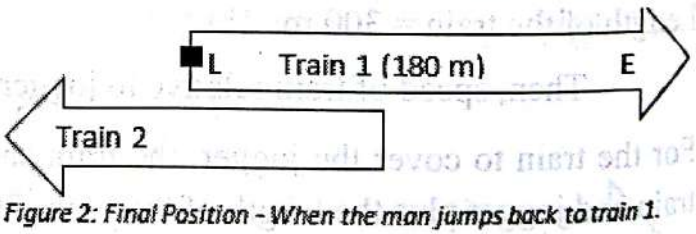

Figure 2: Final Position - When the man jumps back to train 1.

At first, the man jumped from engine (E) of the train 1 to a particular point in train 2.

Now we have to find the distance and hence the time for this point (in train 2) to meet the last coach (L) of train 1.

This point has to cover the full length of Train 1 (180 m) to reach the last coach.

Therefore, the relative distance to be covered by the two trains, d = 180 m.

$$\text{Relative speed} = 72 + 72 = 144 \text{ km/hr} = 144 \times \frac{5}{18} \text{ m/s} = 40\ m/s.$$

$$\text{Time} = \frac{d}{s} = \frac{180}{40} = \mathbf{4.5\ sec.}$$

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3.2 CSIR-NET Exam Questions – Detailed Solution

1. Two locomotives are running towards each other with speeds of 60 and 40 km/hr. An object keeps on flying to and fro from the front tip of one locomotive to the front tip of the other with a speed of 70 km/hr. After 30 minutes, the two locomotives collide and the object is crushed. What distance did the object cover before being crushed?

(a) 50 km **(b) 45 km** **(c) 35 km** **(d) 10 km**

Answer (c) **(CSIR-NET DEC 2014)**

Difficulty rating: 3/5 *Average time for solving: 60 seconds*

Here the movement of the flying object (bird) is not dependent on the movement of the locomotive, unlike as in other moving locomotive problems.

Let us explain it in detail using a pictorial representation.

The given problem can be pictured as follows. (The dark dot represents the flying object.)

Here, **the flight of the flying object does not depend on the flight of the locomotives.**

(Because the speeds of both the locomotives are less than that of the flying object. If the speed of any of the two locomotives were more than that of the flying object, the locomotive would have collided with the object before the collision with the other locomotive.)

Hence, for finding the distance covered by the flying object, the speeds of the two locomotives are insignificant, as their speeds do not contribute to the relative speed of the flying object.

Speed of the flying object = 70 km/hr.

Time taken for the collision to happen = 30 min = $^1/_2$ hr.

Distance travelled by the flying object = Speed × Time = $70 \times \frac{1}{2} =$ **35 km.**

2. To go from engine to the last coach of his train of length 200m, a man jumped from his train to another train moving on a parallel track in the opposite direction, waited till the last coach of his original train appeared and then jumped back. In how much time did he reach the last coach if the speed of each train was 60km/hr?

(a) 5 s **(b) 6 s** **(c) 10 s** **(d) 12s**

Answer (b) **(CSIR-NET JUNE 2014)**

Difficulty rating: 3.5/5 *Average time for solving: 90 seconds*

This question can be pictured as shown below.

Clearly, the length of train 2 does not matter as we are only concerned about the movement of a particular point inside Train 2.

The position of the man is denoted by the dark circle.

At first, the man jumped from engine (E) of the train 1 to a particular point in train 2.

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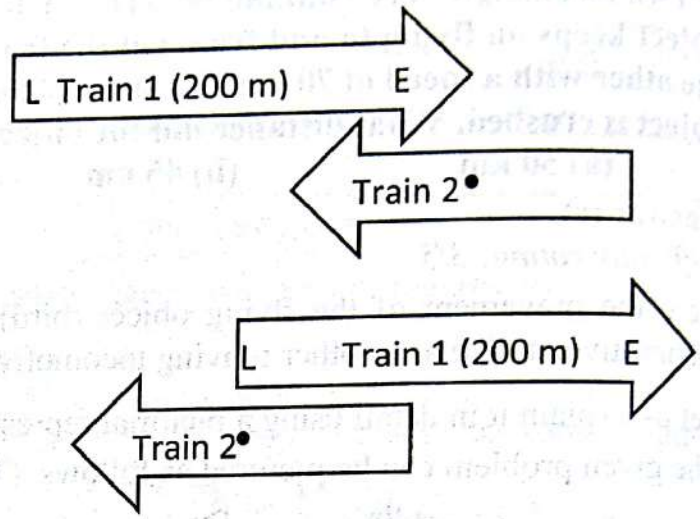
Now we have to find the distance and hence the time taken by this point (in train 2) to meet the last coach (L) of train 1.

This point has to cover the full length of train 1 (200 m) to reach the last coach.

Thus the relative distance to be covered by the two trains, d = 200 m.



Relative speeds of the two trains,

$$s = (60 + 60) \text{ km/hr}$$
$$= 120 \times \frac{1000}{60 \times 60} \text{ m/s}$$
$$= \frac{200}{6} \text{ m/s.}$$

Time $= \frac{d}{s} = \frac{200}{\frac{200}{6}} =$ **6 sec.**

3. There is a train of length 500 m, in which a man is standing at the rear end. At the instant the rear end crosses a stationary observer on a platform, the man starts walking from the rear to the front and the front to the rear of the train at constant speed of 3 km/hr. The speed of the train is 80 km/hr. The distance of the man from the observer at the end of 30 minutes is

(a) 41.5 km **(b) 40.5 km** **(c) 40.0 km** **(d) 41 km**

Answer (b) **(CSIR-NET JUNE 2014)**

Difficulty rating: 4/5 *Average time for solving: 110 seconds*

Here the "rear end" means the back end.

Let us picture the given question as shown below. The dark circle denotes the position of the man in the train and the dark square denotes the position of the observer in the platform.

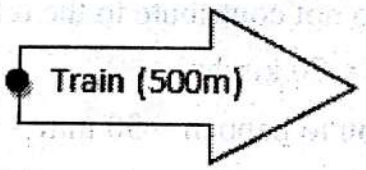


Figure 1: Initial position - When the rear end (man) crosses the observer.

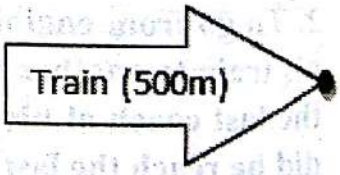


Figure 2: Final position: The instant after 30 minutes

First, consider the man alone.

Speed of man = 3 km/hr

Time = 30 minutes $= \frac{1}{2}$ hr.

Distance travelled by man = Speed × Time $= 3 \times \frac{1}{2} = 1.5$ km $= 1500$ m.

However, in the first 500 m, the man will reach the front end of the train.

In 1000 m, the man will reach rear end of the train again.

In 1500 m, the man will reach the front end of the train again.

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Thus, after 30 minutes, the man is actually at the front end of the train, and that is already **500 m** away from the observer.
(See the change in position of the dark circle in Figure 1 and Figure 2.)

Hence, the only thing remaining is to find out the distance covered by the **front end of the train** from the observer on the platform.

Speed of the train = 80 km/hr.

Time = 30 minutes = $\frac{1}{2}$ hr.

Distance travelled by the train = $80 \times \frac{1}{2}$ = **40 km.**

The distance of the man from the observer on the platform at the end of 30 minutes

= 40 km + 500 m = **40.5 km.**

4. Two trains of lengths 150 m and 250 m pass each other with constant speeds on parallel tracks in opposite directions. The drivers and guards are at extremities of the trains. The time gap between the drivers passing each other and first driver-guard pair passing each other is 30 s. How much later will other driver-guard pair pass by?

(a) 10 s **(b) 20 s** **(c) 30 s** **(d) 50 s**

Answer (b) **(CSIR-NET DEC 2013)**

Difficulty rating: 3.5/5 *Average time for solving: 90 seconds*

The given data can be pictured as shown below.

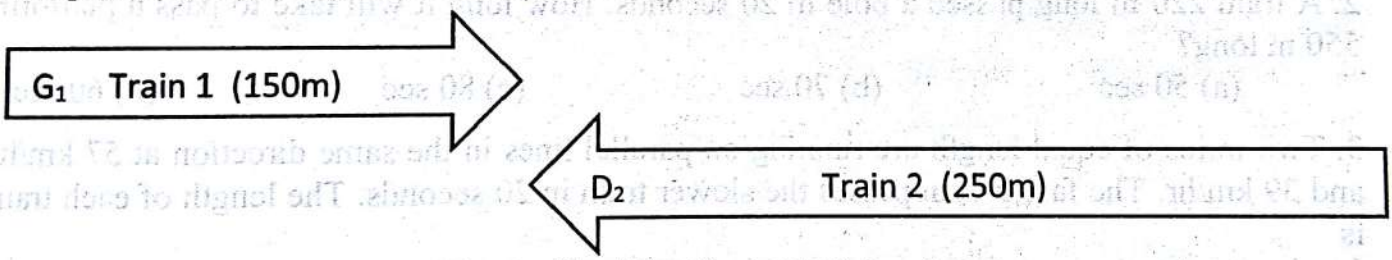

Figure 1: First position: When the two trains meets

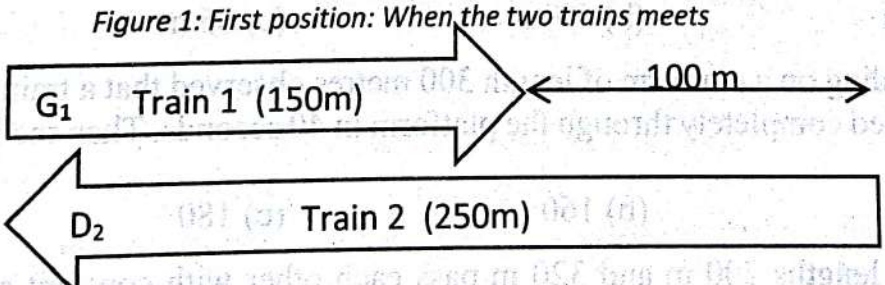

Figure 2: Second position - When the first driver-guard pair meets

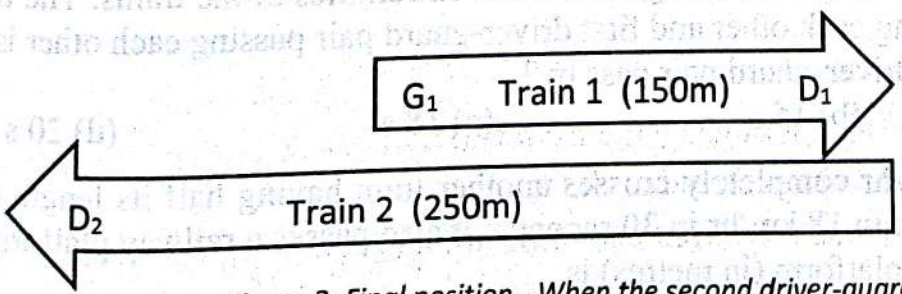

Figure 3: Final position - When the second driver-guard pair meets.

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At t = 0 sec,

D_1 and D_2 meets.

At t = 30 sec,

G_1 and D_2 meets.

For this to happen, as they both have same speed, two trains **together** have to cover 150 m.

Thus, relative speed of two trains $= \frac{\text{Distance}}{Time} = \frac{150}{30} = 5$ m/s.

Now we have to find the time taken for D_1 and G_2 to meet.

For D_1 and G_2 to meet, both trains together will have to cover a distance corresponding to the difference in lengths of the two trains.

Thus, here the distance covered by the two trains together = 100 m.

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}} = \frac{100}{5} = \mathbf{20\ sec}.$$

3.3 Practice Questions

1. Two stations A and B are 220 km apart on a straight line. One train starts from A at 7 am and travels towards B at 20 km/hr speed. Another train starts from B at 10 am and travels towards A at a speed of 20 km/hr. At what time they will meet?

(a) 1 pm (b) 2 pm (c) 3 pm (d) 4 pm

2. A train 220 m long passed a pole in 20 seconds. How long it will take to pass a platform 550 m long?

(a) 50 sec (b) 70 sec (c) 80 sec (d) 60 sec

3. Two trains of equal length are running on parallel lines in the same direction at 57 km/hr and 39 km/hr. The faster train passes the slower train in 20 seconds. The length of each train is

(a) 50 m (b) 60 m (c) 65 m (d) 72 m

4. A person standing on a platform of length 300 metres observed that a train takes 15 seconds to pass him, passed completely through the platform in 40 seconds. Then the length of the train (in metres) is

(a) 120 (b) 160 (c) 180 (d) 300

5. Two trains of lengths 200 m and 320 m pass each other with constant speeds on parallel tracks in opposite directions. The drivers and guards are at extremities of the trains. The time gap between the drivers passing each other and first driver-guard pair passing each other is 25 s. How much later will other driver-guard pair pass by?

(a) 9 s (b) 15 s (c) 18 s (d) 20 s

6. A train travelling at 54 km/hr completely crosses another train having half its length and travelling in opposite direction in 18 km/hr in 30 seconds. It also passes a railway platform in 40 seconds. The length of the platform (in metres) is

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(a) 200 (b) 300 (c) 400 (d) 420

7. Two trains travel in opposite directions at 55 km/hr and 53 km/hr and a man sitting in slower train passes the faster train in 5.5 seconds. The length of the faster train (in metres) is

(a) 140 (b) 162 (c) 165 (d) 168

8. Two trains whose lengths 180 m and 220 m respectively are running in opposite directions with respective speeds of 40 km/hr and 50 km/hr. Time taken by them to cross one another will be (in seconds)

(a) 16 (b) 17 (c) 18 (d) 22

9. In what time (in seconds) will a train 140 metres long cross an electric pole, if its speed is 126km/hr?

(a) 4 (b) 4.25 (c) 5 (d) 12.5

10. How long a train 150 metres long running at the speed of 108 km/hr take to cross a bridge 186 metres in length?

(a) 9.8 sec (b) 11.2 sec (c) 12.42 sec (d) 14.3 sec

Answer Key

1. (b) 2. (b) 3. (a) 4. (c) 5. (b) 6. (a) 7. (c) 8. (a) 9. (a) 10. (b)

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CHAPTER 4

SERIES FORMATION

Average Difficulty Rating: 3.3/5

Average time for each question: 109 sec

Number of questions expected for each exam: 2-4

4.1 THEORY

In mathematical terminology, the terms "sequence" and "series" have slightly different meaning. In simple words, a sequence is a comma-separated list of numbers or objects following a special order, whereas, a series is the sum of terms of a sequence. However, here the term "series formation" implies formation of both sequence and series.

In the questions on this section, usually a sequence of letters, numbers or symbols following some particular predefined rules will be given in the question. By understanding and applying these predefined rules, you are expected to find out the succeeding terms or a missing term in the given sequence.

Sequence can be categorized as:

1) Number Sequence
2) Letter Sequence
3) Symbol Sequence

4.1.1 Number Sequence

Some of the important types of number sequence are

4.1.1.1 Arithmetic sequence

Here, successive numbers are obtained by adding (or subtracting) a fixed number to the previous number. The fixed number is called the common difference of the arithmetic sequence.

Examples:

- 2, 4, 6, 8,....
- 1, 6, 11, 16,....
- 10, 8, 6, 4,....

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ILLUSTRATION 1:

Find the next term in the sequence 4, 7, 10, 13,…

(a) 14 (b) 16 (c) 17 (d) 20

Answer (b)

Clearly, this is an arithmetic sequence with common difference 3.

Therefore, the next term is 13 + 3 = **16.**

4.1.1.2 Geometric Sequence

A geometric sequence is the one in which each successive number is obtained by multiplying (or dividing) a fixed number (can be fraction also) by the previous number. That fixed number is called the common ratio of the geometric sequence.

Examples:

- 3, 6, 12, 24,…
- 2048, 1024, 512, 256, 128, 64,…
- 10, -40, 160, -640,…
- 1024, 640, 400, 250,…

Trick: *A simple trick to identify a geometric sequence is by dividing each pair of successive numbers in that sequence and checking whether we are getting a common ratio.*

ILLUSTRATION 2:

What is the next term in the sequence 1250, 1000, 800, 640,…

(a) 580 (b) 512 (c) 400 (d) 398

Answer (b)

First, let us divide each pair of successive numbers.

That is, $\frac{1000}{1250} = \frac{4}{5}$

$\frac{800}{1000} = \frac{4}{5}$

$\frac{640}{800} = \frac{4}{5}$

Clearly, this is a geometric sequence with common ratio $\frac{4}{5}$.

Let x be the next number.

Then, $\frac{x}{640} = \frac{4}{5}$

$\Rightarrow x = \frac{4 \times 640}{5} =$ **512.**

4.1.1.3 Arithmetico-Geometric Sequence

In an **Arithmetico-Geometric sequence**, each successive term is formed by first adding a fixed number to the previous term and then multiplying (or dividing) it by another fixed number.

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In other words, in this sequence, the difference of successive terms should be in Geometric Progression.

Examples:

- 1, 4, 10, 22, 46…
- 2, 12, 42, 132, 402…

ILLUSTRATION 3:

Find the next term in the sequence 3, 21, 75, 237,…

(a) 687 (b) 721 (c) 723 (d) 729

Answer (c)

Clearly, this is an Arithmetico-geometric sequence.

Here, first the fixed number 4 is added to the previous number. Then the fixed number 3 is multiplied to the resulting number.

That is, $(3 + 4) \times 3 = 21$.

$(21 + 4) \times 3 = 75$.

$(75 + 4) \times 3 = 237$.

So the next number = $(237 + 4) \times 3 =$ **723.**

4.1.1.4 Geometrico-Arithmetic Sequence

The sequence in which each successive term is formed by first multiplying (or dividing) a fixed number to the previous term and then adding it by another fixed number is called a **Geometrico-Arithmetic Sequence**.

Examples:

- 2, 11, 38, 119, 362, …
- 3, 7, 15, 31, 63, …

ILLUSTRATION 4:

Find the next term in the sequence 5, 13, 29, 61 …

(a) 69 (b) 111 (c) 123 (d) 125

Answer (d)

Clearly, this is a geometrico-arithmetric sequence.

Here, first the fixed number 2 is multiplied to the previous number. Then the fixed number 3 is added to the resulting number.

That is, $(5 \times 2) + 3 = 13$.

$(13 \times 2) + 3 = 29$.

$(29 \times 2) + 3 = 61$.

Therefore, the next number = $(61 \times 2) + 3 = 122 + 3 =$ **125.**

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4.1.1.5 Prime Sequence

As we know, prime numbers are numbers which can be exactly divided only by one and that number.

Prime sequence is the ordered set of prime numbers starting with any of the prime numbers.

Examples:

- 7, 11, 13, 17, 19,…
- 29, 31, 37, 41, 43,…

4.1.1.6 Power Sequence

In this sequence, every term will be the n^{th} power of some consecutive numbers. It can be square, cube or any higher powers of a particular sequence of numbers.

Examples:

- 1, 4, 9, 16, 25… – Square sequence
- 1, 8, 27, 64, 125, … – Cube sequence
- 1, 16, 81, 256… – 4^{th} power sequence.

4.1.1.7 Reversal Sequence

In this sequence, reversing each term will give us a sequence, which can be any of the above-mentioned sequence.

Example:

- 31, 51, 71, 91, 12…
- 61, 52, 63, 94, 46, 18,..

ILLUSTRATION 5:

Find the next number in the sequence 72, 46, 521,…

(a) 523 (b) 527 (c) 612 (d) 623

Answer (c)

Reversing each term, we get

27, 64, 125,…

This is the cube sequence starting from the cube of 3.

That is, $3^3, 4^3, 5^3$,…

Therefore, the next number = $6^3 = 216$.

Reverse the number again to obtain the next number in the given sequence.

That is, reversing 216 we get **612**.

4.1.1.8 Two-tier sequence

In a two-tier sequence, the difference of the successive terms can form any of the above-mentioned sequence like arithmetic sequence, prime sequence, power sequence etc.

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ILLUSTRATION 6:

What is the next term in the sequence 1, 4, 10, 19,…

(a) 28 (b) 29 (c) 31 (d) 33

Answer (c)

This is a two-tier arithmetic sequence.

That is, the difference of successive terms is given by,

$4 - 1 = 3$

$10 - 4 = 6$

$19 - 10 = 9$

Thus, the differences of successive terms form an arithmetic sequence 3, 6, 9,…

Clearly, the next term in this arithmetic sequence is 12.

Therefore, the difference of next successive terms is 12.

Let x be the next term in the given sequence.

Then, $x - 19 = 12$

$\Rightarrow x = \mathbf{31}$.

ILLUSTRATION 7:

Find the next number in the sequence 5, 12, 23, 36, 53,…

(a) 55 (b) 72 (c) 73 (d) 74

Answer (b)

This is a two-tier prime sequence.

The difference of successive numbers forms the sequence of prime numbers staring from 5.

That is,

$12 - 5 = 7$.

$23 - 12 = 11$.

$36 - 23 = 13$.

$53 - 36 = 17$.

The next prime in the sequence is 19.

Therefore, the next number in the given sequence is $53 + 19 = \mathbf{72}$.

4.1.1.9 Twin sequence

Twin sequence are the one in which two sequence are packed into one.

In a twin sequence any of the above-mentioned sequence like arithmetic sequence, prime sequence, etc. can occur.

Examples:

- 1, 0, 3, 2, 5, 4, 7,…
- 3, 1, 6, 3, 12, 5, 24,…

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ILLUSTRATION 8:

Find the next number in the sequence 1, 7, 3, 11, 5, 13, 7, 17, 9 , …

(a) 10 (b) 11 (c) 19 (d) 21

Answer (c)

This is a typical twin sequence.

Here, the terms in the odd positions (first, third, fifth etc.) form an arithmetic sequence having common difference 2. That is, 1, 3, 5, 7, 9…

The terms in the even positions (second, fourth, sixth etc.) form a prime sequence starting from 7. That is, 7, 11, 13, 17 …

Now the next number comes in the 10^{th} position (even position). Therefore, it is the member of the prime number sequence.

Hence, the next number is the prime number after 17, That is, **19**.

Some Special Sequences

4.1.1.9.1 Fibonacci sequence

In the Fibonacci sequence, the first two terms are 1, 1 and each subsequent term is the sum of the previous two.
That is, 1, 1, 2, 3, 5, 8, 13, 21,…

4.1.1.9.2 Telescoping Sequence

Unlike other sequence, telescoping sequence does not have a set form. A telescoping sequence is any sequence where nearly every term cancels with a preceding or following term.

The best example for a telescoping sequence is $\frac{1}{1\times2}, \frac{1}{2\times3}, \frac{1}{3\times4}, \ldots$

Here the terms in this sequence can be written as follows.

$$\frac{1}{1\times2} = \frac{1}{1} - \frac{1}{2}$$
$$\frac{1}{2\times3} = \frac{1}{2} - \frac{1}{3}$$
$$\frac{1}{3\times4} = \frac{1}{3} - \frac{1}{4}$$
$$\ldots$$

Thus, the series comprising of the terms of the sequence can be written as,

$$\frac{1}{1\times2} + \frac{1}{2\times3} + \frac{1}{3\times4} + \ldots + = (\frac{1}{1} - \frac{1}{2}) + (\frac{1}{2} - \frac{1}{3}) + (\frac{1}{3} - \frac{1}{4}) + \ldots.$$
$$= \frac{1}{1} + (-\frac{1}{2} + \frac{1}{2}) + (-\frac{1}{3} + \frac{1}{3}) + (-\frac{1}{4} + \frac{1}{4}) + \ldots.$$
$$= 1 + 0 + 0 + 0 + \ldots +$$
$$= 1$$

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The series comprising of the terms of the telescoping sequence is called a telescoping series.

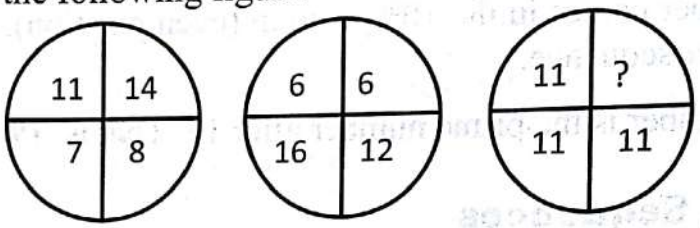
Even though these are the important types of sequence, there are numerous other sequence, which is beyond the scope of this discussion.

There can be questions in which some geometric figures will be given. The geometric figures will be subdivided into a number of parts, each filled with a number except one part. The numbers in the figures follow certain predefined rule. The objective of the question is to identify the predefined rule and find the missing number or letter.

ILLUSTRATION 9:

Find the missing number in the following figure.

(a) 1
(b) 3
(c) 7
(d) 11



Answer (c)

Let x be the missing number.

By observing the first two circles we infer that the sum of the numbers in the circles is 40.

Thus, in the third circle,

$11 + 11 + 11 + x = 40.$

$\Rightarrow x = 40 - 33.$

$\Rightarrow x = 7.$

4.1.2 Letter Sequence

In this type of sequence, a number of letters are arranged in a sequence and we are supposed to find the next letter or letters in the sequence.

The easiest way to solve these questions is by converting the letters into their numerical equivalent.

That is, A = 1, B = 2, C = 3, … ,X = 24, Y = 25, Z = 26.

After converting to numerical equivalent, the sequence become same as a number sequence. Finding the next number in a 'Number Sequence' is already discussed above.

Note that while finding the next number, if there occur a number, say x, greater than 26, subtract 26 from x to get the required number.
In addition, if there occur a negative number, add 26 to it to get the required number.

Trick: For converting an alphabet to the corresponding number or vice-versa, it is convenient to remember one or two equivalents. It saves lot of time.

Say, **remember that J ≡ 10 and T ≡ 20.**

Now any letter can easily be compared to these two letters.

For example, To find the numerical equivalent of V:

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V is the 2nd number from T.

Therefore, V is equivalent to 20 + 2 = 22.

ILLUSTRATION 10:

Find the missing letter

(a) J
(b) K
(c) L
(d) M

C	F	I	L	O
A	C	E	G	I
D	E	F	G	H
Z	X	V	T	R
W	T	Q	N	?

Answer (b)

First, convert every letters to their corresponding numerical equivalent.

Now, every rows of this table forms an arithmetic sequence.

Thus, the row containing the missing number follows an arithmetic sequence with first term 23 and common ratio -3.

3	6	9	12	15
1	3	5	7	9
4	5	6	7	8
26	24	22	20	18
23	20	17	14	?

Let x be the missing number.

Then, 14 + (-3) = x

⇨ x = 11.

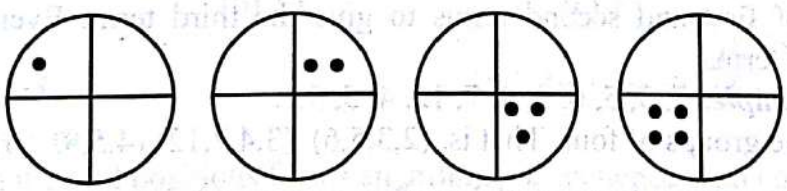
The letter corresponding to 11 is **K**, which is the missing letter in the given question.

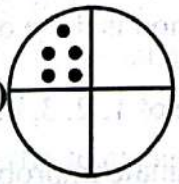
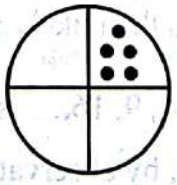
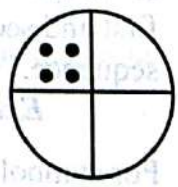
4.1.3 Symbol Sequence

In a symbol sequence, a pattern is given for three or four figures; we have to decide which will be the next figure to complete the given sequence.

ILLUSTRATION 11:

What figure comes next in the following picture sequence?



(a)
(b) 
(c) 
(d) 

Answer (b)

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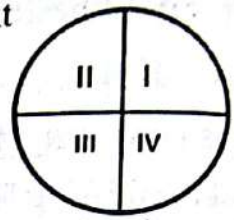
We have to analyse the number and position of the dots to get the next figure.



For convenience, the quadrants are marked as shown in the figure.

Number of dots: The number of dots follows the pattern, 1, 2, 3, 4.

Therefore, in the next figure, it should be 5 dots.

Hence, we can eliminate options (a) and (d) as they both do not possess 5 dots in any of the quadrants.

Position of dots: Now we can analyse the position of dots for remaining options (b) and (c).

The position of dots is shifting through the quadrants in the clock-wise pattern (II), (I), (IV), (III).

Therefore, the next position should be (II).

Hence, **option (b).**

4.1.4 Tips and tricks for identifying the sequence quickly

- If the sequence is increasing or decreasing with a small increase there must be an addition or subtraction involved, possibly an arithmetic sequence.
  Example: 1, 3, 5, 7,…
- If the sequence is increasing or decreasing with a large increase there must be a multiplication or division involved, possibly a geometric sequence.
  Example: 4, 8, 16, 32,…
- If the sequence is increasing or decreasing with a very large increase there must be square or cubes involved, possibly a power sequence.
  Example: 36, 49, 64, 81,…
- If the sequence is not linearly increasing and a pattern is not formed after solving for three terms then there is likely a combination of sequence, That is, a twin sequence.
  Example: 1, 0, 3, 2, 5, 4, 7…
- If a pattern is not formed after solving for 3 terms then the pattern may involve an operation of first and second terms to give the third term. Even it may involve grouping of terms.
  Example: 2, 3, 5, 6, 3, 4, 7, 12, 4, 5, 9…
  Take groups of four. That is, (2,3,5,6), (3,4,7,12),(4,5,9,…)
- If second term cannot be formed from first term or third term cannot be formed from first and second terms then likely an operation is done on another sequence to get this sequence.
  Example: 1, 4, 9, 16,… is a square of 1, 2, 3, 4,…
- For symbol sequence, by observation, eliminate improbable options to save time.

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4.2 CSIR-NET Exam Questions – Detailed Solution

1. What is the next term in the following sequence?

7, 11, 13, 17, 19, 23, 29,…

(a) 37 **(b) 35** **(c) 31** **(d) 33**

Answer (c) **(CSIR-NET DEC 2014)**

Difficulty rating: 2.5/5 *Average time for solving: 50 seconds*

This is the sequence of prime numbers starting from 7.

Remember: Prime number is a number, which has exactly two factors, namely itself and 1.

Therefore, the next prime number after 29 is **31**.

(For more understanding of prime numbers, read the section "Prime numbers" in Chapter 5.)

2. Every month, the price of a particular commodity falls in this order:

1024, 640, 400, 250,…

What is the next value?

(a) 156.25 **(b) Approx. 39** **(c) 64** **(d) 40**

Answer (a) **(CSIR-NET DEC 2014)**

Difficulty rating: 3/5 *Average time for solving: 90 seconds*

Let x be the next value in the sequence.

Let us check the ratio of successive terms.

$\frac{640}{1024} = \frac{5}{8}$ $\frac{400}{640} = \frac{5}{8}$ $\frac{250}{400} = \frac{5}{8}$

Clearly, this is a geometric sequence with the common ratio $\frac{5}{8}$.

Thus $\frac{x}{250} = \frac{5}{8}$

$\Rightarrow x = \frac{5 \times 250}{8}$

$\Rightarrow x = \mathbf{156.25.}$

3. What is the next number in the following sequence?

2, 3, 4, 7, 6, 11, 8, 15, 10,…

(a) 12 **(b) 13** **(c) 17** **(d) 19**

Answer (d) **(CSIR-NET DEC 2014)**

Difficulty rating: 3/5 *Average time for solving: 100 seconds*

This is a twin sequence.

Here, the terms in the odd positions forms an arithmetic sequence with common difference 2.

That is, 2, 4, 6, 8, 10,…

The terms in the even positions forms an arithmetic sequence with common difference 4.

That is, 3, 7, 11, 15,…

Therefore, the next number in the sequence is **19.**

4. Find the missing letter.

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(a) L
(b) Q
(c) N
(d) O

A	?	Q	E
C	M	S	C
E	K	U	A
G	I	W	Y

(CSIR-NET DEC 2014)

Answer (d)

Difficulty rating: 3.5/5

Average time for solving: 120 seconds

Let us tabulate the numerical equivalents of the given letters as shown below.

1	?	17	5
3	13	19	3
5	11	21	1
7	9	23	25

In the first column, the terms 1, 3, 5, 7 forms an arithmetic sequence with common difference 2.

In the second column, the terms 13, 11, 9 forms an arithmetic sequence with common difference -2.

In the third column, the terms 17, 19, 21, 23 forms an arithmetic sequence with common difference 2.

In the fourth column, the terms 5, 3, 1, 25 forms an arithmetic sequence with common difference -2.

Clearly, the columns form arithmetic sequence with common differences 2, -2, 2, -2 respectively.

The missing number is in the second column. For the second column, the common difference is -2.

Thus, the missing number is 15.

The alphabetical equivalent of 15 is **O.**

5. Continue the sequence

2, 5, 10, 17, 28, 41,_, _, _

(a) 58, 77, 100 **(b) 64, 81, 100** **(c) 43, 47, 53** **(d) 55, 89, 113**

Answer (a)

(CSIR-NET DEC 2014)

Difficulty rating: 3/5

Average time for solving: 120 seconds

This is a two-tier sequence. That is the differences of successive terms form another sequence.

Here $5 - 2 = 3$

$10 - 5 = 5$

$17 - 10 = 7$

$28 - 17 = 11$

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$41 - 28 = 13$.

Thus, the differences of successive terms form a sequence of prime numbers starting from 3.

That is, 3, 5, 7, 11, 13, 17, 19, 23…

Therefore, the next three differences should be 17, 19 and 23 respectively.

Hence, $41 + 17 = \mathbf{58}$

$58 + 19 = \mathbf{77}$

$77 + 23 = \mathbf{100}$ are the next terms in our given sequence.

Note that when getting the number 58 itself, we can deduce the answer as option (a) since no other option has 58 in it.

6. What is the next number in the following sequence?

2, 3, 5, 6, 3, 4, 7, 12, 4, 5, 9 …

(a) 10 **(b) 20** **(c) 13** **(d) 6**

Answer (b) **(CSIR-NET JUNE 2014)**

Difficulty rating: 3.5/5 *Average time for solving: 120 seconds*

Consider each block of four terms:

(2, 3, 5, 6,) (3, 4, 7, 12,) (4, 5, 9,__)…

In a block, say (a, b, c, d), $\mathbf{c = a + b}$ and $\mathbf{d = a \times b}$.

That is, the rule is that in a block, the third term is the sum of the first two terms and the fourth term is the product of the first two terms.

Hence, the next number is $4 \times 5 = \mathbf{20}$.

7. Find the missing number in the sequence?

61, 52, 63, 94, __, 18, 001, 121

(a) 46 **(b) 70** **(c) 66** **(d) 44**

Answer (a) **(CSIR-NET JUNE 2014)**

Difficulty rating: 3/5 *Average time for solving: 80 seconds*

This is a reversal sequence.

Reversing 61 as 16, 52 as 25 and so on, we get the sequence as

16, 25, 36, 49…

That is, $4^2, 5^2, 6^2, 7^2$…

Therefore, missing number is the reverse of $8^2 = 64$.

Reversing 64, we get **46** as the missing number.

8. Find the missing letter

(a) P
(b) K
(c) J
(d) L

A	**B**	**C**	**D**
F	**I**	**L**	**O**
K	**P**	**U**	**Z**
P	**W**	**D**	**?**

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Answer (b) **(CSIR-NET JUNE 2014)**

Difficulty rating: 3/5 *Average time for solving: 120 seconds*

We can tabulate the given letters to their numerical equivalent as,

1	2	3	4
6	9	12	15
11	16	21	26
16	23	4	?

In the first row, the terms 1, 2, 3, 4 forms an arithmetic sequence with common difference 1.

In the second row, the terms 6, 9, 12, 15 forms an arithmetic sequence with common difference 3.

In the third row, the terms 11, 16, 21, 26 forms an arithmetic sequence with common difference 5.

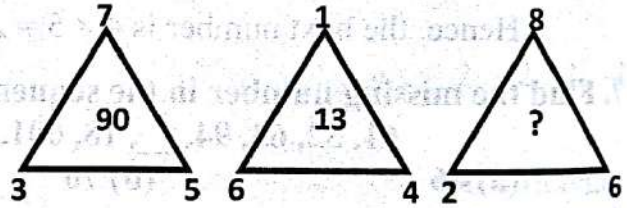
In the fourth row, the terms 16, 23, 4, forms an arithmetic sequence with common difference 7. (Note that $23 + 7 = 30 \equiv 30 - 26 = 4$.)

Clearly, rows form arithmetic sequence with common difference 1,3,5,7 respectively.
Since, the fourth row has common difference 7, the missing number is $4 + 7 = \mathbf{11}$.

The alphabetical equivalent of 11 is **K**.

9. Find the missing number in the triangle?

(a) 16
(b) 96
(c) 50
(d) 80



Answer (d) **(CSIR-NET JUNE 2014)**

Difficulty rating: 3/5 *Average time for solving: 90 seconds*

From the first 2 triangles, we can observe that,

$$90 = (7 \times 5 \times 3) - (7 + 5 + 3)$$
$$13 = (1 \times 4 \times 6) - (1 + 4 + 6)$$

Thus obviously,

$$(8 \times 6 \times 2) - (8 + 6 + 2) = 96 - 16 = \mathbf{80}.$$

10. The next number of the sequence 1, 5, 14, 30, 55,... is

(a) 85 **(b) 90** **(c) 91** **(d) 95**

Answer (c) **(CSIR-NET DEC 2013)**

Difficulty rating: 3/5 *Average time for solving: 120 seconds*

This is a two-tier sequence.

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That is the differences of successive terms form a sequence of perfect squares.
That is,

$1^2 = 1$

$1 + 2^2 = 1 + 4 = 5$

$5 + 3^2 = 5 + 9 = 14$

$14 + 4^2 = 14 + 16 = 30$

$30 + 5^2 = 30 + 25 = 55$

Thus,

$55 + 6^2 = 55 + 36 = \mathbf{91.}$

11. Consider the sequence of ordered sets of natural numbers: {1}, {2,3}, {4,5,6},... What is the last number in the 10th set?

(a) 10 **(b) 19** **(c) 55** **(d) 67**

Answer (c) **(CSIR-NET DEC 2013)**

Difficulty rating: 3.5/5 *Average time for solving: 140 seconds*

The 1st set {1} has **1** element.

The 2nd set {2, 3} has **2** elements, and the last element 3 is the sum of first **two** natural numbers (that is, 1 + 2).

The 3rd set {4, 5, 6} has **3** elements, and the last element 6 is the sum of first **three** natural numbers. (that is, 1 + 2 + 3)

The 4th set {7, 8, 9, 10} has **4** elements, and the last element 10 is the sum of first **four** natural numbers. (that is, 1 + 2 + 3 + 4)

Proceeding the same way,
The 10th set has **10** elements, and the last element is the sum of first **ten** natural numbers.
(that is, 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10)

Remember: Sum of first n natural numbers $= \frac{n(n+1)}{2}$

Thus, sum of first ten natural numbers $= \frac{10(10+1)}{2} = \frac{110}{2} = \mathbf{55.}$

12. What is the next number in the following sequence?

39, 42, 46, 50,...

(a) 52 **(b) 53** **(c) 54** **(d) 55**

Answer (d) **(CSIR-NET DEC 2013)**

Difficulty rating: 3.5/5 *Average time for solving: 120 seconds*

Let x be the next number in the sequence.

In this sequence, we can observe the predefined rule as,

"If a, b, c, d,... is the sequence, then b – a = the first digit of a."

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That is, 39, first digit is 3, hence, $42 - 39 = 3$.

42, first digit is 4, hence, $46 - 42 = 4$.

46, first digit is 4, hence, $50 - 46 = 4$.

Thus, 50, first digit is 5, hence, $x - 50 = 5$.

$\Rightarrow x = \mathbf{55}$.

13. A lucky man finds 6 pots of gold coins. He counts the coins in the first four pots to be 60, 30, 20 and 15 respectively. If there is a definite progression, what would be the numbers of coins in the next two pots?

(a) 10 and 5 **(b) 4 and 2** **(c) 15 and 15** **(d) 12 and 10**

(CSIR-NET JUNE 2013)

Answer (d)

Difficulty rating: 3.5/5

Average time for solving: 120 seconds

The given sequence is 60, 30, 20, 15…

Let x and y be the next two terms.

$\frac{30}{60} = \frac{1}{2}$ $\frac{20}{30} = \frac{2}{3}$ $\frac{15}{20} = \frac{3}{4}$

Hence the ratios we obtained are $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}$

Here the numerators is in the sequence 1, 2, 3, 4,….

And the denominators is in the sequence 2, 3, 4, 5,…

Hence, the next numerator and denominator is 4 and 5 respectively.

Therefore the next ratio is $\frac{4}{5}$

Thus, $\frac{x}{15} = \frac{4}{5}$

$\Rightarrow x = \frac{4 \times 15}{5}$

$\Rightarrow x = 12$

Again, $\frac{y}{12} = \frac{5}{6}$

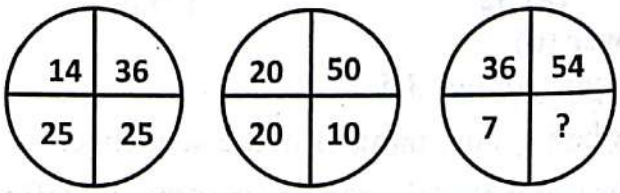
$\Rightarrow y = \frac{5 \times 12}{6}$

$\Rightarrow y = 10$

Hence, coins in the next two pots are **12 and 10** respectively.

14. Find the missing number:

(a) 1
(b) 0
(c) 2
(d) 3



Answer (d) **(CSIR-NET JUNE 2013)**

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Difficulty rating: 3/5 *Average time for solving: 80 seconds*

Let x be the missing number.

From the first two circles,

$14 + 36 + 25 + 25 = 100.$

$20 + 50 + 20 + 10 = 100.$

Thus, we observe that **the sum of numbers in each circle is 100**.

Therefore, for the third circle,

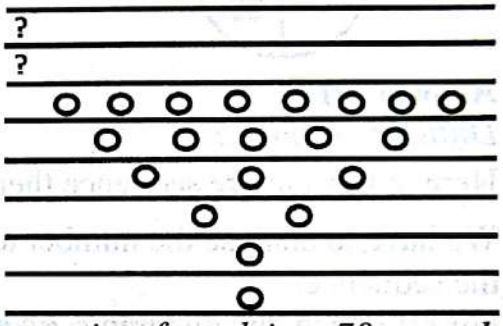
$36 + 54 + 7 + x = 100.$

$\Rightarrow 97 + x = 100.$

$\Rightarrow x = \mathbf{3}.$

15. In the figure, the numbers of circles in the blank rows must be

(a) 12 and 20
(b) 13 and 20
(c) 13 and 21
(d) 10 and 11



Answer (c) **(CSIR-NET JUNE 2013)**

Difficulty rating: 3/5 *Average time for solving: 70 seconds*

The numbers of circles are following the **Fibonacci sequence.**

Note: Fibonacci sequence
The first two numbers are 1,1 and each subsequent number is the sum of the previous two.
*That is, 1, 1, 2, 3, 5, 8, **13, 21**,...*

Thus, the numbers of circles in the blank row are **13 and 21** respectively.

16. What is the value of $\frac{1}{1\times2} + \frac{1}{2\times3} + \frac{1}{3\times4} + \ldots$ to ∞?

(a) $\frac{2}{3}$ **(b) 1** **(c) 2** **(d)** ∞

Answer (b) **(CSIR-NET JUNE 2013)**

Difficulty rating: 3.5/5 *Average time for solving: 120 seconds*

The given series is called the **telescopic series.**

The terms in the telescopic sequence can be written as follows..

$$\frac{1}{1\times2} = \frac{1}{1} - \frac{1}{2}, \quad \frac{1}{2\times3} = \frac{1}{2} - \frac{1}{3}, \quad \frac{1}{3\times4} = \frac{1}{3} - \frac{1}{4} \quad \ldots$$

Thus, the series can be written as,

$$\frac{1}{1\times2} + \frac{1}{2\times3} + \frac{1}{3\times4} + \ldots + = \left(\frac{1}{1} - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \ldots.$$

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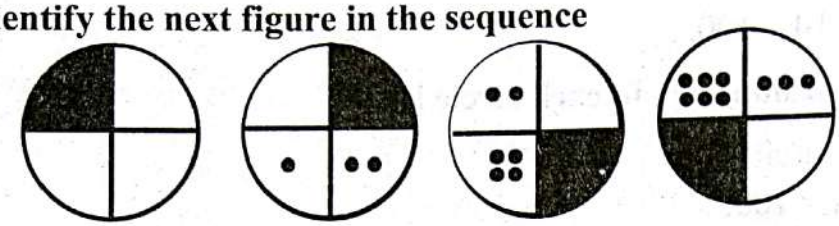
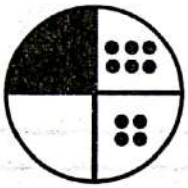
$$= \frac{1}{1} + (-\frac{1}{2} + \frac{1}{2}) + (-\frac{1}{3} + \frac{1}{3}) + (-\frac{1}{4} + \frac{1}{4}) + \ldots.$$

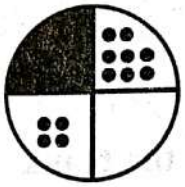
$$= 1 + 0 + 0 + 0 + \ldots +$$

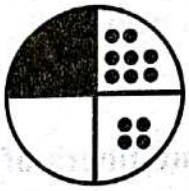
$$= \mathbf{1.}$$

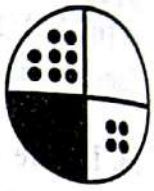
17. Identify the next figure in the sequence



(a) 

(b) 

(c) 

(d) 

Answer (c) **(CSIR-NET JUNE 2013)**

Difficulty rating: 3/5 *Average time for solving: 130 seconds*

Here in this picture sequence there are two varying objects - the dots and the shaded quadrant.

We have to analyse the number and position of both these objects to deduce the next figure in the sequence.

For reference, the quadrants are marked as shown in the figure.

Now we can examine the four figures in the given picture sequence and observe the pattern followed by the dots and the shaded quadrants.

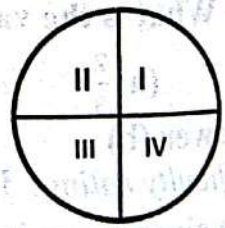
Position of the shaded quadrant:

The shaded quadrants are shifting in clock-wise direction following the pattern II-I-IV-III.

Therefore, in the next figure the shaded quadrant should be on quadrant II.

Since the shaded quadrant is on quadrant III on option (d), it can be eliminated.

Now we can analyse the dots for the remaining options (a), (b) and (c).



Number of dots:

The number of dots is following the pattern, (0,0), (1,2), (2,4), (3,6).

Therefore in the next figure, it should be (4,8) dots.

Since in option (a) the number of dots is (4,6), option (a) can be eliminated.

Now we can analyse the position of dots for the remaining options (b) and (c).

Position of dots:

The position of dots is shifting through the quadrants in clock-wise direction following the pattern (IV,III) , (III,II), (II,I).

Therefore the next position should be (I,IV).

The only option satisfying all these conditions is **option (c).**

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18. If the sum of the next two terms of the series below is x, what is the value of $\log_2 x$?
2, -4, 8, -16, 32, -64, 128,...

(a) 128 **(b) 10** **(c) 256** **(d) 8**

Answer (d) **(CSIR-NET DEC 2012)**

Difficulty rating: 3.5/5 *Average time for solving: 120 seconds*

$\log_2 x = ?$

First we should find x.

Given: x is the sum of next two terms in the given sequence.

The given sequence is **2, -4, 8, -16, 32, -64, 128,...**

This is clearly a geometric sequence with common ratio -2.

Thus, the next two terms is given by:

$128 \times -2 = -256$ and $-256 \times -2 = 512$.

Thus, $x = (-256) + 512 = 256$.

$\log_2 256 = \log_2 2^8$

Note: $\log_a a^b = b$

Hence, $\log_2 2^8 = \mathbf{8}$.

19. What is the next number of the sequence 24, 30, 33, 39, 51,... ?

(a) 57 **(b) 69** **(c) 54** **(d) 81**

Answer (a) **(CSIR-NET DEC 2012)**

Difficulty rating: 3/5 *Average time for solving: 90 seconds*

The predefined rule that holds for this sequence can be observed as

"The difference between two successive numbers is the sum of digits of the first number."

That is, $30 - 24 = 6 = 2 + 4$

$33 - 30 = 3 = 3 + 0$

$39 - 33 = 6 = 3 + 3$

$51 - 39 = 12 = 3 + 9$

Let x be the next number.

Sum of digits of 51 is $5 + 1 = 6$.

Thus $x - 51 = 6$

$\Rightarrow x = 57$

20. What is the next number in this "see and tell" sequence?
1, 11, 21, 1211, 111221,...

(a) 312211 **(b) 1112221** **(c) 1112222** **(d) 1112131**

Answer (a) **(CSIR-NET DEC 2012)**

Difficulty rating: 4/5 *Average time for solving: 130 seconds*

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In this sequence, every number is formed by clubbing the number and the cardinality of distinct numbers occurring together in the previous number. *(Note: Cardinality means number of occurrence.)*

That is, If we start with the first number 1, we have **one 1**. So the next number is **11** (one 1)

In 11, we have, **two 1's**. Therefore, the next number is **21.** (two 1)

In 21, we have **one 2 and one 1**. Therefore, the next number is **1211.** (one 2 one 1)

In 1211, we have **one 1, one 2, two 1**. Therefore, the next number is **111221.**(one 1, one 2, two 1)

Hence, in 111221, we have **three 1, two 2, one 1.**

Therefore, the next number in the sequence is **312211.** (three 1, two 2, one 1)

21. In a sequence $\{a_n\}$, every term is equal to the sum of all its previous terms. If $a_0 = 3$, then $\lim_{n\to\infty} \frac{a_{n+1}}{a_n}$ is

(a) 3 **(b) 2** **(c) 1** **(d) e**

(CSIR-NET DEC 2012)

Answer (b)

Difficulty rating: 4/5

Average time for solving: 110 seconds

Given that, every term is equal to the sum of all its previous terms and $a_0 = 3$.

Hence, $a_1 = a_0 = 3$,

$a_2 = a_0 + a_1 = 3 + 3 = 6$,

$a_3 = a_0 + a_1 + a_2 = 3 + 3 + 6 = 12$,

$a_4 = a_0 + a_1 + a_2 + a_3 = 3 + 3 + 6 + 12 = 24$.

Thus, the sequence is 3, 3, 6, 12, 24…

Thus, after first term, the ratio of $(n+1)^{th}$ to the n^{th} term is 2.

(For e.g., $\frac{12}{6} = 2, \frac{24}{12} = 2$…)

Hence, $\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = 2$.

22. Find the missing letter?

(a) H

(b) L

(c) Z

(d) V

A	EGK	C
?		P
U		R
Q		V
B	OJF	D

(CSIR-NET DEC 2012)

Answer (c)

Difficulty rating: 4.5/5

Average time for solving: 150 seconds

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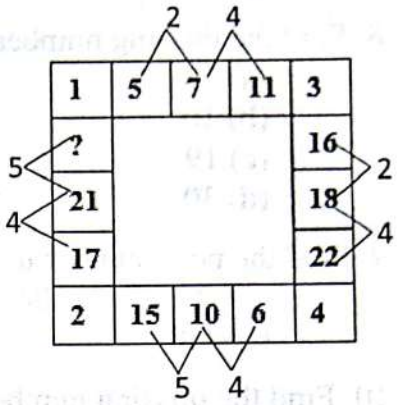
First, we can tabulate the given letters into their equivalent numbers.

Here, the table has 5 rows and 5 columns.

From the Figure it is clear that,
In the 1st row and 5th column, the difference of numbers in the middle are 2 and 4.

Thus, in the 1st column and 5th row, the difference of numbers in the middle should be 5 and 4. (See figure)
Hence, the missing number should be 5 more than 21.

Therefore, 26 is the missing number and **Z** is the letter corresponding to it.



Figure

4.3 PRACTICE QUESTIONS

1. Find the number which can be placed at the sign of interrogation?

3	17	48
4	9	32
6	5	?

(a) 9 (b) 11 (c) 13 (d) 24

2. Find the missing number from the given responses.
256, 32, _ , 4, 2
(a) 6 (b) 8 (c)12 (d) 16

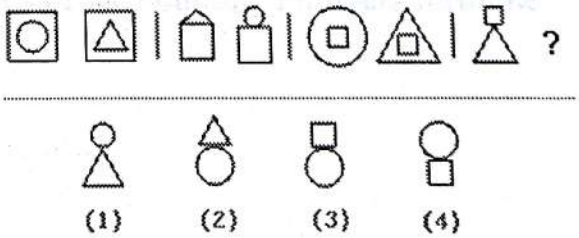
3. Find the missing letter

B	C	D	E
I	L	O	R
P	U	Z	E
W	D	K	?

(a) E
(b) P
(c) R
(d) T

4. Find the next number in the sequence 1.4, 2.2, 3.0, 3.8, ...
(a) 4.6 (b) 4.8 (c) 4.9 (d) 5.0

5. What comes next in the following picture sequence



(a) 1
(b) 2
(c) 3
(d) 4

6. What is the next term in the following sequence?
5, 7, 11, 13, 17, 19, 23, 29,...
(a) 30 (b) 31 (c) 32 (d) 33

7. What is the next term in the following sequence?
1600, 1200, 900,...
What is the next value?

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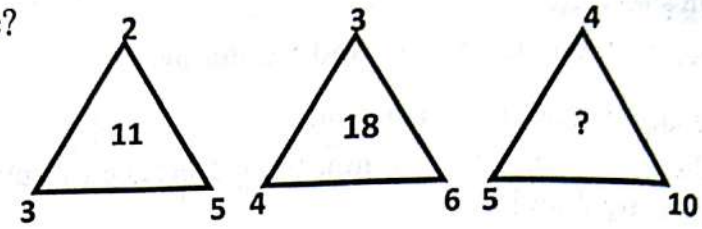
(a) 600 (b) 675 (c) 680.5 (d) 800

8. Find the missing number in the triangle?

(a) 7
(b) 10
(c) 19
(d) 30



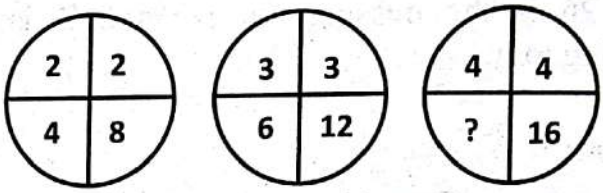
9. Find the next number in the sequence

301, 302, 304, 307, ...

(a) 308 (b) 309 (c) 310 (d) 311

10. Find the missing number:

(a) 4
(b) 6
(c) 8
(d) 12



Answer Key

1. (d) 2. (b) 3. (c) 4. (a) 5. (c) 6. (b) 7. (b) 8. (d) 9. (d) 10. (c)

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CHAPTER 5

NUMBERS

Average Difficulty Rating: 3/5

Average time for each question: 90 sec

Number of questions expected for each exam: 2-4

5.1 THEORY

A number is a mathematical object used for counting and measuring. A group of digits, called numerals, denotes it. Usually, in our everyday life, we use the decimal system comprising ten numbers viz. 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 called zero, one, two,...,nine, respectively.

5.1.1 Different types of numbers:

5.1.1.1 Natural Numbers

The counting numbers are called natural numbers.

Hence 1,2,3,4… are all natural numbers.

5.1.1.2 Whole Numbers

All counting numbers and 0 (zero) together form the set of whole numbers.

Hence 0,1,2,3,4,… are whole numbers.

5.1.1.3 Integers

All counting numbers including zero and the negatives of the counting numbers form the set of integers.

Hence …, -3, -2, -1, 0, 1, 2, 3,… are all integers.

Set of positive integers = {1, 2, 3, 4, 5,…}

Set of negative integers = {-1, -2, -3, -4, -5,…}

Set of all non-negative integers = {0, 1, 2, 3, 4, 5,…}

5.1.1.4 Even Numbers

An integer which is exactly divisible by two (2) is called an even number.

Thus -6, -4, -2, 0, 2, 4, 6, 8, 10,… are all even numbers.

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5.1.1.5 Odd Numbers

An integer which cannot be exactly divisible by two (2) is called an odd number.

Thus -5, -3, -1, 1, 3, 5, 7, 9, 11… are all odd numbers.

5.1.1.6 Prime Numbers

A counting number is called a prime number if it has exactly two positive factors (divisors), **namely itself and 1.**

Example: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89 and 97 are the primes less than 100.

- **Test for finding out whether the given number is prime or not:**

Let p be a given number and let n be the smallest counting number such that $n^2 \geq p$.

Now, test whether p is divisible by any of the prime numbers less than or equal to n.

If yes, then p is not prime. Otherwise, p is prime.

Example: Check whether 67 is a prime number.

We know that, $9^2 > 67$.

Prime numbers less than 9 are 2, 3, 5, and 7.

Clearly, none of them divides 67.

Thus, 67 is a prime number.

- **Prime Factorization**

The prime factors of a positive integer are the prime numbers that divide that integer exactly. The prime factorization of a positive integer is the process of listing of the integer's prime factors, together with their multiplicities.

Example 1: Prime factorization of 240 $= 2 \times 2 \times 2 \times 2 \times 3 \times 5$

$= 2^4 3^1 5^1$.

in which the factors 2, 3, 5 have multiplicities 4, 1, 1 respectively.

Example 2: Prime factorization of 3150 $= 2 \times 3 \times 3 \times 5 \times 5 \times 7$

$= 2^1 3^2 5^2 7^1$.

in which the factors 2, 3 , 5, 7 have multiplicities 1, 2, 2, 1 respectively.

- **Relatively prime numbers**

Two integers are relatively prime if they share no common positive factors (divisors) except 1.

Example: 12 and 25 are relatively prime numbers.

$12 = 2^2 \times 3$, $25 = 5^2$.

Clearly, they share no common positive factor except 1.

5.1.1.7 Composite Number

The natural numbers, which are not prime, are called composite numbers.

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4, 6, 8, 9, 10, 12… are all composite numbers.

Note: 1 is neither prime nor composite.

5.1.1.8 Perfect Number

A perfect number is a natural number, which when you add up all of the factors less than that number will give you that number.

Example: The factors of 28 are 1, 2, 4, 7, and 14.

$1 + 2 + 4 + 7 + 14 = 28$.

Thus, 28 is a perfect number.

5.1.1.9 Perfect Square

A perfect square is a number that can be written as the product of two equal factors.

Example: 36, 49, 64,… are all perfect squares.

Important properties of perfect squares:

- No perfect square ends with 2, 3, 7, 8.
- No perfect square ends with an odd number of zeros.
- **Perfect squares can be recognized by the fact that all of their prime factors have even multiplicities.**

  Example: Prime factorization of $900 = 2^2 3^2 5^2$

  in which multiplicities of the factors 2, 3, 5 are 2, 2, 2 respectively, which are all even. Thus 900 is a perfect square.

5.1.2 Tests of Divisibility

Divisibility by 2

A number is divisible by 2 if the last digit is even. i.e., 0, 2, 4, 6 or 8.

Divisibility by 3

A number is divisible by 3 if the sum of its digits is divisible by 3.

Divisibility by 4

A number is divisible by 4 if the number formed by the last two digits is divisible by 4.

Divisibility by 5

A number is divisible by 5 if its last digit is 0 or 5.

Divisibility by 6

A number is divisible by 6 if the number is divisible by both 2 and 3.

Divisibility by 7

To check whether a given number is divisible by 7, subtract twice the unit digit from the number represented by the remaining digits of the number to obtain an integer.

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Then the original number is divisible by 7 if the obtained integer is divisible by 7.

Example: 343 is divisible by 7, we take the unit digit 3, double it to 6, and then subtract it from 34, then 34 – 6 = 28, which is divisible by 7.

Divisibility by 8
A number is divisible by 8 if the number formed by the last three digits is divisible by 8.

Divisibility by 9
A number is divisible by 9 if the sum of the digits is divisible by 9.

Divisibility by 10
A number is divisible by 10 if it ends in a 0.

Divisibility by 11
A number is divisible by 11 if the difference of the sum of the digits in theeven and odd positions in the number is either 0 or divisible by 11.

Example: 979 is divisible by 11,
Sum of digits in the odd positions = 9 + 9 = 18
Sum of digits in the even positions = 7
Difference = 18 -7 = 11, which is divisible by 11.

5.1.3 GCD and LCM

Factors and Multiples:
If a number, say, 'a' exactly divides another number 'b', we can say that 'a' is a factor of 'b'. Further, 'b' is called a multiple of 'a'.

5.1.3.1 Greatest Common Divisor (GCD) or Highest Common Factor (HCF):

The GCD or HCF of two or more than two numbers is the greatest number that divides each of them exactly.

Method for finding the GCD of two or more numbers:
Factorization Method: Express each one of the given numbers as the product of prime factors (Prime factorize). The product of least powers of common prime factors gives GCD.

Example: Find the GCD of 504 and 264?
Prime factors of $504 = 2 \times 2 \times 2 \times 3 \times 3 \times 7 = 2^3 3^2 7^1$.
Prime factors of $264 = 2 \times 2 \times 2 \times 3 \times 11 = 2^3 3^1 11^1$.
Prime factors common to 504 and 264 $= 2^3 3^1 = 2 \times 2 \times 2 \times 3 = 24$.
Thus, the GCD of 504 and 264 is 24.

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5.1.3.2 Least Common Multiple (LCM)

The LCM of two or more numbers is the smallest number which is exactly divisible by each one of the given numbers.

Method for finding the LCM of two or more numbers:

Factorization Method: Express each one of the given numbers as the product of prime factors (Prime factorize). The product of highest powers of all the prime factors gives their LCM.

Example: Find LCM of 504 and 264

Prime factors of $504 = 2 \times 2 \times 2 \times 3 \times 3 \times 7 = 2^3 3^2 7^1$.

Prime factors of $264 = 2 \times 2 \times 2 \times 3 \times 11 = 2^3 3^1 11^1$.

Product of highest powers of all prime factors of 504 and $264 = 2^3 3^2 7^1 11^1$

$= 2 \times 2 \times 2 \times 3 \times 3 \times 7 \times 11 = 5544$.

Thus, the LCM of 504 and 264 is 5544.

Important Properties of GCD and LCM

- Product of two numbers = Product of their GCD and LCM
- GCD of given numbers always divides their LCM
- GCD of given fractions $= \frac{\text{GCD of Numerators}}{\text{LCM of Denominators}}$
- LCM of given fractions $= \frac{\text{LCM of Numerators}}{\text{GCD of Denominators}}$

ILLUSTRATION 1:

The GCD of two numbers is 24 and their LCM is 1080. If one of the numbers is 120, find the other number?

(a) 13 (b) 202 (c) 216 (d) 256

Answer (c)

Let a and b be the two numbers.

Given a = 120, b= ?

GCD of a and b = 24, LCM of a and b = 1080.

We know that, Product of two numbers = Product of their GCD and LCM.

$\Rightarrow 120 \times b = 24 \times 1080$

$\Rightarrow b = \frac{24 \times 1080}{120}$

$\Rightarrow b = 216$.

5.1.4 BODMAS rule

BODMAS (Bracket - of – Division – Multiplication – Addition – Subtraction) rule depicts the correct sequence in which the operations are to be executed, to find out the value of a given mathematical expression.

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Thus, in simplifying an expression, first of all the brackets must be removed (by solving out the expressions in the brackets) strictly in the order ( ), { }, and [ ].

After removing the brackets, we must use the following operations strictly in the order:
(i) Division (ii) Multiplication (iii) Addition (iv) Subtraction.

ILLUSTRATION 2:

$(42 \div 6 + 9 - 3 \times 2) + (15 \times 5 - 6) =$

(a) 27/11 (b) 43 (c) 79 (d) -5

Answer (c)

By BODMAS rule,

$(42 \div 6 + 9 - 3 \times 2) + (15 \times 5 - 6) = ((42 \div 6) + 9 - (3 \times 2)) + ((15 \times 5) - 6)$

$= ((7 + 9) - 6) + ((15 \times 5) - 6)$

$= (16 - 6) + (75 - 6)$

$= 10 + 69$

$= \mathbf{79.}$

5.1.5 Modulus of a real number:

Modulus of a real number x is defined as

$|x| = x$ if $x > 0$

$|x| = -x$ if $x < 0$

In simple words, modulus function takes the positive value of any real number.

Example 1: $|-10| = 10$.

Example 2: $|15| = 15$.

5.1.6 Quadratic equation

A quadratic equation is an equation of the form $f(x)$: $ax^2 + bx + c = 0$, where a,b,c are all real and $a \neq 0$, and x is an unknown variable. The values of x satisfying $f(x) = 0$ are called its roots.

- In $ax^2 + bx + c = 0$, the expression $D = b^2 - 4ac$ is called its discriminant.
- Let α, β be the roots of the equation, $ax^2 + bx + c = 0$. Then,

$$\textbf{Sum of roots} = (\alpha + \beta) = -\frac{b}{a}$$

$$\textbf{Product of roots} = \alpha\beta = \frac{c}{a}$$

- Whenever the roots of a quadratic equations are given, the quadratic equation is given by

$$x^2 - (\text{Sum of roots})x + \text{Product of roots} = 0.$$

Example: If the roots are given as 3 and 5, then the quadratic equation will be

$x^2 - (3 + 5)x + (3 \times 5) = 0$

$\Rightarrow$ $x^2 - 8x + 15 = 0$.

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5.1.7 Fractions

A fraction represents a part of a whole or, more generally, any number of equal parts. A simple fraction (examples: $\frac{3}{4}$ and 5/7) consists of an integer numerator, displayed above a line (or before a slash), and a non-zero integer denominator, displayed below (or after) that line.

Proper fraction: A fraction whose numerator is less than the denominator.

Examples: $\frac{1}{2}, \frac{2}{3}, \frac{19}{20}$.

Improper fraction: A fraction whose numerator is equal to or greater than the denominator.

Examples: $\frac{4}{3}, \frac{10}{3}, \frac{20}{19}$.

Mixed Fraction: An improper fraction can be expressed as a whole number and a proper fraction. This expression is called mixed fraction.

Examples: $3\frac{1}{2}, 5\frac{7}{8}, 12\frac{3}{17}$.

Method to convert improper fraction into mixed fraction

Step 1: Divide the numerator by the denominator.

Step 2: Write down the whole number answer (Quotient).

Step 3: Then write down the remainder above the denominator.

ILLUSTRATION 3:

Convert $\frac{600}{11}$ to a mixed fraction

Here numerator = 600, denominator = 11.

Step 1: Divide 600 by 11.

Step 2: The quotient is 54 and remainder is 6.

Step 3: $54\frac{6}{11}$ is our required mixed fraction.

5.1.8 Laws of Exponents

Definition of a^m: If 'a' is a natural number, then a^m stands for the product of 'm' factors each equal to 'a'.

- $(\frac{a}{b})^m = \frac{a^m}{b^m}$
- $a^0 = 1$
- $a^{-m} = \frac{1}{a^m}$
- $a^{\frac{1}{m}} = \sqrt[m]{a}$

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Properties of exponents

- $a^m \times a^n = a^{m+n}$
- $\frac{a^m}{a^n} = a^{m-n}$
- $(a^m)^n = a^{mn}$
- $(ab)^m = a^m.b^m$
- For any natural numbers a, b, c, d,
  $a^b > a^c > a^d$, if and only if $b > c > d$.
- For three positive real numbers 'a', 'b', 'c' with the same positive valued exponent x,
  $a^x > b^x > c^x$ if and only if $a > b > c$.

Some important expansions

- $(a + b)^2 = a^2 + 2ab + b^2$
- $(a - b)^2 = a^2 - 2ab + b^2$
- $a^2 - b^2 = (a + b)(a - b)$
- $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$
- $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$
- $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
- $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

ILLUSTRATION 4:

$16^0 + 16^2 + \frac{1}{16^2} + 16^{\frac{1}{2}} + \frac{1}{16^{\frac{1}{2}}}$ equals

(a) 1 (b) $16^{2\frac{1}{2}}$ (c) $256\frac{9}{256}$ (d) $261\frac{65}{256}$

Answer (d)

Here, $16^0 = 1$, $16^2 = 256$, $\frac{1}{16^2} = \frac{1}{256}$, $16^{\frac{1}{2}} = \sqrt{16} = 4$, $\frac{1}{16^{\frac{1}{2}}} = \frac{1}{\sqrt{16}} = \frac{1}{4}$

Thus, $16^0 + 16^2 + \frac{1}{16^2} + 16^{\frac{1}{2}} + \frac{1}{16^{\frac{1}{2}}} = 1 + 256 + \frac{1}{256} + 4 + \frac{1}{4}$

$= \mathbf{261 + \frac{65}{256}}$.

ILLUSTRATION 5:

Evaluate the value of x for which the following equation is satisfied.

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$6^{3.2} \times 36^{1.6} \times 7^{6.4} \times 42^{2.6} = 42^x$

(a) 8.6 (b) 9 (c) 10.2 (d) 12

Answer (b)

Given: $42^x = 6^{3.2} \times 36^{1.6} \times 7^{6.4} \times 42^{2.6}$

Let us try to make the base of LHS and RHS the same.

For that, we have to use the above-mentioned laws of exponents.

⇨ $42^x = 6^{3.2} \times (6^2)^{1.6} \times 7^{6.4} \times 42^{2.6}$

⇨ $42^x = 6^{3.2} \times 6^{3.2} \times 7^{6.4} \times 42^{2.6}$ [Since $(a^m)^n = a^{mn}$]

⇨ $42^x = 6^{3.2+3.2} \times 7^{6.4} \times 42^{2.6}$ [Since $a^m \times a^n = a^{m+n}$]

⇨ $42^x = 6^{6.4} \times 7^{6.4} \times 42^{2.6}$

⇨ $42^x = (6\times7)^{6.4} \times 42^{2.6}$ [Since $a^m \times b^m = (ab)^m$]

⇨ $42^x = 42^{6.4} \times 42^{2.6}$

⇨ $42^x = 42^{6.4+2.6}$ [Since $a^m \times a^n = a^{m+n}$]

⇨ $42^x = 42^9$

⇨ **x = 9.**

ILLUSTRATION 6

Evaluate $\frac{(873+542)^2 + (873-542)^2}{(873\times873+542\times542)}$

(a) 1 (b) 2 (c) 4 (d) 8

Answer (b)

Let 873 and 542 be denoted as a and b respectively.

Then the given expression can be written as $\frac{(a+b)^2+(a-b)^2}{(a\times a)+(b\times b)}$

$= \frac{(a+b)^2+(a-b)^2}{a^2+b^2}$.

We know that $(a + b)^2 = a^2 + 2ab + b^2$ and $(a - b)^2 = a^2 - 2ab + b^2$.

Thus $\frac{(a+b)^2+(a-b)^2}{a^2+b^2} = \frac{(a^2+2ab+b^2)+(a^2-2ab+b^2)}{a^2+b^2}$

$= \frac{a^2+2ab+b^2+a^2-2ab+b^2}{a^2+b^2}$

$= \frac{a^2+b^2+a^2+b^2}{a^2+b^2}$

$= \frac{2(a^2+b^2)}{a^2+b^2}$

$= 2.$

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5.1.9 Last digit of a power

Note: While multiplying two numbers having more than one digit, the last digit of the product is the last digit of the product of the last digit of the two numbers.
For example, 332 × 332 =110224, the last digit of this product can be simply obtained by multiplying 2 × 2 = 4 and taking the last digit of that product, i.e., 4 itself.

The last digit of a number a^b forms a particular sequence or order depending on the unit digit of the number (a) and the power the number is raised to (b).

This sequence can be called the power cycle of a number and thus it depends on its unit digit.

Consider the power cycle of 2:

$2^1 = \mathbf{2}$, $2^5 = 32$,

$2^2 = \mathbf{4}$, $2^6 = 64$,

$2^3 = \mathbf{8}$, $2^7 = 128$,

$2^4 = 16$ $2^8 = 256$

Clearly, we can observe that the unit digit gets repeated after every 4^{th} power of 2.

Hence, we can say that 2 has a power cycle of 2, 4, 6, 8, with frequency 4.

Thus,

2^{4k+1} have last digit as 2.

2^{4k+2} have last digit as 4.

2^{4k+3} have last digit as 8.

2^{4k+4} have last digit as 6.

where, k = 0, 1, 2, 3, …

Note that

4k + 1 means the set of natural numbers, which gives remainder 1 when divided by 4.

4k + 2 means the set of natural numbers, which gives remainder 2 when divided by 4.

4k + 3 means the set of natural numbers, which gives remainder 3 when divided by 4.

4k + 4 means the set of natural numbers, which gives remainder 0 when divided by 4.

Note: For all the numbers ending in 2, the power cycle and its frequency is the same as that of 2.

Thus to find the last digit of a number raised to any power, we need to know the power cycle of digits from 0 to 9 which are given below.

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Last digit	Power Cycle	Frequency
0	0	1
1	1	1
2	2, 4, 8, 6	4
3	3, 9, 7, 1	4
4	4, 6	2
5	5	1
6	6	1
7	7, 9, 3, 1	4
8	8, 4, 2, 6	4
9	9, 1	2

Note that if you can learn this table, you can find the last digit of any number of the form a^b in few seconds.

ILLUSTRATION 7:

What is the last digit of 32^{91}?

(a) 2 (b) 4 (c) 6 (d) 8

Answer (d)

To obtain the last digit of 32^2, 32^3, …, multiply the last digits i.e., 2×2, 4×2, 8×2 etc.
Hence, the last digit is given by,

For 32^1, Last digit = 2,i.e., 2
For 32^2, Last digit = 2 × 2 = 4, i.e., 4
For 32^3, Last digit = 4 × 2 = 8, i.e., 8
For 32^4, Last digit = 8 × 2 = **16,** i.e., 6

For 32^5, Last digit = 6 × 2 = **12,** i.e., 2
For 32^6, Last digit = 2 × 2 = 4, i.e., 4
For 32^7, Last digit = 4 × 2 = 8, i.e., 8
For 32^8, Last digit = 8 × 2 = **16,** i.e., 6

We can observe that the unit digit gets repeated after every 4^{th} power of 32.

Hence, we can say that 32 has a power cycle of 2, 4, 6, 8, with frequency 4.
Therefore, for powers, 1, 5, 9, 13,…., the last digit will be 2.
For powers, 2, 6, 10, 14,…., the last digit will be 4.
For powers, 3, 7, 11, 15,…., the last digit will be 8.
For powers, 4, 8, 12, 16,…., the last digit will be 6.

In the question, the power is 91.

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We have to check, in which of the above series does 91 comes.

91 is a number that gives a remainder 3 when divided by 4.

Therefore, 91 is in the series 3, 7, 11, 15…

Hence, the last digit is **8**.

ILLUSTRATION 8:

What is the last digit of 13^{121}?

(a) 1 (b) 3 (c) 7 (d) 9

Answer (b)

To obtain the last digit of 13^2, 13^3… multiply the last digits i.e., 3×3, 9×3, 7×3 etc.

Hence, the last digit is given by,

For 13^1, Last digit = 3,i.e., 3
For 13^2, Last digit = $3 \times 3 = 9$, i.e., 9
For 13^3, Last digit = $9 \times 3 = 27$, i.e., 7
For 13^4, Last digit = $7 \times 3 = 21$, i.e., 1.

For 13^5, Last digit = $1 \times 3 = 3$,i.e., 3
For 13^6, Last digit = $3 \times 3 = 9$, i.e., 9
For 13^7, Last digit = $9 \times 3 = 27$, i.e., 7
For 13^8, Last digit = $7 \times 3 = 21$, i.e., 1.

Clearly, we can observe that the unit digit gets repeated after every 4^{th} power of 3.

Hence, we can say that 2 has a power cycle of 3, 9, 7, 1, with frequency 4.

Therefore, for powers, 1, 5, 9, 13… the last digit will be 3.

For powers, 2, 6, 10, 14,…., the last digit will be 9.

For powers, 3, 7, 11, 15,…., the last digit will be 7.

For powers, 4, 8, 12, 16,…., the last digit will be 1.

In the question, the power is 121.

We have to check, in which series does 121 comes.

121 is a number that gives a remainder 1 when divided by 4.

Therefore, 121 is in the series 1, 5, 9, 13,…

Hence, the last digit is **3**.

5.1.10 Some important sequences

Arithmetic Sequence

In an arithmetic sequence, each term of a sequence differs from its preceding term by a constant called the common difference.

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The n^{th} term of an Arithmetic Sequence is, $t_n = a + (n-1)d$

Sum of n^{th} terms of an Arithmetic sequence is given by

$S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}$(first term + last term)

where a = first term, d = common difference.

Sum of first n consecutive natural numbers = $(1 + 2 + 3 + \ldots + n) = \frac{n(n+1)}{2}$

Sum of first n consecutive square numbers = $(1^2 + 2^2 + 3^2 + \ldots + n^2) = \frac{n(n+1)(2n+1)}{6}$

Sum of first n consecutive cubic numbers = $(1^3 + 2^3 + 3^3 + \ldots + n^3) = \frac{n^2(n+1)^2}{4}$

Geometric Sequence

In a geometric sequence, every term bears a constant ratio with its preceding term.

The n^{th} term of a geometric series is $t_n = ar^{n-1}$

Sum of n^{th} terms of a Geometric sequence $= \frac{a(r^n-1)}{(r-1)}$

where a is the first term, r is the common ratio and $r > 1$

Sum of terms of an infinite geometric progression $= \frac{a}{1-r}$, $r < 1$.

5.1.11 Fundamental principle of counting

If there are m ways of doing a first thing and for each of them, there are n ways of doing a second thing, then the total number of ways of doing both the things together is $m \times n$.

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5.2 CSIR-NET Exam Questions – Detailed Solution

1. What is the 94th term of the following sequence? 1, 1, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3, 4, 4, 4, 4, 4, 4, 4, 4,....

(a) 8 **(b) 9** **(c) 10** **(d) 11**

Answer (c) **(CSIR NET DEC 2014)**

Difficulty rating: 3.5/5 *Average time for solving: 110 seconds*

Consider each block of identical numbers as one set.
Assume that the 94th term is in the nth set.

Here the first two terms are 1, the next four terms are 2, the next six terms are 3 and goes on in the same manner…

Therefore, for the 1st set, there are 2×1 elements which are both 1.

For the 2nd set, there are 2×2 elements which are all 2.

For the 3rd set, there are 2×3 elements which are all 3.

Proceeding the same way, we can induce that,

For the nth set, there are 2n elements, which are all n.

We can tabulate the above observation as follows.

Set	Number of elements in the set	Cumulative number of elements	All elements in the set are
1	2 × 1 = 2	2	1
2	2 × 2 = 4	6	2
3	2 × 3 = 6	12	3
4	2 × 4 = 8	20	4
5	2 × 5 = 10	30	5
6	2 × 6 = 12	42	6
7	2 × 7 = 14	56	7
8	2 × 8 = 16	72	8
9	2 × 9 = 18	90	9
10	2 × 10 = 20	110	10

From the table, we can observe that the 94th term occur in the 10th set and all the elements in the 10th set are 10.

Hence, the 94th term is **10.**

Alternative solution

By observing the pattern of the first few terms, we can induce that **"in the nth set, there are 2n elements and all the elements are n."**

Thus, our question can be reformulated as
Minimize n for which

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$2+4+6+\ldots.+2n \geq 94$

1. $2(1+2+3+\ldots.+n) \geq 94$
2. $(1+2+3+\ldots.+n) \geq 47.$

Remember: Sum of the first n consecutive natural numbers $= \frac{n(n+1)}{2}$.

Thus, Minimize n for which $\frac{n(n+1)}{2} \geq 47$.

To complete, put the given options in this inequality and find the minimum value for which this inequality is satisfied.

Clearly n = 10 is the minimum value for which the inequality is satisfied.

That is $\frac{10 \times 11}{2} = 55 \geq 47$.

Thus 94 occur in the 10^{th} set, and so the 94^{th} term is **10**.

2. Which of the following numbers is a perfect square?

(a) 1022121 **(b) 2042122** **(c) 3063126** **(d) 4083128**

(CSIR-NET DEC 2014)

Answer (a)

Difficulty rating: 4/5 *Average time for solving: 120 seconds*

Observe the last digit of every perfect square.

0^2=**0**, 1^2=**1**, 2^2=**4**, 3^2=**9**, 4^2=**16**, 5^2=**25**, 6^2=**36**, 7^2= **49**, 8^2=**64**, 9^2=**81**….

Clearly, a perfect square does not end with 2, 3, 7 or 8.

Thus, we can readily eliminate options (b) and (d).

Now we consider the remaining options (a) and (c).

Remember: *Perfect squares can be recognized by the fact that all of their prime factors have even multiplicities. (For further information, read the section "Important properties of perfect squares" in the theory part.)*

Prime factorize options (a) and (c).

$3063126 = 2\times3\times11\times46411$, in which the factors 2, 3, 11, 46411 all have multiplicities 1, which is odd. Thus 3063126 is not a perfect square.

$1022121 = 3^2\times337^2 = (3\times337)^2$, which is clearly a perfect square.

3. The equation $m^2 - 33n + 1 = 0$, where m & n are integers has

(a) no solution **(b) exactly one solution**

(c) exactly two solutions **(d) infinitely many solutions**

(CSIR-NET DEC 2014)

Answer (a)

Difficulty rating: 4/5 *Average time for solving: 100 seconds*

Given: $m^2 - 33n + 1 = 0$.

$\Rightarrow m^2 = 33n - 1$

$\Rightarrow m = \sqrt{33n-1}$

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No such integer n gives such an integer m. (The proof of which is beyond the scope of this discussion)

Hence, **no solution.**

4. We define a function f(N) = sum of digits of N, expressed as decimal number.
E.g. f(137) = 1 + 3 + 7 = 11:
Evaluate $f(2^7 3^5 5^6)$.

(a) 10 **(b) 18** **(c) 28** **(d) 11**

Answer (b) **(CSIR-NET DEC 2014)**

Difficulty rating: 3/5 *Average time for solving: 60 seconds*

Here we are required to find the value of $2^7 3^5 5^6$ and then find its sum of digits.

However, in this case usual multiplication demands much time.

First, we can simplify the expression by finding the powers of 10 in the product.

For that, we can club all the powers of 2 and 5.

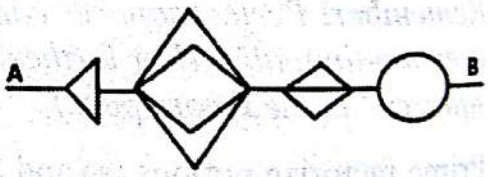
Hence,

$$\begin{aligned} 2^7 3^5 5^6 &= 2 \times 3^5 \times 2^6 \times 5^6 \\ &= 2 \times 3^5 \times (2 \times 5)^6 \\ &= 2 \times 3^5 \times 10^6 \\ &= 486 \times 10^6 \\ &= 486000000. \end{aligned}$$

Thus, $f(2^7 3^5 5^6)$= sum of digits of 486000000

$$\begin{aligned} &= 4 + 8 + 6 + 0 \\ &= \mathbf{18.} \end{aligned}$$

5. A mouse has to go from a point A to B without retracing any part of the path, and never moving backwards. What is the total number of distinct paths that the mouse may take to go from A to B?



(a) 11 **(b) 48**
(c) 72 **(d) 24**

Answer (b) **(CSIR-NET DEC 2014)**

Difficulty rating: 2.5/5 *Average time for solving: 60 seconds*

Between points A and B, there are four junctions with 2,4,3,2 paths respectively.

For the first two paths, there are four paths each and for each of the four paths, there are three paths each and for each of the three paths, there are two paths each.

Then, by the **fundamental principle of counting,**

The total number of distinct paths from A to B = $2 \times 4 \times 3 \times 2 =$ **48.**

6. The sum of first n natural numbers with one of them missed is 42. What is the number that was missed?

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(a) 1 **(b) 2** **(c) 3** **(d) 4**

Answer (c) **(CSIR-NET DEC 2014)**

Difficulty rating: 2.5/5 *Average time for solving: 60 seconds*

Let x be the missed number.

Since x is a positive number, the sum of the first 'n' natural numbers including x should be greater than 42.

By trial and error, find the minimum 'n' for which the sum is more than 42.

***Remember**: Sum of first 'm' natural numbers* $= \frac{m(m+1)}{2}$

We have, $1+2+3+4+5+6+7+8+9 = \frac{9\times10}{2} = 45$

Thus $45 - x = 42$

$\Rightarrow x = 3$.

7. If N, E and T are distinct positive integers such that N × E × T = 2013, then which of the following is the maximum possible sum of N, E and T?

(a) 39 **(b) 2015** **(c) 675** **(d) 671**

Answer (c) **(CSIR-NET DEC 2014)**

Difficulty rating: 2.5/5 *Average time for solving: 80 seconds*

$2013 = 3 \times 671$ (By prime factorization)

Therefore, $2013 = 1 \times 3 \times 671$ is the only possible way in which 2013 can be expressed as the product of three distinct positive integers.

Hence the sum $= 1 + 3 + 671 =$ **675**

Note that here $2013 = 2013 \times 1 \times 1$ is not possible because the number 1 is repeated and hence not distinct positive integers.

Thus **675** is the maximum possible sum.

8. If n is a positive integer, then $n(n+1)(n+2)(n+3)(n+4)(n+5)(n+6)$ is divisible by

(a) 3 but not 7 **(b) 3 and 7** **(c) 7 but not 3** **(d) neither 3 nor 7**

Answer (b) **(CSIR-NET DEC 2014)**

Difficulty rating: 2/5 *Average time for solving: 50 seconds*

Note 1: In every three consecutive integers, there is a multiple of 3.

Note 2: In every seven consecutive integers, there is a multiple of 7.

n, n+1,…, n+6 are 7 consecutive numbers.

Then, clearly, there will be a multiple of 3 and 7 in these numbers.

Therefore, $n(n+1)(n+2)(n+3)(n+4)(n+5)(n+6)$ is divisible by **3 and 7**.

9. The least significant bit of an 8-bit binary number is zero. A binary number whose value is 8 times the previous number has

(a) 12 bits ending with three zeros **(b) 11 bits ending with four zeros**

(c) 11 bits ending with three zeros **(d) 12 bits ending with four zeros**

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Answer (b) **(CSIR-NET DEC 2014)**

Difficulty rating: 3.5/5 *Average time for solving: 90 seconds*

Note 1: Least Significant bit is the last bit of a binary number.

Let us take a typical 8-bit binary number, say 10000010.

Clearly, the least significant bit here is zero.

We have to find the binary number whose value is 8 times the above number.

Note 2: Rules of binary addition: 0+0 =0, 1+1 =10,

Note 3: In any number system, for any number x,

$x + x = 2x$,
$2x + 2x = 4x$,
$4x + 4x = 8x$, *that is 8 times the number x.*

Therefore, to find 8 times the number 10000010, we should perform 3 cycles of the above-mentioned binary addition.

Thus,

10000010 +
10000010
2 times→ 100000100 +
100000100
4 times → 1000001000 +
1000001000
8 times→**10000010000**

That is, **11 bits ending with four zeros.**

10. Binomial theorem in algebra gives $(1 + x)^n = a_0 + a_1x + a_2x^2 + \ldots + a_nx^n$, where $a_0, a_1, \ldots, a_n$ are constants depending on n. What is the sum of $a_0 + a_1 + a_2 + \ldots + a_n$?

(a) 2^n **(b) n** **(c) n^2** **(d) $n^2 + n$**

Answer (a) **(CSIR-NET DEC 2014)**

Difficulty rating: 2.5/5 *Average time for solving: 60 seconds*

Given: $(1 + x)^n = a_0 + a_1x + a_2x^2 + \ldots + a_nx^n$

Put x = 1 in the LHS and RHS of this equation.

⇨ $(1+1)^n = a_0 + a_1 + a_2 + \ldots + a_n$

⇨ $2^n = a_0 + a_1 + a_2 + \ldots + a_n$

⇨ $a_0 + a_1 + a_2 + \ldots + a_n = \mathbf{2^n}$.

11. How many digits are there in $2^{17} \times 3^2 \times 5^{14} \times 7$?

(a) 14 **(b) 15** **(c) 16** **(d) 17**

Answer (d) **(CSIR-NET JUNE 2014)**

Difficulty rating: 2.5/5 *Average time for solving: 60 seconds*

Note that here you are not supposed to multiply all the numbers to get the product.

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First, we can simplify the expression by finding the powers of 10 in the product.
For that, we can club the powers of 2 and 5.

$$2^{17} \times 3^2 \times 5^{14} \times 7 = 2^3 \times 2^{14} \times 5^{14} \times 3^2 \times 7 = (2 \times 5)^{14} \times 2^3 \times 3^2 \times 7 = 10^{14} \times 8 \times 9 \times 7$$
$$= 504 \times 10^{14}$$

That is, 504 followed by 14 zeros.

That is, 14 + 3= **17 digits**.

12. The following sum is:

$$1 + 1 - 2 + 3 - 4 + 5 - 6 \ldots . - 20 = ?$$

(a) 10 **(b) -10** **(c) -11** **(d) -9**

Answer (d) **(CSIR-NET JUNE 2014)**

Difficulty rating: 2.5/5 *Average time for solving: 60 seconds*

For solving this question, you are not expected to add up all the numbers, which is a real tedious job.

In the sum, there are 21 numbers.

Leaving the first number 1, pair all other numbers to find the answer.

$$1 + (1 - 2) + (3 - 4) + (5 - 6) \ldots . + (19 - 20)$$
$$= 1 + \underbrace{(-1) + (-1) \ldots . + (-1)}_{10 \text{ terms}}$$
$$= 1 - 10$$
$$= \mathbf{-9.}$$

13. You are given 100 verbs using which you have to form sentences containing at least one verb, without repeating the verbs, under the condition that the number of verbs (from this set of 100) in any two sentences should not be equal. The maximum number of sentences you can form is

(a) 10 **(b) 13** **(c) 14** **(d) 100**

Answer (b) **(CSIR-NET JUNE 2014)**

Difficulty rating: 3.5/5 *Average time for solving: 110 seconds*

Here proper understanding of the question is very important.

We are given 100 verbs.
We are required to form sentences using these hundred verbs following two conditions.

Condition 1: The number of verbs in any two sentences should not be equal. i.e., if we use only one verb in one sentence, it is not possible to construct another sentence with only one verb. If we form a sentence with only two verbs, then it is not possible to construct another sentence with only two verbs and so on.

Condition 2: If we use a verb in one sentence, we are not allowed to use the same verb in any other sentence.

Let n be the maximum number of sentences we can form satisfying these conditions.

By condition 1:

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1st sentence have 1 verb in it.

2nd sentence have 2 verbs in it.

3rd sentence have 3 verbs in it.

...

nth sentence have n verbs in it.

Thus by the above two conditions, only the number of verbs we use matter and the actual verbs we use does not matter. The problem becomes analogous to maximizing "n" such that sum of first "n" natural numbers is less than or equal to 100.

That is, Maximize n: $1 + 2 + 3 + \ldots + n \leq 100$.

Remember: Sum of first n natural numbers $= \frac{n(n+1)}{2}$.

By trial and error, putting the 4 options, find the maximum value of n for which, $\frac{n(n+1)}{2} \leq 100$.

Putting n = 13, we get $\frac{13 \times 14}{2} \leq 100$

$\Rightarrow 91 \leq 100$

Thus, **n = 13** is the maximum value of n which satisfies the inequality.

14. Suppose $x \Delta y = (x-y)^2$

$x \circ y = (x+y)^2$

$x * y = (x \times y)^{-1}$

$x . y = x \times y$ **+,- and × have their usual meanings.**

What is the value of $\{(197 \circ 315) - (197 \Delta 315)\} . (197 * 315)$?

(a) 1018 **(b) 512** **(c) 2** **(d) 4**

Answer (d) **(CSIR-NET JUNE 2014)**

Difficulty rating: 2.5/5 *Average time for solving: 80 seconds*

Here, if we substitute variables x and y for the numbers in the given question, the question becomes easier to solve and it saves much time.

Thus, consider 197 and 315 as x and y respectively.

Then, the given expression becomes, $\{(x \circ y) - (x \Delta y)\} . (x * y)$

Applying the given definitions, we get,

$\{(x \circ y) - (x \Delta y)\} . (x * y) = \{(x+y)^2 - (x-y)^2\} \times (x \times y)^{-1}$

$= \{x^2 + y^2 + 2xy - x^2 - y^2 + 2xy\} \times \frac{1}{xy} = \frac{4xy}{xy} = 4.$

15. If $A \times B = 24$, $B \times C = 32$, $C \times D = 48$, then $A \times D$

(a) cannot be found **(b) is a perfect square** **(c) is a perfect cube** **(d) is odd**

Answer (b) **(CSIR-NET JUNE 2014)**

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Difficulty rating: 2.5/5 *Average time for solving: 90 seconds*

Normal Solution:

Given:

$A \times B = 24$ -------- (1)

$B \times C = 32$ -------- (2)

$C \times D = 48$ -------- (3)

Equation (1) divided by equation (2) gives

$\frac{A}{C} = \frac{3}{4}$------ (4)

Equation (3) multiplied by equation (4) gives,

$A \times D = 48 \times \frac{3}{4} = 36$

$= 6^2$, **a perfect square.**

Quick Solution:

$(A \times D) = \frac{(A \times B) \times (C \times D)}{B \times C} = \frac{24 \times 48}{32} = 36 = 6^2$, a perfect square.

16. Suppose n is a positive integer. Then $(n^2 + n)(2n + 1)$

(a) may not be divisible by 2

(b) is always divisible by 2 but may not be divisible by 3

(c) is always divisible by 3 but may not be divisible by 6

(d) is always divisible by 6

Answer (d) **(CSIR-NET JUNE 2014)**

Difficulty rating: 3.5/5 *Average time for solving: 90 seconds*

$(n^2 + n)(2n + 1) = n(n+1)(2n+1)$

We know that,

Sum of squares of first n natural numbers $= \frac{n(n+1)(2n+1)}{6}$

Sum of squares of first n natural numbers is always a natural number.

⇨ $\frac{n(n+1)(2n+1)}{6}$ is a natural number.

For $\frac{n(n+1)(2n+1)}{6}$ to be a natural number, n(n+1)(2n+1) should be a multiple of 6.

⇨ **n(n+1)(2n+1) is divisible by 6.**

17. Consider the set of numbers $\{17^1, 17^2, \ldots, 17^{300}\}$. How many of these numbers end with the digit 3?

(a) 60 **(b) 75** **(c) 100** **(d) 150**

Answer (b) **(CSIR-NET JUNE 2014)**

Difficulty rating: 4/5 *Average time for solving: 120 seconds*

Prerequisite: Read the section **"Last digit of a power".**

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To obtain the last digit of 17^2, 17^3,..., just multiply the last digits i.e., 7×7, 9×7, 3×7 etc.

Thus, for 17^1, Last digit = **7**

17^2, Last digit = 7×7 = **49**, i.e., 9

17^3, Last digit = 9×7 = **63**, i.e., 3

17^4, Last digit = 3×7 = **21**, i.e., 1

17^5, Last digit = 1×7 = **7**, i.e., 7

17^6, Last digit = 7×7 = **49**, i.e., 9

17^7, Last digit = 9×7 = **63**, i.e., 3

17^8, Last digit = 3×7 = **21**, i.e., 1.

Note that {7, 9, 3, 1} is the cycle of last digits which gets repeated.

The digit 3 appears once in each cycle of length 4.

In 300, there are $\frac{300}{4} = 75$ cycles of length 4.

Therefore, in the given set, there are **75** numbers with the last digit 3.

18. If a_i, b_i and c_i are distinct, how many terms will the expansion of the product $(a_1 + a_2 + a_3)(b_1 + b_2 + b_3 + b_4)(c_1 + c_2 + c_3 + c_4 + c_5)$ contain?

(a) 12 **(b) 30** **(c) 23** **(d) 60**

Answer (d) **(CSIR-NET DEC 2013)**

Difficulty rating: 3.5/5 *Average time for solving: 90 seconds*

$(a_1 + a_2 + a_3)(b_1 + b_2 + b_3 + b_4)(c_1 + c_2 + c_3 + c_4 + c_5)$

3 terms 4 terms 5 terms

Let p, q, r be the number of a_i, b_i, c_i terms respectively.

We know by the principle of counting that,

If a_i, b_i, c_i are distinct, the number of terms in the product = $p \times q \times r$.

Thus total number of terms = 3×4×5 = **60.**

(For getting convinced, multiply a smaller expansion of the product for say,

$a_1(b_1 + b_2)(c_1 + c_2 + c_3)$

$a_1(b_1 + b_2)(c_1 + c_2 + c_3) = a_1b_1c_1 + a_1b_1c_2 + a_1b_1c_3 + a_1b_2c_1 + a_1b_2c_2 + a_1b_2c_3$ ---- 6 terms

That is, 1 × 2 × 3 = 6 terms.)

19. $4^0 + 4^2 + \frac{1}{4^2} + 4^{\frac{1}{2}} + \frac{1}{4^{\frac{1}{2}}}$ equals

(a) 40 **(b) $4^{2\frac{1}{2}} + 4^{-2\frac{1}{2}}$** **(c) $19\frac{9}{16}$** **(d) $22\frac{9}{16}$**

Answer (c) **(CSIR-NET DEC 2013)**

Difficulty rating: 2/5 *Average time for solving: 90 seconds*

We know that, $4^0 = 1$, $4^2 = 16$, $\frac{1}{4^2} = \frac{1}{16}$, $4^{\frac{1}{2}} = \sqrt{4} = 2$, $\frac{1}{4^{\frac{1}{2}}} = \frac{1}{\sqrt{4}} = \frac{1}{2}$

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Thus, $4^0 + 4^2 + \frac{1}{4^2} + 4^{\frac{1}{2}} + \frac{1}{4^{\frac{1}{2}}} = 1 + 16 + \frac{1}{16} + 2 + \frac{1}{2}$

$= 19 + \frac{1}{16} + \frac{1}{2}$

$= 19 + \frac{9}{16}$. That is, **option (c).**

20. Every time a ball falls to ground, it bounces back to half the height it fell from. A ball is dropped from a height of 1024 cm. The maximum height from the ground to which it can rise after the tenth bounce is

(a) 102.4 cm **(b) 1.24 cm** **(c) 1 cm** **(d) 2 cm**

Answer (c) **(CSIR-NET DEC 2013)**

Difficulty rating: 3.5/5 *Average time for solving: 90 seconds*

If you have difficulty in understanding the given question, try to imagine each statement of the question in real life scenario.

Consider a rubber ball been dropped from a height of 1024 cm.

Surely, the ball will not rise up to the same height after the first bounce.

Given that it bounces back to half the height ($\frac{1}{2} \times 1024$), it was dropped from initially.

After the second bounce, it bounces back to half the height ($\frac{1}{2} \times (\frac{1}{2} \times 1024)$), after the first bounce.

This process continues in the same manner.

We are asked to find the height of the ball after the tenth bounce.

Initial height = 1024 cm

Height after 1st bounce $= \frac{1}{2} \times 1024$

Height after 2nd bounce $= \frac{1}{2} \times (\frac{1}{2} \times 1024) = (\frac{1}{2})^2 \times 1024$

Height after 3rd bounce $= \frac{1}{2} \times (\frac{1}{2} \times \frac{1}{2} \times 1024) = (\frac{1}{2})^3 \times 1024$

Similarly, height after 10th bounce $= (\frac{1}{2})^{10} \times 1024 = \frac{1}{1024} \times 1024 =$ **1 cm.**

21. $(25 \div 5 + 3 - 2 \times 4) + (16 \times 4 - 3) =$

(a) 61 **(b) 22** **(c) 41/24** **(d) 1**

Answer (a) **(CSIR-NET DEC 2013)**

Difficulty rating: 2/5 *Average time for solving: 50 seconds*

Prerequisite: *"BODMAS" rule- The order of operations is "Bracket Of Division, Multiplication, Addition, Subtraction".*

By BODMAS rule,

$(25 \div 5 + 3 - 2 \times 4) + (16 \times 4 - 3) = ((25 \div 5) + 3 - (2 \times 4)) + ((16 \times 4) - 3)$

$= ((5 + 3) - 8) + ((16 \times 4) - 3)$

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$= (8 - 8) + (64 - 3)$

$= \mathbf{61.}$

22. For real numbers x and y, $x^2 + (y - 4)^2 = 0$. Then the value of x + y is

(a) 0 **(b) 2** **(c) $\sqrt{2}$** **(d) 4**

Answer (d) **(CSIR-NET DEC 2013)**

Difficulty rating: 3/5 *Average time for solving: 100 seconds*

The quickest way to solve this question is to find an x and y satisfying the given equation.

Given: $x^2 + (y - 4)^2 = 0$.

Here x^2 and $(y - 4)^2$ are non-negative real numbers and the sum of non-negative real numbers is 0 only when both the numbers are 0.

Therefore $x^2 = 0$ and $(y - 4)^2 = 0$.

⇨ $x = 0, y = 4$.

Thus, $x + y = 0 + 4 = \mathbf{4}$.

23. n is a natural number. If n^5 is odd, which of the following is true?

A. n is odd **B. n^3 is odd** **C. n^4 is even**

(a) A only **(b) B only** **(c) C only** **(d) A and B only**

Answer (d) **(CSIR-NET JUNE 2013)**

Difficulty rating: 2.5/5 *Average time for solving: 60 seconds*

The easiest way to solve this question is by taking an example and checking with the given options.

Take a small natural number say 3, such that its fifth power is odd.

$3^5 = 243$, odd

$3^4 = 81$, odd

$3^3 = 27$, odd

Clearly, only A and B are correct. Hence, **option (d).**

Note that it is impossible to take an even number with its fifth power odd.

24. Suppose you expand the product $(x_1 + y_1)(x_2 + y_2)\ldots(x_{20} + y_{20})$. How many terms will have only one x and rest y's?

(a) 1 **(b) 5** **(c) 10** **(d) 20**

Answer (d) **(CSIR-NET JUNE 2013)**

Difficulty rating: 3/5 *Average time for solving: 100 seconds*

Here we have to pick the terms in the expansion with only one x_i term in it.

Check for smaller products

$(x_1 + y_1)$ ----- 1 term

$(x_1 + y_1)(x_2 + y_2) = (x_1 x_2 + x_1 y_2 + x_2 y_1 + y_1 y_2)$ ----- 2 terms

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Continuing like this, we can induce that if 20 factors are there in the product, there will be 20 terms in the expression with only one x and rest all y's.

25. What is the last digit of 7^{73}?

(a) 7 **(b) 9** **(c) 3** **(d) 1**

Answer (a) **(CSIR-NET JUNE 2013)**

Difficulty rating: 3.5/5 *Average time for solving: 110 seconds*

Prerequisite: *Read the section* "Last digit of a power".

To obtain the last digit of 7^2, 7^3, …, multiply the last digits i.e., 7×7, 9×7, 3×7 etc.

Hence, the last digit is given by,

For 7^1, Last digit = **7,** i.e., 7

For 7^2, Last digit = $7 \times 7 = 49$, i.e., 9

For 7^3, Last digit = $9 \times 7 = 63$, i.e., 3

For 7^4, Last digit = $3 \times 7 = 21$, i.e., 1.

For 7^5, Last digit = $1 \times 7 = 7$, i.e., 7

For 7^6, Last digit = $7 \times 7 = 49$, i.e., 9

For 7^7, Last digit = $9 \times 7 = 63$, i.e., 3

For 7^8, Last digit = $3 \times 7 = 21$, i.e., 1.

Note that {7, 9, 3, 1} is the cycle of last digits which gets repeated.

Therefore, for powers, 1, 5, 9, 13… the last digit will be 7 itself.

For powers, 2, 6, 10, 14… the last digit will be 9.

For powers, 3, 7, 11, 15… the last digit will be 3.

For powers, 4, 8, 12, 16… the last digit will be 1.

In the question, the power is 73.

We have to check in which series does 73 occurs.

Clearly, 73 is a number, which gives a remainder 1 when divided by 4.

Thus, 73 is in the series 1, 5, 9, 13…

Therefore, the last digit is 7.

26. In solving a quadratic equation of the form $x^2 + ax + b = 0$, one student took the wrong value of a and got the roots as 6 and 2; while another student took the wrong value of b and got the roots as 6 and 1. What are the correct values of a and b, respectively?

(a) 7 and 12 **(b) 3 and 4** **(c) -7 and 12** **(d) 8 and 12**

Answer (c) **(CSIR-NET JUNE 2013)**

Difficulty rating: 3/5 *Average time for solving: 90 seconds*

Prerequisite: *Read the theory section* "Quadratic Equations".

For a general quadratic equation, $px^2 + qx + r = 0$,

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Sum of roots $= -\frac{q}{p}$, Product of roots $= \frac{r}{p}$

Here $x^2 + ax + b = 0$

So $p = 1, q = a, r = b$.

Thus, sum of roots $= -a$

Product of roots $= b$

First student took wrong value of a, but correct value of b. Therefore, the product of roots is correct.

Thus, product of roots (b) $= 6 \times 2 = 12$.

Another student took wrong value of b, but correct value of a. Therefore, the sum of roots is correct.

Thus, sum of roots $(-a) = 6 + 1 = 7$

$\Rightarrow -a = 7$

$\Rightarrow a = -7$

Thus the correct values of a and b are -7 and 12.

27. Choose the largest number:

(a) 2^{500} (b) 3^{400} (c) 4^{300} (d) 5^{200}

Answer (b) **(CSIR-NET JUNE 2013)**

Difficulty rating: 3/5 *Average time for solving: 100 seconds*

Prerequisite: *Read the theory section "Laws of exponents".*

For comparing these numbers, either we should make the base same, or the exponent same.

Here in the base, the numbers are 2, 3, 4 and 5.
Obviously it is not easy to make the base same here.

Therefore we can try making the exponent same.

$2^{500}, 3^{400}, 4^{300}, 5^{200} = (2^5)^{100}, (3^4)^{100}, (4^3)^{100}, (5^2)^{100}$

$= 32^{100}, 81^{100}, 64^{100}, 25^{100}$.

Now the exponents are all the same. Therefore, the number with largest base is the largest number.

Hence, $81^{100} = \mathbf{3^{400}}$ is the largest number.

28. Define $a \otimes b$ = LCM (a,b) + GCD (a,b) and $a \oplus b = a^b + b^a$.

What is the value of $(1 \oplus 2) \otimes (3 \oplus 4)$?

Here LCM = least common multiple and GCD = greatest common divisor.

(a) 145 (b) 286 (c) 436 (d) 572

Answer (c) **(CSIR-NET JUNE 2013)**

Difficulty rating: 3/5 *Average time for solving: 100 seconds*

Prerequisite: Read the theory section "GCD and LCM".

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Given: $a \oplus b = a^b + b^a$

Therefore, $1 \oplus 2 = 1^2 + 2^1 = 1 + 2 = 3$,

$3 \oplus 4 = 3^4 + 4^3 = 81 + 64 = 145$

Thus, $(1 \oplus 2) \otimes (3 \oplus 4) = 3 \otimes 145$

Given: $a \otimes b = \text{LCM}(a,b) + \text{GCD}(a,b)$

Thus, $(1 \oplus 2) \otimes (3 \oplus 4) = \text{LCM}(3,145) + \text{GCD}(3,145)$

$= 435 + 1$

$=$ **436.**

29. How many pairs of positive integers have GCD 20 and LCM 600? (GCD = greatest common divisor, LCM = least common multiple)

(a) 4 **(b) 0** **(c) 1** **(d) 7**

Answer (a) **(CSIR-NET JUNE 2013)**

Difficulty rating: 3.5/5 *Average time for solving: 110 seconds*

Prerequisite: *Read the theory section "GCD and LCM".*

We know that for two numbers a and b,

$\text{GCD}(a,b) \times \text{LCM}(a,b) = a \times b$

$\Rightarrow 20 \times 600 = a \times b$

That is, $a \times b = 12000$.

Since a and b have GCD 20, both a and b are multiples of 20.

That is, $a = 20x$, $b = 20y$, where x and y are relatively prime numbers.

Thus, $20x \times 20y = 12000$

$\Rightarrow 400xy = 12000$

$\Rightarrow xy = 30$.

Therefore, possible combinations of x and y are

x	1	2	3	5
y	30	15	10	6

Thus, possible pairs of such positive integers with GCD 20 and LCM 600 are,

$20 \times 1, 20 \times 30 = 20, 600$

$20 \times 2, 20 \times 15 = 40, 300$

$20 \times 3, 20 \times 10 = 60, 200$

$20 \times 5, 20 \times 6 = 100, 120$

Thus, there are **4 pairs of such numbers.**

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30. Which of the following numbers is the largest?

$2^{3^4}, 2^{4^3}, 3^{2^4}, 3^{4^2}, 4^{2^3}, 4^{3^2}$

(a) 2^{3^4} **(b)** 3^{4^2} **(c)** 4^{3^2} **(d)** 4^{2^3}

Answer (a) **(CSIR-NET DEC 2012)**

Difficulty rating: 3.5/5 *Average time for solving: 110 seconds*

We are asked to find the largest among the given six numbers.

For time saving, we can consider only the four given options.

Thus, consider, $\mathbf{2^{3^4}, 3^{4^2}, 4^{3^2}, 4^{2^3}}$

⇨ $2^{81}, 3^{16}, 4^9, 4^8$

Clearly, $4^8 < 4^9$.

So, 4^{2^3} can be eliminated from the above-considered options.

To compare the remaining 3 numbers, $2^{81}, 3^{16}, 4^9$, the easiest way is to make base or exponent of each number the same.

In this case, to make the exponent same is a tedious task.

So we can try to make the bases same.

$$2^{81} = 2 \times 2^{80} = 2 \times 4^{40}$$

Thus, our remaining three numbers become

$2 \times 4^{40}, 3^{16}, 4^9$.

Since both the base and exponents of 4^{40} is greater than both 3^{16} and 4^9, 4^{40} is greater than both the other numbers.

Then surely, 2 times 4^{40} is also greater than both the other numbers.

Hence, $\mathbf{2^{81}}$ is the largest.

31. Which of the following curve just touches the x-axis?

(a) $y = x^2 - x + 1$ **(b)** $y = x^2 - 2x + 2$

(c) $y = x^2 - 10x + 25$ **(d)** $y = x^2 - 7x + 12$

Answer (c) **(CSIR-NET DEC 2012)**

Difficulty rating: 3/5 *Average time for solving: 80 seconds*

Note: *For a curve to touch the x-axis exactly at one point, the condition is that, at y=0, equation must have real and equal roots.*

Now check whether each option satisfy this condition.

Clearly $x^2 - 10x + 25 = 0$

⇨ $(x-5)^2 = 0$

⇨ $x = 5,5$, real and equal roots.

32. Which is the least among the following?

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(a) $0.33^{0.33}$ (b) $0.44^{0.44}$ (c) $\pi^{\frac{-1}{\pi}}$ (d) $e^{\frac{-1}{e}}$

Answer (d) **(CSIR-NET DEC 2012)**

Difficulty rating: 4/5 *Average time for solving: 80 seconds*

The given options can be written as $0.33^{0.33}$, $0.44^{0.44}$, $(\frac{1}{\pi})^{\frac{1}{\pi}}$, $(\frac{1}{e})^{\frac{1}{e}}$.

Here for each options base and exponents are same.

That is, all the options are of the form x^x.

Note: The function x^x is minimum for $x = \frac{1}{e}$. (The proof of this result is by using function derivative test, which is beyond the scope of this discussion)

Thus, $e^{\frac{-1}{e}} = \frac{1}{e^{\frac{1}{e}}}$ is the least among the given options.

33. Consider the following equation: $x^2 + 4y^2 + 9z^2 = 14x + 28y + 42z - 147$ where x, y and z are real numbers. Then the value of $x + 2y + 3z$ is

(a) 7 (b) 14 (c) 21 (d) not unique

Answer (c) **(CSIR-NET DEC 2012)**

Difficulty rating: 3.5/5 *Average time for solving: 120 seconds*

$$x^2 + 4y^2 + 9z^2 = 14x + 28y + 42z - 147$$

Grouping like terms together,

$$(x^2 - 14x) + ((2y)^2 - 28y) + ((3z)^2 - 42z) + 147 = 0$$

Making perfect squares by adding adequate numbers,

$$(x^2 - 14x + 49) + ((2y)^2 - 28y + 49) + ((3z)^2 - 42z + 49) = 0.$$

$$(x - 7)^2 + (2y - 7)^2 + (3z - 7)^2 = 0.$$

The LHS is the sum of 3 non-negative numbers (because every square number is positive)

For the sum of 3 non-negative numbers to be zero, they all should be zero.

i.e., $(x - 7)^2 = 0$, $(2y - 7)^2 = 0$, $(3z - 7)^2 = 0$.

$\Rightarrow$ $(x - 7) = 0$, $(2y - 7) = 0$, $(3z - 7) = 0$.

$\Rightarrow$ $x = 7$, $y = \frac{7}{2}$, $z = \frac{7}{3}$.

Hence, $x + 2y + 3z = 7 + (2 \times \frac{7}{2}) + (3 \times \frac{7}{3}) = 7 + 7 + 7 = \mathbf{21}$.

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5.3 Practice Questions

1. In solving a problem that reduces to a quadratic equation one student makes a mistake only in the constant term of the equation and obtains 8 and 2 for the roots. Another student makes a mistake only in the coefficient of the first degree term and finds -9 and -1 for the roots. The correct equation was:

(a) $x^2 - 10x + 9 = 0$ (b) $x^2 + 10x + 9 = 0$
(c) $x^2 - 10x + 16 = 0$ (d) $x^2 - 8x - 9 = 0$

2. The sum of three prime numbers is 100. If one of them exceeds another by 36, then one of the numbers is

(a) 7 (b) 29 (c) 41 (d) 67

3. The number 111,111,111,111 is divisible by

(a) 9 and 11 (b) 5 and 11 (c) 3 and 9 (d) 3 and 11

4. The product of two numbers is 735 and their H.C.F. is 7. The number of such pairs is

(a) 1 (b) 2 (c) 3 (d) 4

5. Which of the following numbers is a perfect square?

(a) 126736 (b) 137802 (c) 149913 (d) 204048

6. Four metal rods of lengths 78 cm, 104 cm, 117 cm and 169 cm are to be cut into parts of equal length. Each part must be as long as possible. What is the maximum number of pieces that can be cut?

(a) 27 (b) 36 (c) 43 (d) 480

7. Evaluate the value of x for which the following equation is satisfied.

$$5^{4.6} \times 25^{2.3} \times 7^{9.2} \times 35^{2.8} = 35^x$$

(a) 10.2 (b) 10.6 (c) 11 (d) 12

8. We define a function f(N) = sum of digits of N, expressed as decimal number.
E.g. f(137) = 1 + 3 + 7 = 11:Evaluate $f(2^8 3^3 5^6)$.

(a) 5 (b) 9 (c) 14 (d) 15

9. What is the last digit of 38^{38}?

(a) 2 (b) 4 (c) 6 (d) 8

10. Choose the largest number:

(a) 2^{150} (b) 3^{120} (c) 4^{90} (d) 5^{60}

Answer Key

1. (a) 2. (d) 3. (d) 4. (b) 5. (a) 6. (b) 7. (d) 8. (b) 9. (b) 10. (b)

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CHAPTER 6

PROBABILITY AND ARRANGEMENTS

Average Difficulty Rating: 3.2/5

Average time for each question: 100 sec

Number of questions expected for each exam: 1-2

6.1 THEORY

6.1.1 PROBABILITY

6.1.1.1 Preliminaries

Random Experiment: An experiment when conducted repeatedly under the same conditions, if it does not give the same result, it is called the random experiment.

Event: A collection of one or more of the outcomes of a random experiment is called the event. Example: When a dice is thrown, the occurrence of face marked 1 is an event.

Mutually Exclusive Events: Two or more events are said to be mutually exclusive if the occurrence of any one of them excludes the occurrence of all others in a single experiment. Example: When throwing a coin, the occurrence of Head (H) and occurrence of tail (T) are mutually exclusive events. If Head occurs, it prevents the occurrence of Tail.

Union of Events: Let A and B be two events defined in a sample space. The union of events A and B is the collection of all outcomes that belong either to A or to B or to both A and B and is denoted by A UU B.

Intersection of Events: Let A and B be two events defined in a sample space. The intersection of events A and B is the collection of all outcomes that belong to both A and B and is denoted by $A \cap B$.

Sample space or Outcome space: The set of all possible outcomes is called the sample space of a random experiment.

Random Experiment	Sample Space (S)
Toss a coin	S = {Head, Tail}
Toss 2 coins simultaneously	S = {HH, HT, TH, TT}
Roll a dice	S = {1, 2, 3, 4, 5, 6}
Football game	S = {win, lose, draw}

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6.1.1.2 How to approach questions on Probability?

The EVENT is the foundation stone of probability. Hence, the first objective while trying to solve any question in probability is to define the event. The event whose probability is to be determined is described in the question and your task in trying to solve the problem is to define it.

You can define the event either narrowly or broadly. Narrow definitions of events are the building blocks of any probability problem and whenever there is a doubt about a problem, it is better to get into the narrowest form of the event definition.

The difference between the narrow and broad definition of event is explained through the following illustration.

ILLUSTRATION 1:

What is the probability of getting a number greater than 3, in a throw of an unbiased dice having 6 faces?

(a) 1/6 (b) ½ (c) 1/3 (d) 2/3

Answer (b)

The broad definition of the event here is getting a number greater than 3 and this probability is 3/6. However, this event can also be broken down into its basic definitions as:

The event is defined as getting 4 or 5 or 6. The individual probabilities of each of these are 1/6, 1/6 and 1/6 respectively.

Hence the required probability is $1/6 + 1/6 + 1/6 = 3/6 = 1/2$.

Although in this example it seems highly trivial, the narrow event-definition approach is very effective in solving difficult problems on probability.

In general, event definition means breaking up the event to the most basic building blocks, which have to be connected together through the two English conjunctions – AND and OR.

The use of the conjunction AND

Whenever you see AND as the natural conjunction joining two separate parts of the event definition, you can replace the AND by the multiplication sign.

Thus if events A AND B have to occur, and if the probability of their occurrence are P(A) and P(B) respectively, then the probability that A AND B occur is got by connecting P(A) AND P(B). Replacing the AND by multiplication sign, we get the required probability as:
$P(A) \times P(B)$

Example

If the probability of student A being in a particular team is 1/4 and the probability of student B being in the same team is 1/3, then the probability that both the students are in the same particular team is got by

Event Definition: A being in the team AND B being in the team.

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That is, $P(A) \times P(B) = 1/4 \times 1/3 = 1/12$

Note that since we use the conjunction AND in the event definition here, we multiply the individual probabilities that are connected by the conjunction AND.

The use of the conjunction OR

Whenever you see OR as the natural conjunction joining two separate parts of the event definition, you can replace the OR by the addition sign.

Thus if events A OR B have to occur, and if the probability of their occurrence are P(A) and P(B) respectively, then the probability that A OR B occur is got by connecting P(A) OR P(B). Replacing the OR by addition sign, we get the required probability as:
$P(A) + P(B)$.

Example

If the probability of student A being in a particular team is 1/4 and the probability of student B being in the same team is 1/3, then the probability that either of the two students are in the particular team is got by

Event Definition: A being in the team OR B being in the team.

That is, $P(A) + P(B) = 1/4 + 1/3 = 7/12$

Note that since we use the conjunction OR in the event definition here, we add the individual probabilities that are connected by the conjunction OR.

Combination of AND and OR

In examination, questions may not be simple enough to solve directly using any of the two conjunction AND or OR. However, no matter how complicated the problem on probability is, it can be broken down into its narrower parts, which can be connected by ANDs and ORs to get the event definition. Once the event is defined, the probability of each narrow event within the broad event is calculated and all the narrow events are connected by Multiplication (for AND) or by Addition (for OR) to get the final solution.

Example: What is the probability that the sum of the numbers is not less than 11, when two dice are thrown simultaneously?

Event definition: The sum of the numbers is not less than 11 if it is either 11 or 12.

which can be done by
(6 AND 5) OR (5 AND 6) OR (6 AND 6)

$= (1/6 \times 1/6) + (1/6 \times 1/6) + (1/6 \times 1/6)$

$= 1/36 + 1/36 + 1/36 = 3/36$

$= 1/12$

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6.1.1.3 Another approach of Probability

The probability of an event is defined as

$$\text{P(Event)} = \frac{\text{Number of successful outcomes}}{\text{Total number of possible outcomes}}$$

This means that the probability of any event can be obtained by counting the numerator and the denominator independently.

The counting can be done using different methods like physical counting, use of concepts of permutations and combinations etc.

Example: When we roll a dice, we could define an event A as the occurrence of even numbers.

Here,

S = {1, 2, 3, 4, 5, 6}, n(S) = 6, A = {2, 4, 6}, n(A) = Number of outcomes = 3

$$\text{P(Occurrence of even numbers)} = \frac{\text{Number of successful outcomes}}{\text{Total number of possible outcomes}}$$

$$= \frac{n(A)}{n(S)} = \frac{3}{6} = \frac{1}{2}$$

6.1.1.4 Important facts about Probability

- The probability of any event ranges from zero to one. That is, $0 \leq P(\text{Event}) \leq 1$.
- The sum of probabilities of all the events in a sample space is 1. That is,
  $P(E_1) + P(E_2) + P(E_3) + \ldots = 1$.
- For any two events A and B, we have
  $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- For any two mutually exclusive events A and B, we have
  $P(A \cup B) = P(A) + P(B)$.

ILLUSTRATION 2:

Two dice are thrown simultaneously. The probability of obtaining a total score of 7 is:

(a) $\frac{1}{3}$ (b) $\frac{1}{6}$ (c) $\frac{1}{9}$ (d) $\frac{7}{36}$

Answer (b)

When two dice are thrown simultaneously, all the six numbers can be obtained on both the dice.

(1,1), (1,2), (1,3), (1,4), (1,5), (1,6)
(2,1), (2,2), (2,3), (2,4), (2,5), (2,6)
(3,1), (3,2), (3,3), (3,4), (3,5), (3,6)

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(4,1), (4,2), (4,3), (4,4), (4,5), (4,6)
(5,1), (5,2), (5,3), (5,4), (5,5), (5,6)
(6,1), (6,2), (6,3), (6,4), (6,5), (6,6)

Thus, number of total ways = 6×6 = 36 ways.

$$\text{Required Probability} = \frac{\text{Number of possible ways}}{\text{Number of total ways}}$$

Possible ways by which 7 can be obtained = (1,6), (2,5), (3,4), (4,3), (5,2), (6,1).

Thus, Number of possible ways = 6.

$$\text{Required Probability} = \frac{6}{36} = \frac{1}{6}$$

ILLUSTRATION 3:

The probabilities of three mutually exclusive events X, Y and Z are given by 3/7, 4/7 and 2/7 respectively. The statement is:

(a) True (b) False (c) Nothing can be said (d) Could be either

Answer (b)

For any mutually exclusive events, the total probability is the sum of the individual probabilities.

That is, $P(X \cup Y \cup Z) = P(X) + P(Y) + P(Z)$

$$= \frac{3}{7} + \frac{4}{7} + \frac{2}{7}$$

$$= \frac{9}{7} > 1.$$

However, no probability can be greater than 1.

Therefore, the statement is wrong.

6.1.2 PERMUTATIONS AND COMBINATIONS

Factorial Notation: Let n be a positive integer.

Then, factorial n, denoted by n! is defined as:

$n! = n(n-1)(n-2).....3.2.1$ *Note:* We define 0! = 1.

Example: $6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1.$

6.1.2.1 Permutations:

A permutation is an arrangement of all or part of a set of objects with regard to the order of the arrangement. **Note that here the order of arrangements matters.**

Example 1: Suppose we have a set of three letters: A, B, and C. Consider the different ways by which we can **arrange** two letters from that set. Each possible arrangement would be an

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example of a permutation. The complete list of possible permutations would be AB, BA, AC, CA, BC and CB.

Example 2: All permutations made with the letters A, B, C taking all at a time are {ABC, BCA, CAB, CBA, BAC, ACB}

Number of permutations: Number of all permutations of n things, taken r at a time, is given by:

$$^{n}P_{r} = \frac{n!}{(n-r)!}$$

Example 1: $^{5}P_{2} = \frac{5!}{(5-2)!} = \frac{5!}{3!} = \frac{5\times4\times3\times2\times1}{3\times2\times1} = 5\times4 = 20.$

Example 2: $^{8}P_{3} = \frac{8!}{(8-3)!} = \frac{8!}{5!} = \frac{8\times7\times6\times5\times4\times3\times2\times1}{5\times4\times3\times2\times1} = 8\times7\times6 = 336.$

ILLUSTRATION 4:

In how many different ways can the letters of the word 'LOGIC' be arranged in such a way that the vowels always come together?

(a) 40 (b) 48 (c) 24 (d) 120

Answer (b)

The word 'LOGIC' has 5 different letters with two vowels 'O' and 'I'.

Here the required condition is that the vowels should always come together.

That is, we can consider 'OI' as a single letter.

Then, we have to arrange the 4 letters L, G, C, and (OI).

Now, 4 letters can be arranged in 4! = 24 ways.

The vowels (O, I) can be arranged among themselves in 2! = 2 ways. (viz. OI and IO)

Thus, required number of ways = 2 × 24 = 48 ways.

ILLUSTRATION 5:

How many 3-digit numbers can be formed from the digits 1, 3, 5, 7, 9, which are divisible by 5 and none of the digits is repeated?

(a) 3 (b) 9 (c) 12 (d) 15

Answer (c)

Since each desired number is divisible by 5, we must have 5 at the unit place. Therefore, there is only one way of doing it.

The tens place can now be filled by any of the remaining four digits (1, 3, 7, 9). Therefore, there are 4 ways of filling the tens place.

The hundreds place can now be filled by any of the remaining 3 digits. Therefore, there are 3 ways of filling it.

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Therefore, the required number of numbers = $(1 \times 4 \times 3) = 12$.

6.1.2.2 Combinations:

A combination is a selection of all or part of a set of objects, without regard to the order in which objects are selected.

Example: Suppose we have a set of three letters: A, B, and C. Consider the different ways by which we can **select** 2 letters from that set. Each possible selection would be an example of a combination. The complete list of possible selections would be: AB, AC, and BC. Note that AB and BA are considered to be one combination, because the order in which objects are selected does not matter (unlike permutations).

Number of combinations: Number of all combinations of n things, taken r at a time, is given by: $^{n}C_{r} = \frac{n!}{r!(n-r)!}$

Example 1: $^{5}C_{2} = \frac{5!}{2!(5-2)!} = \frac{5!}{2!3!} = 10.$

Example 2: $^{8}C_{3} = \frac{8!}{3!(8-3)!} = \frac{8!}{3!5!} = 56.$

ILLUSTRATION 6:

In how many ways a committee, consisting of 4 men and 3 women can be formed from 7 men and 5 women?

(a) 240 (b) 350 (c) 400 (d) 420

Answer (b)

Here the question is to **select** some objects from a set of objects.

Clearly, we have to use combinations.

4 men can be selected from 7 men by $^{7}C_{4}$ ways.

3 women can be selected from 5 women by $^{5}C_{3}$ ways.

Thus, required number of ways $= {}^{7}C_{4} \times {}^{5}C_{3} = \frac{7!}{4!\,(7-4)!} \times \frac{5!}{3!\,(5-3)!} = \frac{7!}{4!\,3!} \times \frac{5!}{3!\,2!} = 35 \times 10 = 350.$

Tip to understand whether permutation or combination to be used:

A permutation focuses on the arrangement of objects with regard to the order in which they are arranged. A combination in contrast, focuses on the selection of objects without regard to the order in which they are selected.

Hence seeing a question, first check whether the order matters. If the order matters, it is permutation; and combination if otherwise.

If the question is to **arrange** some objects, it is usually permutation. On the other hand, if it is to **select** some objects from a given set of objects, then it is combination.

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Shortcut formulas for quick solving:

- Number of different permutations of n different things taking r different things at a time is equal to ${}^{n}P_{r} = \frac{n!}{(n-r)!}$
- Number of different permutations of n different things (taking all at a time) when p of them are similar of one type, q of them are similar of another type and the remaining n - (p + q) are all different, is given by $\frac{n!}{p!q!}$
- Number of different permutations of n different things, taking all n things at a time is **n!**
- Number of different permutations of n different things, taken r at a time when each thing may be repeated any number of times in each arrangement is $\mathbf{n^r}$.
- Number of different permutations of n different things arranged around a circle is (n-1)! (Here clock-wise and anticlockwise arrangements are considered as different arrangements).
- If clockwise and anticlockwise arrangements are not considered different, number of circular arrangements of n different things = $\frac{1}{2}$**(n-1)!**
- Number of different combinations of n different things taking r different things at a time is equal to ${}^{n}C_{r} = \frac{n!}{r!\,(n-r)!}$
- Number of ways in which (m + n + p) different things can be divided into three groups containing m, n and p things respectively is given by $\frac{(m+n+p)!}{m!n!p!}$

6.2 CSIR-NET Exam Questions – Detailed Solution

1. A code consists of at most two identical letters followed by at most four identical digits. The code must have at least one letter and one digit. How many distinct codes can be generated using letters A to Z and digits 1 to 9?

(a) 936 **(b) 1148** **(c) 1872** **(d) 2574**

Answer (c) **(CSIR-NET DEC 2014)**

Difficulty rating: 3.5/5 *Average time for solving: 120 seconds*

If you have difficulty in understanding the question, try to analyze each phrase separately.

"at most two identical letters" –at most implies that maximum two letters can be used. However, "identical" implies that even if we can use two letters, both of them should be same. That is "AA" or "BB" or "CC" or …"ZZ".

Similarly, for "at most four identical digits" – maximum of 4 identical numbers can be used. That is "1111", "2222", "3333", … "9999".

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Thus, there are 8 different types of codes based on the number of characters on the code.

They are:
1 letter and 1 digit, 1 letter and 2 digit, 1 letter and 3 digit, 1 letter and 4 digit, 2 letter and 1 digit, 2 letter and 2 digit, 2 letter and 3 digit, 2 letter and 4 digit.

Hence, each of the above-mentioned 8 types of codes can have 26×9 distinct codes.

Thus, total number of distinct codes $= 8 \times 26 \times 9 =$ **1872.**

2. Suppose in a box there are 20 red, 30 black, 40 blue and 50 white balls. What is the minimum number of balls to be drawn, without replacement, so that you are certain about getting 4 red, 5 black, 6 blue and 7 white balls?

(a) 140 **(b) 97** **(c) 104** **(d) 124**

Answer (d) **(CSIR-NET JUNE 2014)**

Difficulty rating: 3.5/5 *Average time for solving: 110 seconds*

In these types of questions, we should consider the worst case that could happen.

Given 20 red, 30 black, 40 blue and 50 white balls in a box.

Clearly, the least number of balls is of the red colour among the four different coloured balls.

We want to get 4 red, 5 black, 6 blue and 7 white balls from the box.

Here the worst case is that only after taking all the White, Blue and Black balls, we get the first red ball.

So in total, $50 + 40 + 30 = 120$ balls before the first red ball is drawn.

Now only red balls are remaining.

Therefore, the next four balls can only be red balls, making the total count **124.**

3. Students of a school are divided into 4 groups. What is the probability that three friends get into the same group?

(a) $\frac{3}{4}$ **(b)** $\frac{1}{64}$ **(c)** $\frac{1}{16}$ **(d)** $\frac{1}{3}$

Answer (c) **(CSIR-NET DEC 2013)**

Difficulty rating: 3.5/5 *Average time for solving: 80 seconds*

Let student 1, student 2, student 3 be the three friends.

Let A, B, C, D be four groups.

Let us consider one group, say A.

Group A is one of the 4 groups.

So the probability of student 1 getting into group A $= \frac{1}{4}$

Similarly probability of student 2 getting into group A $= \frac{1}{4}$

Similarly probability of student 3 getting into group A $= \frac{1}{4}$

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Then, probability that the three friends get into group A = $(\frac{1}{4} \times \frac{1}{4} \times \frac{1}{4})$

However, since there are four groups,

Probability that the three friends get into any one of these 4 groups = $4 \times (\frac{1}{4} \times \frac{1}{4} \times \frac{1}{4}) = \frac{1}{16}$

4. How many 9 digit positive integers are there, the sum of squares of whose digits is 2?

(a) 8 (b) 9 (c) 10 (d) 11

(CSIR-NET DEC 2013)

Answer (a)

Average time for solving: 110 seconds

Difficulty rating: 3/5

Here the 1st digit should be 1, and any one of the remaining 8 digits should be 1. All the other entries should be 0.

For example, 100000001 is a nine-digit number, whose sum of squares of digits is 2.

Since, 1st digit is fixed and the other 1 can occupy any of the 8 positions, there can be **8** such numbers.

5. 366 players participate in a knock-out tournament. In each round all competing players pair together and play a match, the winner of each match moving to the next round. If at the end of a round there is an odd number of winners, the unpaired one moves to the next round without playing a match. What is the total number of matches played?

(a) 366 (b) 282 (c) 365 (d) 418

(CSIR-NET DEC 2013)

Answer (c)

Average time for solving: 110 seconds

Difficulty rating: 3.5/5

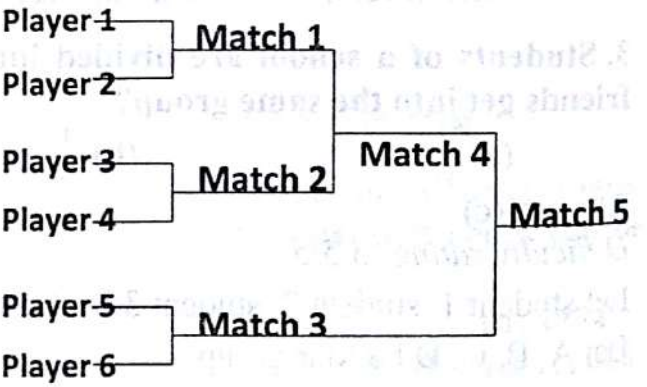
We can check this question for lesser number of players.

That is, instead of 366 players, we will check for say, 6 players.

For 6 players, there are exactly 5 matches as shown in the figure.

That is, for n players, there will be n-1 matches.

Hence, for 366 players, there will be 365 matches.

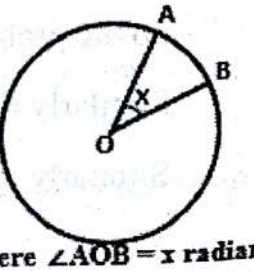


6. A point is chosen at random from a circular disc shown in the figure. What is the probability that the point lies in the sector OAB?

(a) $\frac{2x}{\pi}$ (b) $\frac{x}{\pi}$ (c) $\frac{x}{2\pi}$ (d) $\frac{x}{4\pi}$

Answer (c)

(CSIR-NET JUNE 2013)



(where ∠AOB = x radians)

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Difficulty rating: 3/5 *Average time for solving: 100 seconds*

Remember: Area of a sector = $\frac{n}{360}\pi r^2$,

where, n = Degree of the central angle of the sector.

Here n = x

To find the probability, it is enough to find the share of the sector OAB in the full circle.

Thus, Probability that the point lies in the sector OAB = $\frac{\text{Area of sector OAB}}{\text{Area of Circle}} = \frac{(\frac{\pi r^2 x}{360})}{\pi r^2} = \frac{x}{360} = \frac{x}{2\pi}$

7. Two integers are picked at random from the first 15 positive integers without replacement. What is the probability that the sum of the two numbers is 20?

(a) $\frac{3}{4}$ **(b)** $\frac{1}{21}$ **(c)** $\frac{1}{105}$ **(d)** $\frac{1}{20}$

Answer (b) **(CSIR-NET JUNE 2013)**

Difficulty rating: 3/5 *Average time for solving: 100 seconds*

Here 2 positive integers are taken from the total 15 positive integers.

$$\text{Probability} = \frac{\text{Number of succesful outcomes}}{\text{Total number of possible outcomes}}$$

Total number of possible outcomes = Number of ways in which two positive integers can be selected from 15 positive integers = $15C_2 = \frac{15 \times 14}{2} = 105$.

The possible pairs for which the sum is 20 is (5,15), (6,14), (7,13), (8,12), (9,11).
That is, there are 5 such pairs.

Therefore, number of successful outcomes = 5.

Thus Probability = $\frac{5}{105} = \frac{1}{21}$.

8. During a summer vacation, of 20 friends from a hostel, each wrote a letter to each of all others. The total number of letters written was

(a) 20 **(b) 400** **(c) 200** **(d) 380**

Answer (d) **(CSIR-NET JUNE 2013)**

Difficulty rating: 2.5/5 *Average time for solving: 80 seconds*

Out of the 20 persons, consider any one person.

This person sends one letter each to all other persons.
Therefore, he/she sends 19 letters.

There are 20 persons.
Therefore, 20 persons can send letters to all the other 19.

Thus, 20 × 19 = **380** letters in total.

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9. What is the number of distinct arrangements of the letters of the word UGCCSIR so that U and I cannot come together?

(a) 2520 **(b) 720** **(c) 1520** **(d) 1800**

Answer (d) **(CSIR-NET DEC 2012)**

Difficulty rating: 3.5/5 *Average time for solving: 110 seconds*

We are asked to find the distinct arrangements that U and I do not come together.

For that, it is easier, if we find the distinct arrangements that U and I come together and subtract it from the total number of distinct arrangements.

Number of distinct arrangements of the word UGCCSIR = $\frac{7!}{2!}$ = 2520

(The 2! in the denominator is because the letter C is occurring two times.)

Considering U and I together, there are six letters: G, C, C, S, R, and (UI).

Thus, number of distinct arrangements so that U & I come together = $\frac{6! \times 2!}{2!}$ = 720.

(The 2! in the numerator is because U and I can come together in two ways - UI and IU.)

No. of distinct arrangements such that U and I do not come together = 2520 – 720 = **1800.**

10. Suppose there are socks of N different colours in a box. If you take out one sock at a time, what is the maximum number of socks that you have to take out before a matching pair is found? Assume that N is an even number.

(a) N **(b) N + 1** **(c) N – 1** **(d) N/2**

Answer (a) **(CSIR-NET DEC 2012)**

Difficulty rating: 3/5 *Average time for solving: 80 seconds*

Let N = 4.

That is, there are socks of 4 different colours in the box, say Red, Blue, Green and White.

Let us consider the worst case that can happen for the given case.

Let the first choice be a red sock, the second a blue sock the third a green sock and the fourth a white sock.

As the fifth choice, we are sure to get one among the above four colours. (Since, there are no other colours.)

Thus, for 4 different colours, we get a matching pair after 4 takes.

Therefore, for the case of N socks we get a matching pair after **N** takes.

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6.3 PRACTICE QUESTIONS

1. The probabilities of three mutually exclusive events A, B and C are given by 1/3, 3/4 and 1/6 respectively. The statement:
(a) is true (b) is false (c) nothing can be said (d) could be either

2. What is the minimum number of cards you need to uncover from the top of a well-shuffled deck of 52 playing cards, to ensure that you have two cards of a suit?
(a) 41 (b) 15 (c) 7 (d) 5

3. There are 10 participants in a seminar. A list is to be prepared showing the sequence in which the 10 participants will speak. In how many ways can the list be prepared if one particular speaker "P" speak before another particular speaker "Q".
(a) 5! (b) $2 \times 5!$ (c) $5 \times 9!$ (d) 9!

4. A question paper had 5 questions. Each question could only be answered as True (T) or False (F). Each candidate answered all the questions. Yet, no two candidates wrote the answers in an identical sequence. How many different sequences of answers are possible?
(a) 20 (b) 32 (c) 40 (d) 64

5. In how many different ways can be the letters of the word 'INDIA' be arranged?
(a) 50 (b) 60 (c) 72 (d) 120

6. How many words can be formed from the letters of the word 'PIGEON' so that the vowels always come together?
(a) 6 (b) 144 (c) 216 (d) 720

7. In how many ways a committee, consisting of 6 men and 3 women can be formed from 8 men and 6 women?
(a) 450 (b) 560 (c) 600 (d) 625

8. How many two digit even numbers can be composed from nine digits 1,2,3,…9?
(a) 50 (b) 81 (c) 45 (d) 36

9. Suppose in a box there are 15 red, 25 green, 20 blue and 40 yellow balls. What is the minimum number of balls to be drawn, without replacement, so that you are certain about getting 5 red, 7 green, 3 blue and 4 yellow balls?
(a) 5 (b) 75 (c) 90 (d) 110

10. In a tournament, 9 teams play league matches. If each team plays against every other team once only then how many matches are played?
(a) 36 (b) 45 (c) 54 (d) 66

Answer Key

1.(b) 2.(d) 3.(d) 4.(b) 5.(b) 6.(b) 7.(b) 8.(d) 9.(c) 10.(a)

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CHAPTER 7

DISTANCES AND DIRECTIONS

Average Difficulty Rating: 3/5

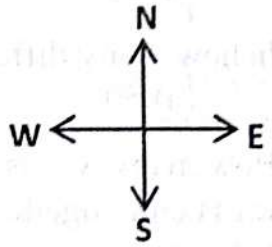
Average time for each question: 105 sec

Number of questions expected for each exam: 2-4

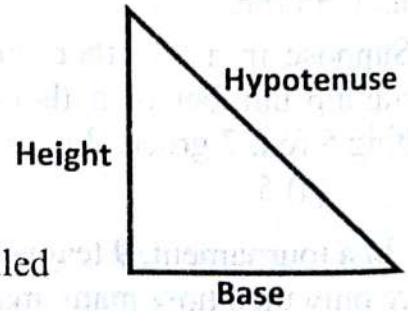
7.1 THEORY

7.1.1 Direction Sense Test

Points to remember



- A person facing towards North, on taking left turn will face towards West and on taking the right turn towards the East. To make it simple, *remember that right turn means clockwise rotation and left turn means anticlockwise rotation.*
- Each turn mentioned in the question is equivalent to an angle rotation of 90°.
- In order to determine the shortest straight distance between the two given points the Pythagoras theorem should be used.



Pythagoras Theorem
In a right triangle,

$(\text{Hypotenuse})^2 = (\text{Base})^2 + (\text{Height})^2$

The numbers which satisfy this relation, are called Pythagorean triplets.

Example: (3, 4, 5), (5, 12, 13), (6, 8, 10), (7, 24, 25), (8, 15, 17), (9, 40, 41), (11, 60, 61), (12, 35, 37), (15, 20, 25) etc.

Note: *If you see a triangle with sides exactly as any of these Pythagorean triplets, you can be sure that the triangle is a right triangle.*

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ILLUSTRATION 1:

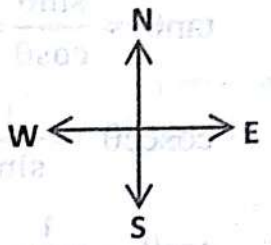
Sobin went 40 km to the North, then turning to East he went 50 km, then turning to the right he went 20 km and then he turns to right again and went 50 km. How far was he from his starting point?

(a) 5 km (b) 10 km (c) 20 km (d) 25 km

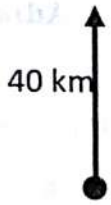
Answer (c)

We can analyse each statement and try to picture each of them.

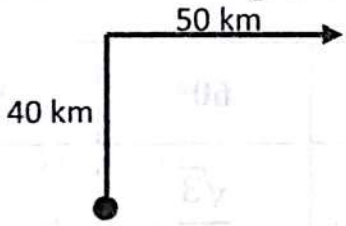
The dark circle denotes the starting point and the dark arrow denotes the corresponding position of Sobin.



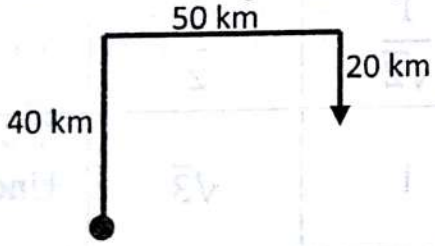
Given 1: "Sobin went 40 km to the North"



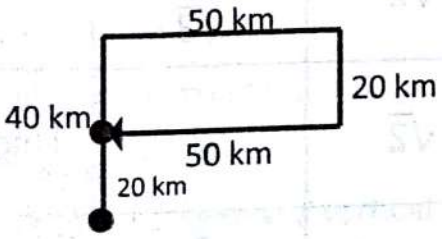
Given 2: "then turning to East he went 50 km"



Given 3: "then turning to the right he went 20 km"



Given 4: "then he turns to right again and went 50 km"



Clearly, Sobin is 20 km from his starting point.

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Trigonometric ratios

$$\sin\theta = \frac{\text{Opposite side}}{\text{Hypotenuse}}$$

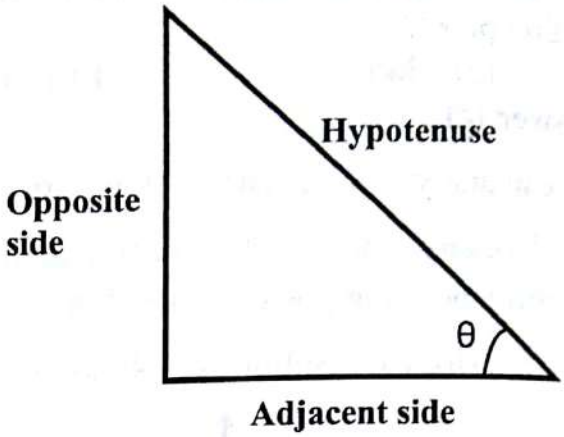
$$\cos\theta = \frac{\text{Adjacent side}}{\text{Hypotenuse}}$$

$$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{\text{Opposite side}}{\text{Adjacent side}}$$

$$\text{cosec}\theta = \frac{1}{\sin\theta} = \frac{\text{Hypotenuse}}{\text{Opposite side}}$$

$$\sec\theta = \frac{1}{\cos\theta} = \frac{\text{Hypotenuse}}{\text{Adjacent side}}$$

$$\cot\theta = \frac{1}{\tan\theta} = \frac{\text{Adjacent side}}{\text{Opposite side}}$$



Trigonometric measures of angles

Angle	0°	30°	45°	60°	90°
sin θ	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
cos θ	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
tan θ	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	Undefined
cosec θ	Undefined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
sec θ	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	Undefined
cot θ	Undefined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

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7.1.2 Heights and Distances

Problems under this section usually involves the use of trigonometry in finding the height of a building or object, mostly will involve a triangle, with at least an angle given. There can be questions in which some right-angled triangles are involved and applying Pythagoras theorem will do the trick in most of those problems. A good hold of solving the quadratic equations is also required to solve some of the problems. The primary step to solve any of this question is to picture the given situation.

ILLUSTRATION 2:

A person standing on the bank of a river observes that the angle subtended by a tree on the opposite bank is 60°, when he retires 60 metres from the riverbank; he finds the angle to be 30°. Then the breadth of the river is:

(a) 20 m (b) 30 m (c) 40 m (d) 60 m

Answer (b)

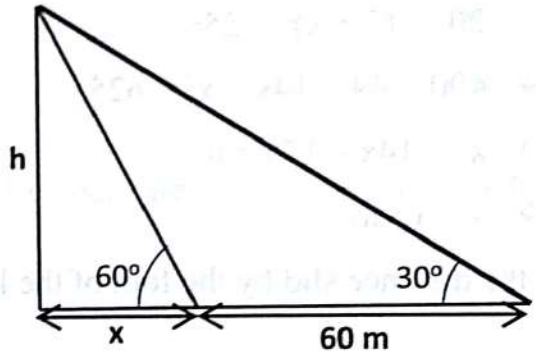
Let x be the breadth of the river and h be the height of the tree.

Then, consider the triangle with one angle 60°.

$$\tan 60^\circ = \frac{\text{Opposite side}}{\text{Adjacent side}} = \frac{h}{x}$$

$$\Rightarrow \sqrt{3} = \frac{h}{x}$$

$$\Rightarrow h = \sqrt{3}x \quad \text{---- (1)}$$



Now, consider the triangle with one angle 30°.

$$\tan 30^\circ = \frac{\text{Opposite side}}{\text{Adjacent side}} = \frac{h}{x+60}$$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{x+60}$$

$$\Rightarrow x + 60 = \sqrt{3}h$$

By (1), $x + 60 = 3x$

$$\Rightarrow 2x = 60$$

$$\Rightarrow x = 30.$$

Thus the breadth of the river = 30 m.

ILLUSTRATION 3:

A 25 m ladder is placed against a vertical wall of a building. The foot of the ladder is 7 m from the base of the building. If the top of the ladder slips 4 m, then the foot of the ladder will slide:

(a) 9 m (b) 15 m (c) 5 m (d) 8 m

Answer (d)

The given question can be pictured as shown.

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AB denotes the building.

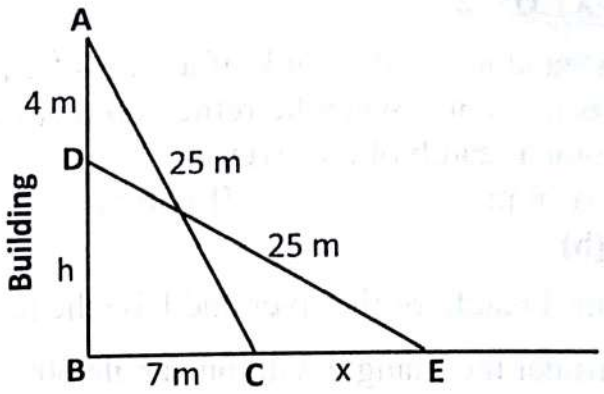
AC and DE denotes the position of the ladder at the two instances.

Let the distance slid by the foot of the ladder be x metres.

Clearly, ΔABC and ΔDBE are right-angled triangles.

Consider ΔDBE.

By Pythagoras theorem,



$(h + 4)^2 + 7^2 = 25^2$.

⇨ $h^2 + 8h + 16 + 49 = 625$.

⇨ $h^2 + 8h - 560 = 0$.

⇨ $h = 20$ m.

Now consider ΔDBE,

$20^2 + (7 + x)^2 = 25^2$.

⇨ $400 + 49 + 14x + x^2 = 625$.

⇨ $x^2 + 14x - 176 = 0$

⇨ $x = 15$ m.

Thus the distance slid by the foot of the ladder = 15 m.

7.1.3 Latitude-Longitude problems

Latitude-longitude system is a geographic coordinate system that enables every location on the Earth to be specific by a set of numbers.

Latitude is the geographic coordinate that specifies the north-south position of a point on Earth's surface. Latitude varies from 0^o to 90^o North and South at the poles. The 0^o latitude is called the equator.

Example: $45^oN, 50^oS$

Longitude is the geographic coordinate that specifies the east-west position of a point on Earth's surface. Longitude varies from 0^o at Greenwich to 180^o East and West.

Example: $45^oE, 50^oW$

Note that North Pole and South Pole are 180^olatitude apart, East, and West are at right angles with North and South.

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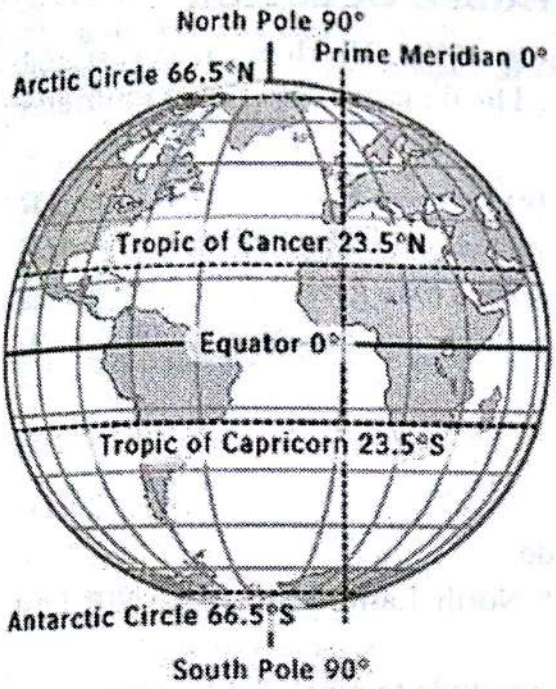

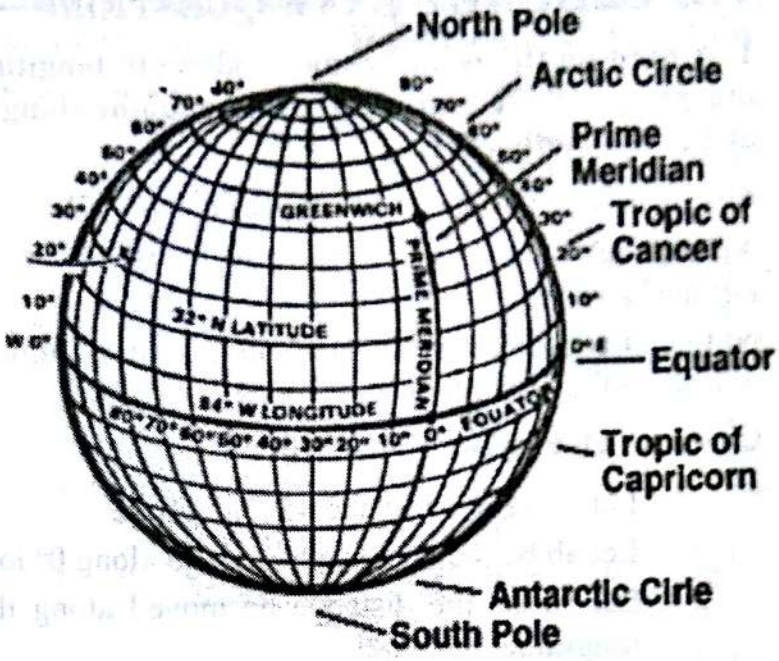

Degree and radian measure

Weight is measured in Kilograms and Pounds. Likewise, there are two units to measure angles: degrees and radians.

$360^\circ = 2\pi$ radian.

⇨ $\mathbf{1^\circ = \frac{\pi}{180}}$ **radian.**

In almost all cases, especially cases involving π (and if the answer options are not in degrees), it is better to use radians for calculation.

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7.2 CSIR-NET Exam Questions – Detailed Solution

1. A man on the equator moves along 0° longitude up to 45° N. He then turns east and moves up to 90° E, and turns to the equator along 90° E. The distance covered in multiples of Earth's radius R is

(a) $(\frac{\sqrt{3}}{4}\pi)$ **R** **(b)** $(\frac{\pi}{2}+\frac{\pi}{4\sqrt{2}})$ **R** **(c)** $(\frac{\pi}{2}+\frac{\pi}{2\sqrt{2}})$ **R** **(d)** $(\frac{\pi}{4}+\frac{\pi}{\sqrt{2}})$ **R**

Answer (c) **(CSIR-NET JUNE 2014)**

Difficulty rating: 4/5 *Average time for solving: 100 seconds*

Note that coordinate with N or S in it is the latitudinal coordinate and coordinate with E or W is longitudinal coordinate.

Given R is the radius of the earth.

Let r be radius on the 45° N latitude.

Let ab be the distance he moved along 0° longitude.

Let bc be the distance he moved along the 45° North Latitude to reach 90° East longitude.

Let cd be the distance he moved along 90° East Longitude to reach the Equator.

We have to find D = ab + bc + cd.

Clearly ab = cd.

We know that **Arc length = Radius × Angle.**

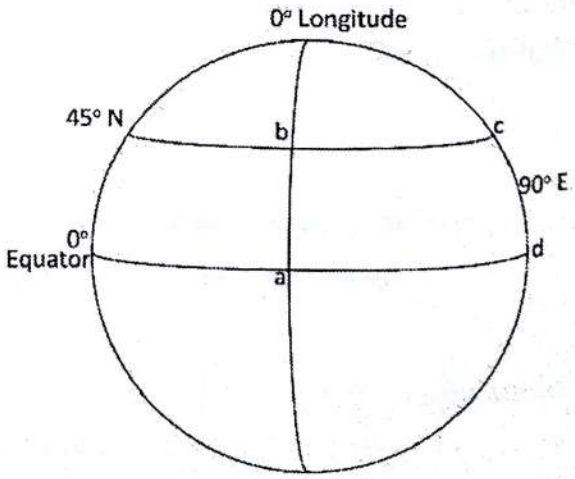
Since the answer options are all in terms of π, we have to convert the degree measure into radians.

Remember: $1^\circ = \frac{\pi}{180}$ radian.

$$ab = R \times 45^\circ = R \times 45 \times \frac{\pi}{180} = \frac{\pi R}{4}$$

$$bc = r \times 90^\circ = r \times 90 \times \frac{\pi}{180} = \frac{\pi r}{2}$$

Now, bc is in terms of r. We should convert it into an expression in term of R.



Here, $\sin 45^\circ = \frac{r}{R}$

$$\frac{1}{\sqrt{2}} = \frac{r}{R}$$

That is, $r = \frac{R}{\sqrt{2}}$

Hence, $bc = \frac{\pi R}{2\sqrt{2}}$

Thus, D = ab + bc + cd

$$= \frac{\pi R}{4} + \frac{\pi R}{2\sqrt{2}} + \frac{\pi R}{4} = \frac{\pi R}{2} + \frac{\pi R}{2\sqrt{2}} = (\frac{\pi}{2} + \frac{\pi}{2\sqrt{2}})R.$$

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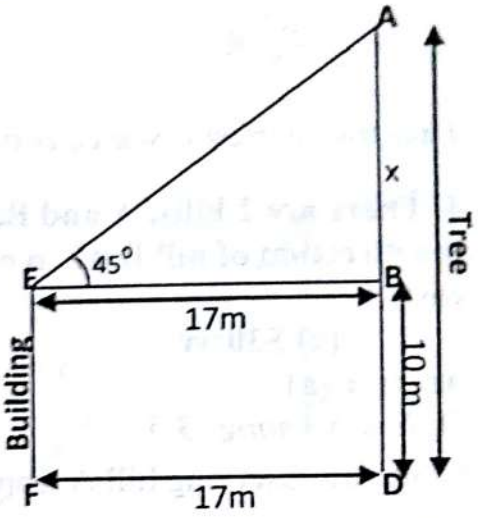
2. An archer climbs to the top of a 10 m high building and aims at a bird on top of a tree 17 m away. The line of sight from the archer to the bird makes an angle of 45° to the horizontal. What is the height of the tree?

(a) 17 m **(b) 27 m** **(c) 37 m** **(d) 47 m**

Answer (b) **(CSIR-NET JUNE 2014)**

Difficulty rating: 2.5/5 *Average time for solving: 70 seconds*

Note that the height of the archer here is considered as negligible.



The given question can be pictured as shown.

In ΔABE,

$$\tan 45^\circ = \frac{\text{Opposite side}}{\text{Adjascent side}} = \frac{x}{17}$$

$$\Rightarrow 1 = \frac{x}{17}$$

$$\Rightarrow x = 17.$$

Height of the tree = 10 + x = 10 + 17 = **27 m.**

3. Two points A and B on the surface of the Earth have the following latitude and longitude co-ordinates.

A:30°N, 45°E **B:30°N, 135°W**

If R is the radius of the Earth, the length of the shortest path from A to B is

(a) $\sqrt{3}\frac{\pi R}{2}$ **(b)** $\frac{\pi R}{3}$ **(c)** $\frac{\pi R}{6}$ **(d)** $\frac{2\pi R}{3}$

Answer (a) **(CSIR-NET DEC 2013)**

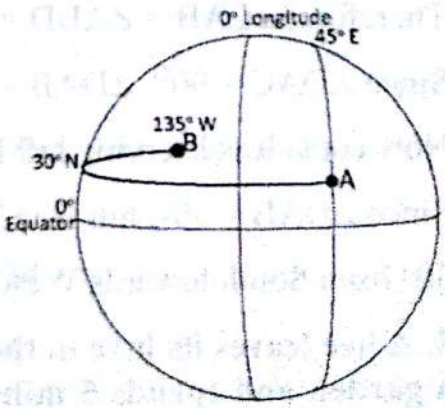
Difficulty rating: 4/5 *Average time for solving: 100 seconds*

The given question data can be pictured as shown in the figure.
For the given latitude and longitude, the points A and B are plotted on the figure and the shortest path is marked by the darkened curve.

The points A and B are lying on the same latitude, i.e., 30° N.

Therefore, the distance between A and B is the longitudinal distance between A and B.

The shortest path is along the 30° N latitude.



Given R is the radius of the earth.

Let r be the radius along 30° N latitude.

Consider the 30° N latitude circle.

Clearly, A and B are 180° apart from each other.

For 360° circumference, $2\pi r$ is the distance.

Therefore, for 180°, πr is the distance between A and B.

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However, we have to get the distance with respect to the radius of the Earth (R).
For that first, we should express r in terms of the radius of earth, R.

We know that, $\sin 60^\circ = \frac{r}{R}$

$\Rightarrow \frac{\sqrt{3}}{2} = \frac{r}{R}$

$\Rightarrow r = \frac{\sqrt{3}}{2} R$

Thus the shortest distance between A and B $= \pi r = \pi \frac{\sqrt{3}}{2} R = \sqrt{3} \frac{\pi R}{2}$.

4. There are 2 hills, A and B, in a region. If hill A is located N30°E of hill B, what will be the direction of hill B when observed from hill A? (N30°E means 30° from north towards east).

(a) S30°W **(b) S60°W** **(c) S30°E** **(d) S60°E**

Answer (a) **(CSIR-NET DEC 2013)**

Difficulty rating: 3/5 *Average time for solving: 50 seconds*

Consider observing hill A from hill B.

Let us try to picture the data as shown below.

The points A and B respectively denote Hill A and Hill B.

Construct a line parallel to the North-South axis.

Given: Hill A is located N30°E of hill B.

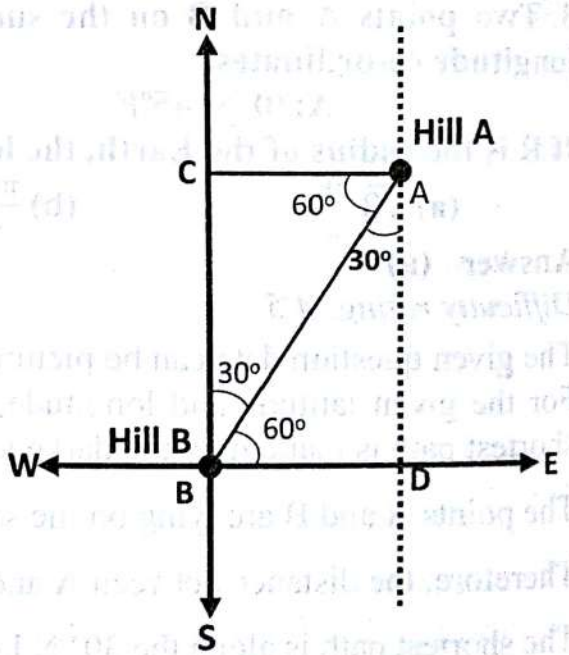
N30°E means 30° from north towards east.

That is, $\angle CBA = 30^\circ$.

Since $\angle CBD = 90^\circ$, $\angle ABD = 60^\circ$.

The north-south axis and the constructed line are parallel, and AB is a transversal.

For a pair of parallel lines and transversal, interior opposite angles are equal.



Therefore, $\angle CAB = \angle ABD = 60^\circ$.

Since $\angle DAC = 90^\circ$, $\angle DAB = 30^\circ$.

Now consider observing hill B from hill A.

Since $\angle DAB = 30^\circ$, hill B is 30° from South towards West.

30° from South towards West is represented as **S30°W.**

5. A bee leaves its hive in the morning and after flying for 30 minutes due south reaches a garden and spends 5 minutes collecting honey. Then it flies for 40 minutes due west

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and collects honey in another garden for ten minutes. Then it returns to the hive taking the shortest route. How long was the bee away from its hive? (Assume that the bee flies at constant speed)

(a) 85 min **(b) 155 min** **(c) 135 min** **(d)less than 1 hour**

Answer (c) **(CSIR-NET JUNE 2013)**

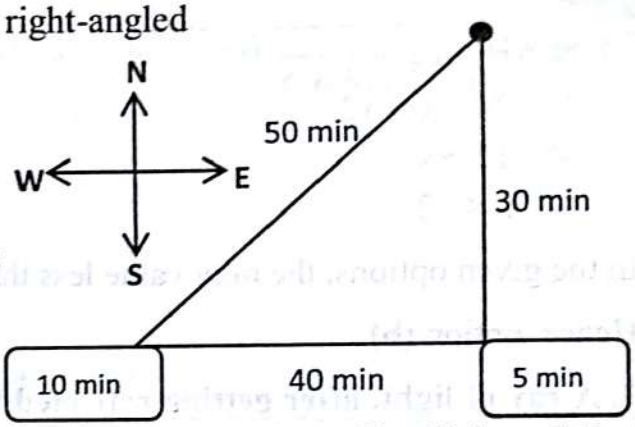
Difficulty rating: 2.5/5 *Average time for solving: 80 seconds*

The given data can be pictured as shown in the figure.

The bee flies first 30 minutes south and then 40 minutes west.

Then the bee takes the shortest route to return to its hive.

Clearly, the shortest route is the hypotenuse of a right-angled triangle with sides 30 and 40 (See figure).



By Pythagoras theorem,

The shortest route $= \sqrt{30^2 + 40^2}$

$= \sqrt{900 + 1600} = \sqrt{2500} = 50$

The bee also spends 5 minutes and 10 minutes respectively in the two gardens.

Obviously, the time the bee was away from its hive is the sum of the time of its flight and time it spends in each garden.

Thus the required time = 30 + 5 + 40 + 10 + 50 = **135 min.**

6. A bird perched at the top of 12 m high tree sees a centipede moving towards the base of the tree from a distance equal to twice the height of the tree. The bird flies along a straight line to catch the centipede. If both move at the same speed, at what distance from the base of the tree will the centipede be picked up by the bird?

(a) 16 m **(b) 9 m** **(c) 12 m** **(d) 14 m**

Answer (b) **(CSIR-NET JUNE-2013)**

Difficulty rating: 3/5 *Average time for solving: 80 seconds*

Normal Solution:

The given situation can be pictured as shown in the figure.

Let A and C be the initial positions of bird and centipede respectively.

Let D be the point at which the bird catches the centipede.

Let x be the distance from the base of the tree, when the bird catches the centipede.

Therefore, DC = 24 - x

Since they both move at the same speed, AD = DC = 24 – x

In ΔABD,

$12^2 + x^2 = (24-x)^2$

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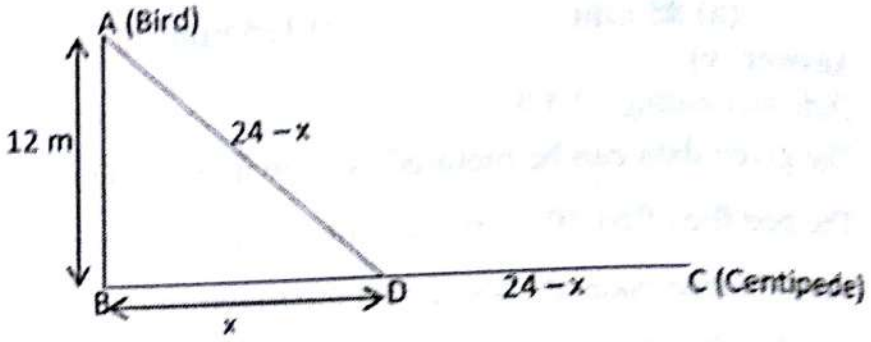
$\Rightarrow \quad 144 + x^2 = 576 - 48x + x^2$

$\Rightarrow \quad 48x = 432$

$\Rightarrow \quad x = 9$ **m.**



Figure

Quick Solution:

Consider the right triangle ABD in the above figure.

Clearly, AD > AB (Since, Hypotenuse of a right triangle always greater than the other two sides)

Thus, AD > 12

$\Rightarrow \quad 24 - x > 12$

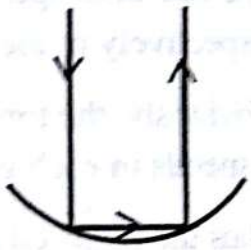
$\Rightarrow \quad 12 > x$

$\Rightarrow \quad x < 12$

In the given options, the only value less than 12 is **9.**

Hence, option **(b).**

7. A ray of light, after getting reflected twice from a hemispherical mirror of radius R (see the figure), emerges parallel to the incident ray. The separation of the original incident ray and the final reflected ray is



(a) R **(b) R√2** **(c) 2R** **(d) R√3**

Answer (b) **(CSIR-NET JUNE 2013)**

Difficulty rating: 3.5/5 *Average time for solving: 90 seconds*

(Unlike other questions, this question needs some prerequisite knowledge of physics.)

Here ray AB is parallel to ray CD.

Therefore ∠ABC + ∠BCD = 180°.

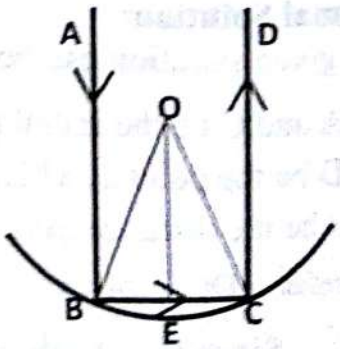
By the law of reflection,

∠ABO = ∠OBC = 45°

$$\cos 45^\circ = \frac{\text{Adjacent side}}{\text{Hypotenuse}} = \frac{BE}{R}$$

$$\Rightarrow \quad \frac{1}{\sqrt{2}} = \frac{BE}{R}$$

$$\Rightarrow \quad BE = \frac{R}{\sqrt{2}}$$



Hence BC = 2BE = **R√2.**

8. A crow is flying along a horizontal circle of radius R at a height R above the horizontal

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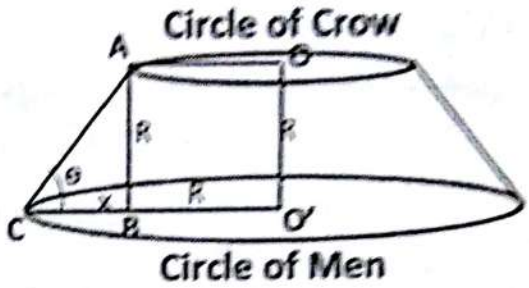
ground. Each of a number of men on the ground found that the angular height of the crow was fixed angle (< 45°) when it was closest to him. Then all these men must be on a circle on the ground with a radius

(a) R + R sinθ (b) R + R cosθ (c) R + R tanθ (d) R + R cotθ

Answer (d) **(CSIR-NET JUNE 2013)**

Difficulty rating: 3.5/5 *Average time for solving: 90 seconds*

The given data can be pictured as shown.



We are asked to find the radius of the circle of men.

Given that, R is the radius of the Circle of crow.

Let R + x be the radius of the Circle of men.

Therefore, if we can find the value for x, we can find the radius of circle of men.

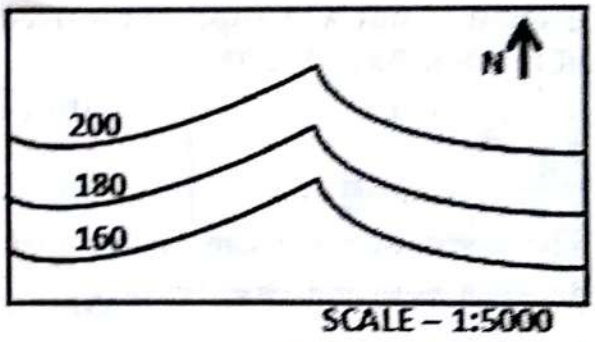
Consider ΔABC,

$$\cot\theta = \frac{BC}{AB} = \frac{x}{R}$$

$$\Rightarrow \quad x = R\cot\theta$$

Hence, radius of circle of men = R + x = **R + R cotθ.**

9. The map given shows contour lines which connect points of equal ground surface elevation in a region. Inverted 'V' shaped portions of contour lines represent a valley along which a river flows. What is the downstream direction of the river?



(a) North (b) South

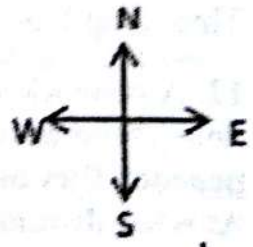
(c) East (d) West

Answer (b) **(CSIR-NET JUNE 2013)**

Difficulty rating: 3/5 *Average time for solving: 40 seconds*

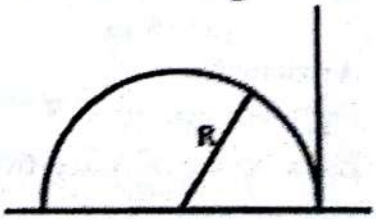
It is clear from the map itself that the upstream direction is North.

Clearly, the height difference in the graph shows that the river flows from upstream to downstream.



Hence, the downstream direction is **South.**

10. A semi-circular arch of radius R has a vertical pole put on the ground together with one of its legs. An ant on the top of the arch founds the angular height of the tip of the pole to be 45°. The height of the pole is



(a) $\sqrt{2}$ R (b) $\sqrt{3}$ R (c) $\sqrt{4}$ R (d) $\sqrt{5}$ R

Answer (c) **(CSIR-NET DEC 2012)**

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Difficulty rating: 3.5/5 *Average time for solving: 90 seconds*

Construct OD, DB, and AD as shown in the figure.

Let us explain some confusing terms in the question with the help of the figure.

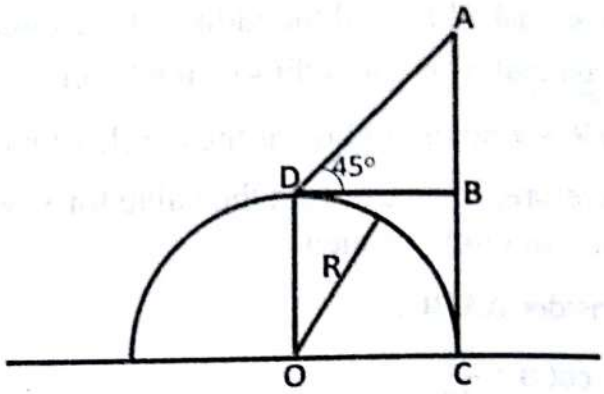
"Ant on top of the arch" means the point D in the figure.

"Angular height" is $\angle ADB = 45^\circ$.

Height of the pole, AC = ?

Clearly from the figure, OC = DB = R

OD = CB = R



Now consider the right-angled triangle ΔABD

$$\tan 45^\circ = \frac{\text{Opposite side}}{\text{Adjascent Side}} = \frac{AB}{DB}$$

$$\Rightarrow 1 = \frac{AB}{R}$$

$$\Rightarrow AB = R$$

Height of the pole, AC = AB + BC = R + R = 2R = $\sqrt{4}$ **R.**

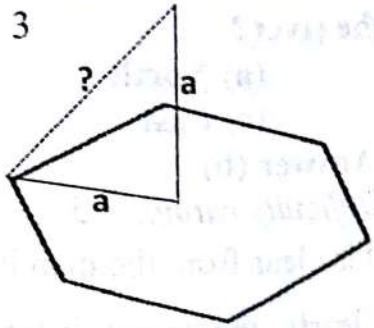
11. A vertical pole of length 'a' stands at the center of a horizontal regular hexagonal ground of side a. A rope that is fixed taut in between a vertex on the ground and the tip of the pole has a length

(a) a **(b) $\sqrt{2}$a** **(c) $\sqrt{3}$a** **(d) $\sqrt{6}$a**

Answer (b) **(CSIR-NET DEC 2012)**

Difficulty rating: 3/5 *Average time for solving: 80 seconds*

The given question can be pictured as shown. (Visualize it in 3 dimensions to understand clearly.)



Since, the angle between the pole and the vertex is 90°, we can use Pythagoras theorem to find the length of the rope.

Thus, length of the rope = $\sqrt{a^2 + a^2} = \sqrt{2}a$.

12. A peacock perched on the top of a 12 m high tree spots a snake moving towards its hole at the base of the tree from a distance equal to thrice of the height of the tree. The peacock flies towards the snake in a straight line and they both move at the same speed. At what distance from the base of the tree will the peacock catch the snake?

(a) 16 m **(b) 18 m** **(c) 14 m** **(d) 12 m**

Answer (a) **(CSIR-NET DEC 2012)**

Difficulty rating: 3/5 *Average time for solving: 70 seconds*

Let x be the distance from the base of the tree where the peacock catch the snake.

The question can be pictured as shown in the figure.

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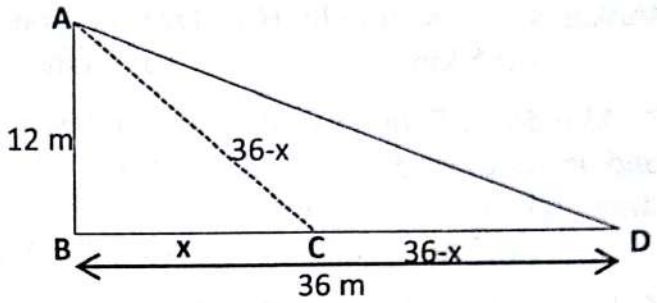
Since BC = x, CD = 36 – x.

Since the peacock and the snake moves at same speed, AC = CD = 36 – x.

In ΔABC, by Pythagoras Theorem,



$12^2 + x^2 = (36-x)^2$

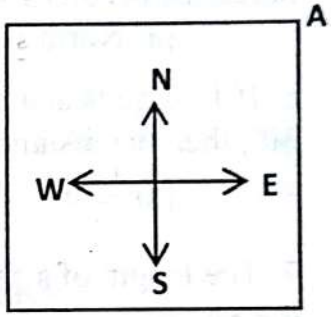
⇨ $144 + x^2 = 36^2 - (2 \times 36 \times x) + x^2$

⇨ $144 = 1296 - 72x$

⇨ $72x = 1152$

⇨ $x = \mathbf{16\ m.}$

13. Suppose a person A is at the North-East corner of a square (see the figure). From that point he moves along the diagonal and after covering $\frac{1}{3}$rd portion of the diagonal, he goes to his left and after sometimes he stops, rotates 90° clockwise and moves straight. After a few minutes he stops, rotates 180° anticlockwise. Towards which direction he is facing now?

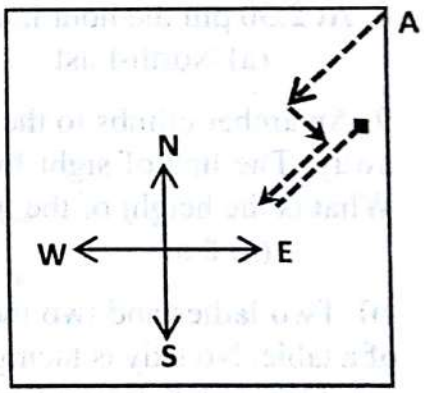


(a) North-East
(b) North-West
(c) South-East
(d) South-West

Answer (a) **(CSIR-NET DEC 2012)**

Difficulty rating: 3/5 *Average time for solving: 40 seconds*

The question can be pictured as shown.



The dark arrowed position shows the final position of the person. Clearly, he is facing to the **North-East** direction.

7.3 Practice Questions

1. Mathew standing on the bank of river finds that the angle of elevation of the top of a tower on opposite side bank is 45°. Which of the following statement is correct?

(a) Breadth of the river is half of the height of the tower.
(b) Breadth of the river and the height of the tower are equal.
(c) Breadth of the river is twice the height of the tower.
(d) None of these

2. A direction pole was situated on the crossing. Due to an accident, the pole turned in such a manner that the pointer which was showing East started showing South. One traveller went to the wrong direction thinking it to be west. In what direction actually was he traveling?

(a) South (b) East (c) West (d) North

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3. Sonu starts from home in early morning and walks straight for 10 km facing the Sun. Then he takes a right turn and walks for 4 km. Then he turns right again and walks for 1 km and then turns left and walks for 2 km. Then he turns right, travels 4 km and then turns right and travels for 6 km straight. How far is he from the starting point?

(a) 5 km (b) 6 km (c) 8 km (d) 12 km

4. Alin drives 7 km towards South and turns to the left and drives 5 km. Then, she turns again and drives towards left hand and drives 7 km. How far is she from her starting point. In which direction would she be driving?

(a) 5 km South (b) 3 km West (c) 5 km North (d) 6 km North

5. In a meeting, the map of a village was placed in such a manner that southeast becomes west, northeast becomes south and so on. What will north become?

(a) North (b) South east (c) North west (d) West

6. If Jose aims at a bird on the top of a pole of height 11metres with an angle of elevation of 30°, then the distance of the pole from Jose is

(a) $\frac{11}{\sqrt{3}}$ m (b) $11\sqrt{3}$ m (c) 11 m (d) $\sqrt{3}$ m

7. The length of a shadow of a vertical tower is $\sqrt{3}$ times the height. The angle of elevation of the Sun is

(a) 30° (b) 45° (c) 60° (d) 90°

8. At 2:30 pm the hour hand will point in which direction.

(a) North-East (b) South-West (c) East (d) North-West

9. An archer climbs to the top of a 8 m high building and aims at a bird on top of a tree 10 m away. The line of sight from the archer to the bird makes an angle of 45° to the horizontal. What is the height of the tree?

(a) 8 m (b) 10 m (c) 18 m (d) 20 m

10. Two ladies and two men are playing cards and are seated at North, East, South and West of a table. No lady is facing East. The persons sitting opposite to each other are not of the same sex. One man is facing South. Which directions are the ladies facing?

(a) East and West (b) South and East
(c) North and West (d) North and East

Answer Key

1.(b) 2.(d) 3.(a) 4.(c) 5.(b) 6.(b) 7.(a) 8.(a) 9.(c) 10.(c)

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CHAPTER 8

FINDING THE X

Average Difficulty Rating: 2.9/5

Average time for each question: 93 sec

Number of questions expected for each exam: 1-2

8.1 THEORY

8.1.1 Preliminaries

This section is an extension of the "Linear equations" which most of you might have studied in your high school math class. Therefore, we can start with a discussion of two of the basic algebraic methods for solving a pair of linear equations.

Algebraic methods of solving a pair of linear equations

1. Substitution method:

ILLUSTRATION 1:

Solve the following pair of equations

$$7x - 5y = 2 \qquad (1)$$
$$x + 2y = 3 \qquad (2)$$

Solution:

Step 1: We pick either of the equations and write one variable in terms of the other.

It is easier if we first consider the equation in which the coefficient of x or y is 1.

Here in equation (2), the coefficient of x is 1.

Therefore, consider equation (2)

$$x + 2y = 3$$

$$\Rightarrow x = 3 - 2y \text{ -------(3)}$$

Step 2: Substitute the above value of x in equation (1). We get

$$7(3 - 2y) - 5y = 2$$

$$\Rightarrow 21 - 14y - 5y = 2$$

$$\Rightarrow 19 = 19y$$

$$\Rightarrow y = 1$$

Step 3: Substituting this value of y in equation (3)

$$x = 3 - 2 = 1$$

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Thus, $x = 1, y = 1$ is the solution of the given pair of equations.

2. Elimination method:

ILLUSTRATION 2:

The ratio of incomes of two persons is 9 : 7 and the ratio of their expenditure is 4 : 3. If each of them manages to save Rs.2000 per month, find their monthly incomes.

Solution:

Let us denote the incomes of the two persons by 9x and 7x and their expenditures by 4y and 3y respectively. Then the equations formed in the situation is given by:

$9x - 4y = 2000$ ----------(1)

$7x - 3y = 2000$ ----------(2)

Step 1: Multiply equation (1) by 3 and equation (2) by 4 to make the coefficients of y equal.

Then we get the equations:

$27x - 12y = 6000$ ---------(3)

$28x - 12y = 8000$ ---------(4)

Step 2: Subtract equation (3) from equation (4) to eliminate y, because the coefficients of y are the same.

So, we get

$(28x - 27x) - (0) = 8000 - 6000$

$\Rightarrow x = 2000$

Step 3: Substituting this value of x in equation (1), we get

$9(2000) - 4y = 2000$

$\Rightarrow 4y = 18000 - 2000$

$\Rightarrow 4y = 16000$

$\Rightarrow y = 4000$

Thus, the solution of the equations is $x = 2000, y = 4000$.

Therefore, the monthly incomes of the persons are Rs.18,000 and Rs.14,000 respectively.

8.1.2 Daily life problems

These are the problems which depicts everyday life situations. You should analyse the situation and express it in equations to get the required answer. All these questions demands you to express them in one or two variables, say x and y. A naive outline for approaching these problems is given below.

Step 1: Analyse the question and decide whether to use one variable or two variables ("x" or "x" and "y")

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Step 2: Express the situation in terms of the above variable(s).

Step 3: Solve the equation to find the value of the variable(s).

Tip: Always assign variable "x" for what you require as the final answer.

ILLUSTRATION 3:

A farmer divides his cows among his four sons so that one son gets one- half the herd, a second son, one-fourth, a third son, one-fifth and the fourth son, 7 cows. Then how many cows did the farmer had?

(a) 80 (b) 100 (c) 140 (d) 180

Answer (c)

Assume that the farmer had x cows.

Given: "one son gets one- half", that is $\frac{x}{2}$.

Given: "a second son, one-fourth", that is $\frac{x}{4}$.

Given: "a third son, one-fifth", that is $\frac{x}{5}$.

Given: The fourth son got 7 cows.

Clearly, the sum of the cows got by each son gives the total number of cows, x.

Hence, $\frac{x}{2}+\frac{x}{4}+\frac{x}{5}+7=x$

$\Rightarrow \frac{10x+5x+4x}{20}+7=x$

$\Rightarrow 7=x-\frac{19x}{20}$

$\Rightarrow \frac{x}{20}=7$

$\Rightarrow x=140.$

Hence, the farmer had 140 cows.

ILLUSTRATION 4:

A candidate should score 45% marks of the total marks to pass the exam. He gets 520 marks and fails by 20 marks. The total marks in the exam are:

(a) 1000 (b) 1100 (c) 1200 (d) 1400

Let x be the total marks in the exam.

Given 1: "A candidate should score 45% marks of the total marks to pass the exam."

That is he should score $\frac{45}{100}x$ marks to pass the exam.

Given 2: "He gets 520 marks."

Given 3: He fails by 20 marks.

From *Given 2* and *Given 3*, we can infer that, 520 marks is 20 less than the pass mark,

Combining *Given 1*, 520 marks is 20 less than $\frac{45}{100}x$.

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Thus, putting our observations into an equation, we get,

$$520 = \frac{45}{100}x - 20.$$

$$\Rightarrow 540 = \frac{45}{100}x$$

$$\Rightarrow x = \frac{54000}{45}$$

$$\Rightarrow x = 1200.$$

Thus, the total marks in the exam is 1200.

8.1.3 Age problem

Age Problem is one of the most important type in the daily life problems.

In the age problem, statements regarding age of one or more people will be given. You should analyse those statements to find the age of a particular person asked in the question.

ILLUSTRATION 5:

The present age of a father is 3 years more than three times the age of his son. Three years hence, father's age will be 10 years more than twice the age of the son. Find the present age of the father?

(a) 30 years (b) 33 years (c) 35 years (d) 40 years

Answer (b)

We are asked to find the present age of the father.

Assume the present age of the father to be x and the present age of the son to be y.

Let us analyse each of the statements.

Given: Three times the age of the son. That is 3y.

Given: **The present age of a father is 3 years more than three times the age of his son.**

That is, x is 3 years more than 3y

$\Rightarrow x = 3y + 3$----------(1)

Now, analyse the second statement.

Given: **Three years hence, father's age will be 10 years more than twice the age of son.**

Note: "Three years hence" means "after three years".

Thus, after three years the father's age will be x + 3 and son's age will be y +3

Therefore, x + 3 will be 10 years more than twice of y + 3.

That is, $x + 3 = 2(y+3) + 10$

$\Rightarrow x + 3 = 2y + 16$ ----------(2)

Substitute value of x of equation (1) in equation (2)

$\Rightarrow 3y + 3 + 3 = 2y + 16$

$\Rightarrow y = 10$

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Now, substitute this y of equation (1) to get the value of x.

That is, $x = 3 \times 10 + 3$

$\Rightarrow x = 33.$

Hence, present age of the father = **33 years.**

8.2 CSIR-NET Exam Questions – Detailed Solution

1. 20% of students of a particular course get jobs within one year of passing. 20% of the remaining students get jobs by the end of second year of passing. If 16 students are still jobless, how many students had passed the course?

(a) 32 **(b) 64** **(c) 25** **(d) 100**

Answer (c) **(CSIR-NET DEC 2014)**

Difficulty rating: 3/5 *Average time for solving: 100 seconds*

Let x be the number of students who passed the course.

Given 1: "20% of students get jobs within one year of passing."

That is, $\frac{20}{100}x$.

Given 2: "20% of the remaining students get jobs by the end of second year of passing."

"the remaining" = $\frac{100-20}{100}x = \frac{80}{100}x$

"20% of the remaining" = $\frac{20}{100} \times (\frac{80}{100}x)$

Given 3: "16 jobless students."

Here the total number of students is the sum of the above three sets of students.

Thus, by the given statements,

$\Rightarrow \frac{20}{100}x + (\frac{20}{100} \times (\frac{80}{100}x)) + 16 = x$

$\Rightarrow \frac{20}{100}x + \frac{16}{100}x + 16 = x$

$\Rightarrow 36x + 1600 = 100x$

$\Rightarrow 64x = 1600$

$\Rightarrow x = 25$

Thus, the number of students who passed the course = **25.**

2. Three years ago, the difference in the ages of two brothers was 2 years. The sum of their present ages will double in 10 years. What is the present age of the elder brother?

(a) 6 **(b) 11** **(c) 7** **(d) 9**

Answer (b) **(CSIR-NET JUNE 2014)**

Difficulty rating: 2.5/5 *Average time for solving: 90 seconds*

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Let x be the present age of the elder brother and y be the present age of younger brother.

Given 1: **"Three years ago, the difference in the ages of two brothers was 2 years."**

That is, 3 years ago, the age of the elder brother was $x - 3$ and that of the younger brother was $y - 3$.

$$(x - 3) - (y - 3) = 2$$

$$\Rightarrow (x - y) = 2 \text{ ----------(1)}$$

After 10 years, the age of the brothers are $x + 10$ and $y + 10$ respectively.

Given 2: **The sum of their present ages will double in 10 years.**

That is, $2(x + y) = (x + 10 + y + 10)$

$$\Rightarrow 2x + 2y = x + y + 20$$

$$\Rightarrow x + y = 20 \text{----------(2)}$$

Solving equations (1) and (2), we get,

$$\Rightarrow 2x = 22.$$

$$\Rightarrow x = 11.$$

Hence, the present age of the elder brother = **11.**

3. In a room, we have one grandfather, two fathers, two sons, and a grandson. The age of one father is seven times the age of his son. The age of the other father is twice his son's age. Assuming that there are only 3 people in the room and the grandfather is 70 years old, how old is the grandson?

(a) 1 **(b) 2** **(c) 5** **(d) cannot be determined**

Answer (c) **(CSIR-NET DEC 2013)**

Difficulty rating: 3/5 *Average time for solving: 90 seconds*

According to the question, there are 3 people in the room and they are son, his father and his grandfather.

Observations:

(1) Grandfather's age is 70.

(2) "The age of one father is seven times the age of his son."

(3) "The age of the other father is twice his son's age."

Case 1: If Observation (3) is about grandfather-father duo,

$$\text{Age of Father} = \frac{1}{2} \times \text{Age of Grandfather}$$

$$= \frac{1}{2} \times 70 = 35 \text{ years.}$$

$$\text{Age of Son} = \frac{1}{7} \times \text{Age of Father}$$

$$= \frac{1}{7} \times 35 = 5 \text{ years.}$$

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Case 2: If Observation (2) is about grandfather-father duo, father's age will be 10 and by observation (3), son's age will be 5.

In either cases, son's age is 5. (option c)

Note: Though father's age being 10 sounds absurd, logically it is correct according to given problem, and that is why it is mentioned here.

4. In an enclosure there were both crows and cows. If there are 30 heads and 100 legs, what fraction of them are crows?

(a) 1:3 **(b) 1:4** **(c) 1:10** **(d) 3:10**

Answer (a) **(CSIR-NET DEC 2013)**

Difficulty rating: 2.5/5 *Average time for solving: 90 seconds*

Let x and y be the number of crows and cows respectively.

Number of heads = Number of crows + Number of cows

Thus, $x + y = 30$ ------- (1)

We know that a crow has 2 legs and a cow has 4 legs.

Thus, all the crows together have 2x legs and all the cows together have 4y legs.

Therefore, $2x + 4y = 100$ ------- (2)

Solving equations (1) and (2), we get $x = 10, y = 20$.

$$\text{Required fraction} = \frac{\text{Number of crows}}{\text{Total number}} \quad \frac{x}{x+y} = \frac{10}{10+20} = \frac{1}{3} = \mathbf{1:3.}$$

5. Three fishermen caught fishes and went to sleep. One of them woke up, took away one fish and $\frac{1}{3}$rd of the remainder as his share, without others' knowledge. Later, the three of them divided the remainder equally. How many fishes were caught?

(a) 58 **(b) 19** **(c) 76** **(d) 88**

Answer (b) **(CSIR-NET DEC 2013)**

Difficulty rating: 3.5/5 *Average time for solving: 100 seconds*

Let x be the number of fishes caught.

When one fisherman took 1 fish, the remaining is (x-1) fishes.

Again, he took $\frac{1}{3}^{rd}$ of this (x-1) fishes.

Thus, the remaining is $\frac{2}{3}$ (x-1) fishes.

Since they divided this $\frac{2}{3}$ (x-1) fishes among 3 persons, it should be a multiple of 3.

That is, $\frac{2}{3}$ (x-1) = 3k, where k is an integer.

Now put each option for x in this equation to see which one will satisfy an integer value for k.

Clearly, $\frac{2}{3}$ (19-1) = 3k

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⇨ $\frac{2}{3} \times 18 = 3k$

⇨ $12 = 3k$

⇨ $k = 4$, an integer.

Hence **19** fishes were caught.

(For the other options, we cannot find an integer k satisfying the equation.)

6. A king ordered that a golden crown be made for him from 8 kg of gold and 2 kg of silver. The goldsmith took away some amount of gold and replaced it by an equal amount of silver and the crown when made, weighed 10 kg. Archimedes knew that under water gold lost 1/20th of its weight, while silver lost 1/10th. When the crown was weighed under water, it was 9.25 kg. How much gold was stolen by the goldsmith?

(a) 0.5 kg **(b) 1 kg** **(c) 2 kg** **(d) 3 kg**

Answer (d) **(CSIR-NET JUNE 2013)**

Difficulty rating: 3.5/5 *Average time for solving: 120 seconds*

Given 1: **"The goldsmith took away some amount of gold."**

Assume that, the gold smith took x kg gold from total amount of 8 kg gold.

That is, the weight of gold becomes 8 - x.

Given 2: **"replaced it by an equal amount of silver."**

Then, x kg of silver was added to the total amount of 2 kg silver.

That is, the weight of silver becomes 2 + x.

Lost weight under water:

Given 3: **"The crown was weighed under water and it was 9.25 kg."**

Thus the crown lost $10 - 9.25 = 0.75$ kg $= \frac{3}{4}$ kg.

Gold lost $1/20^{th}$ of its weight.

Silver lost $1/10^{th}$ of its weight.

Hence, lost weight under water can be equated as:

$$[(8 - x) \times \frac{1}{20}] + [(2 + x) \times \frac{1}{10}] = \frac{3}{4}$$

⇨ $\frac{8 - x + 4 + 2x}{20} = \frac{3}{4}$

⇨ $\frac{12 + x}{20} = \frac{3}{4}$.

⇨ $48 + 4x = 60$.

⇨ $4x = 12$.

⇨ $x = 3$.

Hence, the amount of gold stolen = **3 kg.**

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7. 25% of 25% of a quantity is x% of the quantity, where x is

(a) 6.25% **(b) 12.5%** **(c) 25%** **(d) 50%**

Answer (a) **(CSIR-NET DEC 2012)**

Difficulty rating: 2.5/5 *Average time for solving: 60 seconds*

Let y be the quantity.

Given: **"25% of 25% of the quantity"**

That is, $\frac{25}{100} \times \frac{25}{100} \times y$.

Given: **"x% of the quantity"**

That is $\frac{x}{100} \times y$.

Hence by the given condition,

$$\frac{25}{100} \times \frac{25}{100} \times y = \frac{x}{100} \times y$$

$\Rightarrow$ **x = 6.25%.**

8.3 PRACTICE QUESTIONS

1. A mother's age is four times as much as the sum of the ages of her three children, but after six years her age will be only double the sum of their ages. The age of the mother is

(a) 30 years (b) 40 years (c) 55 years (d) 60 years

2. 18 years ago, Jibin was three times as old as his son. But now, he is only twice as old now. What is Jibin's age today (in years)?

(a) 36 (b) 45 (c) 54 (d) 72

3. Ramiz gave one-fourth of the amount he had to Ajesh. Ajesh in turn gave half of what he received from Ramiz to Shan. If the difference between the remaining amount with Ramiz and the amount received by Shan is Rs.1000, how much money did Shan receive from Ajesh?

(a) Rs.80 (b) Rs.100 (c) Rs.200 (d) Rs.450

4. In an election, one of the two candidates gets 35% votes and loses by 90 votes. Total number of votes is:

(a) 300 (b) 400 (c) 600 (d) 1000

5. The pig population in community A increases at 30% per year while that in B increases at 60% per year. If the present populations of A and B are equal, the ratio of the number of the pigs in B to that in A after 2 years will be

(a) 256:169 (b) 16:13 (c) 3:16 (d) 1:2

6. One says, "Give me a hundred, friend! I shall then become twice as rich as you". The other replies, "If you give me ten, I shall be six times as rich as you". Tell me what is the amount of their (respective) capital? (Question from the ancient Indian Mathematical text "Bijaganitha".)

(a) Rs.40, Rs.170 (b) Rs.30, Rs.170

(c) Rs.50, Rs.175 (d) Rs.50, Rs.180

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7. A man divides Rs.11 lakhs among his 4 sons, 5 daughters and 2 nephews. If each daughter receives five times as much as each nephew, and each son receives seven times as much as each nephew, how much does each son receive?

(a) Rs.60,000 (b) Rs.1,00,000 (c) Rs.1,20,000 (d) Rs.1,40,000

8. On Children's day, sweets were to be equally distributed among 170 children in a school. Actually on the Children's day, 20 students were absent and therefore each child got 2 sweets extra. Total how many sweets were available for distribution?

(a) 1480 (b) 2500 (c) 2550 (d) 2700

9. There are two examination rooms X and Y. If 10 students are sent from X to Y, then the number of students in each room is the same. If 15 candidates are sent from Y to X, then the number of students in X is double the number of students in Y. The number of students in room Y is :

(a) 45 (b) 55 (c) 65 (d) 200

10. The students of a class are made to stand in rows. If 3 students are extra in a row, there would be 1 row less. If 3 students are less in a row, there would be 2 rows more. Find the number of students in the class.

(a) 64 (b) 24 (c) 36 (d) 48

Answer Key

1. (d) 2. (d) 3. (c) 4. (a) 5. (a) 6. (a) 7. (d) 8. (c) 9. (c) 10. (c)

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CHAPTER 9

AVERAGE

Average Difficulty Rating: 3/5

Average time for each question: 94 sec

Number of questions expected for each exam: 1-2

9.1 THEORY

9.1.1 Arithmetic Mean:

In simple language, an average is the sum of a list of numbers divided by the number of elements in the list.

$$\textbf{Average or Arithmetic mean} = \frac{\textbf{Sum of elements}}{\textbf{Total number of elements}}$$

Example: The average of four observations $\{50, 55, 70, 85\} = \frac{50+55+70+85}{4} = \frac{260}{4} = 65.$

Although the arithmetic mean is not the only "mean", it is by far the most commonly used. Therefore, if the term "mean" is used without specifying whether it is the arithmetic mean, the geometric mean, or some other mean, it is assumed to refer to the arithmetic mean.

9.1.2 Important facts about Arithmetic mean:

- If each of the given element is increased or decreased by x, then their average is also respectively increased or decreased by x.

  Example: The average of 45, 55, 80 is 60. If 20 is added to each of the elements the new set is 65, 75, 100 whose average is 80. Clearly 80 is 20 more than the initial average 60.

- If each of the given elements is multiplied by x, then their average gets multiplied by x.

- If each of the given elements is divided by a nonzero x, then their average gets divided by x.

- If x_1 and x_2 are the arithmetic means of two samples of sizes n_1 and n_2 respectively, then the arithmetic mean z of the distribution combining the two samples can be calculated as

$$z = \frac{n_1x_1 + n_2x_2}{n_1 + n_2}$$

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- If $x_1, x_2, \ldots, x_k$ are the arithmetic means of k samples of sizes $n_1, n_2, n_3, \ldots, n_k$ respectively, then the arithmetic mean z of the distribution combining the k samples can be calculated as

$$z = \frac{n_1x_1 + n_2x_2 + \cdots + n_kx_k}{n_1 + n_2 + \cdots + n_k}$$

- The value of the arithmetic mean of two elements is always in between the value of the two elements.

9.1.3 Geometric Mean:

The geometric mean is defined as the n^{th} root of the product of n numbers.

That is, to find the geometric mean of a set of 'n' numbers, multiply the 'n' numbers and then take the n^{th} root of the product.

If $x_1, x_2, \ldots, x_n$ are the 'n' elements, then

Geometric mean $= \sqrt[n]{x_1x_2 \ldots x_n}$

Example 1: Geometric mean of two numbers, say 3 and 27 is the square root of their product; i.e., $\sqrt{3 \times 27} = \sqrt{81} = 9$.

Example 2: Geometric mean of three numbers, say 4, 6 and 9 is the cube root of their product; i.e., $\sqrt[3]{4 \times 6 \times 9} = \sqrt[3]{216} = 6$.

Note: Geometric mean is always less than or equal to the Arithmetic mean.

9.1.4 Median:

Median is the middle value that separates the higher half from the lower half of the set of observations.

To find the median, we arrange the observations in the increasing (or descending) order. If there is an odd number of observations, the median is the middle value. If there is an even number of observations, the median is the average of the two middle values.

Example 1: To find the median of the set of observations {10, 15, 12, 3, 55}, arrange the set of observations in the increasing order as {3, 10, 12, 15, 55}.

Since there are 5 observations (odd), the median is the 3^{rd} term, that is 12.

Example 2: To find the median of the set of observations {5, 20, 10, 35}, arrange the set of observations in the increasing order as {5, 10, 20, 35}.

Since there are 4 observations (even), the median is the average of 2^{nd} and 3^{rd} terms.

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That is, Median = $\frac{10+20}{2} = 15$.

9.1.5 Mode:

The mode is the most frequently appearing value in the set of observations.

Example: The mode of the set of observations {1, 5, 4, 5, 7, 2, 2, 5}

Here 5 is the most frequently appearing value.

Hence, 5 is the mode.

9.1.6 Standard Deviation:

Standard deviation is a measure that is used to quantify the amount of variation or dispersion of a set of data values. A standard deviation close to 0 indicates that the data points tend to be very close to the mean of the set, while a high standard deviation indicates that the data points are spread out over a wider range of values.

Variance for N observations is defined as:

$$\textbf{Variance} = \sum \frac{1}{N}(x_i - \textbf{Mean})^2$$

$$\textbf{Standard Deviation} = \sqrt{\textbf{Variance}}$$

Example: Consider the set of observations {1, 3, 5, 7}

Here the Mean = $\frac{1+3+5+7}{4} = \frac{16}{4} = 4$.

$$\text{Variance} = \frac{(1-4)^2 + (3-4)^2 + (5-4)^2 + (7-4)^2}{4}$$

$$= \frac{9+1+1+9}{4}$$

$$= 5.$$

Standard Deviation = $\sqrt{\text{Variance}} = \sqrt{5}$

ILLUSTRATION 1:

The average marks of three batches of students having 70, 80 and 50 students respectively are 60, 50 and 46. The average marks of all the 200 students, taken together is

(a) 51.5 (b) 52 (c) 52.5 (d) 53

Answer (c)

The data given in the table can be denoted using the following table.

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	Batch 1	Batch 2	Batch 3
Number of students	70	80	50
Average Marks	60	50	46

We know that if x_1, x_2, and x_3 are the arithmetic means of three samples of sizes n_1, n_2 and n_3 respectively, then the arithmetic mean z of the distribution combining the three samples can be calculated as

$$z = \frac{n_1x_1 + n_2x_2 + n_3x_3}{n_1 + n_2 + n_3}$$

Thus, Average marks of 200 students $= \frac{(70\times60)+(80\times50)+(50\times46)}{70+80+50} = \frac{10500}{200} = \mathbf{52.5.}$

ILLUSTRATION 2:

Of the three numbers, the average of the first and second is greater than the average of the second and third by 20. What is the difference between the first and third of the three numbers?

(a) 15 (b) 20 (c) 30 (d) 40

Answer (d)

Let a, b, c be the three numbers.

The average of first and second number (i.e. a and b) $= \frac{a+b}{2}$.

The average of second and third number (i.e. b and c) $= \frac{b+c}{2}$.

Given that the average of the first and second is greater than the average of the second and third by 20.

$\Rightarrow \frac{a+b}{2} = \frac{b+c}{2} + 20.$

$\Rightarrow \frac{a+b}{2} - \frac{b+c}{2} = 20.$

$\Rightarrow \frac{a+b-b-c}{2} = 20$

$\Rightarrow a - c = 40.$

Thus, the difference between the first and third number = **40.**

ILLUSTRATION 3:

The average age of a cricket team reduces by 3 when two senior players aged 37 and 38 are replaced by 2 youngsters of equal age. What is the age of these players?

(a) 20 (b) 21 (c) 22 (d) 24

Answer (b)

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As you know, in a cricket team there are 11 players.

At first, let x be the average age of the cricket team.

Then the sum of ages of the cricket team = 11x.

Given: Two senior players aged 37 and 38 are excluded from the team.

Consequently, now the total age = $11x - 37 - 38 = 11x - 75$.

Given: 2 youngsters of equal age are included in the team.

Let y be the ages of each of the youngsters.

Consequently, now the new total age of the team = $11x - 75 + 2y$ -------(1)

Given: Average age of the team reduces by 3 due to this replacing.

Consequently the new average age = $x - 3$.

Again the new total age of the team = $11(x - 3)$ -------(2)

Thus, both equation (1) and equation (2) gives the new total age of the team.

Hence,

$11x + 2y - 75 = 11x - 33$

$\Rightarrow 2y = 42$

$\Rightarrow y = 21$.

Hence, the age of each of the youngsters is 21.

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9.2 CSIR-NET Exam Questions – Detailed Solution

1. Weights (in kg) of 13 persons are:
70, 72, 74, 76, 78, 80, 82, 84, 86, 88, 90, 92, 94
Two new persons having weights 100 kg and 79 kg join the group. The average weight of the group increases by
(a) 0 kg (b) 1 kg (c) 1.6 kg (d) 1.8 kg
(CSIR-NET DEC 2014)
Answer (b)
Difficulty rating: 2.5/5 *Average time for solving: 90 seconds*

Normal solution

Average of 13 persons $= \frac{(70+72+74+76+78+80+82+84+86+88+90+92+94)}{13} = \frac{1066}{13} = 82.$

Since two new persons are added, the number of observations increases to 15.

Thus, New average $= \frac{1066+100+79}{15} = \frac{1245}{15} = 83.$

Increase in average = New average – Old average = 83-82 = **1 kg.**

Quick solution

Since the middle term (7^{th} term) is 82 and the rest 12 terms are symmetrically distributed around it, the average of this 13 persons are 82. (Symmetrically distributed means six in left and six in right and the difference between the numbers are the same (2)).

Now, two persons with weights 100 kg and 79 kg respectively are added to the group.

Person with 100 kg contribute 100 – 82 = 18 more to the new average.

Person with 79 kg contribute 82 - 79 = 3 less to the new average.

Therefore, there combined contribution is 18 – 3 = 15 kg more to the new average.

Total number of persons in new group = 15.

Increase in average $= \frac{15}{15} =$ **1 kg.**

2. Students in a group A obtained the following marks: 40, 80, 70, 50, 60, 90, 30. Students in group B obtained 40, 80, 35, 70, 85, 45, 50, 75, 60 marks. Define:
Dispersion (D) = (maximum marks - minimum marks), and
Relative Dispersion (RD) = $\frac{\text{Dispersion}}{\text{Mean}}$
Then (a) RD of group A = RD of group B (b) RD of group A > RD of group B
(c) RD of group A < RD of group B (d) D of group A < D of group B
Answer (b) **(CSIR-NET JUNE 2014)**
Difficulty rating: 2/5 *Average time for solving: 100 seconds*

Given: Dispersion (D) = (maximum marks - minimum marks),

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Relative Dispersion (RD) $= \frac{\text{Dispersion}}{\text{Mean}}$

Group A

Dispersion (D) $= 90 - 30 = 60.$

Mean $= \frac{40+80+70+50+60+90+30}{7} = \frac{420}{7} = 60.$

RD $= \frac{60}{60} = 1.$

Group B

Dispersion (D) $= 85 - 35 = 50.$

Mean $= \frac{40+80+35+70+85+45+50+75+60}{9} = \frac{540}{9} = 60.$

RD $= \frac{50}{60} < 1.$

Thus, **RD of Group A > RD of Group B.**

Hence, **option (b).**

3. What is the arithmetic mean of $\frac{1}{1\times2}, \frac{1}{2\times3}, \frac{1}{3\times4}, \frac{1}{4\times5}, \ldots, \frac{1}{100\times101}$?

(a) 0.01 **(b)** $\frac{1}{101}$ **(c) 0.00111…** **(d)** $\frac{\frac{1}{49\times50}+\frac{1}{50\times51}}{2}$

Answer (b) **(CSIR-NET DEC 2013)**

Difficulty rating: 3.5/5 *Average time for solving: 100 seconds*

Prerequisite: Read theory section "Telescopic series" in Chapter 4.)

This is the **telescopic sequence.**

We can write the terms in the telescopic sequence as

$$\frac{1}{1\times2} = \frac{1}{1} - \frac{1}{2}$$

$$\frac{1}{2\times3} = \frac{1}{2} - \frac{1}{3}$$

$$\frac{1}{3\times4} = \frac{1}{3} - \frac{1}{4}$$

$\ldots.$

$$\frac{1}{100\times101} = \frac{1}{100} - \frac{1}{101}$$

Clearly, there are 100 terms.

Hence, the arithmetic mean of these numbers is given by,

$$\frac{\frac{1}{1\times2}+\frac{1}{2\times3}+\frac{1}{3\times4}+\frac{1}{4\times5}+\cdots+\frac{1}{100\times101}}{100} = \frac{1-\frac{1}{2}+\frac{1}{2}-\frac{1}{3}+\frac{1}{3}-\frac{1}{4}+\cdots+\frac{1}{100}-\frac{1}{101}}{100}$$

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$$= \frac{1-\frac{1}{101}}{100} = \frac{\frac{100}{101}}{100} = \frac{100}{101} \times \frac{1}{100} = \frac{1}{101}.$$

4. Suppose the sum of the seven positive numbers is 21. What is the minimum possible value of the average of the squares of these numbers?

(a) 63 **(b) 21** **(c) 9** **(d) 7**

(CSIR-NET DEC 2012)

Answer (c)

Difficulty rating: 3.5/5 *Average time for solving: 90 seconds*

Note: *Minimum possible value of the sum of seven positive numbers is the sum of the set of average of these seven numbers repeated seven times.*

Let x be the average of these 7 numbers.

Given, x+x+x+x+x+x+x = 21.

⇨ 7x = 21.

⇨ x = 3.

Therefore, the numbers are 3,3,3,3,3,3,3.

The minimum possible value of the average of squares of these numbers $= \frac{9+9+9+9+9+9+9}{7}$

$= 9.$

5. Let $A = \frac{1^{13}+2^{13}+3^{13}+\cdots 100^{13}}{100}$, $B = \frac{1^{13}+3^{13}+5^{13}+\cdots 99^{13}}{50}$, $C = \frac{2^{13}+4^{13}+6^{13}+\cdots 100^{13}}{50}$

Which of the following is true?

(a) B < C < A **(b) A < B < C** **(c) B < A < C** **(d) C < A < B**

(CSIR-NET DEC 2012)

Answer (c)

Difficulty rating: 3.5/5 *Average time for solving: 90 seconds*

Consider B and C.

Both have the same denominator and in the numerator of B, each of the terms is less than the corresponding term in the numerator of C.

(That is, $1^{13} < 2^{13}$, $3^{13} < 4^{13}$,... ,$99^{13} < 100^{13}$.)

Thus, **B < C.**

Now, we try to establish some relation between A, B, C.

$$B + C = \frac{1^{13}+3^{13}+5^{13}+\cdots 99^{13}}{50} + \frac{2^{13}+4^{13}+6^{13}+\cdots 100^{13}}{50} = \frac{1^{13}+2^{13}+3^{13}+\cdots 100^{13}}{50} = 2A$$

Thus, $A = \frac{B+C}{2}$.

That is, A is the Arithmetic mean of B and C.

Remember: The value of Arithmetic mean is always in between the numbers.

Hence, **B < A < C.**

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9.3 PRACTICE QUESTIONS

1. If each number in a set of ten numbers is increased by 20, the arithmetic mean of the original ten numbers:

(a) remains the same
(b) is increased by 20
(c) is increased by 200
(d) is increased by 10

2. A student obtained 75%, 80% and 85% in three subjects. If the mark of another subject is added, then his average cannot be less than

(a) 60% (b) 65% (c) 80% (d) 90%

3. The batting average for 25 innings of a cricketer is 52 runs. His highest score differs his lowest score by 144 runs. If these two innings are excluded, the average of the remaining 23 innings is 48 runs. The highest score of the player is

(a) 166 (b) 168 (c) 170 (d) 172

4. The average of nine numbers is x and the average of three of these is y. If the average of remaining is z, then :

(a) $3x = y + z$ (b) $3x = y + 2z$ (c) $3x = 2y + z$ (d) None of these

5. The average age of husband, wife and their child 1 years ago was 23 years and that of wife and the child 2 years ago was 15 years. The present age of the husband is:

(a) 35 years (b) 38 years (c) 36 years (d) 40 years

6. The average of X_1, X_2 and X_3 is 25. Twice the sum of X_1 and X_3 is 40. What is the value of X_2?

(a) 30 (b) 40 (c) 45 (d) 55

7. The average of four consecutive even numbers is 49. The largest of these numbers is

(a) 48 (b) 50 (c) 52 (d) 54

8. The arithmetic mean of a set of 50 numbers is 38. If two numbers, namely, 45 and 55, are discarded, the mean of the remaining set of numbers is:

(a) 36.5 (b) 37 (c) 37.2 (d) 37.5

9. The mark of a student was wrongly entered as 87 instead of 70. Due to that the average marks for the class got increased by half. The number of students in the class is:

(a) 30 (b) 32 (c) 34 (d) 36

10. The average score of a cricketer for 20 matches is 42.4runs. If the average for the first 13 matches is 55, then find the average for the last 7 matches

(a) 18 (b) 17 (c) 19 (d) 20

Answer Key

1. (b) 2. (a) 3. (c) 4. (b) 5. (b) 6. (d) 7. (c) 8. (d) 9. (c) 10. (c)

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CHAPTER 10

MONETARY PROBLEMS

Average Difficulty Rating: 3.1/5

Average time for each question: 107 sec

Number of questions expected for each exam: 1-2

10.1 THEORY

Monetary problems are those related to money. This section includes Simple Interest, Compound Interest, profit and loss etc.

10.1.1 Simple Interest:

Interest computed only on principal and not on principal plus interest incurred in the previous period(s) is called Simple Interest (S.I.).

Interest $I = \frac{P \times R \times T}{100}$

where P is the principal amount, R is the rate percentage, T is the time period.

Also $A = P + I$ where A is the total amount accrued at the end of the period.

Or $\mathbf{A = P(1 + \frac{RT}{100})}$

Shortcuts:

- Given principal (P), rate of interest (R), if we are asked to find the time period in which the principal gets doubled, just calculate $\frac{100}{R}$ which will be the required time period.
- Given principal (P), rate of interest (R), if we are asked to find the time period in which the principal gets tripled, just calculate $\frac{200}{R}$ which will be the required time period.

ILLUSTRATION 1:

A sum of Rs.2455 lent at simple interest at 10% per annum will become double in

(a) 8 years (b) $8\frac{1}{2}$ years (c) 9 years (d)10 years

Answer (d)

Normal Solution:

The formula for Simple interest is given by

$$I = \frac{P \times R \times T}{100}$$

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Here P = Rs.2455, R = 10 %.

Note that here the interest is calculated annually.

Given that the principal will be doubled.

Therefore, the amount A = 2 × 2455 = Rs.4910.

We know that A = P + I.

That is, I = A – P.

⇨ I = 4910 – 2455 = Rs. 2455.

Applying the formula, we get

$2455 = \frac{2455 \times 10 \times T}{100}$

⇨ $T = \frac{100}{10}$ = **10 years.**

Quick Solution:

Given principal P = 2455, rate of interest R = 10%,

Time period in which the principal gets doubled = $\frac{100}{R} = \frac{100}{10}$ = **10 years.**

10.1.2 Compound Interest:

Compound interest (C.I) is the interest added to the principal of a deposit or loan so that the added interest also earns interest from then on. This addition of interest to the principal is called compounding.

$$A = P\left(1 + \frac{R}{100}\right)^n$$

where P is the principal amount, R is the rate percentage, n is the time period.
A is the sum of the principal and the interest.

Thus, the formula for interest can be given as

I = A – P

$= P\left(\left(1 + \frac{R}{100}\right)^n - 1\right)$

Compound interest can be calculated annually, half-yearly, quarterly or even daily.

The formula is all the same, but the rate percentage(R) and time period (n) should be altered accordingly.

ILLUSTRATION 2:

A sum of money lent at compound interest amounts to Rs.1254 in 2 years and to Rs.1405 in 3 years. Find the rate of interest.

(a) 5% (b) 10% (c) 12% (d) 15%

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Answer (c)

Clearly, the interest is compounded annually.

Let x be the principal amount.

The formula for compound interest is given by

$$A = P\left(1 + \frac{R}{100}\right)^n$$

Then, $1254 = x\left(1 + \frac{R}{100}\right)^2$ ------ (1)

$$1405 = x\left(1 + \frac{R}{100}\right)^3 \text{ ------ (2)}$$

We have to find the rate of interest (R).

Dividing (2) by equation (1) to eliminate x.

$$\frac{1405}{1254} = 1 + \frac{R}{100}$$

$$\Rightarrow 1.12 = \frac{100+R}{100}$$

$$\Rightarrow 112 = 100 + R$$

$$\Rightarrow R = 112 - 100 = 12$$

10.1.3 Profit and Loss:

Cost Price (C.P.): The price at which an article is purchased is called cost price (C.P.)

Sale Price (S.P.): The price at which an article is sold is called sale price (S.P.) or selling price.

Marked Price (M.P.): The price put on the article by the seller or manufacturer is called marked price (M.P). Normally marked price is the price we see as the M.R.P or maximum retail price.

Profit or Gain: If Sale price is greater than Cost price then the seller is said to have a profit or gain.

Loss: If Cost price is greater than Sale price then the seller is said to have incurred a loss.

10.1.3.1 Important Formulas:

Profit or Gain = S.P. – C.P.

Loss = C.P. – S.P.

$$\text{Profit \% or Gain\%} = \left(\frac{SP-CP}{CP}\right)\times 100 = \left(\frac{Gain}{CP}\right)\times 100$$

$$\text{Loss\%} = \left(\frac{CP-SP}{CP}\right)\times 100 = \left(\frac{Loss}{CP}\right)\times 100$$

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ILLUSTRATION 3:

Sreeram bought 12 kg. of sugar at the rate of Rs.15 per kg. and 20 kg. of sugar at the rate of Rs.17 per kg. He mixed the two varieties and sold the mixture at the rate of R.20 per kg. What was his total gain by doing so?

(a) < 25% gain (b) < 25% loss (c) > 25% gain (d) no loss, no gain

Answer (a)

Given: Cost price of 1 kg. sugar of first type = Rs.15.

Then, Cost price of 12 kg. sugar of first type = 12 × 15 = Rs.180.

Given: Cost price of 1 kg. sugar of second type = Rs.17.

Then, Cost price of 20 kg. sugar of second type = 20 × 17 = Rs.340.

Total Cost price (CP) = Rs.180 + Rs.340 = Rs.520.

Given: Selling price of 1 kg. sugar = Rs.20

Total Selling Price (SP) = Selling price of 12 + 20 = 32 kg sugar = 32 × 20 = Rs.640.

Gain = S.P. – C.P. = 640 – 520 = Rs.120.

$$\text{Gain \%} = \frac{\text{Gain}}{\text{CP}} \times 100 = \frac{120}{520} \times 100 \approx 23\%.$$

Hence, **option (a).**

10.2 CSIR-NET EXAM QUESTIONS – DETAILED SOLUTION

1. A bank offers a scheme wherein deposits made for 1600 days are doubled in value, the interest being compounded daily. The interest accrued on the deposit of Rs.1000/- over the first 400 days would be Rs.

(a) 250 (b) 183 (c) 148 (d) 190

Answer (d) **(CSIR-NET DEC 2014)**

Difficulty rating: 4.5/5 *Average time for solving: 150 seconds*

Here the interest is being compounded daily.

Formula for compound interest is given by,

$$\mathbf{A = P(1 + \frac{r}{100})^n}$$

where, A =Amount, P = Principal, r = Rate per day, n = Number of days.

For the principal P = Rs.1000 and n = 1600 days, the amount is doubled.

That is, $2000 = 1000\,(1 + \frac{r}{100})^{1600}$ ------- (1)

For the principal P = Rs.1000, n = 400 days, let x be the amount.

Then, $x = 1000(1 + \frac{r}{100})^{400}$

Taking 4th power on both LHS and RHS to make the equation identical with equation (1).

That is, $x^4 = 1000^4\,(1 + \frac{r}{100})^{1600}$ --------- (2)

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Dividing equations (2) by equation (1).

$$\frac{x^4}{2000} = \frac{1000^4}{1000}$$

$\Rightarrow x^4 = 2 \times 10^{12}$.

Taking 4th root,

$\Rightarrow x = 2^{\frac{1}{4}} \times 10^3$

$\Rightarrow x = 1.1892 \times 10^3$

$\Rightarrow x = 1189.2$

Thus Amount = Rs.1189.2

Interest = Amount – Principal = 1189.2 – 1000 = 189.2 ≈ **190.**

2. A person sells two objects at Rs.1035/- each. On the first object, he suffers a loss of 10% while on the second he gains 15%. What is his net loss/gain percentage?

(a) 5% gain (b) < 1% gain (c) < 1% loss (d) no loss, no gain

Answer (b) **(CSIR-NET DEC 2014)**

Difficulty rating: 2.5/5 *Average time for solving: 100 seconds*

Let us analyse each statement.

Given 1: "A person sells two objects at Rs.1035/- each."

That is, Selling price of the first object = Rs.1035.

Selling price of the second object = Rs.1035.

Let Cost price of the first and second object be x and y respectively.

Remember: Gain% = $(\frac{SP-CP}{CP})\times 100$

Loss% = $(\frac{CP-SP}{CP})\times 100$

Given 2: "On the first object, he suffers a loss of 10%."

Thus, $10 = (\frac{x-1035}{x})\times 100$

$\Rightarrow 10x = 100x - 103500$

$\Rightarrow 90x = 103500$

$\Rightarrow x = 1150.$

Given 3: "On the second object, he gains 15%."

Thus, $15 = (\frac{1035-y}{y})\times 100$

$\Rightarrow 15y = 103500 - 100y$

$\Rightarrow 115y = 103500$

$\Rightarrow y = 900.$

Total CP = x + y = 1150 + 900 = Rs.2050

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Total SP = 1035 + 1035 = Rs.2070

Since, SP > CP, the seller attains a gain.

$$\text{Net Gain \%} = \left(\frac{2070-2050}{2050}\right)\times 100 = \frac{2000}{2050} \approx 0.98 < \mathbf{1.}$$

3. You get 20% returns on your investment annually, but also pay a 20% tax on the gain. At the end of 5 years, the net gain made by you (as percentage of the capital) is approximately

(a) 0 **(b) 16** **(c) 80** **(d) 100**

Answer (c) **(CSIR-NET JUNE 2014)**

Difficulty rating: 3/5 *Average time for solving: 80 seconds*

Let Rs.100 be the invested money.

Given 1: "20% returns on your investment annually."

Thus, Gain in one year = Rs.20.

Given 2: "pay a 20% tax on the gain."

Thus, Net gain in one year after paying tax $= 20 \times \frac{100-20}{100} = 20 \times \frac{80}{100} =$ Rs.16

Net gain at the end of 5 years = ?

Remember: $I = \frac{PRT}{100}$

Here, P = 100, R = 16, T = 5.

Thus, $I = \frac{100\times 16\times 5}{100} = 80.$

Thus, net gain in 5 years after paying tax = **Rs.80.**

Note that even though the question does not clearly mention whether to use simple interest or compound interest, we should assume it simple interest because if we proceed using compound interest formula the value we obtain will be more than 100, which is not there in any of the given options.

4. After giving 20% discount on the marked price to a customer, the seller's profit was 20%. Which of the following is true?

(a) Sale price $= \frac{(\text{Marked price + Cost price})}{2}$

(b) Sale price $< \frac{(\text{Marked price + Cost price})}{2}$

(c) $\frac{2(\text{Marked price + Cost price})}{3} >$ **Sales price** $> \frac{(\text{Marked price + Cost price})}{2}$

(d) Sale price $> \frac{2(\text{Marked price + Cost price})}{3}$

Answer (b) **(CSIR-NET JUNE 2014)**

Difficulty rating : 3.5/5 *Average time for solving: 120 seconds*

Let Sale Price, **SP = Rs.100.**

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Given 1: This Sale Price is the price after giving 20% discount on the Marked Price (MP).

Thus SP = MP - $\frac{20}{100}$ MP

⇨ $100 = \frac{100MP - 20\,MP}{100}$

⇨ 10000 = 80MP

⇨ **MP = Rs.125**

Given 2: The seller's profit was 20%.

Remember: Profit = SP – CP

⇨ 20 = 100 – CP

⇨ **CP = Rs.80**

Thus 80,100,125 is a set of prices (CP, SP and MP respectively) which satisfies all the conditions in the question.

Therefore, now it is sufficient to check the options on these values.

Clearly, $100 < \frac{125+80}{2}$

⇨ **Sale Price** $< \frac{\text{(Marked price + Cost price)}}{2}$ holds.

5. A merchant buys equal numbers of shirts and trousers and pays Rs.38000. If the cost of 3 shirts is Rs.800 and that of a trouser is Rs.1000, then how many shirts were bought?

(a) 60 (b) 30 (c) 15 (d) 10

Answer (b) **(CSIR-NET JUNE 2014)**

Difficulty rating: 2.5/5 *Average time for solving: 100 seconds*

Since, the merchant buys equal number of shirts and trousers,

Let x be the number of shirt and x itself be the number of trousers.

Cost of 3 shirts = Rs.800

Cost of 1 shirt = $\frac{800}{3}$

Cost of 1 trouser = 1000.

Given: Total Cost Price = Rs.38000.

Thus, $\frac{800}{3}$ x + 1000 x = 38000

⇨ $\frac{800x + 3000x}{3} = 38000$

⇨ $3800x = 3 \times 38000$

⇨ **x = 30.**

Thus, the merchant bought 30 shirts.

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6. A fruit vendor buys 120 Shimla apples at 4 for Rs.100, and 120 Golden apples at 6 for Rs.100. She decides to mix them and sell at 10 for Rs.200. She will make

(a) no profit, no loss (b) a loss of 4% (c) a gain of 4% (d) a loss of 10%

(CSIR-NET DEC 2013)

Answer (b)

Difficulty rating: 2.5/5

Average time for solving: 100 seconds

Given: Cost price of 4 Shimla Apples = Rs.100.

Thus cost Price of 1 Shimla Apple $= \frac{100}{4} =$ Rs.25.

Then cost price of 120 Shimla Apples = 120 × 25 = Rs.3000.

Given: Cost price of 6 Golden Apples = Rs.100.

Thus, Cost Price of 1 Shimla Apple $= \frac{100}{6}$

Cost price of 120 Golden Apples $= 120 \times \frac{100}{6} =$ Rs.2000.

Total Cost price (CP) = **Rs.5000**

Given: Selling price of 10 Apples = Rs.200

Thus, Selling price of 1 Apple $= \frac{200}{10} =$ Rs. 20

Selling price of 240 Apples (SP) = 240 × 20 = **Rs.4800**

Remember: Loss % $= \frac{CP-SP}{CP} \times 100$

$= \frac{5000-4800}{5000} \times 100 = \frac{200}{5000} \times 100 = 4\%.$

Thus, the fruit vendor suffers a loss of 4%.

7. A student buys a book from an online shop at 20% discount. His friend buys another copy of the same book in a book fair for Rs.192 paying 20% less than his friend. What is the full price of the book?

(a) Rs.275 (b) Rs.300 (c) Rs.320 (d) Rs.392

(CSIR-NET DEC 2013)

Answer (b)

Difficulty rating: 3.5/5

Average time for solving: 100 seconds

Normal Solution

Let x be the full price of the book.

Given 1: The student got 20% discount at online shop.

Thus, Price at online shop $= x - \frac{20}{100}x = \frac{80}{100}x = \frac{4}{5}x$

Given 2: Rs.192 is 20% less than the online price.

Thus, Price at book fair = Online price $- \frac{20}{100} \times$ (Online Price)

$\Rightarrow 192 = \frac{4}{5}x - (\frac{20}{100} \times \frac{4}{5}x)$

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$\Rightarrow 192 = \frac{4}{5}x(1 - \frac{20}{100})$

$\Rightarrow 192 = \frac{4}{5} x \left(\frac{100 - 20}{100}\right)$

$\Rightarrow 192 = \frac{4}{5} x \times \frac{80}{100}$

$\Rightarrow 192 = \frac{16}{25} x$

$\Rightarrow x = \frac{192 \times 25}{16} = \text{Rs.}300.$

Thus, the full price of the book is **Rs.300.**

Alternate Logical Solution

In monetary problems, even if we have the formula for profit or loss percentage, try to do the problem logically, which saves whole lot of time. Let us discuss the same for the solution of this problem.

Let x be the full price of the book.

Given 1: The student got 20% discount at online shop.

The student got 20% discount online ≡ Student bought book at 80% of the full price = $\frac{80}{100}x$.

Thus, **$\frac{80}{100}x$ is the online price.**

Given 2: Rs.192 is 20% less than the online price.

That is, 192 = 20% less than the online price.

$\Rightarrow$ 192 = (100 - 20)% of the online price.

$\Rightarrow$ 192 = 80% of the online price.

$\Rightarrow 192 = \frac{80}{100} \times (\frac{80}{100}x)$

$\Rightarrow 192 = \frac{64}{100}x$

$\Rightarrow x = 300$

Thus, the full price of the book is **Rs.300.**

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10.3 PRACTICE QUESTIONS

1. Anjli saved Rs.250 in buying a Sari on sale. If she spent Rs.2500 for the Sari, she saved about:

(a) 8% (b) 9% (c) 10% (d) 11%

2. If 18 apples and 30 oranges cost as much as 30 apples and 18 oranges which of the following conclusions is correct?

(a) Orange and apple have identical prices (b) Orange's price is double that of apple
(c) No conclusion can be drawn (d) Apple is cheaper than orange

3. Five times A's money added to B's money is more than Rs.51. Three times A's money minus B's money is Rs.21. If a represents A's money in dollars and b represents B's money in rupees, then:

(a) $a > 9, b > 6$ (b) $a > 9, b < 6$ (c) $a > 9, b = 6$ (d) $2a = 3b$

4. If the price of a television set is increased by 20%, then by what approximate percentage should the new price be reduced to bring the price back to the original level?

(a) 17 % (b) 24 % (c) 30 % (d) 35%

5. Malu owns a home worth Rs.10,000. She sells it to Mary at a 10% profit based on the worth of the house. Mary sells the house back to Malu at a 10% loss. Then:

(a) Malu comes out even. (b) Malu makes Rs. 1100 on the deal.
(c) Malu makes Rs. 900 on the deal. (d) Malu loses Rs.900 on the deal.

6. The cost price of 30 articles is the same as the selling price of x articles. If the profit is 20%, then the value of x is:

(a) 15 (b) 16 (c) 18 (d) 25

7. The price of LED television is marked at Rs.20,000. If successive discounts of 20%, 10% and 5% be allowed, then at what price does a customer buy it?

(a) Rs.15000 (b) Rs.13680 (c) Rs.16000 (d)Rs.8945

8. How much time (in years) will it take for an amount of Rs. 760 to yield Rs. 114 as interest at 5% per annum of simple interest?

(a) 3.5 (b) 3 (c) 4.5 (d) 5

9. A sum invested at 10% simple interest per annum grows to Rs.900 in 5 years. The same amount at 12% simple interest per annum in 8 years will grow to:

(a) Rs.900 (b) Rs.950 (c) Rs.1176 (d)Rs.1256

10. Compound interest for a certain sum for 2 years at 10% per annum is Rs. 630. The simple interest (in Rs.) on the same sum for thrice the time at half the rate percent per annum is :

(a) 400 (b) 900 (c) 600 (d) 800

Answer Key

1. (b) 2. (a) 3. (a) 4. (a) 5. (b) 6. (d) 7. (b) 8. (b) 9. (c) 10. (b)

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CHAPTER 11

GEOMETRY

Average Difficulty Rating: 3.5/5

Average time for each question: 102 sec

Number of questions expected for each exam: 1-3

11.1 THEORY

Geometry is the branch of mathematics that deals with the properties, measurement and relations of points, lines, angles, surfaces and solids.

Geometry is such a vast topic and therefore it is almost an impossible endeavour to gather or compile all the concepts pertaining to this in a single book. Therefore, we will be discussing only the relevant concepts with respect to the type of questions that are posed in the CSIR-NET examination. With a proper understanding of these concepts, you will be able to attempt any geometrical questions in the CSIR-NET examination.

Geometry can be divided into plane geometry and solid geometry.

11.1.1 Plane Geometry

A plane is any two-dimensional surface such as circles and squares. Plane geometry is all about figures in a plane.

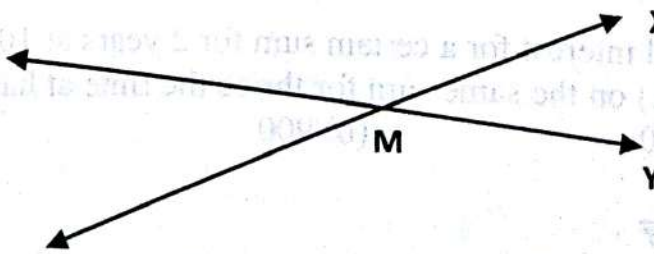
Let us discuss some important aspects in plane geometry.

1. Line

Types of lines

- **Intersecting lines**

Two or more lines that meet at a point are called intersecting lines. The common point is called the point of intersection.



Example: Here the lines X and Y intersect at the point M.

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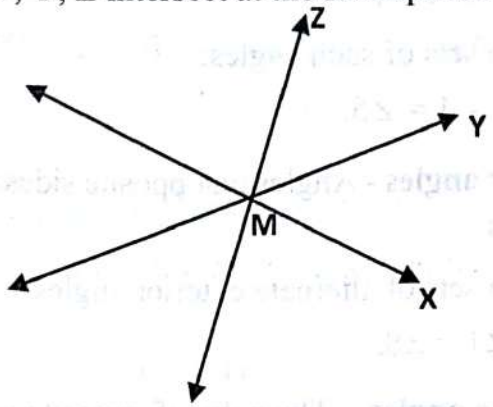
▪ Concurrent Lines

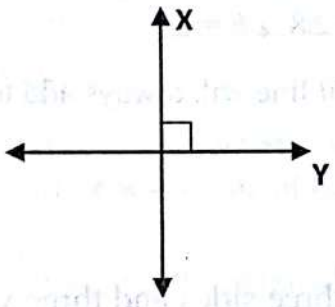
Three or more lines in a plane are concurrent if they intersect at a single point.

Example: Here the three lines X, Y, Z intersect at the same point M.



▪ Perpendicular Lines

A line is said to be perpendicular to another line if the two lines intersect at a right angle (90° angle).



Example: Here the lines X and Y are perpendicular lines.

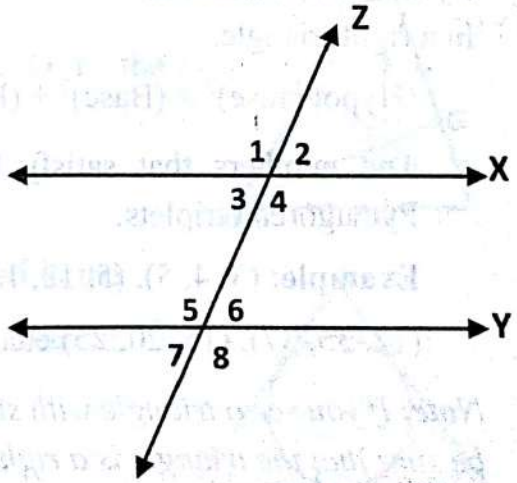
▪ Parallel Lines

Two lines in a plane that do not intersect or touch at any point are called parallel lines.

Parallel lines and the transversal

A transversal is a line that intersects two or more other lines. When it intersects parallel lines, the following angles are congruent (equal).

In the figure, X and Y are parallel lines and Z is the transversal line.



Corresponding angles - Angles that are in the same position on each line.

In the figure, there are four such sets of angles:

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That is, $\angle 1 = \angle 5$, $\angle 3 = \angle 7$, $\angle 2 = \angle 6$, $\angle 4 = \angle 8$.

Alternate interior angles - Angles on the opposite sides of the transversal and on the interior of the parallel lines.

Here, there are two sets of such angles:

That is, $\angle 3 = \angle 6$, $\angle 4 = \angle 5$.

Alternate exterior angles - Angles on opposite sides of the transversal and on the exterior of the parallel lines.

Here, there are two sets of alternate exterior angles:

That is, $\angle 7 = \angle 2$, $\angle 1 = \angle 8$.

Vertically opposite angles – The pairs of opposite angles formed between each line and the transversal.

Here, there are four such sets of angles:

That is, $\angle 1 = \angle 4$, $\angle 2 = \angle 3$, $\angle 5 = \angle 8$, $\angle 6 = \angle 7$.

Note: Angles on one side of a straight line will always add to 180 degrees [Linear Pair].

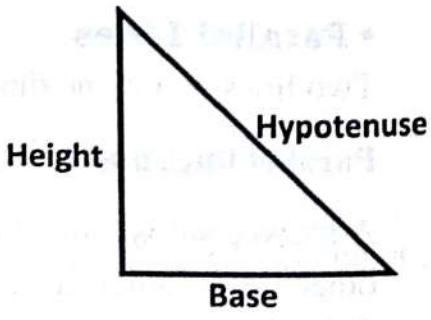
2. Triangle

A triangle is a closed figure with three sides and three vertices.

Types of triangles

- **Right triangles**

A triangle with a right angle (That is, 90°) is called a right triangle.



Pythagoras Theorem

In a right triangle,

$(\text{Hypotenuse})^2 = (\text{Base})^2 + (\text{Height})^2$

The numbers that satisfy this relation are called Pythagorean triplets.

Example: (3, 4, 5), (5, 12, 13), (7, 24, 25), (8, 15, 17), (9, 40, 41), (11, 60, 61), (12, 35, 37), (15, 20, 25) etc.

Note: *If you see a triangle with sides exactly as any of these Pythagorean triplets, you can be sure that the triangle is a right triangle.*

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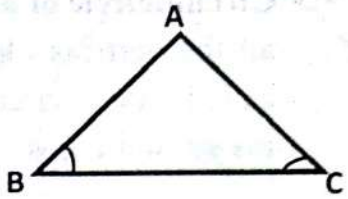
- **Isosceles triangles**

Triangles with two equal sides are called isosceles triangles.

The angles opposite to the equal sides are also equal.

Here sides AB = AC. Also ∠ABC = ∠ACB.

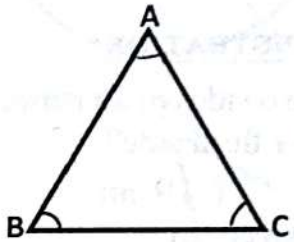


- **Equilateral Triangles**

Triangles with three equal sides are called equilateral triangles.

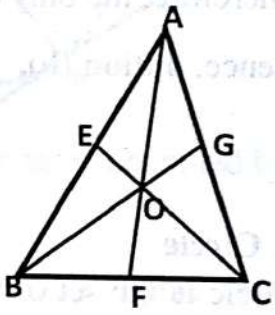
For equilateral triangles, each of the three angles are also equal to 60°.

Here, AB = BC = AC, and ∠BAC = ∠ABC = ∠BCA = 60°.



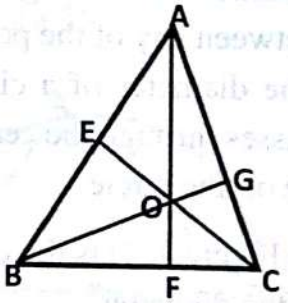
Important facts about triangles

- Sum of the interior angles of a triangle is 180° (Angle Sum Property)
- The sum of any two sides of a triangle is greater than the third side.
- The diagonal of a square divides the square into two isosceles right triangles.
- **Median** – A median of a triangle is a line segment from one vertex to the mid-point on the opposite side of the triangle.

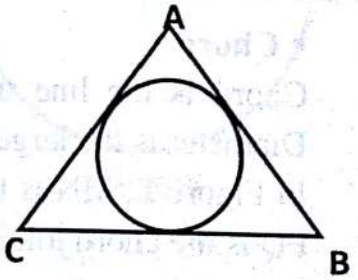


- **Centroid** - A centroid of a triangle is the point where the three medians of the triangle meet.
  Here AF, BG, CE are the medians and O is the centroid of the triangle.
- **Orthocenter** - In a triangle, the orthocenter is the point of intersection of the altitudes.

  Here, AF, BG and CE are the altitudes and O is the orthocenter.
  Note: Orthocenter can also be outside the triangle.



- **Incircle** - The incircle is the largest circle contained in the triangle; it touches (is tangent to) the three sides. The center of the incircle is called the triangle's incenter.



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- **Circumcircle of a triangle** is a circle, which passes through all the vertices of the triangle. The center of this circle is called the circumcenter and its radius is called the circumradius.

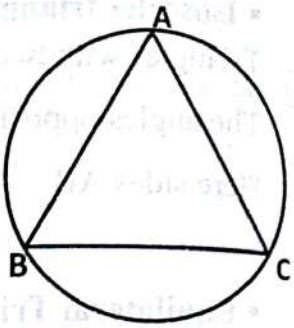

ILLUSTRATION:

If two sides of an isosceles triangle are 9 cm and 25 cm respectively. What would be the length of its third side?

(a) 9 cm (b) 25 cm (c) 34 cm (d) $\sqrt{25^2 - 9^2}$ cm

Answer (b)

For an isosceles triangle, length of two of its sides should be equal.

Hence, the length of the third side should be 9 cm or 25 cm.

However, we know that the sum of any two sides of a triangle is greater than the third side.

If the length of the third side is 9 cm, then the sum of 9 and 9 is not greater than 25.

Thus, the length of the third side cannot be 9 cm.

Therefore, the only option remaining is 25 cm.

Hence, **option (b).**

3. Circle

Circle is the set of all points in a plane that are at a given distance from a given point, the centre. The distance between any of the points and the centre is called the radius. The diameter of a circle is any straight line segment that passes through the center of the circle and whose endpoints lie on the circle.

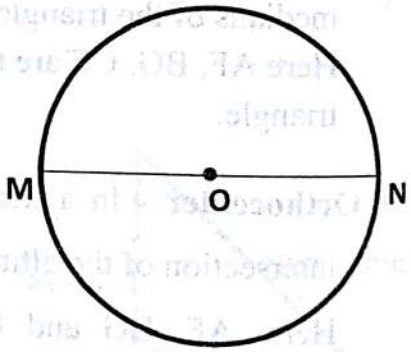

Figure 1

In Figure 1, O is the center, ON or OM the radius, and MN is the diameter.

- **Chord**

Chord is the line that joins two points on a circle. Diameter is the largest chord.

In Figure 2, MN is the chord joining points M and N. PQ is the chord joining points P and Q.

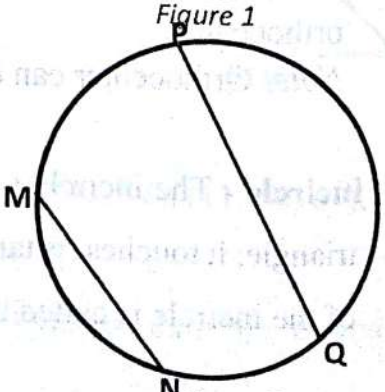

Figure 2

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▪ **Arc**

An arc of a circle is two points on the circle and the continuous part of the circle between the two points.

If the arc measures x°, the arc length is $\frac{x}{360^o}(2\pi r)$.

In Figure 3, the continuous part between points P and Q is the arc.

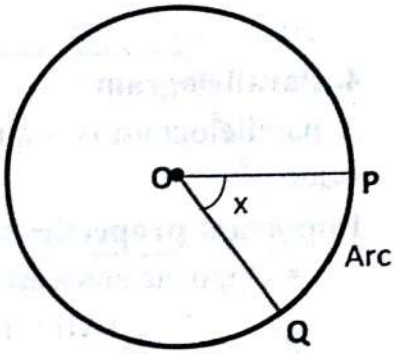

Figure 3

▪ **Tangent**

A tangent to a circle is a line that touches the circle at exactly one point, never entering the circle's interior.

In Figure 4, TP and TQ are the tangents at the points P and Q respectively.

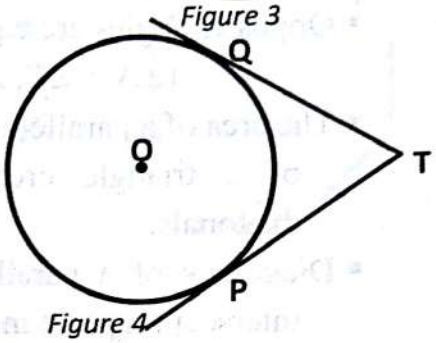

Figure 4

Important circle theorems

1. A tangent and radius meet at right angles.
   (In Figure 5, OP is perpendicular to PT.)

2. Two tangents from a common exterior point to a circle are equal in length. (In Figure 5, TP = TQ.)

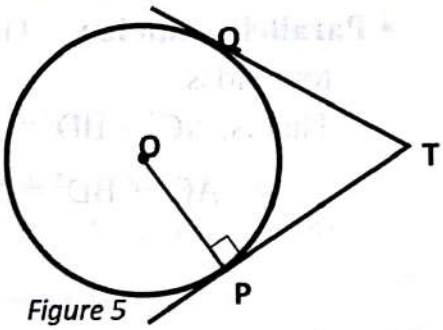

Figure 5

3. The angle subtended by an arc at the center of a circle is twice the angle subtended by the same arc at the circumference of the circle. (See figure 6)

4. Angles subtended by an arc in the same segment of a circle are equal. (See figure 7)

5. The angle subtended in a semicircle is a right angle (90°). (See figure 8)

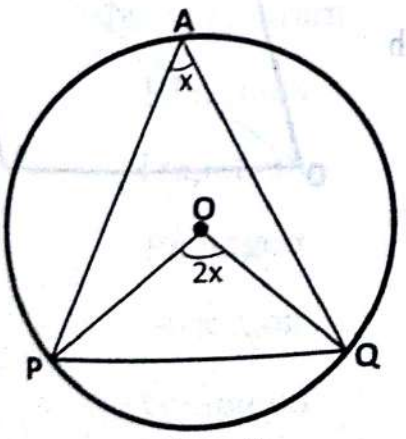

Figure 6

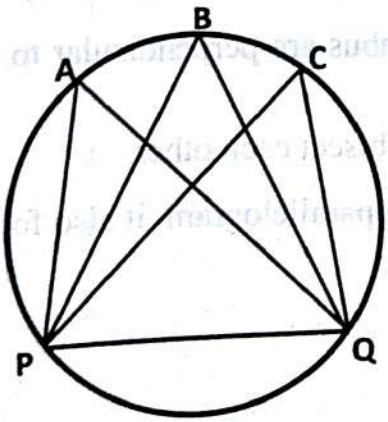

Figure 7

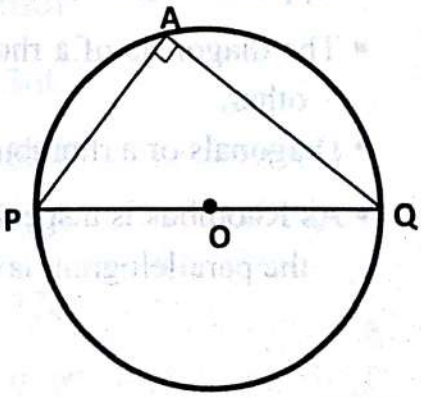

Figure 8

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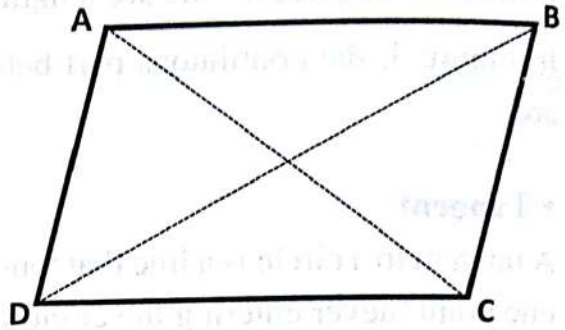
4. Parallelogram

A parallelogram is a quadrilateral (closed figure with four sides) with two pairs of parallel sides.



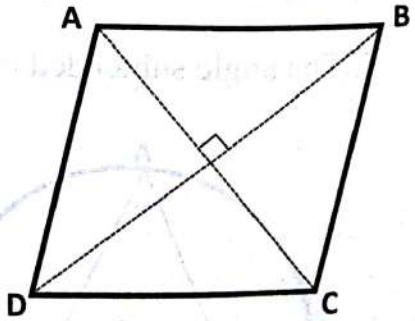
Important properties of Parallelogram

- Opposite sides are parallel and equal.
  $(AB = DC, AD = BC)$
- Opposite angles are equal.
  $(\angle A = \angle C, \angle B = \angle D)$
- The area of a parallelogram is twice the area of a triangle created by one of its diagonals.
- Diagonals of a parallelogram bisect each other. That is, they are divided in their intersection point into two.
- **Parallelogram law** – The sum of squares of diagonals is equal to the sum of squares of four sides.
  That is, $AC^2 + BD^2 = AB^2 + BC^2 + DC^2 + AD^2$
  $\Rightarrow$ $AC^2 + BD^2 = 2(AB^2 + BC^2)$

5. Rhombus

A parallelogram with all sides equal is called a rhombus.

Important properties of Rhombus



- Opposite sides are parallel and all sides are equal.
  $(AB = DC = AD = BC)$
- Opposite angles are equal. $(\angle A = \angle C, \angle B = \angle D)$
- The diagonals of a rhombus are perpendicular to each other.
- Diagonals of a rhombus bisect each other.
- As Rhombus is a special parallelogram, it also follows the parallelogram law.

6. Other important polygons

A closed plane figure with three or more sides is called a polygon.

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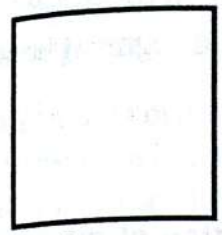
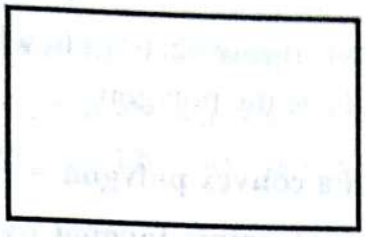
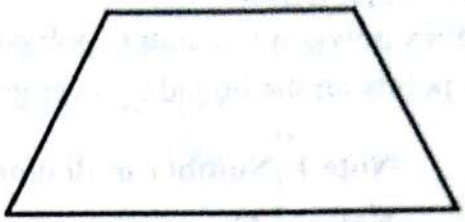
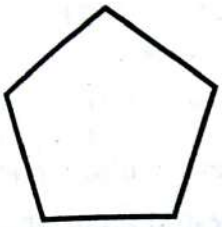
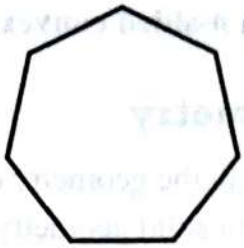

Square


Rectangle


Trapezium


Pentagon

Hexagon


Heptagon

Octagon

Sum of interior angles of a polygon

Sum of interior angles of a polygon can be evaluated using the formula:

Sum of interior angles = (n-2)180°

where n is the number of sides of the polygon.

Example: Sum of interior angle of any quadrilateral (square, rectangle etc.)

$= (4 - 2)180° = 360°.$

Thus, sum of interior angles of important polygons can be tabulated as follows.

Polygon	Number of sides	Sum of interior angles
Triangle	3	180°
Square	4	360°
Rectangle	4	360°
Parallelogram	4	360°
Rhombus	4	360°
Trapezium	4	360°
Pentagon	5	540°
Hexagon	6	720°
Heptagon	7	900°
Octagon	8	1080°

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Convex polygon

Convex polygon is a simple polygon (not self-intersecting) in which no line segment between two points on the boundary ever goes outside the polygon.

Note 1. Number of diagonals of a convex polygon $= \frac{n(n-3)}{2}$

Note 2. The maximum number of points formed by intersection of all pairs of diagonals of an n-sided convex polygon $= {}^{n}C_{4}$.

11.1.2 Solid Geometry

Solid Geometry deals with the geometry of three-dimensional figures. Sphere, Cube, Cone etc. are examples of figures in solid geometry. All the figures in solid geometry and their formulas are discussed in the next chapter.

11.2 CSIR-NET Exam Questions – Detailed Solution

1. What is the maximum number of whole laddoos having diameter of 6 cm that can be packed in a box whose inner dimensions are 24 × 18 × 17 cm^3?

(a) 24 **(b) 30** **(c) 33** **(d) 36**

Answer (c) **(CSIR-NET DEC 2014)**

Difficulty rating: 4/5 *Average time for solving: 120 seconds*

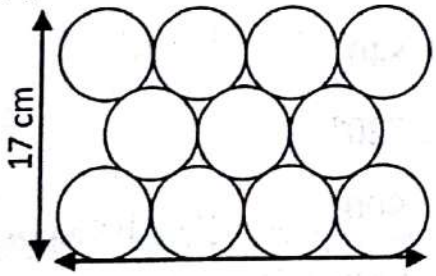
Note: Hexagonal Closed Packing

The most efficient packing for spheres is hexagonal closed packing. In hexagonal closed packing, the layers of spheres are packed so that spheres in alternating layers overlie one another.

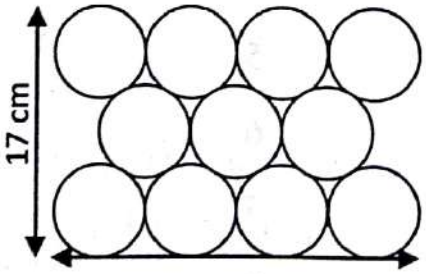
The below explanation demands your imagination for proper understanding, as the three dimensional figures like spheres and cuboids cannot be easily represented in a two dimensional paper.

Consider the 24 × 17 plane as the base layer.

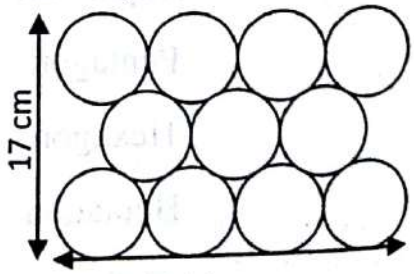
In the base layer, by hexagonal closed packing, arrange 11 laddoos as shown in the figure. (Even though the breadth is 17 centimetres, if the spheres in the middle row are placed in the gaps of the other two rows, as shown in the figure, we can form 3 rows.)



Layer 1 (Base layer)



Layer 2



Layer 3

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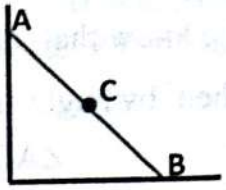
Now the height is 18 cm, so we can accommodate 3 such rows each of 11 laddoos. (See Figure)

Thus, 11 + 11 + 11 = **33 laddoos** can be arranged in the box this way.

2. A ladder rests against a wall as shown. The top and bottom ends of the ladder are marked A and B. The base B slips. The central point C of the ladder falls along



(a) a parabola
(b) the arc of a circle
(c) a straight line
(d) a hyperbola

Answer (b) **(CSIR-NET DEC 2014)**

Difficulty rating: 4/5 *Average time for solving: 100 seconds*

The movement of the ladder is pictured in Figure 1 where the darks dots show the movement of point C.

Thus, clearly the central point C falls along the **arc of a circle** (Figure 2).

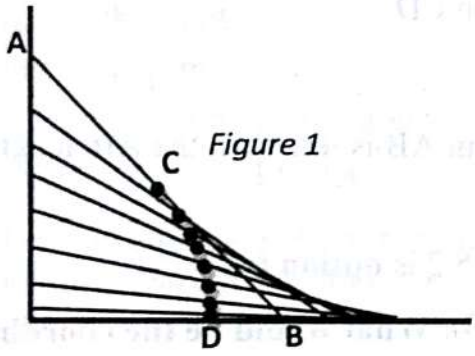


Figure 1

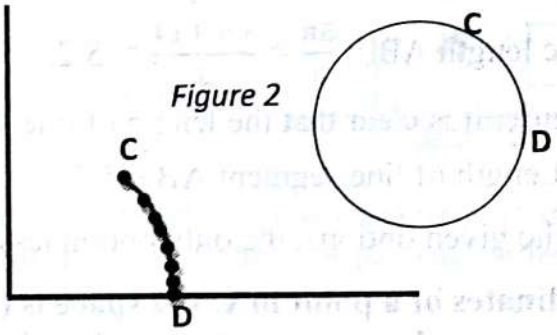
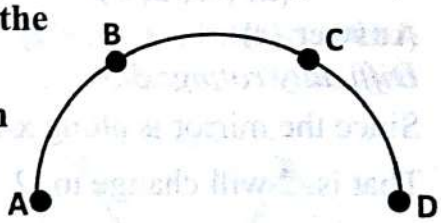


Figure 2

3. On a semi-circle of diameter 10 m drawn on a horizontal ground are standing 4 boys A,B, C and D with distances AB = BC = CD. The length of the line segment joining A and B is



(a) 5 m **(b) 6 m** **(c) 7 m** **(d)** $\frac{5\pi}{3}$ **m**

Answer (a) **(CSIR-NET JUNE 2014)**

Difficulty rating: 3.5/5 *Average time for solving: 90 seconds*

Normal Solution:

Note: The question is not to find the arc length AB but to find the length of the line segment AB.

Let O be the center of the semicircle.

Join O with the four points A, B, C, D to get OA, OB, OC, OD respectively (See Figure).

By Angle sum property, $\angle AOB + \angle BOC + \angle COD = 180^{\circ}$.

Given: "4 boys A, B, C and D standing with distances AB = BC = CD."

Here, the distance means the arc length, which is the length of the curved region between A and B or B and C or C and D.

Thus, arc length AB = arc length BC = arc length CD.

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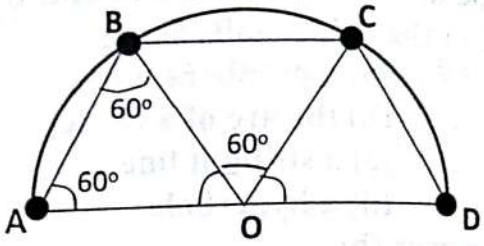
Since, OA = OB, ΔAOB is an isosceles triangle.

Remember: For an isosceles triangle, opposite angles of the equal sides are equal.

That is, ∠ABO = ∠BAO.

We know that, ∠AOB = 60°.



Then, by angle sum property,

$$\angle ABO + \angle BAO = 120^\circ.$$

$$\Rightarrow \quad \angle ABO = \angle BAO = \angle AOB = 60^\circ.$$

That is, ΔAOB is an equilateral triangle.

Thus, length of the line segment, AB = OA = **5 m.**

Quick Solution:

Total perimeter of the semicircle = $\pi \times 5 = 5\pi$.

Given: Arc length AB = Arc length BC = Arc length CD.

Thus Arc length AB = $\frac{5\pi}{3} \approx \frac{5 \times 3.14}{3} \approx 5.2$.

From figure, it is clear that the length of line segment AB is less than the Arc length AB.

That is, Length of line segment AB < 5.2.

Now in the given options, the only option less than 5.2 is **option (a).**

4. Coordinates of a point in x, y, z space is (1, 2, 3). What would be the coordinates of its reflection in a mirror along the x-z plane?

(a) (-1,-2,-3) **(b) (1,-2,-3)** **(c) (1,-2,3)** **(d) (-1,2,3)**

Answer (c) **(CSIR-NET JUNE 2014)**

Difficulty rating: 3/5 *Average time for solving: 50 seconds*

Since the mirror is along x-z plane, only the coordinate of y will change into its negative.

That is, 2 will change to -2.

Hence, **(1,-2, 3).**

5. What is the length of the longest rod that can be put in a hemispherical bowl of radius 10 cm such that no end of the rod is outside the bowl? (Assume that the rod has negligible thickness.)

(a) $10\sqrt{2}$ **(b) $10\sqrt{3}$** **(c) $10\sqrt{4}$** **(d) $10\sqrt{5}$**

Answer (c) **(CSIR-NET JUNE 2014)**

Difficulty rating: 3/5 *Average time for solving: 80 seconds*

Given: Radius of the hemispherical bowl = 10 cm.

The length of the largest rod, which can be put in the hemispherical bowl of negligible width, will be the diameter itself.

That is, Length of the longest rod = Length of the diameter = 20 cm = 10×2 cm = **$10\sqrt{4}$ cm.**

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6. A rectangular area of sides 9 and 6 units is to be covered by square tiles of sides 1, 2 and 5 units. The minimum number of tiles needed for this is

(a) 3 **(b) 11** **(c) 12** **(d) 15**

Answer (c) **(CSIR-NET JUNE 2014)**

Difficulty rating: 3/5 *Average time for solving: 90 seconds*

The area of the rectangular area to be covered = $9 \times 6 = 54$ units.

Here, note that **the square tiles cannot be broken.**

The squares tiles with sides 1, 2 and 5 units have surface area of 1, 4, and 25 units respectively.

Thus, all the tiles in the arrangement with minimum number of tiles should have one of these surface areas and the surface areas of tiles should sum up to 54 units.

The arrangement with the minimum number of tiles is as shown in the figure.

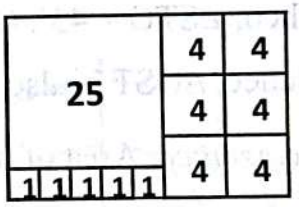
Here $25 + 4 + 4 + 4 + 4 + 4 + 4 + 1 + 1 + 1 + 1 + 1 = 54$ units.

That is, 1 tile with 5 units

6 tiles with 2 units

5 tiles with 1 unit.

is the arrangement with minimum number of tiles.



Hence, in total $1 + 6 + 5 =$ **12 tiles.**

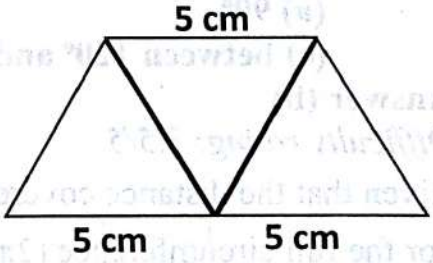
7. Three identical equilateral-triangular plates of side 5 cm each are placed together such that they form a trapezium. The length of the longer of the two parallel sides of this trapezium is

(a) $5\sqrt{\frac{3}{4}}$ **cm** **(b)** $5\sqrt{2}$ **cm** **(c) 10 cm** **(d)** $10\sqrt{3}$ **cm**

Answer(c) **(CSIR-NET JUNE 2014)**

Difficulty rating: 3/5 *Average time for solving: 70 seconds*

The given data can be pictured as shown in the figure.

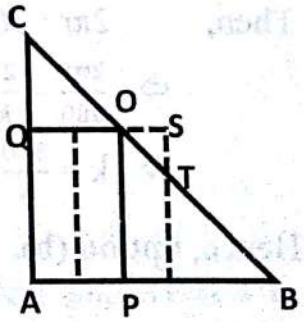


Clearly, two sides of the equilateral triangle with length of sides 5 cm each form the longer parallel side.

Thus the length of the longer of the two parallel sides

$= 5 + 5 =$ **10 cm.**

8. Consider a right-angled triangle ABC where AB = AC = 3. A rectangle APOQ is drawn inside it, as shown, such that the height of the rectangle is twice its width. The rectangle is moved horizontally by a distance 0.2 as shown schematically in the diagram (not to scale). What is the value of the ratio $\frac{\text{Area of } \Delta ABC}{\text{Area of } \Delta OST}$ **?**



(a) 625 **(b) 400** **(c) 225** **(d) 125**

Answer (c) **(CSIR-NET JUNE 2014)**

Difficulty rating: 3.5/5 *Average time for solving: 120 seconds*

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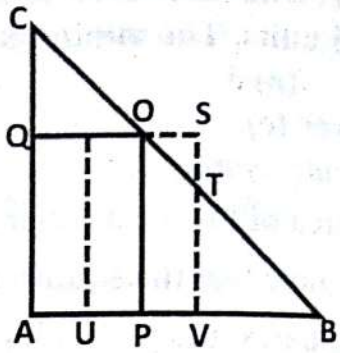
Mark the points U and V as shown in the figure.



Since AB = AC = 3, ΔABC is an isosceles triangle.

Remember: For an isosceles triangle, angles opposite to the equal sides are also equal.

Given ∠CAB = 90°.

Hence by angle sum property,

∠ACB = ∠ABC = 45°.

PV = OS = 0.2

Since QS and AB are parallel and BC is a transversal, ∠SOT = ∠ABC = 45°. (Alternate interior angles)

Also, ∠OST = 90°.

Then, ∠STO = 45°. (Angle sum property)

Hence, ΔOST is also an isosceles triangle with sides OS = ST = 0.2

Remember: Area of triangle = $\frac{1}{2}$ × Base × Height

Thus,

$$\text{Area of } \Delta ABC = \frac{1}{2} \times 3 \times 3 = 4.5$$

$$\text{Area of } \Delta OST = \frac{1}{2} \times 0.2 \times 0.2 = 0.02$$

$$\text{Hence, } \frac{\text{Area of } \Delta ABC}{\text{Area of } \Delta OST} = \frac{4.5}{0.02} = \frac{450}{2} = \mathbf{225.}$$

9. A cart wheel rolls along a straight line. If the distance covered is equal to the diameter of the wheel, what is the angle through which the wheel has turned?

(a) 90° **(b) between 90° and 120°**

(c) between 120° and 150° **(d) between 150° and 180°**

Answer (b) **(CSIR-NET DEC 2013)**

Difficulty rating: 3.5/5 *Average time for solving: 100 seconds*

Given that the distance covered is the diameter (d) of the wheel.

For the full circumference (2πr) to be covered, the angle to be covered is 360°.

Let k be the angle corresponding to the distance 2r.

Then, $2\pi r : 360^\circ = 2r : k$

$$\Rightarrow \frac{2\pi r}{360} = \frac{2r}{k}$$

$$\Rightarrow k = \frac{360}{\pi} \approx \frac{360}{3.14} \approx \mathbf{114.65^\circ.}$$

Hence, **option (b).**

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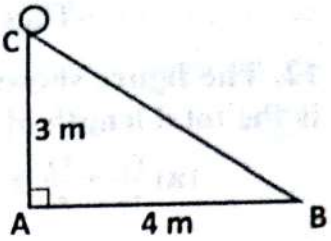
10. As shown in the diagram, a sphere is placed on the top of an incline. It rolls down the incline without slipping in exactly 50 turns. The radius of the sphere is



(a) $(\frac{5}{\pi})$ cm (b) $(\frac{5}{\pi})$ m (c) $(\frac{10}{\pi})$ cm (d) 10 cm

Answer (a) **(CSIR-NET DEC 2013)**

Difficulty rating: 3/5 *Average time for solving: 90 seconds*

Let r be the radius of the sphere.

When it rolls down one complete turn, it will cover the circumference of the circle ($2\pi r$).

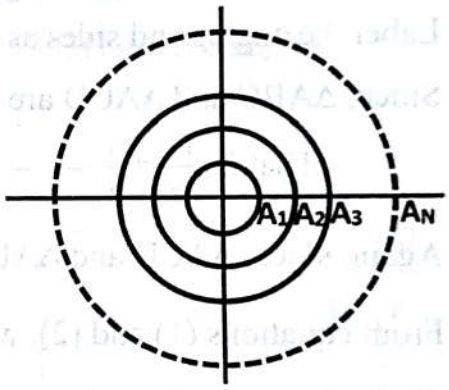
Thus for 50 complete turns, the distance covered, BC = $50 \times 2\pi r$.

By Pythagoras theorem, BC = 5 m.

Therefore, $5 = 50 \times 2\pi r$

$$\Rightarrow r = (\frac{5}{100\pi})\ m = \frac{5}{100\pi} \times 100\ m = (\frac{5}{\pi})\ cm.$$

11. A set of concentric circles of integer radii 1, 2,…,N is shown in the figure. An ant starts at point A_1, goes round the first circle, returns to A_1, moves to A_2, goes round the second circle, returns to A_2, moves to A_3 and repeats this until it reaches A_N . The distance covered by the ant is



(a) N(N + 1) **(b) 2Nπ+ N**

(c) π(N+1)N + N-1 **(d) π(N-1)N + N-1**

Answer (d) **(CSIR-NET DEC 2013)**

Difficulty rating: 3.5/5 *Average time for solving: 100 seconds*

Distance covered by the ant =

Distance covered along circular paths + Distance covered along straight line from A_1 to A_N

Distance covered along circular paths:

Remember: Sum of first n consecutive natural numbers = $\frac{n(n+1)}{2}$

Here the radius of circles are respectively 1, 2, 3,…..,N – 1

Thus, distance covered along circular paths = $2\pi1 + 2\pi2 + 2\pi3 + \ldots + 2\pi(N-1)$

$$= 2\pi(1 + 2 + 3 + \ldots + (N-1))$$

$$= 2\pi\frac{((N-1)(N))}{2}$$

$$= \pi N(N-1).$$

Distance between each of the successive A_i's are 1.

That is, A_1 to $A_2 = 1$, A_2 to $A_3 = 1$, … , A_{N-1} to $A_N = 1$.

Thus, distance covered along straight line from A_1 to $A_N = 1 + 1 + \ldots + 1$(N-1 times) = **N – 1.**

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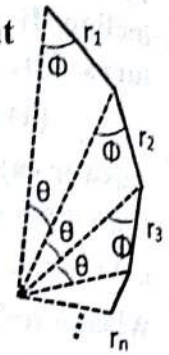
Thus distance covered by the ant = $\pi N(N-1) + N - 1$.

12. The figure shows an infinite series of triangles, in which $r_1 > r_2 > r_3$. What is the total length of the solid line segments in the figure?



(a) $\frac{r_1}{r_2} + \frac{r_2}{r_3} + \ldots$ (b) $\frac{{r_1}^2}{r_1 - r_2}$ (c) $\frac{{r_2}^2}{r_1 + r_2}$ (d) $\frac{r_1 - r_2}{{r_1}^2}$

Answer (b) **(CSIR-NET DEC 2013)**

Difficulty rating: 4/5 *Average time for solving: 130 seconds*

Prerequisite: *Knowledge of 10^{th} grade geometry is advisable for better understanding.*

Consider each triangle with sides $r_1, r_2, \ldots$

In each of these triangles one angle is θ and other angle is ϕ.

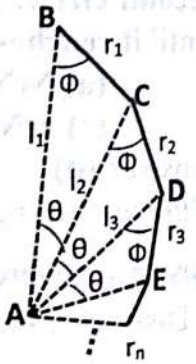
Then by angle sum property, the third angle should also be the same, say ψ for each triangle.

Since all the 3 corresponding angles are equal, these triangles are similar.

Label the angles and sides as shown in the figure.

Since, ΔABC and ΔACD are similar, their corresponding sides are proportional.

That is, $\frac{r_1}{r_2} = \frac{l_2}{l_3}$ ----------(1)



Again, since, ΔACD and ΔADE are similar, we have, $\frac{r_2}{r_3} = \frac{l_2}{l_3}$ ----------(2)

From equations (1) and (2), we have, $\frac{r_1}{r_2} = \frac{r_2}{r_3}$

Similarly comparing other triangles, we get,

$$\frac{r_1}{r_2} = \frac{r_2}{r_3} = \frac{r_3}{r_4} = \frac{r_4}{r_5} = \ldots.$$

$$\Rightarrow \frac{r_2}{r_1} = \frac{r_3}{r_2} = \frac{r_4}{r_3} = \frac{r_5}{r_4} = \ldots.$$

Thus, $r_1, r_2, r_3, \ldots$ are an **infinite geometric sequence** with first term r_1 and common ratio $\frac{r_2}{r_1}$

Remember: Sum of infinite geometric progression = $\frac{a}{1-r}$

Thus, sum of sides $r_1, r_2, \ldots = \frac{r_1}{1 - \frac{r_2}{r_1}} = \frac{r1^2}{r_1 - r_2}$

13. A circle of radius 7 units lying in the fourth quadrant touches the x-axis at (10,0). The center of the circle has coordinates

(a) (7,7) (b) (-10,7) (c) (10,-7) (d) (7,-7)

Answer (c) **(CSIR-NET DEC 2013)**

Difficulty rating: 3/5 *Average time for solving: 100 seconds*

The given question can be pictured as shown.

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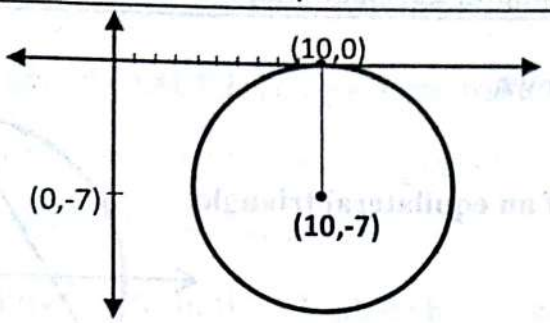
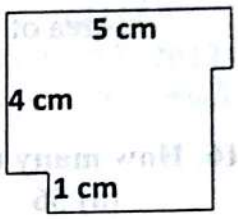


Clearly, the center of the circle is **(10,-7).**

14. What is the perimeter of the given figure, where adjacent sides are right angles to each other?



(a) 20 cm **(b) 18 cm**

(c) 21 cm **(d) Cannot be determined**

Answer (a) **(CSIR-NET DEC 2013)**

Difficulty rating: 3.5/5 *Average time for solving: 110 seconds*

Extend the given figure to get a square of edge 5 cm (See figure 1).

Since, adjacent sides are right angles to each other, ABCD and EFGH are rectangles.

Remember: Opposite sides of rectangles are equal.

Thus, AB = DC, HE = GF, EF = HG.

The perimeter of the given figure

= PA + AB + BC + CH + HE + EF + FQ + QP,

Given, PA = 4 cm, BC =1 cm, QP = 5 cm.

Also, DC + CH + HG = 5 cm,

⇨ AB + CH + EF = 5 cm.

Again, GF + FQ = 5 cm,

⇨ HE + FQ = 5 cm.

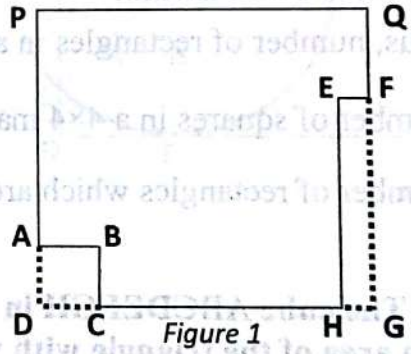


Figure 1

The perimeter of the given figure = PA + AB + BC + CH + HE + EF + FQ + QP

= PA + BC + QP + (AB + CH + EF) + (HE + FQ)

= 4 + 1 + 5 + 5 + 5

= **20.**

15. There is an equilateral triangle in the XY - plane with its center at the origin. The distance of its sides from the origin is 3.5 cm. The area of its circumcircle in cm^2 is

(a) 38.5 **(b) 49**

(c) 63.65 **(d) 154**

Answer (d) **(CSIR-NET JUNE 2013)**

Difficulty rating: 3.5/5 *Average time for solving: 100 seconds*

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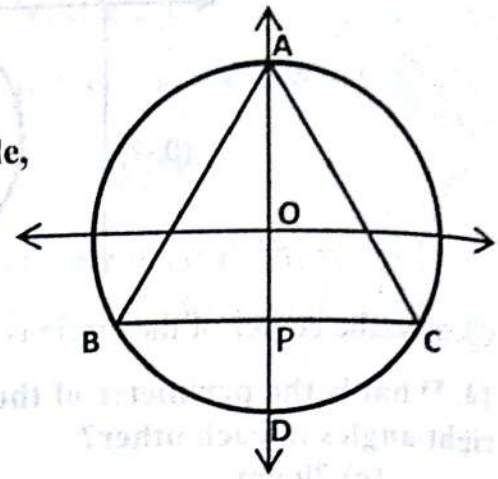
The given data can be pictured as shown.



Given OP = 3.5 cm.

Remember: **For the circumcircle of an equilateral triangle,**

$$\mathbf{OD = 2 \times OP.}$$

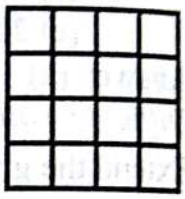
Thus, OD = 7 cm.

Here the radius of the circumcircle = OD = 7cm.

$$\text{Area of circumcircle} = \pi \times 7^2 = \frac{22}{7} \times 7^2$$

$$= \mathbf{154\ cm^2.}$$

16. How many rectangles (which are not squares) are there in the figure?



(a) 56 (b) 70 (c) 86 (d) 100

Answer (b) **(CSIR-NET JUNE 2013)**

Difficulty rating: 3/5 *Average time for solving: 100 seconds*

Remember: Number of squares in an n×n matrix $= 1^2 + 2^2 + \ldots + n^2 = \frac{n(n+1)(2n+1)}{6}$

Number of rectangles in an n×n matrix $= [\frac{n(n+1)}{2}]^2$

Here it is a 4×4 matrix.

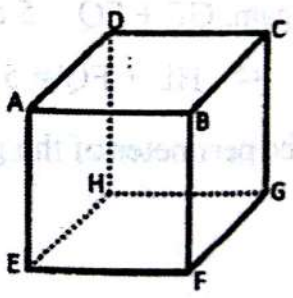
Thus, number of rectangles in a 4×4 matrix $= [\frac{4(4+1)}{2}]^2 = 10^2 = 100.$

Number of squares in a 4×4 matrix $= \frac{4(4+1)(8+1)}{6} = 30.$

Number of rectangles which are not squares = Number of rectangles – Number of squares

$$= 100 - 30 = \mathbf{70.}$$

17. The cube ABCDEFGH in the figure has each edge equal to a. The area of the triangle with vertices A, C and F is

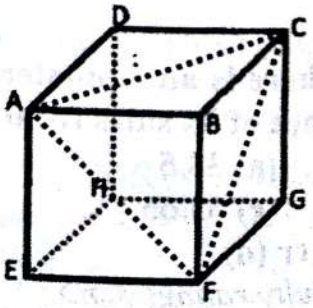


(a) $\frac{\sqrt{3}}{4}\mathbf{a^2}$ **(b)** $\frac{\sqrt{3}}{2}\mathbf{a^2}$ **(c)** $\sqrt{3}\,\mathbf{a^2}$ **(d)** $2\sqrt{3}\,\mathbf{a^2}$

Answer (b) **(CSIR-NET DEC 2012)**

Difficulty rating: 3.5/5 *Average time for solving: 100 seconds*

Join points A, C, F to form the required triangle.



Consider right-angled triangle ΔABF,

By Pythagoras theorem,

$$AF = \sqrt{AB^2 + BF^2} = \sqrt{a^2 + a^2} = \sqrt{2a^2} = \sqrt{2}a.$$

Similarly, from right-angled triangles ΔABC, ΔCBF, we get

$$AC = \sqrt{2}\,a,\ CF = \sqrt{2}\,a$$

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Thus, our required triangle ΔACF is an equilateral triangle with side $\sqrt{2}a$.

Remember: Area of an equilateral triangle $= \frac{\sqrt{3}}{4}(\text{side})^2$.

Thus area of $\Delta ACF = \frac{\sqrt{3}}{4}(\sqrt{2}a)^2 = \frac{\sqrt{3}}{2}a^2$.

18. A circle of radius 5 units in the XY plane has its center in the first quadrant touches the X-axis and has a chord of length 6 units on the Y-axis. The coordinates of its center are

(a) (4,6) **(b) (3,5)** **(c) (5,4)** **(d) (4,5)**

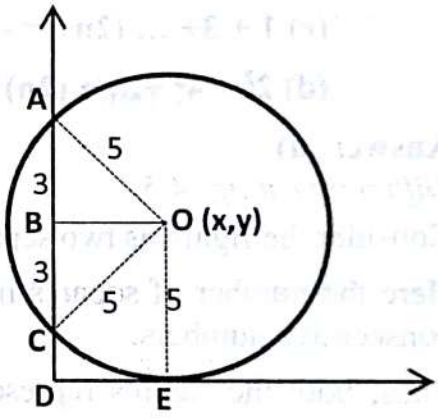
Answer (d) **(CSIR-NET DEC 2012)**

Difficulty rating: 3/5 *Average time for solving: 110 seconds*

The given question can be pictured as shown below.

We have to find the coordinates of the centre O.

That is, we have to find the values of x and y. (See figure)



From the figure, it is clear that

AC = 6 units.

Radius = OA = OC = OE = 5 units.

Construct OB perpendicular to AC.

Then AB = BC = 3 units.

Since OE = 5 units, we obtain that the y coordinate of the center O is 5. (y coordinate is the distance from the x axis.)

Therefore, it remains to find the x coordinate of the center O. (x coordinate is the distance from the y axis.)

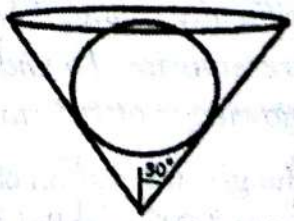
That is, we have to find the length of OB or DE.

In ΔABO, by Pythagoras theorem,

$$OB = \sqrt{5^2 - 3^2} = \sqrt{25 - 9} = \sqrt{16} = 4.$$

Thus the center (O) of the circle = (4,5).

19. A conical vessel with semi-vertical angle 30° and height 10.5 cm has a thin lid. The radius of the sphere in cm is



(a) 3.5 **(b) 5** **(c) 6.5** **(d) 7**

Answer (a) **(CSIR-NET DEC 2012)**

Difficulty rating: 3.5/5 *Average time for solving: 90 seconds*

Join AC and construct a perpendicular OD from the center O of the sphere. Clearly OD = OA = r (Radius of the sphere)

Given AC = 10.5 cm, $\angle OCD = 30^\circ$.

Consider ΔODC

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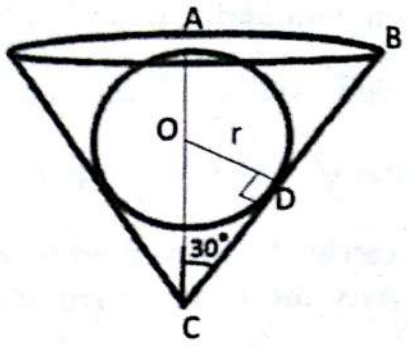
$\angle ODC = 90^\circ$, $\angle OCD = 30^\circ$.



$$\sin 30^\circ = \frac{\text{Opposite side}}{\text{Hypotenuse}} = \frac{r}{OC}$$

$$\Rightarrow \frac{1}{2} = \frac{r}{OC}$$

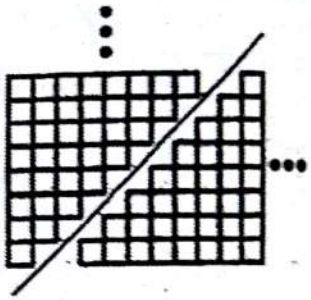
$$\Rightarrow OC = 2r.$$

$AC = AO + OC = r + 2r = 3r.$

Thus, $10.5 = 3r$.

$\Rightarrow$ **$r = 3.5$ cm.**

20. What does this diagram demonstrate?



(a) $1 + 2 + 3 + \ldots + n = \frac{n(n+1)}{2}$

(b) $1^2 + 2^2 + 3^2 + \ldots + n^2 = \frac{n(n+1)(2n+1)}{6}$

(c) $1 + 3 + \ldots (2n + = -1) = n^2$

(d) $2^2 + 4^2 + \ldots + (2n)^2 = \frac{2n(n+1)(2n+1)}{3}$

Answer (a) **(CSIR-NET DEC 2012)**

Difficulty rating: 4/5 *Average time for solving: 80 seconds*

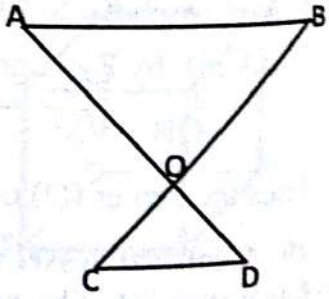
Consider the figure as two separate figures divided by a straight line.

Here the number of squares in both the figures can be corresponded to the sum of first 'n' consecutive numbers.

Thus, both the figures represents the formula for finding the sum of first 'n' consecutive numbers, which is given by

$$1 + 2 + 3 + \ldots + n = \frac{n(n+1)}{2}.$$

21. If AB is parallel to CD and AO = 2OD, then the area of the triangle OAB is bigger than the area of the triangle OCD by a factor of



(a) 2 **(b) 3** **(c) 4** **(d) 8**

Answer (c) **(CSIR-NET DEC 2012)**

Difficulty rating: 3.5/5 *Average time for solving: 90 seconds*

Prerequisite: *To understand this explanation clearly, a prerequisite knowledge of 10^{th} standard geometry is advisable.*

The given question can be pictured as shown.

Since AB is parallel to CD and BC is a transversal,

$\angle C = \angle B$, $\angle D = \angle A$ (Alternate interior angles)

$\angle COD = \angle AOB$ (Vertically Opposite angles)

Thus, ΔAOB and ΔCOD are similar triangles.

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By the property of similar triangles, $\frac{\text{Area of AOB}}{\text{Area of COD}} = \frac{(AO)^2}{(OD)^2}$

Given: AO = 2OD.

Therefore, $\frac{\text{Area of AOB}}{\text{Area of COD}} = \frac{(2OD)^2}{(OD)^2} = \frac{4(OD)^2}{(OD)^2} = \frac{4}{1} = 4.$

22. Four lines are drawn on a plane with no two parallel and no three concurrent. Lines are drawn joining the points of intersection of the previous four lines. The number of new lines obtained in this way is

(a) 3 **(b) 5** **(c) 12** **(d) 2**

Answer (a) **(CSIR-NET DEC 2012)**

Difficulty rating: 4/5 *Average time for solving: 140 seconds*

Remember: Three or more lines in a plane are concurrent if they intersect at a single point.

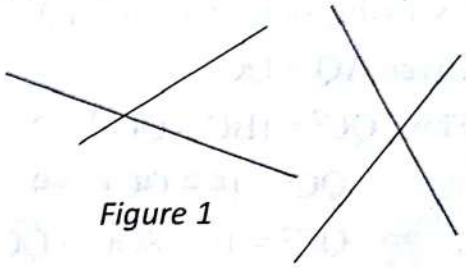


Figure 1

Here it is given that no two parallel and no three concurrent.

We can picture such a situation.

Let these lines be the required 4 lines (See Figure 1).

Extend them so that each of them intersect each other.

The 6 points of intersection is marked by circles (See Figure 2).

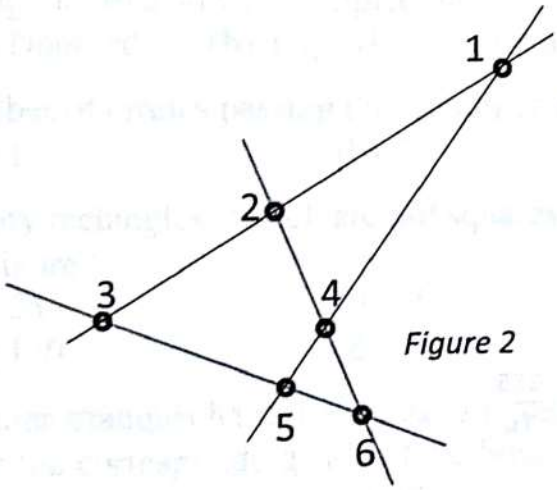


Figure 2

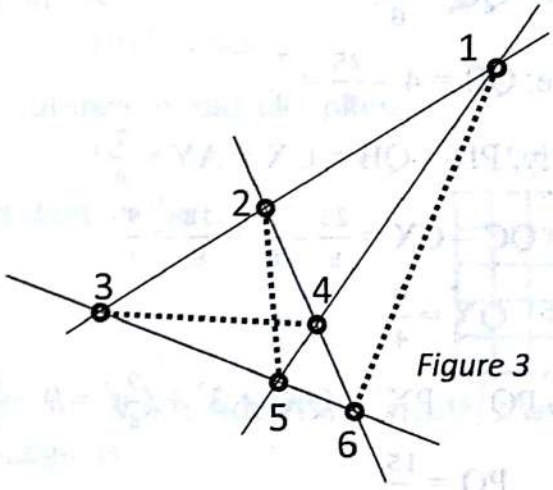


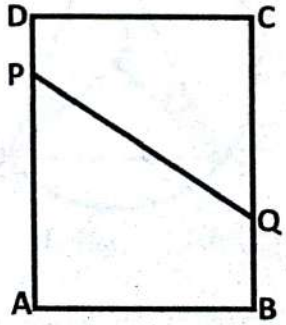
Figure 3

Clearly only three more lines can be joined between these 6 points.

That is, we can join the lines connecting 3 – 4, 2 – 5, 1 – 6. (See Figure 3.)

The dotted lines represent those three lines. Hence **option (a).**

23. A rectangular sheet ABCD is folded in such a way that vertex A meets vertex C, thereby forming a line PQ. Assuming AB = 3 and BC = 4, find PQ. Note that AP = PC and AQ = QC.



(a) $\frac{13}{4}$ **(b)** $\frac{15}{4}$ **(c)** $\frac{17}{4}$ **(d)** $\frac{19}{4}$

Answer (b) **(CSIR-NET DEC 2012)**

Difficulty rating: 4/5 *Average time for solving: 130 seconds*

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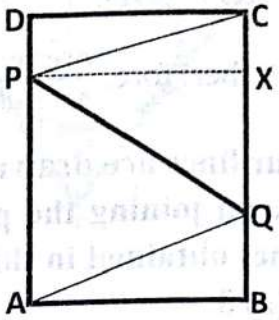
Given: AB = 3 and BC = 4, AP = PC, AQ = QC.

Here we are asked to find PQ.



Construct PX perpendicular to BC.

Then consider the right-angled triangle PXQ.

By Pythagoras theorem, $PQ^2 = PX^2 + QX^2$.

Given AB = 3.

Clearly, AB = **PX = 3** (See figure)

Thus, it remains to find QX.

Consider right-angled triangle ABQ,

By Pythagoras theorem, $AQ^2 = QB^2 + AB^2$.

Given AQ = QC.

Thus, $QC^2 = (BC - QC)^2 + 3^2$

$\Rightarrow QC^2 = (4 - QC)^2 + 9$

$\Rightarrow QC^2 = 16 - 8QC + QC^2 + 9$ *[Remember: $(a-b)^2 = a^2 - 2ab + b^2$]*

$\Rightarrow 8QC = 25$

$\Rightarrow QC = \frac{25}{8}$.

Hence, $QB = 4 - \frac{25}{8} = \frac{7}{8}$.

Clearly, $PD = QB = CX = AY = \frac{7}{8}$.

$QX = QC - CX = \frac{25}{8} - \frac{7}{8} = \frac{18}{8} = \frac{9}{4}$

$\Rightarrow$ **$QX = \frac{9}{4}$.**

Thus, $PQ^2 = PX^2 + QX^2 = 3^2 + (\frac{9}{4})^2 = 9 + \frac{81}{16} = \frac{225}{16}$

$PQ = \frac{15}{4}$.

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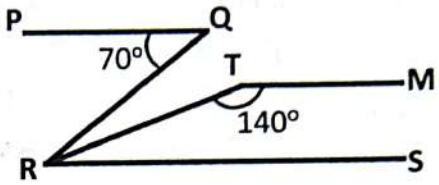
11.3 Practice Questions

1. A rectangular area of sides 11 and 10 units is to be covered by square tiles of sides 1, 3 and 4 units. The minimum number of tiles needed for this is
(a) 9 (b) 12 (c) 25 (d) 26

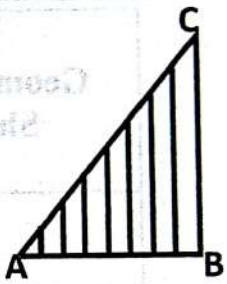
2. In the given figure, PQ || TM || RS. If $\angle PQR = 70^\circ$, $\angle RTM = 140^\circ$, then the $\angle QRT$ =
(a) 20° (b) 25° (c) 30° (d) 35°



3. A box measures 3 cm × 4 cm × 12 cm. The longest rod that can be placed in it has the length
(a) $\sqrt{153}$ cm (b) $\sqrt{159}$ cm (c) 13 cm (d) 15 cm

4. In a triangle, if the three altitudes are equal, then the triangle must be:
(a) Right (b) Isosceles (c) Obtuse (d) Equilateral

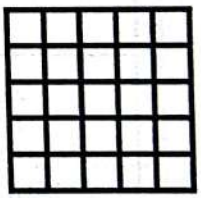
5. Side AC of right triangle ABC is divided into 8 parts. Seven line segments parallel to BC are drawn to AB from the points of division. If BC = 10, then the sum of the lengths of the seven line segments is
(a) 33 (b) 34 (c) 35 (d) 45



6. If the diagonal of a square is tripled, then its area will be
(a) Doubled (b) Tripled (c) 4 times (d) 9 times

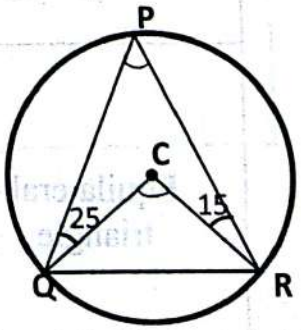
7. The number of circles passing through three non-collinear points in a plane is
(a) 1 (b) 2 (c) 3 (d) 4

8. How many rectangles (which are not squares) are there in the following figure?
(a) 25 (b) 55
(c) 170 (d) 225



9. Two similar triangles have areas 392 cm^2 and 200 cm^2. If one side of the smaller triangle is 10 cm, then the corresponding side of the bigger triangle is
(a) 12 cm (b) 13 cm
(c) 14 cm (d) 15 cm

10. In the given figure, if C is the centre of the circle and $\angle PQC = 25^\circ$ and $\angle PRC = 15^\circ$, then $\angle QCR$ is equal to
(a) 40° (b) 60° (c) 80° (d) 120°



Answer Key

1. (d) 2. (c) 3. (c) 4. (d) 5. (c) 6. (d) 7. (a) 8. (c) 9. (c) 10. (c)

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CHAPTER 12

MENSURATION AND QUANTITATIVE COMPARISONS

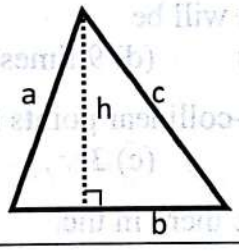
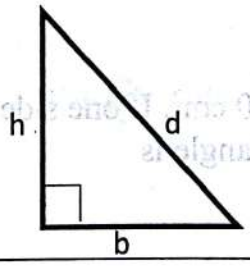
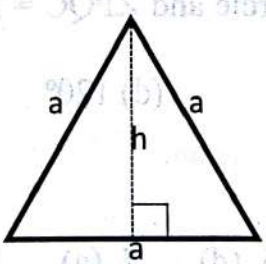
Average Difficulty Rating: 3.2/5

Average time for each question: 103 sec

Number of questions expected for each exam: 2-4

12.1 THEORY

Triangles

Geometrical Shape	Figure	Perimeter	Area
Any Triangle		$a + b + c = 2s$	1. $\frac{1}{2}$ Base × Height 2. $\sqrt{s(s-a)(s-b)(s-c)}$
Right-angled triangle		$b + h + d$	$\frac{1}{2}$ Base × Height
Equilateral triangle		$3a$	1. $\frac{1}{2}ah$ 2. $\frac{\sqrt{3}}{4}a^2$

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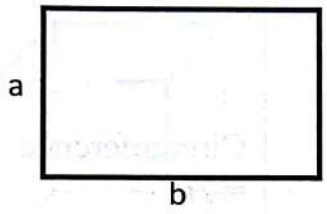
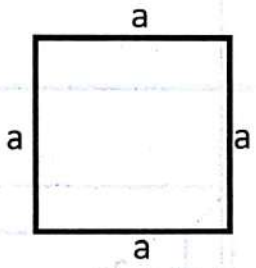
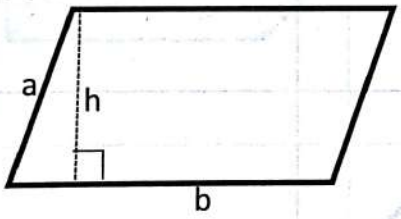
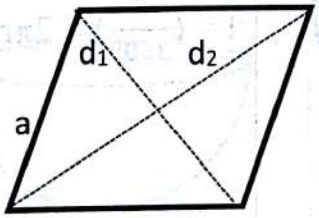
Some important quadrilaterals

Geometrical Shape	Figure	Perimeter	Area
Rectangle		2(a + b)	ab
Square		4a	a^2
Parallelogram		2(a +b)	bh
Rhombus		4a	$\frac{1}{2} d_1 \times d_2$

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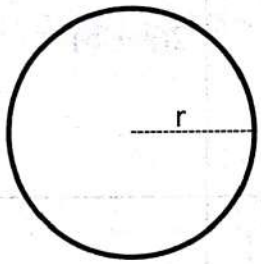
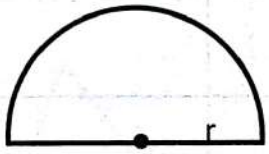
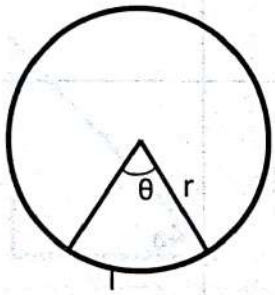
Circular geometrical shapes

Geometrical Shape	Figure	Perimeter	Area
Circle		Circumference $= 2\pi r$	πr^2
Semicircle		$\pi r + 2r$	$\frac{1}{2}\pi r^2$
Sector of a circle		$l + 2r$ where $l = (\frac{\theta}{360^o}) \times 2\pi r$	$(\frac{\theta}{360^o}) \times \pi r^2$

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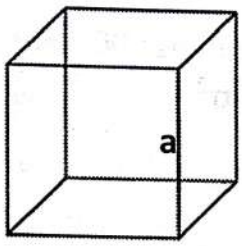
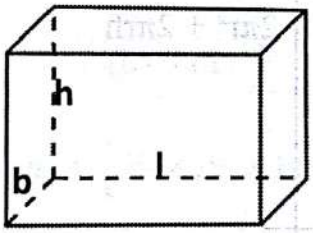
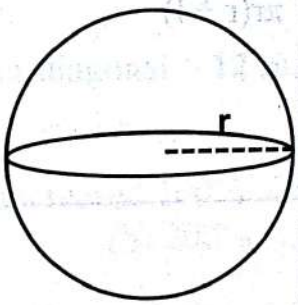
Solid geometrical Shapes

Geometrical Shape	Figure	Surface Area	Volume
Cube		$6a^2$	a^3
Cuboid		$2(lb + lh + bh)$	lbh
Sphere		$4\pi r^2$	$\frac{4}{3}\pi r^3$

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Solid geometrical Shapes

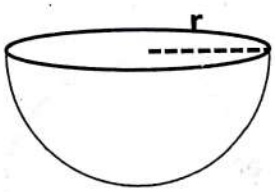
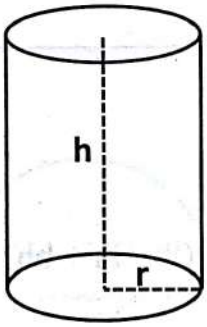
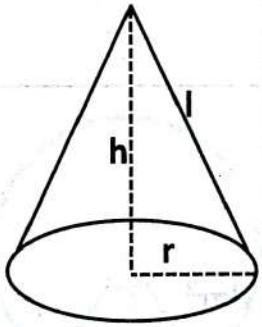
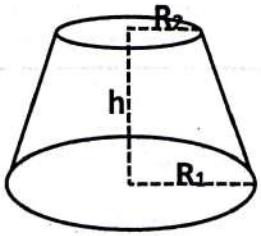
Geometrical Shape	Figure	Surface Area	Volume
Hemi-sphere		$3\pi r^2$	$\frac{2}{3}\pi r^3$
Cylinder		$2\pi r^2 + 2\pi rh$	1. Base Area × Height 2. $\pi r^2 h$
Right-Circular Cone		$\pi r(r + l)$	$\frac{1}{3}\pi r^2 h$
Frustum		$\Pi(R_1 + R_2)\times \sqrt{(R_1 - R_2)^2 + h^2}$	$\frac{\pi}{3}h({R_1}^2 + {R_1}^2 + R_1R_2)$

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ILLUSTRATION 1:

If a bicycle wheel makes 5000 revolutions in moving 11 km, then the diameter of the wheel is:

(a) 110 cm (b) 70 cm (c) 120 cm (d) 100 cm

Answer (b)

Thus, distance covered by the bicycle for making 1 revolution

= Circumference of the bicycle wheel = $2\pi r$.

Thus, distance covered by the bicycle for making 5000 revolutions = $5000 \times 2\pi r$.

Given that, distance covered in 5000 revolutions = 11 km = 11×10^5 cm.

⇨ $5000 \times 2\pi r = 11 \times 10^5$

⇨ $\frac{22}{7} r = 110$

⇨ $r = 35$ cm.

⇨ Diameter = $2r = 2 \times 35 =$ **70 cm.**

ILLUSTRATION 2:

If the area of a rhombus is 108 m^2 and the length of one diagonal is 18 m, the length of the other diagonal is

(a) 8 m (b) 9 m (c) 12 m (d) 16 m

Answer (c)

Remember: Area of rhombus = $\frac{1}{2} d_1 \times d_2$, where d_1, d_2 are diagonals

Given : $d_1 = 18$ m, $d_2 = ?$

Area of the given rhombus = 108 m^2.

⇨ $108 = \frac{1}{2} \times 18 \times d_2$

⇨ $d_2 = 12$.

⇨ Length of the other diagonal = **12 m.**

ILLUSTRATION 3:

If the radius of a circle is increased 100%, the area is increased:

(a) 100% (b) 200% (c) 300% (d) 400%

Answer (c)

Let r be the radius of the given circle.

The area of the circle is πr^2.

Given that, the radius is increased by 100%.

Therefore, the new radius is $r + \frac{100}{100} r = 2r$.

Thus the new area is $\pi(2r)^2 = 4\pi r^2$.

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Therefore, the area increased from πr^2 to $4\pi r^2$.

Thus, the increase in area = $3\pi r^2 = \frac{300}{100}\pi r^2 = 300\%$ of πr^2.

Hence, the area is increased 300%.

ILLUSTRATION 4:

A large watermelon weights 20 kg with 96 % of its weight being water. It is allowed to stand in the sun and some of the water evaporates so that now only 80% of its weight is water. The weight of the watermelon after exposure to sun will be

(a) 2 kg (b) 3 kg (c) 4 kg (d) 5.6 kg

Answer (c)

In 20 kg watermelon, 96% water is present.

⇨ Amount of water = $\frac{96}{100} \times 20 = 19.2$ kg.

Therefore, the amount of non-water matter = 20 – 19.2 = 0.8 kg.

Let x be the weight of watermelon after exposure to sun.

Given: After exposure to sun, some water evaporates.

That is, the mass of watermelon and water has decreased.

Now it contains 80% water and the rest 20% non-water matter.

However, **the amount of non-water matter has not changed.**

Therefore, the amount of the non-water matter is 0.8 kg itself.

Thus, 20% corresponds to 0.8 kg of non-water matter.

Hence, $\frac{20}{100} \times x = 0.8$

⇨ $x = \frac{80}{20} = 4$.

Thus, the weight of the watermelon after exposure to sun will be 4 kg.

ILLUSTRATION 5:

Which of the following figures has the longest perimeter?

(a) A square of side 20 cm (b) A circle of radius 14 cm

(c) A rectangle of sides 24 cm and 15 cm (d) A rhombus of side 20 cm

Answer (b)

Let us find perimeters of each of the given figures.

Consider (a)

Remember: Perimeter of square = 4a, where a is the length of the side.

Thus perimeter of the given square = $4 \times 20 = 80$ cm^2.

Consider (b)

Remember: Perimeter of rectangle = 2(length + breadth)

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Thus perimeter of the given rectangle = $2(24 + 15) = 78$ cm^2.

Consider (c)

Remember: Perimeter of circle = $2\pi r$, where r is the radius of the circle.

Thus perimeter of the given circle = $2\times\frac{22}{7}\times 14 = 88$ cm^2.

Consider (d)

Remember: Perimeter of rhombus = 4a, where a is the length of the side.

Thus perimeter of the given rhombus = $4 \times 20 = 80$ cm^2.

Evidently, the circle of radius 14 cm has the longest perimeter.

12.2 CSIR-NET Exam Questions – Detailed Solution

1. What is the area of the triangle bounded by the lines y = 2x, y = -2x and y = 6?

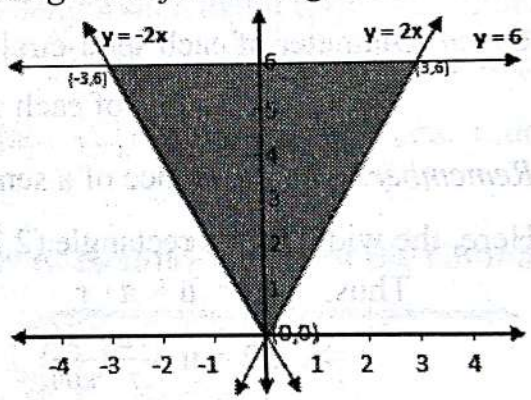
(a) 38 **(b) 18** **(c) 12** **(d) 24**

Answer (b) **(CSIR-NET DEC 2014)**

Difficulty rating: 3/5 *Average time for solving: 120 seconds*

The given region can be plotted as shown.

The shaded portion is the required triangle with base (b) = 6 units and height (h) = 6 units.



$$\text{Area of the triangle} = \frac{1}{2} \times \text{Base} \times \text{Height}$$

$$= \frac{1}{2} \times 6 \times 6$$

$$= \mathbf{18\ units.}$$

2. A cubical piece of wood was filed to make it into the largest possible sphere. What fraction of the original volume was removed?

(a) more than 3/4 **(b) 1/2**

(c) slightly less than 1/2 **(d) slightly more than 1/2**

Answer (c) **(CSIR-NET DEC 2014)**

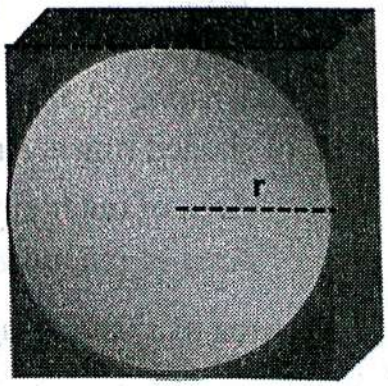
Difficulty rating: 3.5/5 *Average time for solving: 120 seconds*

$$\text{Fraction of the original volume removed} = \frac{\text{Portion Removed}}{\text{Original Volume}}$$

The portion removed = Volume of cube – Volume of sphere.

Let r be radius of the sphere.

Then, edge (a) of the cube = 2r. [See the Figure]



Original volume = Volume of cube = $a^3 = (2r)^3 = 8r^3$.

Volume of sphere = $\frac{4}{3}\pi r^3$.

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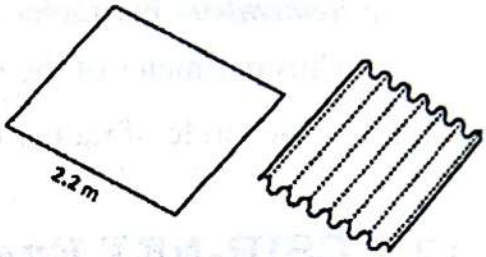
Thus portion removed = $8r^3 - \frac{4}{3}\pi r^3 = [8 - (\frac{4}{3} \times \frac{22}{7})]\, r^3 = [8 - \frac{88}{21}]\, r^3 = \frac{80}{21}r^3$.

Fraction of the original volume removed = $\frac{80}{21}r^3/8r^3 = \frac{80}{21 \times 8} = \frac{10}{21}$

which is clearly **less than** $\frac{1}{2}$

Hence, **option (c).**

3. A 2.2 m wide rectangular steel plate is corrugated as shown in the diagram. Each corrugation is a semi-circle in cross section having a diameter of 7 cm. What will be the width of the steel after it is corrugated?



(a) 1.4 m (b) 1.6 m (c) 0.7 m (d) 1.1 m

Answer (a) **(CSIR-NET DEC 2014)**

Difficulty rating: 3.5/5 *Average time for solving: 110 seconds*

Note that the corrugated figure shown is just a representation and does not represent the exact number of semi-circles.

Let n be the number of semi-circles formed when the plate is corrugated.

Given: Diameter of each semi-circle = 7 cm.

Therefore, Radius of each semi-circle = $\frac{7}{2}$ cm = $\frac{7}{200}$ m.

Remember: Circumference of a semicircle = $\pi \times r$.

Here, the width of the rectangle (2.2 m) is n times the circumference of each semicircle.

Thus, $2.2 = n \times \pi \times r$

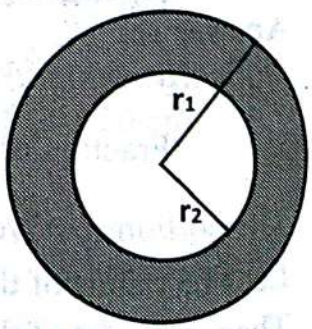
$\Rightarrow 2.2 = n \times \frac{22}{7} \times \frac{7}{200}$

$\Rightarrow n = \frac{2.2 \times 200}{22} = 20$

The width of the corrugated plate is the sum of diameters of all these 20 semi-circles.

Thus, width of corrugated plate = $20 \times \frac{7}{100}$ = **1.4 m.**

4. The areas of inner circle and the shaded ring are equal. The radii r_1 and r_2 are related by



(a) $r_1 = r_2$ (b) $r_1 = r_2\sqrt{2}$ (c) $r_1 = r_2\sqrt{3}$ (d) $r_1 = 2r_2$

Answer (b) **(CSIR-NET DEC 2014)**

Difficulty rating: 2.5/5 *Average time for solving: 90 seconds*

Area of shaded ring = Area of outer circle – Area of inner circle

Area of outer circle = πr_1^2.

Area of inner circle = πr_2^2.

Thus, Area of shaded ring = $\pi r_1^2 - \pi r_2^2 = \pi(r_1^2 - r_2^2)$.

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5. A rectangle of length d and breadth $\frac{d}{2}$ is revolved once completely around its length and once around its breadth. The ratio of volumes swept in the two cases is

(a) 1:1 **(b) 1:2** **(c) 1:3** **(d) 1:4**

Answer (b) **(CSIR-NET DEC 2014)**

Difficulty rating: 3.5/5 *Average time for solving: 120 seconds*

Remember: Volume of cylinder = $\pi r^2 h$.

As the rectangle is revolved around the length, it forms a cylinder with radius $\frac{d}{4}$ and height d.

Thus, volume $V_1 = \pi r^2 h = \pi \times (\frac{d}{4})^2 \times d = \frac{\pi d^3}{16}$

As the rectangle is revolved around the breadth, it forms a cylinder with radius $\frac{d}{2}$ and height $\frac{d}{2}$.

Thus, volume $V_2 = \pi r^2 h = \pi \times (\frac{d}{2})^2 \times \frac{d}{2} = \frac{\pi d^3}{8}$

Ratio of volumes $= \frac{V_1}{V_2} = \frac{\frac{\pi d^3}{16}}{\frac{\pi d^3}{8}} = \frac{1}{2} =$ **1:2.**

Note that here we can also revolve the rectangle around the length such that it forms a cylinder with radius $\frac{d}{2}$ and height $\frac{d}{2}$ and revolve it around the breadth such that it forms a cylinder with radius d and height $\frac{d}{2}$. However, proceeding with those values will also yield us the same ratio as the final answer.

6. A long ribbon is wound around a spool up to a radius R. Holding the tip of the ribbon, a boy runs away from the spool with a constant speed maintaining the unwound portion of the ribbon horizontal. In 4 minutes, the radius of the wound portion becomes $\frac{R}{\sqrt{2}}$. In what further time, it will become $\frac{R}{2}$

(a) $\sqrt{2}$ min **(b) 2 min** **(c) $2\sqrt{2}$ min** **(d) 4 min**

Answer (b) **(CSIR-NET DEC 2014)**

Difficulty rating: 4/5 *Average time for solving: 110 seconds*

At first, the radius of the wound portion is R.

Therefore, the area covered by the ribbon initially, $A_1 = \pi R^2$.

After 4 minutes, the radius of the wound portion is $\frac{R}{\sqrt{2}}$.

Therefore, the area after 4 minutes, $A_2 = \pi(\frac{R}{\sqrt{2}})^2 = \frac{\pi R^2}{2} = \frac{A_1}{2}$.

When the radius of the wound portion becomes $\frac{R}{2}$, the area, A_3 will be $\pi(\frac{R}{2})^2 = \frac{\pi R^2}{4} = \frac{A_2}{2}$.

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Thus for the area, A_1 to get halved, it took 4 minutes.

Therefore, for A_2 to get halved, it will take **2 more minutes.**

7. The area (in m^2) of a triangular park of dimension 50 m, 120 m and 130 m is

(a) 3000 **(b) 3250** **(c) 5550** **(d) 7800**

Answer (a) **(CSIR-NET DEC 2014)**

Difficulty rating: 2.5/5 *Average time for solving: 90 seconds*

Normal Solution

Remember: Area of a triangle with three different sides a, b, c $= \sqrt{s(s-a)(s-b)(s-c)}$

where $s = \frac{a+b+c}{2}$

Here $s = \frac{50+120+130}{2} = 150.$

Area of triangle $= \sqrt{150(150-50)(150-120)(150-130)} = \sqrt{150 \times 100 \times 30 \times 20}$

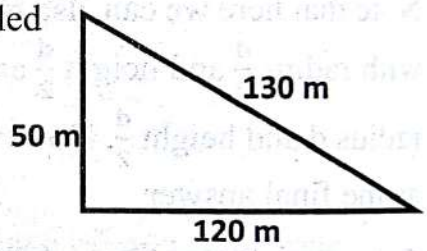
$= \sqrt{9000000} =$ **3000 m^2.**

Quick Solution

Clearly, 50 m, 120 m, 130 m are dimensions of a right-angled triangle, since it obeys Pythagoras theorem.

((50, 120, 130) is a Pythagorean triplet.)

That is, $50^2 + 120^2 = 2500 + 14400 = 16900 = 130^2$.

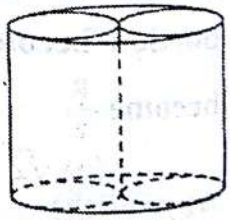


Remember: Area of right-angled triangle $= \frac{1}{2} \times$ Base $\times$ Height.

Thus, Area of the given triangle $= \frac{1}{2} \times 120 \times 50 =$ **3000 m^2.**

8. Two identical cylinders of the same height as a bigger hollow cylinder were put vertically into the latter to fit exactly into it. The volume of the bigger cylinder is V. The volume of each of the smaller cylinder is

(a) $\frac{V}{8}$ **(b)** $\frac{V}{4}$ **(c)** $\frac{V}{2}$ **(d) V**



Answer (b) **(CSIR-NET JUNE 2014)**

Difficulty rating: 3/5 *Average time for solving: 80 seconds*

Remember: Volume of the cube $= a^3$.

Let "r" and "h" be respectively the radius and height of bigger cylinder. Then each of the smaller cylinders are of the radius $\frac{r}{2}$ and of the same height.

Volume of bigger cylinder, $V = \pi \times r^2 \times h$.

Volume of each of the smaller cylinder $= \pi \times (\frac{r}{2})^2 \times h = \frac{V}{4}$

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9. A cubic cavity of edge 20μm is filled with a fluid with a cubic solid of edge 2μm. What percent of the cavity volume is occupied by the fluid?

(a) 10.0 **(b) 20.0** **(c) 90.0** **(d) 99.9**

Answer (d) **(CSIR-NET JUNE 2014)**

Difficulty rating: 3/5 *Average time for solving: 90 seconds*

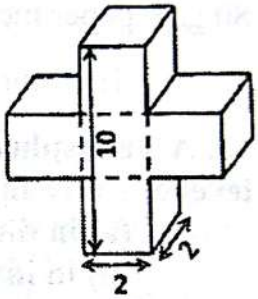
Volume of cubic cavity = $20^3 \mu m^3$

Volume of cubic solid = $2^3 \mu m^3$

Percentage of volume of cavity occupied by the solid = $\frac{2^3}{20^3} \times 100 = 0.1\%$

Percentage of volume of cavity occupied by the fluid = $100 - 0.1 =$ **99.9%** (approx.)

10. The diagram shows 2 perpendicularly inter-grown prismatic crystals (twins) of identical shape and size. What is the volume of the object shown (units are arbitrary)?



(a) 60 **(b) 65** **(c) 72** **(d) 80**

Answer (c) **(CSIR-NET JUNE 2014)**

Difficulty rating: 3/5 *Average time for solving: 100 seconds*

Observe that the two crystals are of cuboidal (rectangular box) shape and the dimension of the cuboids are 10, 2, 2 units respectively.

However, the two crystals have a common area in the centre, which is a cube of edge 2 units.

Remember: Volume of cuboid = length × breadth × height

Volume of cube = $edge^3$

Thus, volume of given figure = Volume of 2 cuboids – Volume of cube

$= 2(10 \times 2 \times 2) - 2^3 = 80 - 8 =$ **72.**

11. In 450 g of pure coffee powder, 50 g of chicory is added. A person buys 100 g of the mixture and adds 5 g of chicory to that. What would be the round off percentage of chicory in this final mixture?

(a) 10 **(b) 5** **(c) 14** **(d) 15**

Answer (c) **(CSIR-NET JUNE 2014)**

Difficulty rating: 3/5 *Average time for solving: 110 seconds*

The given data can be tabulated as shown below.

	Coffee	Chicory	Total
Original mixture	450	50	500
100 g of original mixture	90	10	100
Final mixture (5g chicory added)	90	15	105

Percentage of chicory in the final mixture = $\frac{\text{Amount of chicory in final mixture}}{\text{Total amount of the final mixture}} \times 100$

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$= \frac{15}{105} \times 100 = 14.29 \approx \mathbf{14.}$

12. 80 gsm paper is cut into sheets of 200 mm × 300 mm size and assembled in packets of 500 sheets. What will be the weight of a packet? (gsm =g/m^2)

(a) 1.2 kg (b) 2.4 kg (c) 3.6 kg (d) 4.8 kg

Answer (b) **(CSIR-NET JUNE 2014)**

Difficulty rating: 3/5 *Average time for solving: 110 seconds*

Remember:1 m = 1000 mm, 1 kg = 1000 g.

Surface area of one paper = 200 mm × 300 mm = 0.2 m × 0.3 m = 6×10^{-2} m^2

Surface area of 500 papers = $500 \times 6 \times 10^{-2}$ m^2 = 30 m^2

80 gsm paper means paper in which 1 m^2 area constitute 80 g weight.

Therefore, for surface area of 30 m^2, the weight = 80 × 30 = 2400 g = **2.4 kg.**

13. A hemispherical bowl is being filled with water at a constant volumetric rate. The level of water in the bowl increases

(a) in direct proportion of time.
(b) in inverse proportion of time.
(c) faster than direct proportion of time.
(d) slower than direct proportion of time.

Answer (d) **(CSIR-NET DEC 2013)**

Difficulty rating: 3/5 *Average time for solving: 110 seconds*

If it were a cuboidal (rectangular box) or a cubical vessel, the level of water would have increased in direct proportion of time.

However, for a hemispherical bowl, the radius of the bowl also increases as the level of water increases.

As the radius increases, the area of cross-section also increases with the level of water increases.

Therefore, while going up, the speed of the increase in the level of water gets slower and slower.

Hence, the level of water increases slower than the direct proportion of time, **option (d).**

14. Equal masses of two liquids of densities 6 kg/m^3 and 4 kg/m^3 are mixed thoroughly. The density of the mixture is

(a) 4.8 kg/m^3 (b) 5.0 kg/m^3 (c) 5.2 kg/m^3 (d) 5.4 kg/m^3

Answer (a) **(CSIR-NET DEC 2013)**

Difficulty rating: 3/5 *Average time for solving: 90 seconds*

Normal Solution:

Here there are two liquids, say Liquid 1 and Liquid 2.

Let us find the density of the mixture from the combined volume and combined mass of Liquid 1 and Liquid 2.

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Liquid 1

Mass = m Density = d_1 Volume $v_1 = \frac{m}{d_1} = \frac{m}{6}$

Liquid 2

Mass = m Density = d_2 Volume $v_2 = \frac{m}{d_2} = \frac{m}{4}$

Mixture

Mass M = m + m = 2m

Volume $V = v_1 + v_2 = \frac{m}{d_1} + \frac{m}{d_2} = \frac{m(10)}{24}$

Density $D = \frac{M}{V} = \frac{48}{10} = \mathbf{4.8\ kg/m^3}$

Quick Solution:

Note: density of resultant mixture, $d = \frac{2d_1d_2}{d_1+d_2}$.

where d_1 is density of the liquid 1 and d_2 is the density of the liquid 2.

Thus, $d = \frac{2d_1d_2}{d_1+d_2} = \frac{2\times6\times4}{6+4} = \frac{48}{10} = \mathbf{4.8\ kg/m^3}$.

15. A circle circumstances identical close-packed circles of unit diameter as shown. What is the total area of the shaded portion?

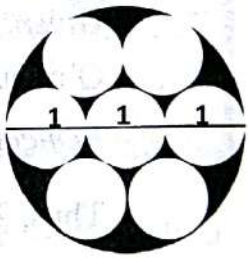
(a) 2 **(b) 2π** **(c) $\frac{1}{2}$** **(d) $\frac{\pi}{2}$**

Answer (d) **(CSIR-NET DEC 2013)**

Difficulty rating: 3/5 *Average time for solving: 110 seconds*

Given: Diameter of each smaller circle = 1 unit.

Then, Diameter of bigger circle = 1+1+1 = 3 units. (See figure)



Remember: Area of circle = πr^2

Area of the shaded portion = Area of bigger circle – (Area of 7 smaller circles)

$$= \pi\left(\frac{3}{2}\right)^2 - 7\left(\pi\left(\frac{1}{2}\right)^2\right) = \frac{9}{4}\pi - \frac{7}{4}\pi = \frac{\pi}{2}$$

16. What does the diagram establish? Note: The diagram is a circle inside a square

(a) $\pi > 3$ **(b) $\pi \geq 2\sqrt{2}$**

(c) $\pi < 4$ **(d) π is closer to 3 than to 4.**

Answer (c) **(CSIR-NET DEC 2013)**

Difficulty rating: 3/5 *Average time for solving: 90 seconds*

Let r be the radius of the circle.

Then each side of the square will be 2r.

Area of circle = πr^2 Area of square = $(\text{side})^2 = (2r)^2 = 4r^2$

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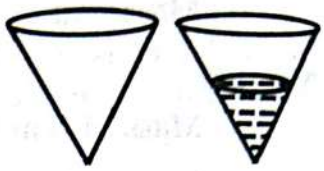
Since the circle is fully inside the square, the area of circle is less than the area of square.

⇨ $\pi r^2 < 4r^2$

⇨ $\pi < 4$.

17. The capacity of the conical vessel shown is V. It is filled with water up to half its height. The volume of water in the vessel is



(a) $\frac{V}{2}$ **(b)** $\frac{V}{4}$ **(c)** $\frac{V}{8}$ **(d)** $\frac{V}{16}$

Answer (c) **(CSIR-NET JUNE 2013)**

Difficulty rating: 3/5 *Average time for solving: 100 seconds*

Remember: Volume of a cone $= \frac{1}{3}\pi r^2 h$.

Here the vessel is of the shape of a cone.

Thus, Volume of the empty vessel $V = \frac{1}{3}\pi r^2 h$.

Note: ***For a cone, the radius is proportional to height.***

Hence, when the height is halved $(\frac{h}{2})$, the radius also is halved, That is, $(\frac{r}{2})$

Thus, Volume of water in the vessel $V' = \frac{1}{3}\pi (\frac{r}{2})^2 \frac{h}{2} = \frac{1}{8}(\frac{1}{3}\pi r^2 h) = \frac{V}{8}$.

18. A 16.2 m long wooden log has a uniform diameter of 2 m. To what length the log should be cut to obtain a piece of 22 m^3 volume?

(a) 3.5 m **(b) 7.0 m** **(c) 14.0 m** **(d) 22.0 m**

Answer (b) **(CSIR-NET JUNE 2013)**

Difficulty rating: 3/5 *Average time for solving: 90 seconds*

Given: Radius r = 1 m, Volume V = 22 m^3 Height h = ?

Obviously, the wooden log is of cylindrical shape.

Remember: Volume of cylinder $V = \pi r^2 h$.

Thus, $22 = \frac{22}{7} 1^2 h$

⇨ h = 7 m. Hence, **option (b).**

19. The distance between two oil rigs is 6 km. What will be the distance between these rigs in maps of 1 : 50000 and 1 : 5000 scales, respectively?

(a) 12 cm and 1.2 cm **(b) 2 cm and 12 cm**

(c) 120 cm and 12 cm **(d) 12 cm and 120 cm**

Answer (d) **(CSIR-NET JUNE 2013)**

Difficulty rating: 3/5 *Average time for solving: 90 seconds*

Normal Solution:

Since in all the answer options the unit of length is cm, we should first convert 6 km into cm.

Remember: 1 km = 1000 m = 1000 × 100 cm.

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Thus, 6 km = 6 × 1000 × 100 cm = 600000 cm

1 : 50000 map means 1 cm = 50000 cm in the map

Similarly for 1:5000 map, 1 cm = 5000 cm in the map.

Therefore, Distance between rigs in the map of 1 : 50000 = $\frac{600000}{50000}$ = **12 cm**.

Distance between rigs in the map of 1 : 5000 = $\frac{600000}{5000}$ = **120 cm**. Hence, **option (d).**

Quick Solution:

Given two maps 1: 50000 and 1: 5000 respectively.

In addition, each answer options have two values.

Intuitively, second map is "bigger" or is more zoomed in, by the scales given.

So definitely, in each answer options, first value should be smaller than the second value.

Hence, readily we can eliminate options (a) and (c)

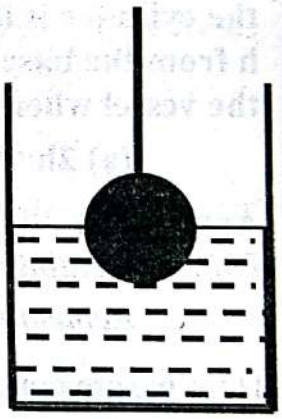
The only options left are (b) and (d).

Now, second map is 10 times bigger than the first, so will the values be.

Hence, option (b) can be eliminated as there is no proportionality of 10 between 2 and 12.

Thus, the obvious left out option is **(d).**

20. A sphere of iron of radius $\frac{R}{2}$ fixed to one end of a string was lowered into water in a cylindrical container of base radius R to keep exactly half the sphere dipped. The rise in the level of water in the container will be



(a) $\frac{R}{3}$ **(b)** $\frac{R}{4}$ **(c)** $\frac{R}{8}$ **(d)** $\frac{R}{12}$

Answer (d) **(CSIR-NET JUNE 2013)**

Difficulty rating: 3.5/5 *Average time for solving: 110 seconds*

Let h be the rise in the level of water in the container.

Since, exactly half of the sphere is dipped, the portion dipped in the water, is actually a hemisphere.

Here, radius of the hemisphere = $\frac{R}{2}$

Remember: Volume of a hemisphere = $\frac{2}{3}\pi r^3$.

Thus, Volume of hemisphere dipped in water = $\frac{2}{3}\pi(\frac{R}{2})^3 = \frac{2}{3}\pi\frac{R^3}{8} = \frac{\pi R^3}{12}$

Volume of water increased = Volume of hemisphere dipped in water.

Volume of water increased is of cylindrical shape with height h and radius R.

Thus, Volume of water increased = $\pi R^2 h$.

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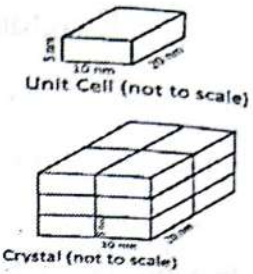
Hence, $\pi R^2 h = \frac{\pi R^3}{12}$

$\Rightarrow$ $\mathbf{h = \frac{R}{12}}$

21. A crystal grows by stacking of unit cells of 10 × 20 × 5 nm size as shown in the diagram. How many unit cells will make a crystal of 1 cm^3 volume?



(a) 10^6 **(b) 10^9** **(c) 10^{12}** **(d) 10^{18}**

Answer (d) **(CSIR-NET JUNE 2013)**

Difficulty rating: 3/5 *Average time for solving: 110 seconds*

Note: 1 nm = 10^{-9} m = 10^{-7}cm

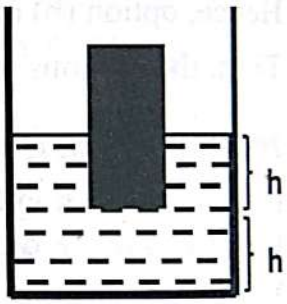
Volume of one unit cell = $(10 \times 10^{-7}) \times (20 \times 10^{-7}) \times (5 \times 10^{-7}) = 1000 \times 10^{-21} = 10^{-18}$ cm^3.

Let x be the number of unit cells which make a crystal of 1 cm^3 volume.

Then, $x \times 10^{-18} = 1$.

$\Rightarrow$ $\mathbf{x = 10^{18}}$.

22. A solid cylinder of basal area A was held dipped in water in a cylindrical vessel of basal area 2A vertically such that a length h of the cylinder is immersed. The lower tip of the cylinder is at a height h from the base of the vessel. What will be the height of the water in the vessel when the cylinder is taken out?



(a) 2h **(b) $\frac{3}{2}$ h** **(c) $\frac{4}{3}$ h** **(d) $\frac{5}{4}$ h**

Answer (b) **(CSIR-NET JUNE 2013)**

Difficulty rating: 3.5/5 *Average time for solving: 110 seconds*

Remember: Volume of cylinder = Base Area × Height.

Here we are concerned about the volumes of 2 cylinders, one the volume of the water filled in the cylindrical vessel and other the volume of the portion of solid cylinder dipped in water.

Volume of the water filled in the cylindrical vessel = 2A × 2h = 4Ah.

Volume of the portion of solid cylinder dipped in water = Ah.

Volume of the water in the vessel when the cylinder is taken out = 4Ah – Ah = 3Ah.

Let H be the new height of the water when the cylinder is taken out.

Thus, volume of the water in the vessel when the cylinder is taken out = 2AH.

$\Rightarrow$ 3Ah = 2AH

$\Rightarrow$ $\mathbf{H = \frac{3}{2}h}$.

23. Of all the triangles that can be inscribed in a semi-circle of radius R with the diameter as one side, the biggest one has the area

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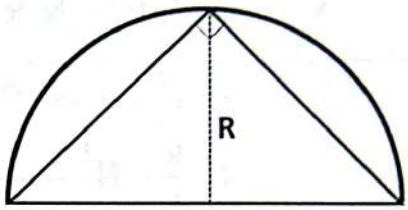
(a) R^2 (b) $R^2\sqrt{R}$ (c) $R^2\sqrt{3}$ (d) $2R^2$

Answer (a) **(CSIR-NET JUNE 2013)**

Difficulty rating: 3/5 *Average time for solving: 100 seconds*

Note 1: Inside the semi-circle, the triangle with diameter as the one side is always a right angled triangle.

Note 2: The triangle with largest area in a semi-circle will be an isosceles triangle.



From Note 1 and Note 2, we can get that the ΔABC is the biggest triangle that can be inscribed in the semicircle.

Area of a triangle $= \frac{1}{2}$ Base × Height.

Clearly the base is the diameter 2R and height is the radius R.

Thus, area of biggest triangle $= \frac{1}{2} \times 2R \times R = \mathbf{R^2}$.

24. A daily sheet calendar of the year 2013 contains sheets of 10 × 10 cm size. All the sheets of the calendar are spread over the floor of a room of 5m × 7.3m size. What percentage of the floor will be covered by these sheets?

(a) 0.1 (b) 1 (c) 10 (d) 100

Answer (c) **(CSIR-NET JUNE 2013)**

Difficulty rating: 3.5/5 *Average time for solving: 110 seconds*

2013 is a non-leap year, so there are 365 days.

Given: The surface area covered by calendar sheets of 1 day = **10 cm × 10 cm.**

Then, the surface area covered by calendar sheets of 365 days = $365 \times 10 \times 10 = 36500\ cm^2$
$= 3.65\ m^2$

The surface area of the floor = $5\ m \times 7.3\ m = 36.5\ m^2$

Percentage of the floor covered by sheets $= \frac{3.65}{36.5} \times 100 = \mathbf{10\%}$.

25. Suppose we make N identical smaller spheres from a big sphere. The total surface area of the smaller spheres is X times the total surface area of the big sphere, where X is

(a) $\sqrt{N}$ (b) 1 (c) $N^{1/3}$ (d) N^3

Answer (c) **(CSIR-NET DEC 2012)**

Difficulty rating: 3.5/5 *Average time for solving: 120 seconds*

Remember:

1. Volume of sphere $= \frac{4}{3}\pi(\text{radius})^3$
2. Surface area of sphere $= 4\pi(\text{radius})^2$

Let V be the volume of bigger sphere and v_i be the volume of each small sphere.

$V = \frac{4}{3}\pi R^3$, where R is the radius of bigger sphere.

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$v_i = \frac{4}{3}\pi r^3$, where r is the radius of each smaller spheres.

When we make N identical smaller spheres from a big sphere, the volume of bigger sphere is N times the volume of each smaller spheres.

That is, $V = N \times v_i$

$\Rightarrow \frac{4}{3}\pi R^3 = N \times \frac{4}{3}\pi r^3$

$\Rightarrow (\frac{R}{r})^3 = N$

$\Rightarrow \frac{R}{r} = N^{\frac{1}{3}}$ ---------(1)

Now consider the **surface areas** of bigger and smaller spheres.

Let A be the surface area of bigger sphere and a_i be the surface area of each of the smaller spheres.

Surface area of bigger sphere, $A = 4\pi R^2$

Surface area of each smaller spheres, $a_i = 4\pi r^2$

Total Surface area of smaller spheres, $a = N \times 4\pi r^2$

Given $a = X \times A$, and we are required to find the value of X.

Thus, $N \times 4\pi r^2 = X \times 4\pi R^2$

$\Rightarrow (\frac{R}{r})^2 = \frac{N}{X}$

$\Rightarrow \frac{R}{r} = (\frac{N}{X})^{\frac{1}{2}}$ ---------(2)

From equations (1) and (2), $N^{\frac{1}{3}} = (\frac{N}{X})^{\frac{1}{2}}$

$\Rightarrow \sqrt{X} = N^{\frac{1}{2}\frac{1}{3}} = N^{\frac{1}{6}} = N^{\frac{1}{3}}$

26. A granite block of 2 m × 5 m × 3 m size is cut into 5 cm thick slabs of 2 m × 5 m size. These slabs are laid over a 2 m wide pavement. What is the length of the pavement that can be covered with these slabs?

(a) 100 m **(b) 200 m** **(c) 300 m** **(d) 500 m**

Answer (c) **(CSIR-NET DEC 2012)**

Difficulty rating: 3/5 *Average time for solving: 100 seconds*

Number of granite slabs $= \frac{\text{Surface area of granite block}}{\text{Surface area of small slabs}} = \frac{2 \times 5 \times 3}{0.05 \times 2 \times 5} = \frac{30}{0.5} = 60.$

These 60 slabs are laid in the pavement.

Since 2m is the width of the pavement, the dimension 2m of each of the small slabs will exactly fit the width of the pavement.

Therefore, the length of the pavement is formed by the 5m in the dimension of each slab.

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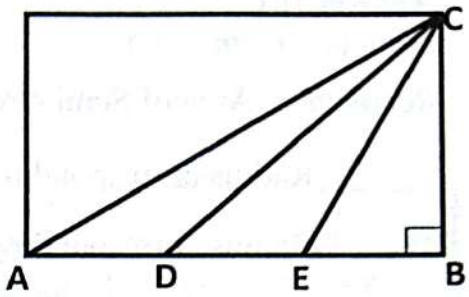
Hence, the total length of pavement covered with granite slabs = 60 × 5 = **300 m.**

27. In the figure, $\angle ABC = \frac{\pi}{2}$, AD = DE = EB. What is the ratio of the area of triangle ADC to that of triangle CDB?



(a) 1:1 **(b) 1:2**
(c) 1:3 **(d) 1:4**

Answer (b) **(CSIR-NET DEC 2012)**

Difficulty rating: 3.5/5 *Average time for solving: 110 seconds*

Remember: Area of triangle $= \frac{1}{2} \times$ Base × Height.

Thus, Area of $\Delta CDB = \frac{1}{2} \times DB \times BC$.

$\frac{\text{Area of } \Delta ADC}{\text{Area of } \Delta CDB} = ?$

Area of ΔADC = Area of ΔABC – Area of ΔDBC

$= (\frac{1}{2} \times AB \times BC) - (\frac{1}{2} \times DB \times BC)$

$= (\frac{1}{2} \times BC)(AB - DB)$ [Since $(\frac{1}{2} \times BC)$ is common in both the terms.]

$= (\frac{1}{2} \times AD \times BC)$ [Since AD = AB – DB]

Hence, $\frac{\text{Area of } \Delta ADC}{\text{Area of } \Delta CDB} = \frac{\frac{1}{2} \times AD \times BC}{\frac{1}{2} \times DB \times BC} = \frac{AD}{DB} = \frac{AD}{2AD} = \frac{1}{2}$

The fraction $\frac{1}{2}$ corresponds to the ratio **1:2**.

28. A string of diameter 1 mm is kept on a table in the shape of a close flat spiral. i.e., a spiral with no gap between turns. The area of the table occupied by the spiral is 1 m^2. Then the length of the string is

(a) 10 m **(b) 10^2 m** **(c) 10^3 m** **(d) 10^6 m**

Answer (c) **(CSIR-NET DEC 2012)**

Difficulty rating: 3.5/5 *Average time for solving: 90 seconds*

Given: Area of spiral = 1 m^2.

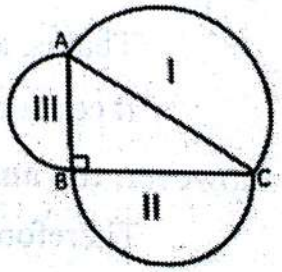
Width of string = Diameter of string = 1 mm = 10^{-3} m.

Area of spiral = Length of string × Width of string.

⇨ 1 = Length of string × 10^{-3}.

⇨ Length of string $= \frac{1}{10^{-3}} =$ **10^3 m.**

29. In the figure, $\angle ABC = \frac{\pi}{2}$. I, II, III are the areas of semicircles on the sides opposite to angles B, A and C respectively. Which of the following is always true?



(a) $II^2 + III^2 = I^2$ **(b) II + III = I**

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(c) $II^2 + III^2 > I^2$ **(d) II + III < I**

Answer (b) **(CSIR-NET DEC 2012)**

Difficulty rating: 3/5 *Average time for solving: 100 seconds*

Remember: Area of Semi circle $= \frac{\pi r^2}{2}$

Radius corresponding to I $= \frac{AC}{2}$

Radius corresponding to II $= \frac{BC}{2}$

Radius corresponding to III $= \frac{AB}{2}$

Thus,

$I = \frac{\pi}{2}(\frac{AC}{2})^2 = \frac{\pi}{8} AC^2$

$II = \frac{\pi}{2}(\frac{BC}{2})^2 = \frac{\pi}{8} BC^2$

$III = \frac{\pi}{2}(\frac{AB}{2})^2 = \frac{\pi}{8} AB^2$

Then, $II + III = \frac{\pi}{8} BC^2 + \frac{\pi}{8} AB^2 = \frac{\pi}{8}(BC^2 + AB^2)$

By Pythagoras Theorem, $AB^2 + BC^2 = AC^2$

Thus, $II + III = \frac{\pi}{8} AC^2 = I$

Hence, **II + III = I.**

30. Cucumber contains 99% water. Ramesh buys 100 kg of cucumbers. After 30 days of storing, the cucumbers lose some water. They now contain 98% water. What is the total weight of cucumbers now?

(a) 99 kg **(b) 50 kg** **(c) 75 kg** **(d) 2 kg**

Answer (b) **(CSIR-NET DEC 2012)**

Difficulty rating: 3.5/5 *Average time for solving: 100 seconds*

In 100 kg cucumber, 99% water is present.

⇨ Amount of water $= \frac{99}{100} \times 100 = 99$ kg.

Therefore, the amount of non-water matter = 100 – 99 = 1 kg.

Let x be the weight of cucumbers after 30 days.

Given: After 30 days of storing, the cucumbers lose some water.

That is, after 30 days, the mass of cucumber and water has decreased.

It contains 98% water and the rest 2% non-water matter.

However, **the amount of non-water matter has not changed.**

Therefore, the amount of the non-water matter is 1 kg itself.

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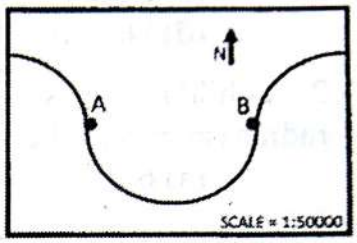
Thus, 2% corresponds to 1 kg of non-water matter.

Hence, $\frac{2}{100} \times x = 1$

$\Rightarrow x = \frac{100}{2} = \mathbf{50\ kg.}$

31. The map given shows a meandering river following a semi-circular path, along which two villages are located at A and B. The distance between A and B along the east-west direction in the map is 7 cm. What is the length of the river between A and B in the ground?



(a) 1.1 km **(b) 3.5 km** **(c) 5.5 km** **(d) 11 km**

Answer (c) **(CSIR-NET DEC 2012)**

Difficulty rating: 3.5/5 *Average time for solving: 110 seconds*

Given: The distance between A and B along the east-west direction in the map = 7 cm. (Figure 1).

Length of river between A and B in the ground is the circumference (perimeter) of the semi-circle between A and B.

Remember: Circumference (Perimeter) of semicircle = πr.

Here radius = $\frac{1}{2}$ × Distance between A and B along east-west direction.

$= \frac{1}{2} \times 7 = \frac{7}{2}$ cm.

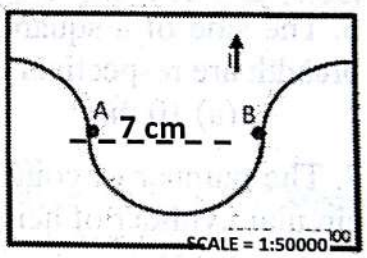


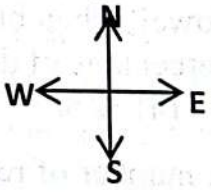
Figure 1

Hence, Circumference of semi-circle $= \frac{22}{7} \times \frac{7}{2} = 11$ cm.

Here the scale of the map is given as 1:50000.

1 : 50000 map means 1 cm = 50000 cm in the map.

Thus, the actual distance is 50000 times larger than that in the map.



Hence, length of river between A and B in the ground = 11 × 50000

= 550000 cm

= 5.5 km.

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12.3 Practice Questions

1. Of the following statements, the one that is incorrect is:
   (a) Doubling the base of a given rectangle doubles the area.
   (b) Doubling the altitude of a triangle doubles the area.
   (c) Doubling the radius of a given circle doubles the area.
   (d) Doubling the diagonal of a square quadruples the area.

2. A child reshapes a cone makeup of clay of height 24 cm and radius 6 cm into a sphere. The radius (in cm) of the sphere is
   (a) 6 (b) 12 (c) 24 (d) 48

3. A horse is placed for grazing inside a square field 12 m long and is tethered to one corner by a rope with a length which is two-third of the length of the field. The approximate area horse can graze is:
   (a) 48 m^2 (b) 50 m^2 (c) 51 m^2 (d) 54 m^2

4. How many cubes of 10 cm edge can be put in a cubical box of 1 m edge?
   (a) 10 (b) 100 (c) 1000 (d) 10000

5. The diameter of a sphere is 18 cm. It is melted and drawn into a wire of diameter 0.4 cm. The length of the wire is
   (a) 24.3 m (b) 121.5 m (c) 243 m (d) 250 m

6. The side of a square whose area is three times the area of a rectangle whose length and breadth are respectively 12 m and 4 m, is
   (a) 10 m (b) 11 m (c) 12 m (d) 14 m

7. The number of coins of radius 0.8 cm and thickness 0.2 cm to be melted to make a right circular cylinder of height 10 cm and base radius 4 cm is:
   (a) 580 (b) 620 (c) 780 (d) 1250

8. A towel, when bleached, was found to have lost 30% of its length and 20% of its breadth. The percentage of decrease in area is:
   (a) 32% (b) 44% (c) 48% (d) 50%

9. The number of revolutions of a wheel of diameter 60 cm makes in travelling a distance of 132 m, is:
   (a) 70 (b) 90 (c) 120 (d) 160

10. An equilateral triangle, a square and a circle have equal perimeters. If the areas of triangle, square and circle is T,S and C respectively, then:
   (a) $T < C < S$ (b) $S < T < C$ (c) $C < S < T$ (d) $T < S < C$

Answer Key

1. (c) 2. (a) 3. (b) 4. (c) 5. (c) 6. (c) 7. (d) 8. (b) 9. (a) 10. (d)

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CHAPTER 13

DATA INTERPRETATION

Average Difficulty Rating: 2.9/5

Average time for each question: 103 sec

Number of questions expected for each exam: 1-2

13.1 THEORY

Data interpretation is the process through which inferences are drawn about the data available for analysis. In other words, the process of drawing inferences and conclusions through the interpretation of data is what data interpretation is all about.

In the case of questions related to data interpretation, data will be given in the form of any of the below mentioned tools followed by questions pertaining to the same. These questions are intended to test the ability of the student to interpret the information presented and to select the appropriate data for answering a question.

Tips to solve data interpretation questions

- Read the data very carefully. Even the minutest word must not be overlooked since many a times even a single word or phrase could become critical.
- The easiest way to solve a data analysis question is to weed out the incorrect options from the given options. While doing so, stop the weeding out process at once you get the correct option. Do not try to check the next option if you get one correct answer. Always remember that time is very important.
- If there are more than one graphs or charts or tables, understand the relationship between them clearly before you proceed to solve the questions asked.
- Answer only the questions asked. Do not answer or calculate things which have not been asked for.
- The given data may be insufficient for interpreting some answer options. Avoid those options at once.

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For the data interpretation questions, data will be represented in any of the data presentation tools like tables, pie-charts, bar graphs etc. Therefore we can have an overview of the different types of questions using these data presentation tools.

13.1.1 Table

A table is a display of data arranged into rows and columns. Almost any quantitative information can be organized into a table. A table consists of horizontal rows and vertical columns. The heading for each row and column helps the reader understand the data and the units used for the same.

ILLUSTRATION 1:

A school has two divisions of Class X students, namely X A and X B. The results of half-yearly and annual examinations are shown in the table given below.

Result	Number of Students	
	Division A	Division B
Students failed in both Exams	9	8
Students failed in half-yearly but passed in Annual Exams	8	10
Students passed in half-yearly but failed in Annual Exams	13	14
Students passed in both Exams	20	23

Based on the above table which of the following statements is false?

(a) Total number of students in Class X is 105.
(b) Number of students in Division B is 55.
(c) Number of students in Class X who passed half yearly exam is 70.
(d) Number of students in Class X who passed annual exam is 68.

Answer (d)

Let us check each of the options.

Consider (a)

Since the classification of the students based on their results and divisions form independent groups, so the total number of students in the class:

$$= 9 + 8 + 13 + 20 + 8 + 10 + 14 + 23 = 105.$$

Hence, true.

Consider (b)

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Again, by the same above argument the number of students in Division B is the sum of all entries in the column corresponding to Division B.

Thus, Number of students in Division B = 8 + 10 + 14 + 23 = 55.

Hence, true.

Consider (c)

Number of students who passed half yearly exam =

(Number of students who passed in half-yearly but failed in Annual Exams)

+ (Number of students who passed in both Exams)

Thus, Number of students who passed half-yearly exam = 13 + 14 + 20 + 23 = 70.

Hence, true.

Consider (d)

Number of students who passed annual exam =

(Number of students who failed in half-yearly but passed in Annual Exams)

+ (Number of students who passed in both Exams)

Thus, Number of students who passed annual exam = 8 + 10 + 20 + 23 = 61.

Hence, false.

13.1.2 Bar chart

A bar chart or bar graph is a chart with rectangular bars with lengths proportional to the values that they represent. The bars can be plotted vertically or horizontally. Bar charts are used for plotting discrete (or discontinuous) data; that is data which has discrete value and is not continuous. Bar graphs consist of an axis and a series of labelled horizontal or vertical bars that show different values for each bar. The numbers along a side of the bar graph are called the scale. The important point to note about bar graphs is their bar length or height—the greater their length or height, the greater their value.

ILLUSTRATION 2:

The bar graph given above gives the sales of books (in 1000 numbers) from 6 branches of a publication company during two consecutive years 2014 and 2015.

Study the graph and choose the wrong option from the following?

(a) B4 sold more books in 2014 than that of B1 in 2015

(b) B3 sold more books than B2 in 2014.

(c) For B2, books sold in 2014 is more than that in 2015.

(d) Total books sold in 2015 is more than that sold in 2014.

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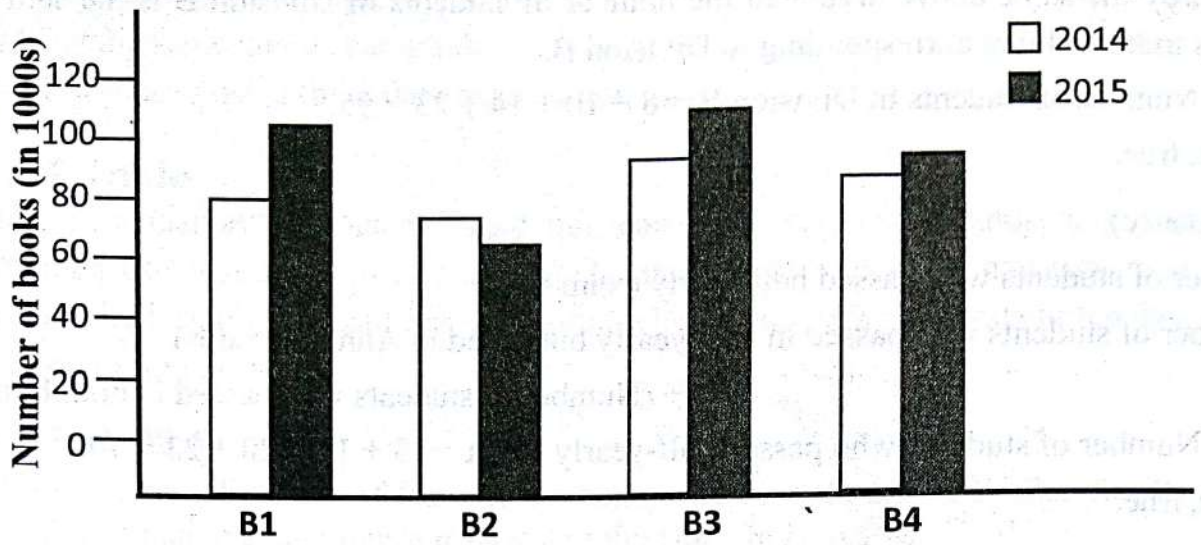

Answer (a)

Let us check each of the options.

Consider (a)

Wrong. Since B1 sold more than 100 books in 2015 and B4 sold less than 100 books in 2014.

Consider (b)

Correct. Since B3 sold more than 80 books in 2014 and B2 sold less than 80 books in 2014.

Consider (c)

Correct. Since B2 sold more than 70 books in 2014 and B2 sold less than 70 books in 2014.

Consider (d)

Correct. Clear from the graph as 3 out of 4 bars of 2015 have greater length than their corresponding bar in 2014.

13.1.3 Line graph

A line graph is a useful data presentation tool for showing a long series of data. Line graphs are also useful for comparing several different series of data in the same graph. Line graphs display data in two dimensions. We call the dimensions the x-axis and the y-axis.

By convention the dependent variable or y variable is on the vertical axis and the independent or x variable is on the horizontal axis. When reading a line graph, you will notice that rise and falls in the line show how one variable is affected by the other.

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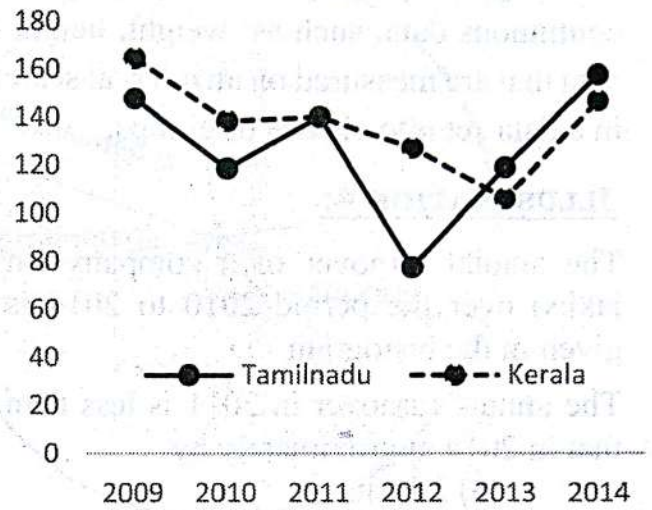
ILLUSTRATION 3:

The given graph shows the top mark obtained in CSIR-NET examination of a particular subject in different years. Study the graph and choose the correct statement among the following?



(a) Top-scorer in CSIR-NET examination was in Kerala during all the given years.

(b) Top score was shared by both states in the year 2013.

(c) Increase in top marks of Tamil Nadu during 2012-2013 is greater than the decrease in top marks of Kerala during the same period.

(d) Tamil Nadu had more top scorers than Kerala.

Answer (c)

Let us check each of the options.

Consider (a)

Wrong. Since Top scorer in 2013 and 2014 is in Tamil Nadu.

Consider (b)

Wrong. Since the graph does not coincide in a point in the year 2013.

Consider (c)

Correct. Since the uphill slope of the graph of Tamil Nadu during 2012-13 is larger than the downhill slope of the graph of Kerala during the same period.

Consider (d)

Wrong. Visibly, from the graph, Kerala has 4 and Tamil Nadu has 3 top scorers.

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13.1.4 Histograms

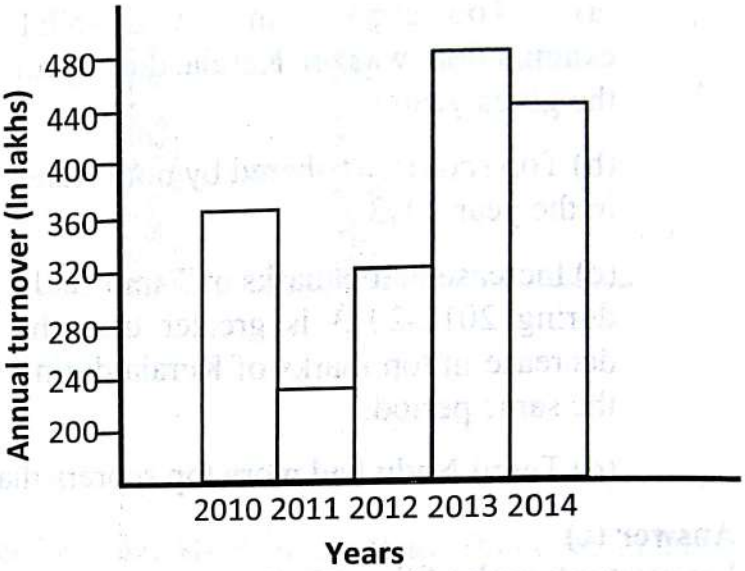
A histogram is a graphical display of data using bars of different heights. It shows a result of continuous data, such as: weight, height etc. It is used to summarize discrete or continuous data that are measured on an interval scale. A histogram divides up the range of possible values in a data set into classes or groups.

ILLUSTRATION 4:

The annual turnover of a company (in lakhs) over the period 2010 to 2014 is given in the histogram.

The annual turnover in 2011 is less than that in 2014 approximately by

(a) 2 lakhs
(b) 20 lakhs
(c) 2 crores
(d) 20 crores



Answer (c)

The annual turnover of 2014 is about 440 lakhs (4.4 crores).

The annual turnover of 2011 is about 240 lakhs (2.4 crores).

Hence the difference = 4.4 crores – 2.4 crores = **2 crores.**

13.1.5 Pie charts

A pie chart is a circular chart divided into sectors. In a pie chart, the arc length of each sector is proportional to the quantity it represents. When a pie chart is formed from a data, it breaks up a whole into its parts. The share of each part in a pie chart is proportionate to its share of the whole data.

As we know, a circle contains 360 degrees and the total value of the pie chart is 100%. The following formula is used for the calculation of the angle of a sector of the pie chart:

$$\text{Angle of a sector} = \frac{\text{Value of the sector}}{\text{Total value}} \times 360^{\circ}.$$

ILLUSTRATION 5:

Which cannot be concluded from the following figures?

(a) The expenditure towards entertainment is least.
(b) The total expenditure is same in both the years.
(c) The family spends more on clothes.

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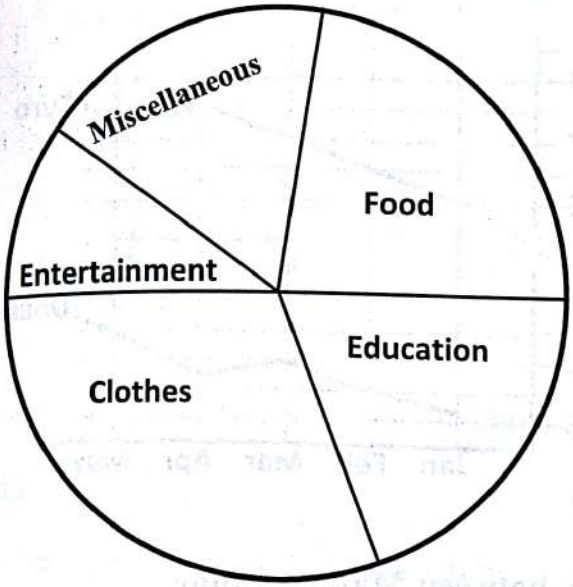
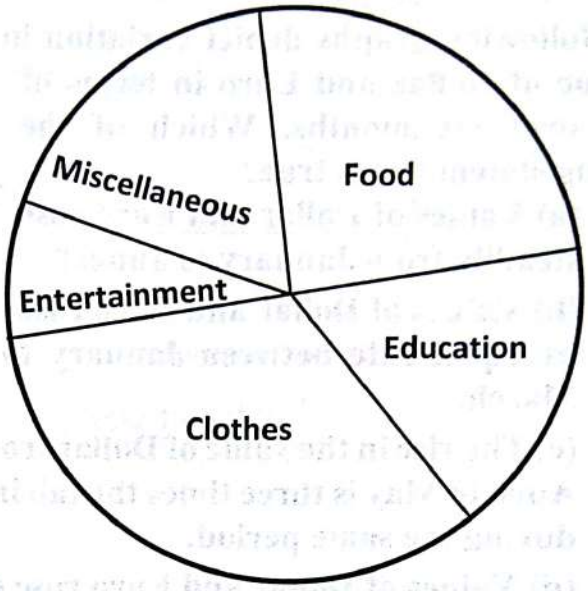
(d) Each year's expenditure on food is almost 25%.



Year 2014



Year 2015

Answer (b)

We can check each of the options to obtain the one which cannot be concluded.

Consider (a)

This can be concluded as the central angle and hence the region corresponding to entertainment is the smallest in both the pie charts.

Consider (b)

This **cannot be concluded** as the total expenditure may or may not be the same in the two pie charts.

Consider (c)

This can be concluded as the central angle and hence the region corresponding to clothes is the largest in both the pie charts.

Consider (d)

This can be concluded as the region of food covers about quarter of the circle in both the pie charts.

Other than the above-mentioned tools, there are different other data presentation tools. Scatter plot is one of them. The scatterplot is widely used to present measurements of two or more related variables. It is particularly useful when the variables of the y-axis are thought to be dependent upon the values of the variable of the x-axis (usually an independent variable).

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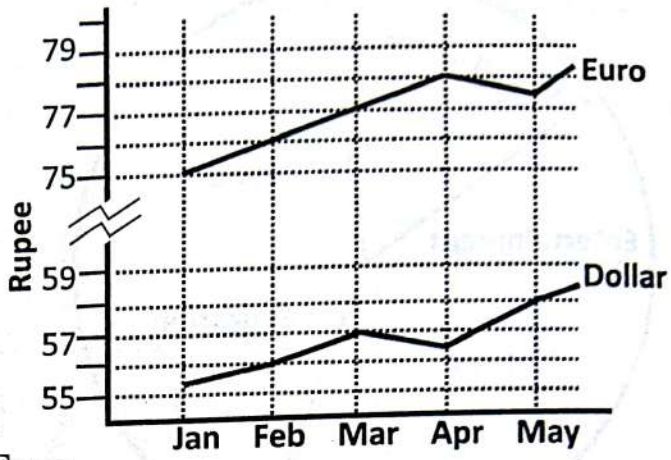
13.2 CSIR-NET Exam Questions – Detailed Solution

1. The following graphs depict variation in the value of Dollar and Euro in terms of Rupee over six months. Which of the following statements is true?



(a) Values of Dollar and Euro rose steadily from January to June.

(b) Values of Dollar and Euro rose by equal rate between January to March.

(c) The rise in the value of Dollar from April to May is three times the fall in Euro during the same period.

(d) Values of Dollar and Euro rose equally between May and June.

Answer (c) **(CSIR-NET DEC 2014)**

Difficulty rating: 2.5/5 *Average time for solving: 100 seconds*

For any graphical analysis question, weeding out the wrong options is the best method to solve. Therefore, let us check each of the answer options.

Consider (a)

From January to June, the graph of Dollar and Euro is not a straight line.

Hence, wrong.

Consider (b)

From January to March, Dollar rose from 55.5 to 57. That is, a difference of Rs.1.5

Euro rose from 75 to 77. That is, a difference of Rs.2

Thus, not equal rates. Hence, wrong.

Consider (c)

From April to May, Dollar rose from 56.5 to 58. That is, a difference of Rs.1.5

Euro fall from 78 to 77.5. That is, a difference of Rs.0.5

Obviously, $1.5 = 3 \times 0.5$. **Hence, correct.**

Consider (d)

From May to June,

Dollar rose from 58 to 58.5. That is, a difference of Rs.0.5

Euro rose from 77.5 to 78.5. That is, a difference of Rs.1

$0.5 \neq 1$. Hence, wrong.

Hence, **option (c).**

2. Which of the following figures best shows that y is inversely proportional to x?

(a) Graph (1) **(b) Graph (2)** **(c) Graph (3)** **(d) Graph (4)**

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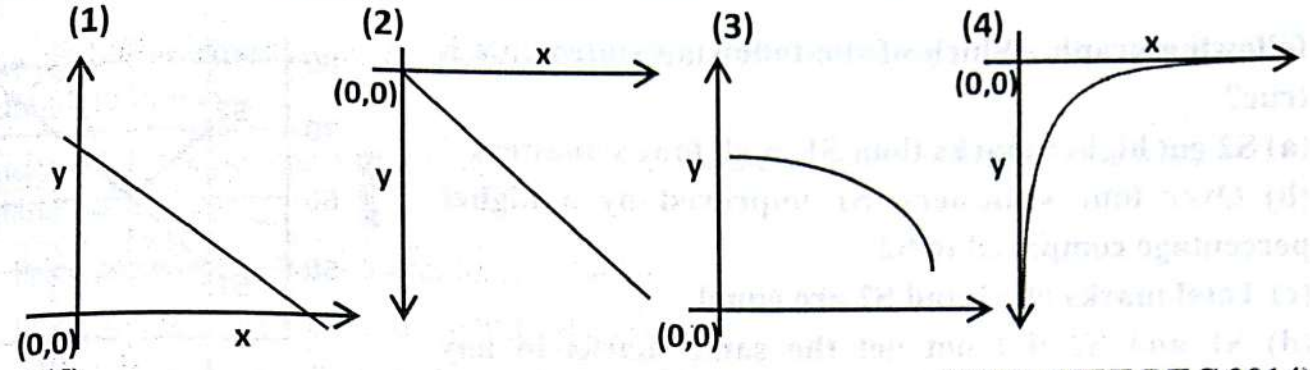


Answer (d) **(CSIR-NET DEC 2014)**

Difficulty rating: 4/5 *Average time for solving: 90 seconds*

y is inversely proportional to x means y is directly proportional to $\frac{1}{x}$.

That is, $y \alpha \frac{1}{x}$

⇨ $y = \frac{k}{x}$, where k is a constant.

⇨ $xy = k$ ------(1)

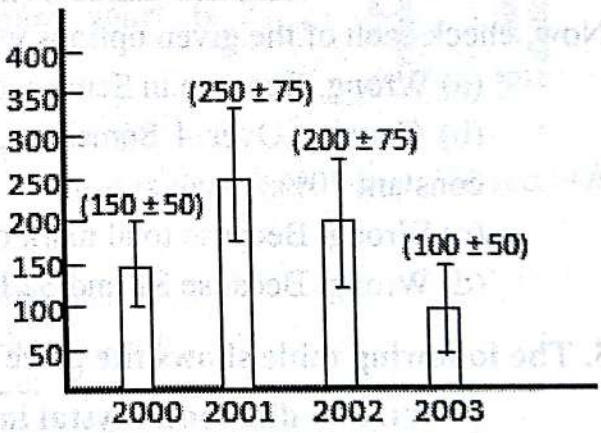
This is the equation of a hyperbola, which means the first two options are irrelevant, as they do not resemble a hyperbola.

Now check the given figures to get the best option, which satisfies this equation (1).

Hence, **option (d).**

3. Average yield of a product in different years is shown in the histogram. If the vertical bars indicate variability during the year, then during which year was the percent variability over the average of the year least?

(a) 2000 (b) 2001 (c) 2002 (d) 2003



Answer (b) **(CSIR-NET DEC 2014)**

Difficulty rating: 2/5 *Average time: 90 seconds*

Percent variability over the average of each year is given as,

In 2000, $\frac{50}{150} \times 100 = 33.33\%$

In 2001, $\frac{75}{250} \times 100 = 30\%$

In 2002, $\frac{75}{200} \times 100 = 37.5\%$

In 2003, $\frac{50}{100} \times 100 = 50\%$

Thus, the least percentage is in **2001.**

4. Marks obtained by two students S1 and S2 in a four-semester course are plotted in the

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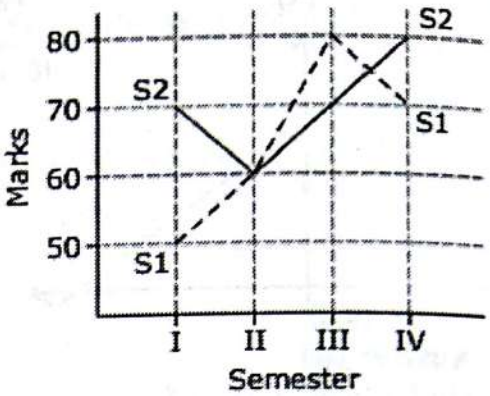
following graph. Which of the following statements is true?



(a) S2 got higher marks than S1 in all four semesters

(b) Over four semesters, S1 improved by a higher percentage compared to S2

(c) Total marks of S1 and S2 are equal

(d) S1 and S2 did not get the same marks in any semester

Answer (b) **(CSIR-NET JUNE 2014)**

Difficulty rating: 2.5/5 *Average time for solving: 120 seconds*

We can tabulate the data obtained from the given graph as shown below.

Student / Semester	S1	S2
1	50	70
2	60	60
3	80	70
4	70	80
Total	**260**	**280**
Average	**65**	**70**

Now, check each of the given options with reference to this table.

(a) Wrong. Because in Semester 3, S2 has lower marks than S1.

(b) Correct. Over 4 Semesters, S1 improved from 50% to 65%, while S2 was at constant 70%.

(c) Wrong. Because total mark of S1 is lower than that of S2.

(d) Wrong. Because S1 and S2 has same mark in Semester 2.

5. The following table shows the price of diamond crystals of a particular quality:

Wt. of a diamond crystal in carat	Price per carat (in lakhs)
1	4
2	8
3	12
4	16

What will be the price (in lakh Rs.) of a 2.5 carat diamond crystal?

(a) 10 **(b) 20** **(c) 25** **(d) 50**

Answer (c) **(CSIR-NET JUNE 2014)**

Difficulty rating: 3.5/5 *Average time for solving: 70 seconds*

The table gives the **price per carat** of diamond crystals with weight 1, 2, 3 and 4 carat respectively.

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Here the "price per carat" of the diamond crystal should not be wrongly understood as "price" of the diamond crystal.

Obviously, the price per carat of 2.5 carat diamond should be in between that of 2 carat and 3 carat diamond.

Given: Price per carat of 2 carat diamond = 8 lakh.

Given: Price per carat of 3 carat diamond = 12 lakh.

Thus **price per carat** of 2.5 carat diamond = 10 lakh.

That is, price of one carat of 2.5 carat diamond = 10 lakh.

Hence, price of a 2.5 carat of 2.5 carat diamond = 2.5 x 10 = 25 lakh.

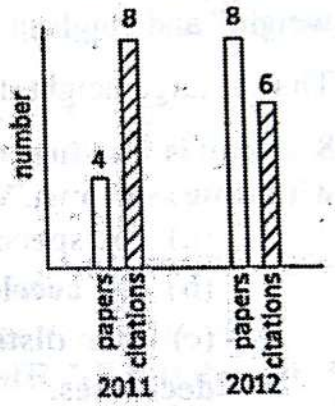
6. The plot depicts the number of research publications of a scientist along with the number of citations. Which of the following statements is not correct?

(a) In the year 2012, 50% more papers were published but citations decreased by 25%

(b) In the year 2012, 100% more papers were published but citations were 75% of the number of papers in that year.

(c) The papers published in the year 2011 are only 33.33% of the total number of papers in both years.

(d) The total number of citations for both years is 16.66% more than the total number of papers.



Answer (a)

(CSIR-NET DEC 2013)

Difficulty rating: 2.5/5 *Average time for solving: 130 seconds*

From the graph, we get the following data:

In 2011: Number of papers = 4. Number of citations = 8.

In 2012: Number of papers = 8. Number of citations = 6.

Percentage increase in papers = $(\frac{8-4}{4}) \times 100 = 100\%$.

Percentage decrease in citations = $(\frac{8-6}{8}) \times 100 = 25\%$.

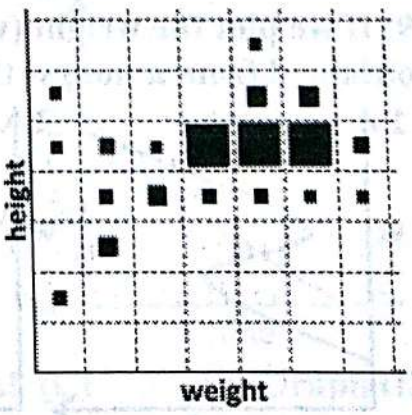
Now checking each option, with respect to the above data.

Evidently, **statement (a)** is wrong. Hence, **option (a).**

7. The distribution of heights and weights in a population is shown in a 2-parameter scatter plot. The size of the square is proportional to the number of persons having particular combination of weight and height. Which statement best describes the trend in the population?

(a) Height and weight are strongly correlated.

(b) Height and weight are anti-correlated.



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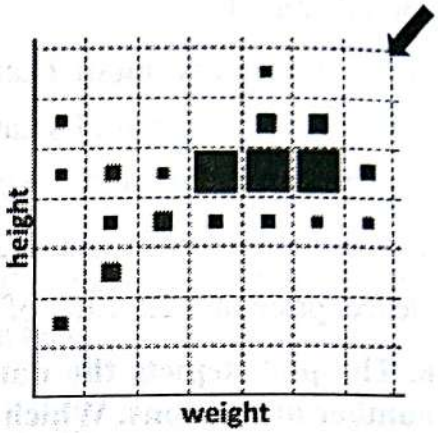
(c) Large heights do not imply proportionately large weights.
(d) Height and weight are independent characteristics.

Answer (c) **(CSIR-NET DEC 2013)**

Difficulty rating: 3.5/5 *Average time for solving: 120 seconds*

By checking each option, we can get the answer.

Given: The size of the square is proportional to the number of persons having particular combination of weight and height.



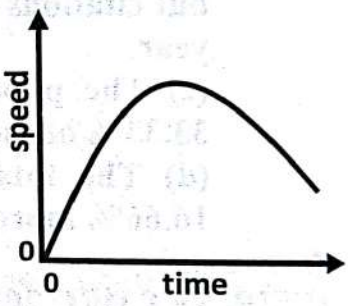
Thus, no square for a particular combination implies that there is no person with that particular combination of height and weight.

As indicated with the dark arrow in the figure, for the highest height and weight, there is no square.

This implies that there is no person possessing both "highest weight" and "highest height".

That is, large height does not imply large weight. Hence, **option (c).**

8. A car is moving along a straight track. Its speed is changing with time as shown. Which of the following statements is correct?



(a) The speed is never zero.
(b) The acceleration is zero once on the path.
(c) The distance covered initially increases and then decreases.
(d) The car comes back to its initial position once.

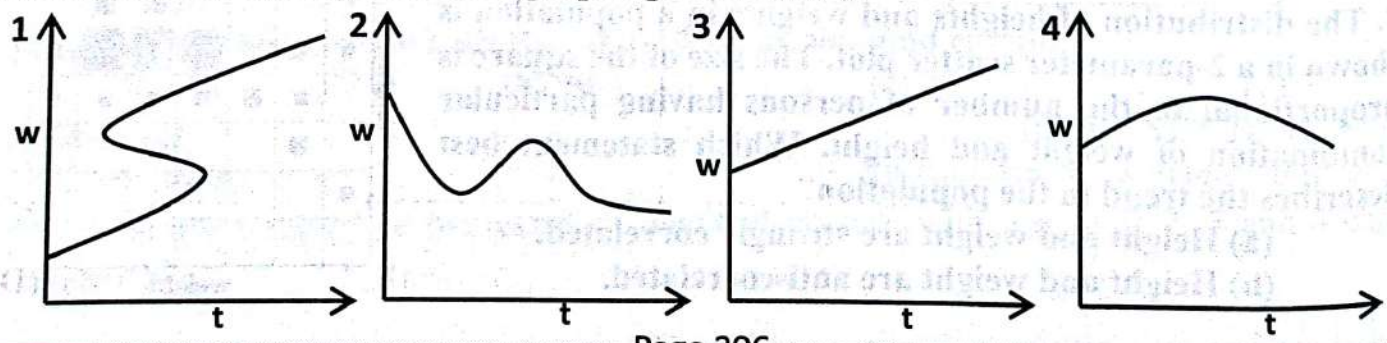
Answer (b) **(CSIR-NET DEC 2013)**

Difficulty rating: 3/5 *Average time for solving: 110 seconds*

We can check each of the options.

(a) Wrong. Because at time t = 0, the speed is zero.
(b) Correct. Only, at the peak point, the acceleration is zero.
(c) Wrong. Because in a speed-time graph, distance never decreases
(d) Wrong. Because from a speed-time graph, nothing can be said about the position of the car

9. If we plot the weight (w) versus age (t) of a child in a graph, the one that will never be obtained from amongst the four graphs given below is



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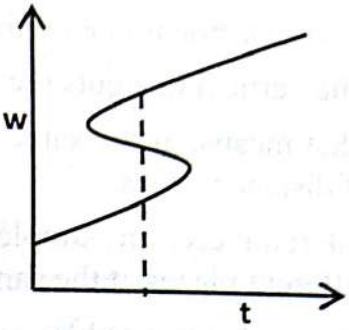
(a) 1 **(b) 2** **(c) 3** **(d) 4**

Answer (a) **(CSIR-NET JUNE 2013)**

Difficulty rating: 2.5/5 *Average time for solving: 80 seconds*

Consider graph (1)



Draw a vertical line on the graph as shown.

The vertical line cuts the graph at three distinct points.

That means, at the same instant of time, three distinct values of weight exists, which is impossible.

Thus, **graph 1** is impossible to obtain.

Hence, **option (a).**

10. In a customer survey conducted during Monday to Friday, of the customers who asked for child care facilities in super markets, 23% were men and the rest, women. Among them, 19.9% of the women and 8.8% of the men were willing to pay for the facilities.

A. What is the ratio of the men to women customers who wanted child care facilities?

B. If the survey had been conducted during the weekend instead, how will the result change?

With the above data,

(a) Only A can be answered. **(b) Only B can be answered.**

(c) Both A and B can be answered. **(d) Neither A nor B can be answered.**

Answer (a) **(CSIR-NET JUNE 2013)**

Difficulty rating: 2.5/5 *Average time for solving: 100 seconds*

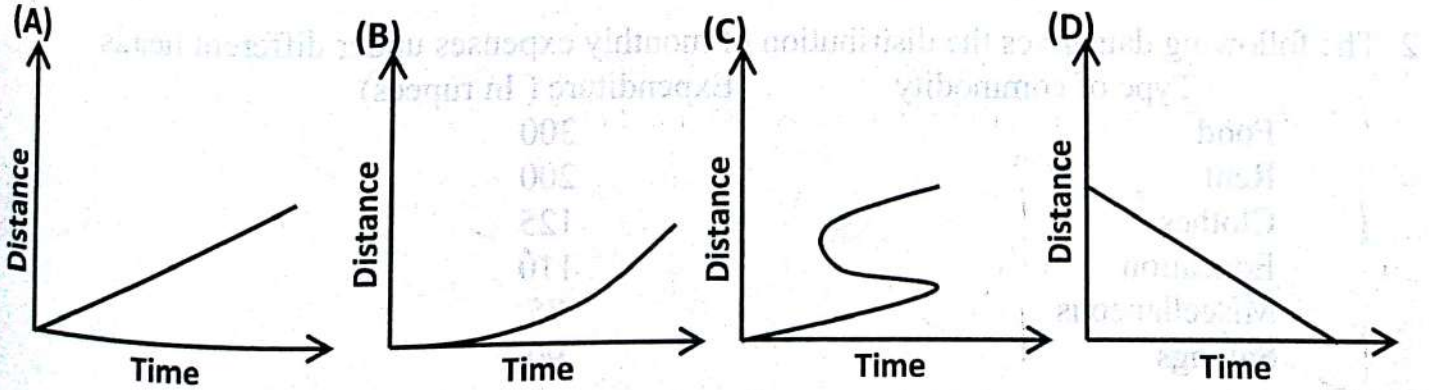
Clearly **A can be answered** since the percentage of men is 23 and that of women is 77.

Thus, the ratio of the men to women customers who wanted childcare facilities is 23:77.

However, no information about the trend in the weekend is given in the question.

Hence, **B cannot be answered**.

11. A student observes the movement of four snails and plots the graphs of distance moved as a function of time as given in figures (A), (B), (C) and (D). Which one of the following is not correct?



(a) Graph (A) **(b) Graph (B)** **(c) Graph (C)** **(d) Graph (D)**

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Answer (c) **(CSIR-NET DEC 2012)**

Difficulty rating: 2.5/5 *Average time for solving: 80 seconds*

Consider graph (C).

Draw a vertical line on the graph as shown in the figure.

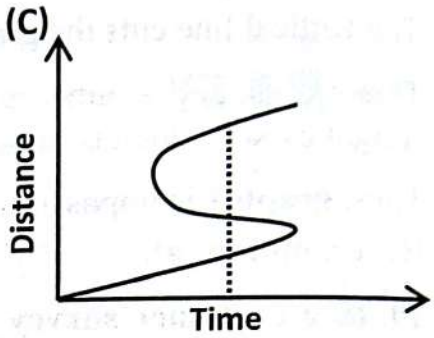
The vertical line cuts the graph at three distinct points.

That means, at the same instant of time, three distinct values of distance exists.

Alternatively, in simple words, the snail exists at three different places at the same instant of time!

Obviously impossible.

Thus, **graph (C)** is not correct.



13.3 PRACTICE QUESTIONS

1. In an office, the distribution of work hours is as shown in the following table

No. of staff members	Number of hours worked
5	0-19
12	20-24
25	25-29
40	30-34
15	35-39
8	40-45

Which among the following options is correct?

(a) At least 5 staff members worked more than 44 hours.
(b) The number of staff members working 30 hours or more is 60.
(c) The percentage of those who worked 35 or more hours is less than 25.
(d) The number of staff members working less than 25 hours is more than staff members working 35 hours or more.

2. The following data gives the distribution of monthly expenses under different heads

Type of commodity	Expenditure (In rupees)
Food	300
Rent	200
Clothes	125
Education	110
Miscellaneous	75
Savings	90

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A pie chart is drawn for the above data, the angle subtended at the centre for the item 'Rent' will be

(a) 20° (b) 80° (c) 85° (d) 120°

3. The age distribution of workers in a factory is as follows

Age (In years)	Number of workers
(20 – 28)	45
(28 – 36)	95
(36 – 44)	100
(44 – 52)	42
(52 – 60)	18

If 15% of the total strength starting from lowest age-group is retrenched and 20% of the total strength from the highest age groups is given premature retirement, then the age limits of workers retained in the factory is

(a) 20 – 36 (b) 28 – 44 (c) 28 – 52 (d) 36 – 52

4. The total number of sixes hit in successive IPLs are given below.

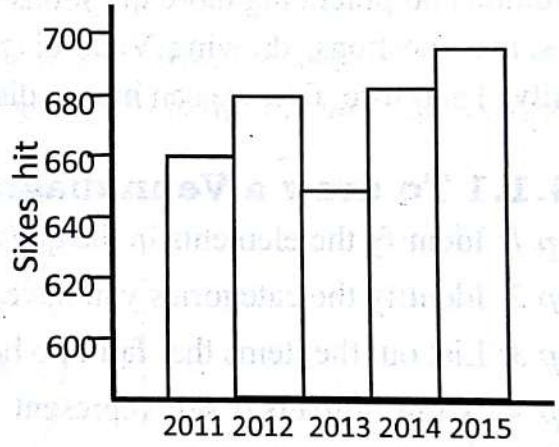
Which of the following is a wrong statement?

(a) Second largest number of sixes were hit in the year 2014.

(b) The least number of sixes were not hit in 2011.

(c) Number of sixes hit in 2015 is more than double the number hit in 2013.

(d) The difference in number of sixes in 2011 and 2013 is less than 40.



Answer Key

1.(c) 2. (b) 3. (b) 4. (c)

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CHAPTER 14

OBSERVATIONAL ABILITY

Average Difficulty Rating: 3/5

Average time for each question: 85 sec

Number of questions expected for each exam: 2-3

14.1 THEORY

Analytical and observational type questions are those that can only be worked out with your observational and analytical skills. For the same reason, there is no particular theory for this section. All we can do is to get a command over these kinds of tricky questions through practice. It should be noted that most of the questions in this section can be done in less than a minute and practicing more questions will help you acquire that speed.

In some questions, drawing Venn-diagrams can help you to visualize the given condition easily. Therefore, first we can have a discussion on how to draw a Venn-diagram.

14.1.1 To draw a Venn diagram:

Step 1: Identify the elements in the question.

Step 2: Identify the categories you have to put these items into.

Step 3: List out the items that fall into both categories, each category, and neither.

Step 4: Draw universal set, represent it as a rectangle. Mark number of elements in the universal set.

Step 5: Draw circles corresponding to each category you identified inside the universal set. Label each circle according to the category you want it to represent. Enter number of items common to each category in the overlapping portion of circles with the respective labels. Insert other details in other part of the circle. Also mark number of items not belonging to any (= number of objects in universal set - sum of items in categories) in the space outside of these circles and inside universal set (if required).

ILLUSTRATION 1:

In a class of 50 students, 18 take Chorus, 26 take Band, and 2 take both Chorus and Band. How many students in the class are not enrolled in either Chorus or Band?

(a) 10 (b) 16 (c) 6 (d) 8

Answer (d)

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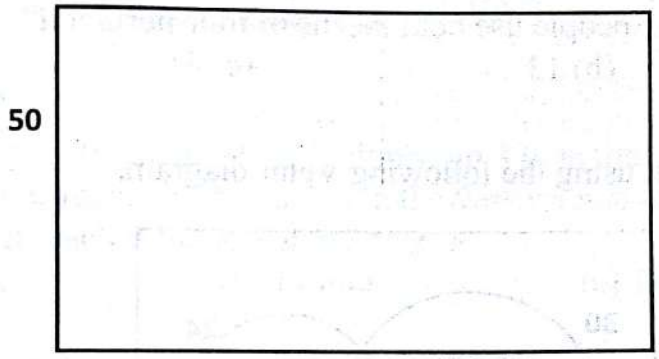
Step 1: We have 50 objects with us.

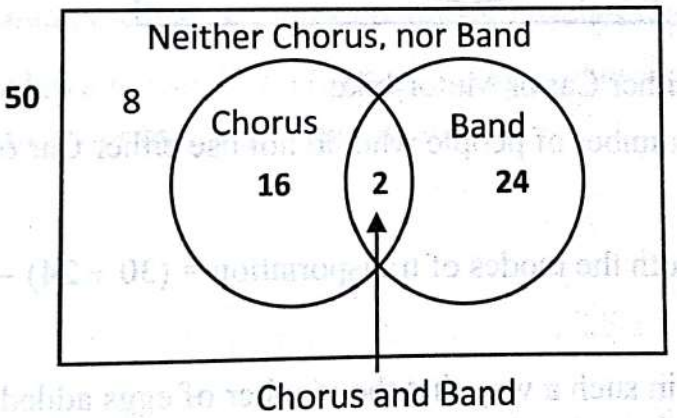
Step 2: Categories are Chorus and Band.

Step 3: Total: 50
Chorus: 18, Band: 26
Chorus and Band: 2
Chorus alone: $18 - 2 = 16$
Band alone: $26 - 2 = 24$.

Step 4: Draw universal set and mark total number of elements



Step 5: Draw circles corresponding to each category that was identified and name them. Fill in the number of people in each category, that which belongs to both of them in the common portion of the circle.



Clearly, 8 students neither belongs to Chorus, nor belongs to Band.

ILLUSTRATION 2:

If m students use a total of m pages of stationery in m days, then 100 students will require 100 pages of stationery in

(a) 100 days (b) m/100 days (c) 100/m day (d) m days

Answer (d)

m students uses m pages in m days.

Suppose m = 1. Then,

1 student uses 1 page in 1 day.

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2 students use 2 pages in **1** day.

Similarly,

100 students use 100 pages in **1** day.

1 is the value we took for 'm'.

Therefore, if m students use a total of m pages of stationery in m days, then 100 students will require 100 pages of stationery in **m** days.

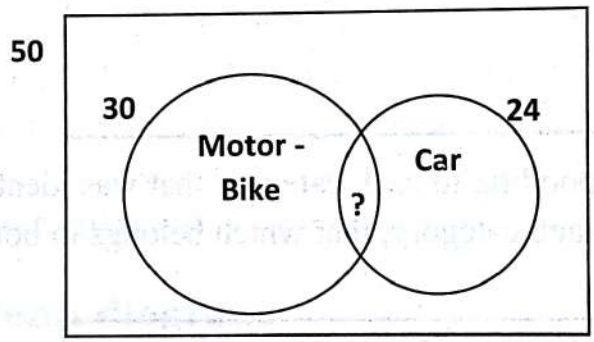
ILLUSTRATION 3:

In a housing colony of 50 people, 30 people use motor-bike, and 24 use car and 9 use neither car nor motor-bike. How many people use both means of transportation?

(a) 4 (b) 13 (c) 41 (d) 54

Answer (b)

The given data can be pictured using the following Venn-diagram.



Number of people who use either Car or Motor-bike

= Total number of people - Number of people who do not use either Car or Bike.

$= 50 - 9 = 41.$

Number of people who use both the modes of transportation $= (30 + 24) - 41 = 54 - 41 =$ **13.**

ILLUSTRATION 4:

A man fills basket with eggs in such a way that the number of eggs added on each successive day is the same as the number already present in the basket. This way the basket gets completely filled in 10 days. After how many days the basket was $1/4^{th}$ full.

(a) 4 (b) 5 (c) 8 (d) 9

Answer (c)

Given: Number of eggs added each day = Number of eggs already present in the basket.

For understanding the pattern of the progression, we can start with one egg.

That is suppose that at first one egg was already present in the basket.

The man would add one more egg in the basket.

Therefore, after the first day the number of eggs becomes 2.

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Similarly, in the second day it becomes 4.

In the third day, it becomes 8 and so on…

Thus, the number of eggs in the basket on a particular day is double the number of eggs in the previous day.

Given: In the 10^{th} day, the basket gets completely filled.

The number of eggs in the 10^{th} day is double the number of eggs in the 9^{th} day.
Thus on the 9^{th} day, the basket was $1/2^{th}$ full.

The number of eggs in the 9^{th} day is double the number of eggs in the 8^{th} day.
Thus on the **8^{th} day**, the basket was $1/4^{th}$ full.

ILLUSTRATION 5:

An insect climbs up a 10 m high pole. It climbs up 3 m in the first minute but slips down 2 m in the second minute, again climbs up 3 m in the third minute and slips down 2 m in the fourth minute and so on. It reaches the top of the pole in

(a) 15 min (b) 17 min (c) 19 min (d) 20 min

Answer (c)

The insect has to climb a distance of 10 m.

Distance covered by insect in every two minutes = 3 m – 2 m = 1 m.

Therefore, distance covered by the insect in the first 18 minutes = 9 × 1 m = 9 m.

Now in the *19^{th} minute*, the insect climbs up 3m and reaches the top of the pole.

Note that we do not have to consider what happens in the 20^{th} minute as the insect has covered the distance it had to cover by the 19^{th} minute itself.

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14.2 CSIR-NET Exam Questions – Detailed Solution

1. Three volumes of a Hindi book, identical in shape and size, are next to each other in a shelf, all up-right, so that their spines are visible, left to right: I, II, III. A worm starts eating from the outside front cover of volume I, and eats its way horizontally to the outside back cover of volume III. What is the distance travelled by the worm, if each volume is 6 cm thick?

(a) 6 cm (b) 12 cm (c) 18 cm (d) a little more than 18 cm

(CSIR-NET DEC 2014)

Answer (a)

Difficulty rating: 3.5/5

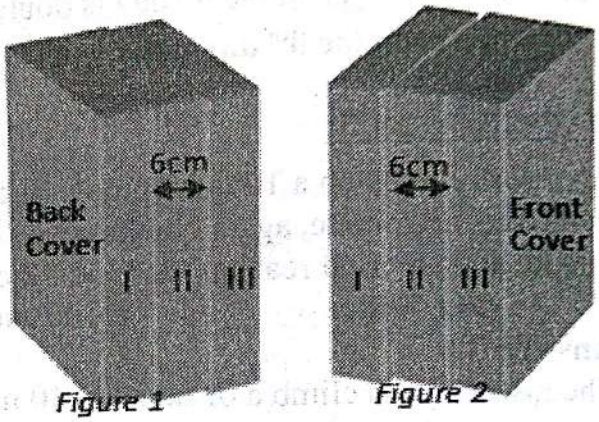
Average time for solving: 110 seconds

Imagine that we are observing the only three books which are kept up-right in a shelf.

Looking from the left, we see the outside **back cover of volume I** (Figure 1).

Looking from the right, we see the outside **front cover of volume III** (Figure 2).

Given that, the worm starts eating from the **outside front cover of volume I**, and eats its way horizontally to the **outside back cover of volume III**.



Therefore, the only distance the worm covered is the thickness of Volume II, that is, **6 cm.**

2. Two platforms are separated horizontally by a distance A and vertically by a distance B. They are to be connected by a staircase having identical steps. If the minimum permissible length is a, and the maximum permissible step height is b, the number of steps the staircase can have is?

(a) $\geq B/b$ (b) $\leq A/a$ (c) $\geq B/b$ and $\leq A/a$ (d) $\leq B/b$ and $\geq A/a$

Answer (c)

(CSIR-NET DEC 2014)

Difficulty rating: 3.5/5

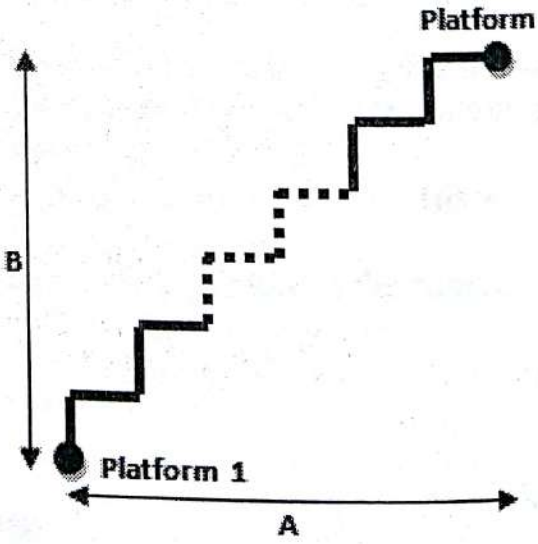
Average time for solving: 110 seconds

The given data can be pictured as shown in the figure.

Since, the minimum permissible length is a, the horizontal length of each step is more than or equal to a.

That is, let the horizontal length of each step be (a+h), where h is a non-negative number.

Since, the maximum permissible step height is b, the vertical height of each step is less than or equal to b.



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That is, let the vertical height of each step be (b-k), where k is a non-negative number.

Let n be the number of steps.

Then $n(a+h) = A$

$\Rightarrow n = \frac{A}{(a+h)}$

$\Rightarrow \mathbf{n \leq \frac{A}{a}}$, since h is a non-negative number.

Also $n(b-k) = B$

$\Rightarrow n = \frac{B}{(b-k)}$

$\Rightarrow \mathbf{n \geq \frac{B}{b}}$, since k is a non-negative number.

3. Two solid iron spheres are heated to 100°C and then allowed to cool. One has the size of a football, the other has the size of a pea. Which sphere will attain the room temperature (constant) first?

(a) The bigger sphere **(b) The smaller sphere**

(c) Both spheres will take same time **(d) It will depend on the room temperature**

Answer (a) **(CSIR-NET DEC 2014)**

Difficulty rating: 3/5 *Average time for solving: 50 seconds*

More the surface area, more will it be exposed to the room temperature.

Hence, more quickly it will attain the room temperature.

The bigger sphere has the larger surface area.

Thus, **the bigger sphere** will attain room temperature first.

4. A sphere is made up of very thin concentric shells of increasing radii (leaving no gaps). The mass of an arbitrary chosen shell is

(a) equal to the mass of the preceding shell **(b) proportional to its volume**

(c) proportional to its radius **(d) proportional to its surface area**

Answer (b) **(CSIR-NET DEC 2014)**

Difficulty rating: 3.5/5 *Average time for solving: 60 seconds*

We know that, Mass = Density × Volume.

Since the shell is of the same substance, the density is all the same.

Thus mass depends only on **volume of the shell**. Hence, **option (b).**

5. A 4 × 4 magic square is given. How many 2 × 2 squares are there in it whose elements add up to 34?

1	15	14	4
12	6	7	9
8	10	11	5
13	3	2	16

(a) 6 **(b) 9**

(c) 4 **(d) 5**

Answer (d) **(CSIR-NET JUNE 2014)**

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Difficulty rating: 2.5/5 *Average time for solving: 100 seconds*

1	15	14	4
12	6	7	9
8	10	11	5
13	3	2	16

Count the number of squares whose sum of elements is 34.

The squares are marked in the figure.

Clearly, there are **5** squares satisfying the given condition.

6. In the growing years of a child, the height increases as the square root of the age while the weight increases in direct proportion to the age. The ratio of the weight to the square of the height in this phase growth

(a) is constant **(b) reduces with age** **(c) increases with age**

(d) is constant only if the weight and height at birth are both zero

Answer (d) **(CSIR-NET JUNE 2014)**

Difficulty rating: 3.5/5 *Average time for solving: 100 seconds*

Let h, w, a be respectively height, weight and age of the child.

Given: $h \propto \sqrt{a}$

$\Rightarrow h = k_1\sqrt{a}$ ---------(1)

Given: $w \propto a$

$\Rightarrow w = k_2 a$ ----------(2)

$$\frac{w}{h^2} = \frac{k_2}{k_1^{\,2}} = \text{a constant.}$$

However, at birth, $\frac{w}{h^2}$ is not this constant as there is some weight and height at birth.

Hence, option **(d).**

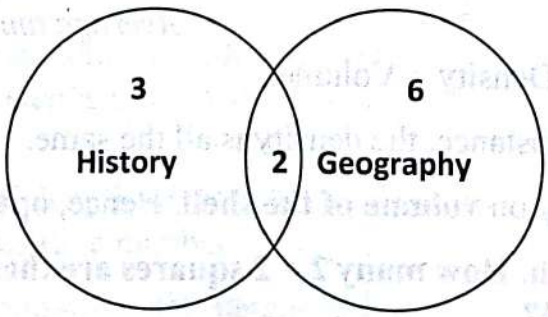
7. In a class of 10 students, 3 failed in History, 6 failed in Geography and 2 failed in both. How many passed in both the subjects?

(a) 1 **(b) 2** **(c) 3** **(d) 0**

Answer (c) **(CSIR-NET DEC 2013)**

Difficulty rating: 2.5/5 *Average time for solving: 60 seconds*

The given data can be pictured as the following Venn-diagram.



Number of students passed in both subjects

= Total number - Number of students failed in at least one subject.

Number of students failed in at least one subject = 6 + 3 – 2 = 7.

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Thus number of students passed in both subjects = 10 – 7 = **3.**

8. Amoebae are known to double in 3 min. Two identical vessels A and B, respectively contain one and two amoebae to start with. The vessel B gets filled in 3 hours. When will A get filled?

(a) 3 hours (b) 2 hours and 57 minutes (c) 3 hours 3 minutes (d) 6 hours

Answer (c) **(CSIR-NET DEC 2013)**

Difficulty rating: 3/5 *Average time for solving: 80 seconds*

Let the starting time t = 0.

At time t = 0 min:

Number of amoebas in Vessel A = 1.

Number of amoebas in Vessel B = 2.

Number of Amoebas is doubled in every 3 minutes.

Thus, at time t = 3 min:

Number of amoebas in Vessel A = 2.

Number of amoebas in Vessel B = 4.

Thus after 3 min, number of amoebas in vessel A will be same as that in vessel B at 0 minutes.

Vessel B takes 3 hours to get filled.

Hence, Vessel A will take **3 hours and 3 minutes** to get filled.

9. A cylinder of radius 1 cm and height 1 cm is broken into three pieces. Which of the following MUST be true?

(a) At least one piece has volume equal to 1 cm^3.

(b) At least two pieces have equal volumes.

(c) At least one piece has volume less than 1 cm^3.

(d) At least one piece has volume greater than 1 cm^3.

Answer (d) **(CSIR-NET DEC 2013)**

Difficulty rating: 3/5 *Average time for solving: 100 seconds*

Volume of the cylinder = $\pi r^2 h = \pi \times 1^2 \times 1 = \pi \approx 3.14$.

Now checking each options,

(a) Wrong. Because 1.1, 1.03, 1.01 is a possible set of volumes.

(b) Wrong. Because 1.1, 1.03, 1.01 is a possible set of volumes.

(c) Wrong. Because 1.1, 1.03, 1.01 is a possible set of volumes.

(d) Correct. If all pieces have volume less than 1 cm^3, the volume of the cylinder will be less than 3 cm^3.

Hence, **option (d).**

10. If a + b + c + d + e = 10 (all positive numbers), then the maximum value of a × b × c × d × e is

(a) 12 (b) 32 (c) 48 (d) 72

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Answer (b) **(CSIR-NET DEC 2013)**

Difficulty rating: 3/5 *Average time for solving: 90 seconds*

Note: *$a \times b \times c \times d \times e$ is maximum, for that set of {a, b, c, d, e), such that the maximum of {a, b, c, d, e} is minimum.*

Here, maximum of {a, b, c, d, e} is minimum for {2, 2, 2, 2, 2}.

That is, $a \times b \times c \times d \times e = 2 \times 2 \times 2 \times 2 \times 2 =$ **32.**

11. A farmer gives 7 full, 7 half-full and 7 empty bottles of honey to his three sons and asks them to share these among themselves such that each of them gets the same amount of honey and the same number of bottles. In how many ways can this be done? (Bottles cannot be distinguished otherwise, they are sealed and cannot be broken).

(a) 0 **(b) 1** **(c) 2** **(d) 3**

Answer (c) **(CSIR-NET DEC 2013)**

Difficulty rating: 3.5/5 *Average time for solving: 120 seconds*

There are three sons and each of them should get same amount of honey and same number of bottles.

Total amount of honey = 7 Full + 7 Half-full = 10.5 units

Hence each son should get $\frac{10.5}{3} = 3.5$ units of honey.

Total number of bottles = **7 full, 7 half-full and 7 empty bottles = 21 bottles.**

Hence, each son should get 7 bottles.

For that, the only **two** arrangements possible are,

	Full	Half-full	Empty	Honey	Bottles
Son 1	3 Bottles	1 Bottle	3 Bottles	3.5 units	7
Son 2	2 Bottles	3 Bottles	2 Bottles	3.5 units	7
Son 3	2 Bottles	3 Bottles	2 Bottles	3.5 units	7
Total	7	7	7	10.5 units	

	Full	Half-full	Empty	Honey	Bottles
Son 1	1 Bottle	5 Bottles	1 Bottle	3.5 units	7
Son 2	3 Bottles	1 Bottle	3 Bottles	3.5 units	7
Son 3	3 Bottles	1 Bottle	3 Bottles	3.5 units	7
Total	7	7	7	10.5 units	

12. A stream of ants goes from a point A to point B and returns to A along the same path.

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All the ants move at a constant speed and from any given point, 2 ants pass per second one way. It takes 1 minute for an ant to go from A to B. How many returning ants will an ant meet in its journey from A to B?

(a) 120 **(b) 60** **(c) 240** **(d) 180**

Answer (c) **(CSIR-NET JUNE 2013)**

Difficulty rating: 3.5/5 *Average time for solving: 110 seconds*

Let AB be the line segment through which the ants move.

To understand this question clearly, divide AB into 60 points (corresponding to the 60 seconds).

Given that, from any given point 2 ants pass per second one way.

Therefore, if the ant is not moving, it would have seen 120 returning ants.

However, since the ant is also moving at the same speed, it will see 2 more ants when it is moving from each point to the other.

Hence, it will see **240** ants in total.

13. A large tank filled with water is to be emptied by removing half of the water present in it every day. After how many days will there be closest to 10% water left in the tank?

(a) One **(b) Two** **(c) Three** **(d) Four**

Answer (c) **(CSIR-NET JUNE 2013)**

Difficulty rating: 2.5/5 *Average time for solving: 60 seconds*

Percentage of water left after each day

After Day 1 = 50 %

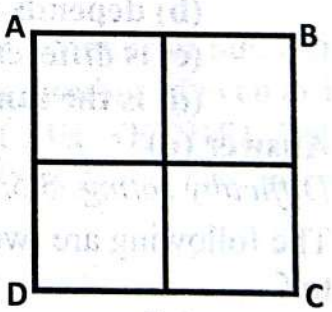
After Day 2 = 25 %

After Day 3 = 12.5 %

After Day 4 = 6.25 %

Clearly, 12.5 is more close to 10 than 6.25. Hence, **option (c).**

14. An ant goes from A to C in the figure crawling only on the lines and taking the least length of path. The number of ways in which it can do so is

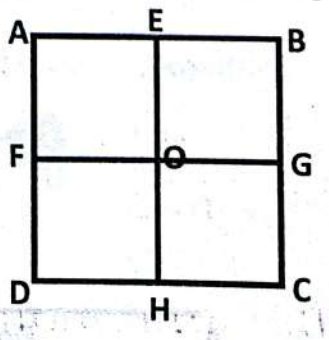


(a) 2 **(b) 4** **(c) 5** **(d) 6**

Answer (d) **(CSIR-NET JUNE 2013)**

Difficulty rating: 2.5/5 *Average time for solving: 80 seconds*

For convenience, label the middle points in between A, B, C, D as E, F, G, H and central point as O (See figure).



Here the distance of A, B, C, D to their corresponding middle points (E, F, G, H) are all the same.

Therefore, we can see that there are **6** different ways, namely:

ABC

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ADC
AEHC
AFGC
AEOGC
AFOHC

Evidently, these are the only ways. Hence, **option (d).**

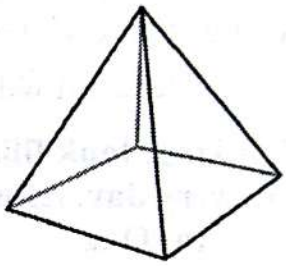
15. A square pyramid is to be made using a wire such that only one strand of wire is used for each edge. What is the minimum number of times that the wire has to be cut in order to make the pyramid?

(a) 3 **(b) 7** **(c) 2** **(d) 1**

Answer (d) **(CSIR-NET JUNE 2013)**

Difficulty rating: 2.5/5 *Average time for solving: 80 seconds*

The shape of a square pyramid is as shown in the figure.



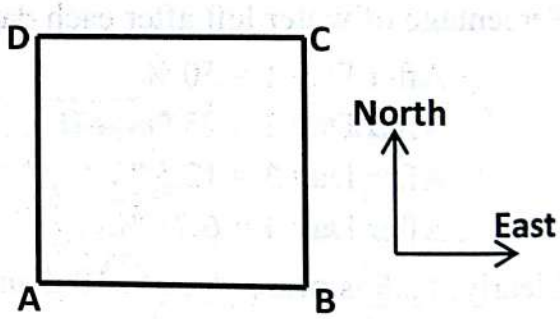
The question is analogous to finding the minimum number of times the pen has to be taken from the paper in order to trace this figure on a paper without retracing any path.

Clearly, the pen should be taken from the paper at least once.

Hence, the wire has to be cut a minimum of **1** time.

16. A person has to cross a square field by going from A to C. The person is only allowed to move towards the east or towards the north or use a combination of these movements. The total distance travelled by the person

(a) depends on the length of each step
(b) depends on the total number of steps
(c) is different for different paths
(d) is the same for all paths



Answer (d) **(CSIR-NET JUNE 2013)**

Difficulty rating: 3.5/5 *Average time for solving: 100 seconds*

The following are two among the many possible ways in which the person can move from A to C.

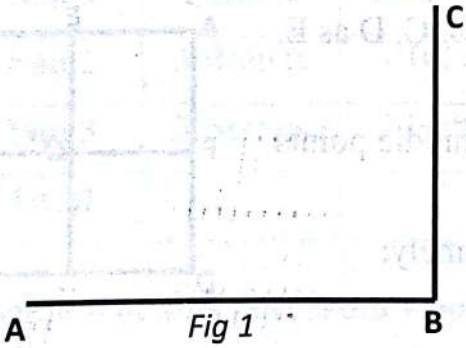


Fig 1

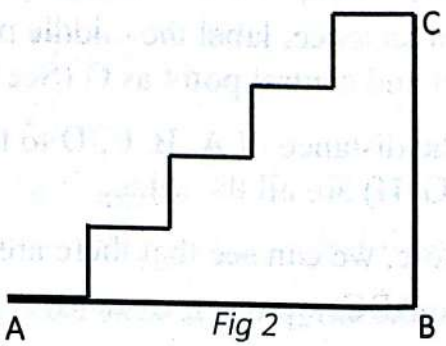


Fig 2

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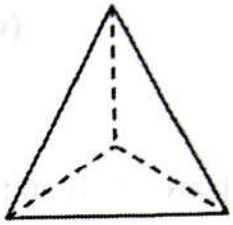
Clearly, by these two ways, the distance travelled is the same as the horizontal lengths in Figure 2 adds to the length AB in Fig 1 and vertical lengths in Fig 2 adds up to the length BC.

Similarly, any other path has the same length as in Figure 1.

Thus, total distance travelled by the person is the same for all paths.

17. A wire of length 6 m is used to make a tetrahedron of each edge 1 m, using only one stand of wire for each edge. The minimum number of times the wire has to be cut is



(a) 2 **(b) 3** **(c) 1** **(d) 0**

Answer (c) **(CSIR-NET DEC 2012)**

Difficulty rating: 2.5/5 *Average time for solving: 80 seconds*

Solving this question is analogous to tracing the path of this figure with a pen on a paper and checking the minimum number of times we take the pen from the paper to complete the figure, without retracing any path.

Clearly, the pen has to be taken at least once. Hence, **option (c).**

18. If 100 cats catch 100 mice in 100 minutes, then how long will take for 7 cats to catch 7 mice?

(a) 100 / 7 minutes **(b) 100 minutes** **(c) 49 / 100 minutes** **(d) 7 minutes**

Answer (b) **(CSIR-NET DEC 2012)**

Difficulty rating: 3/5 *Average time for solving: 60 seconds*

100 cats catch 100 mice in 100 minutes.

That is, 1 cat catches 1 mouse in **100** minutes.

Thus, 7 cats catch 7 mice in **100** minutes.

19. The cities of a country are connected by intercity roads. If a city is directly connected to an odd number of other cities, it is called an odd city. If a city is directly connected to an even number of other cities, it is called an even city. Then which of the following is impossible?

(a) There are even number of odd cities **(b) There are odd number of odd cities**

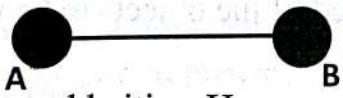
(c) There are even number of even cities **(d) There are odd number of even cities**

Answer (b) **(CSIR-NET DEC 2012)**

Difficulty rating: 3.5/5 *Average time for solving: 130 seconds*

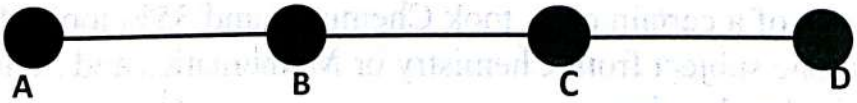
We can check each option and deduce the impossible one.

(a)



Here the two cities A and B are odd cities. Hence even number of odd cities is possible.

(c)



Here, B and C are connected to two cities.

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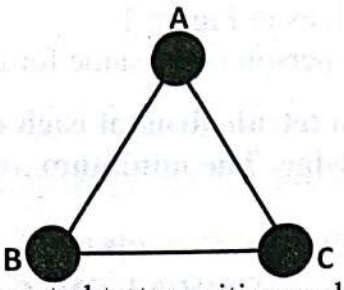
Therefore, B and C are even cities.

Hence, even number of even cities is possible.

(d)



Here, A, B and C are connected to two cities each.

Therefore, all three of them are even cities.

Hence, odd number of even cities is possible.

Thus, three out of four possibilities among the options are possible.

Hence, the remaining **option (b)** is impossible and hence the answer.

Quick Solution (For Mathematics Students)
By first theorem of graph theory, in a graph, there will always be even number of odd vertices.

Hence, odd number of odd cities is impossible. Hence, option **(b).**

20. In a museum there were old coins with their respective years engraved on them, as follows.

A. 1837 AD B. 1907 AD C. 1947 AD D. 200 BC

Identify the fake coin(s)

(a) coin A (b) coin D (c) coin A and B (d) coin C

Answer (b) **(CSIR-NET DEC 2012)**

Difficulty rating: 2.5/5 *Average time for solving: 60 seconds*

At the time of 200 BC, nobody knew that it was **"BC"**, as the Calendar was not divided to BC (Before Christ) and AD (After Christ) before the birth of Christ.

Thus, **200 BC** is the fake coin. Hence, **option (b).**

14.3 Practice Questions

1. With the use of the three different weights, namely, 1 gms, 3 gms and 9 gms, how many objects of different weights can be weighed, if the objects to be weighed and the given weights may be placed in either pan of the scale?

(a) 9 (b) 11 (c) 13 (d) 15

2. 50% of the students of a certain class took Chemistry and 35% took Mathematics. If each student took at least one subject from Chemistry or Mathematics and 30 took both, then total number of students in the class is:

(a) 200 (b) 240 (c) 250 (d) 320

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3. A monkey tries to ascend a greased pole 24 metres high. He ascends 3 metres in first minute and slips down 2 metre in the alternate minute. If he continues to ascend in this fashion, how long does he take to reach the top?

(a) 22nd minute (b) 24th minute (c) 43rd minute (d) 48th minute

4. The six faces of a cube are coloured black, brown, green, red, yellow and blue.

(i) Red is opposite of Black
(ii) Green is between red and black
(iii) Blue is adjacent to yellow
(iv) Brown is adjacent to blue
(v) Red is at the bottom

Which colour is opposite of brown?

(a) Yellow (b) Red (c) Green (d) Blue

5. Kebin earns Rs.30 on the first day and spends Rs.20 on the next day. He again earns Rs.30 on the third day and spends Rs.20 on the fourth day. If he continues to save like this, how soon will he have Rs.100 in hand?

(a) On 8th day (b) On 10th day (c) On 15th day (d) On 20th day

6. A long cylinder has an axially placed three-bladed fan spinning inside it. Bullets are shot through the cylinder at a constant rate. If the number of blades is increased to six, the number of bullets

(a) missing the blades is halved (b) missing the blades is reduced by one-fourth
(c) hitting the blades remains the same (d) hitting the blades is doubled

7. Deepa sells to the first customer half her stock of apples and half an apple, half an apple and half her stock of remaining apples to the second customer. She repeats the same to the third and fourth customers. She finds that she has now 15 apples left. How many had she at first?

(a) 125 (b) 260 (c) 250 (d) 155

8. Majid has four coins, of Rupee 1, 2, 5 and 10 denomination. The number of different sums of money she can form from them is

(a) 16 (b) 15 (c) 12 (d) 8

9. A man born in the first half of nineteenth century was x years old in the year x^2. He was born in:

(a) 1806 (b) 1812 (c) 1825 (d) 1849

10. A cuboid has six sides of different colours. The red side is opposite to black. The blue side is adjacent to white. The brown side is adjacent to blue. The red side is face down. Which one of the following would be the opposite to brown?

(a) Red (b) Black (c) White (d) Blue

Answer Key

1. (c) 2. (a) 3. (c) 4. (a) 5. (c) 6. (d) 7. (c) 8. (b) 9. (a) 10. (c)

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CHAPTER 15

LOGICAL PUZZLES

Average Difficulty Rating: 3.4/5

Average time for each question: 120 seconds

Number of questions expected for each exam: 1

15.1 THEORY

Logical puzzles are one of the most interesting set of questions in the CSIR-NET exams. Just like the detective Sherlock Holmes, who solves the puzzling questions pertaining to a crime by deducing from the observations, these questions too demands one to deduce the correct option from the observations that he/she makes from the given question.

Step 1: Write down all the observations from the given question.
Step 2: Analyse the observations one by one.
Step 3: Eliminate the impossible options for each observations.
Step 4: Continue this process until you weed out three of the options so that you are left out with the obvious correct option.

Note that while analysing, there may be some observations, which are irrelevant with respect to the given question. Avoid them to save time.

Visualizing scenario in real world

In logical puzzles, it is very beneficial to visualize the situation in the real world. It helps us to have better understanding of the question and enables us to analyse the statements easily.

Let us illustrate it through the following illustration.

ILLUSTRATION 1:

Nikhil is taller than Jeswin. Jeswin is shorter than Bintu but taller than Shameer. Nikhil is not shorter than Bintu. Bintu and Jeswin are not fat. Everyone are of different heights. Right arrangement in order of increasing height is:

(a) Jeswin, Shameer, Bintu, Nikhil
(b) Shameer, Jeswin, Bintu, Nikhil
(c) Shameer, Bintu, Jeswin, Nikhil
(d) Shameer, Jeswin, Nikhil, Bintu

Answer: (b)

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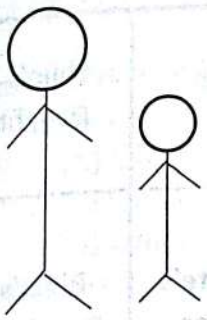
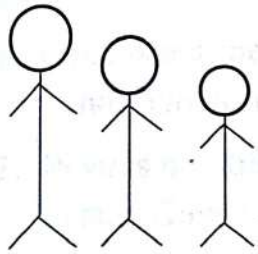
Given 1: Nikhil is taller than Jeswin.



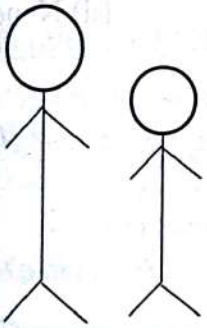
Nikhil and Jeswin

Given 2: Jeswin is shorter than Bintu but taller than Shameer.



Bintu, Jeswin and Shameer

Given 3: Nikhil is not shorter than Bintu



Nikhil and Bintu

Given 4: Bintu and Jeswin are not fat.

Not relevant with respect to height.

Combining information in these three, we can conclude that Shameer (the shortest), Jeswin, Bintu and Nikhil is the right order.

ILLUSTRATION 2:

Annette, Vincy and Jennifer are daughters of Martin, Bibin and Shijo. Four of these people are playing badminton doubles. Shijo's daughter and Martin are partners. Annette's father and Bibin's daughter are also partners. There are not any father-daughter combinations. Who is Annette's father?

(a) Martin (b) Bibin (c) Shijo (d) Cannot be determined from the data.

Answer (c)

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Let us try to visualize the given data in real life scenario.

Given: There are 6 people, Annette, Vincy, Jennifer, Martin, Bibin and Shijo.

Given: Four of these people are playing badminton doubles.

Given: Shijo's daughter and Martin are partners.

Given: Annette's father and Bibin's daughter are also partners.

Since there are no father-daughter combination, Annette's father is not Bibin.

Therefore, the only remaining option is Martin and Shijo.

However, Martin is playing in the other court. (See figure)

Therefore, the only remaining option is Shijo.

Martin	Shijo's Daughter
Annette's Father	Bibin's Daughter

ILLUSTRATION 3:

A boy and a girl are talking.

"I am a boy" - said the first child.

"I am a girl" - said the second child.

At least one of them is lying. Then who is lying?

(a) First child (b) Second child (c) Both (d) None

Answer (c)

Observations:

(1) Given: A boy and girl are talking.

⇨ One child is a boy and the other is a girl.

(2) Given: At least one of them is lying.

Now, we can deduce the answer from the observations:

By observation (2), at least one person should lie.

If the first child is lying, then the first child is a girl.

Then, if the second child is telling the truth, both the children are girls.

This is a contradiction to observation (1).

Hence, the second child is lying.

Similar is the case if the second child is lying.

Thus, both the children are lying.

ILLUSTRATION 4:

There are three people (Cima, Devu and Gowri), who are either Professor, Advocate, or Doctor. The Professor always tells the truth, the Advocate always lies, and the Doctor can either lie or tell the truth.

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Gowri says: "Devu is a Advocate."
Cima says: "Gowri is a Professor."
Devu says: "I am the Doctor."

Then who is who?

(a) Cima – Doctor, Devu – Advocate, Gowri – Professor.
(b) Cima – Professor, Devu – Advocate, Gowri – Doctor.
(c) Cima – Advocate, Devu – Professor, Gowri – Doctor.
(d) Cima – Advocate, Devu – Doctor, Gowri – Professor.

Answer (a)

Observations:

1) Professor always tells the truth.
2) Advocate always lie.

By 1), Cima is not the Professor, since if she is, then Gowri would also be the Professor.
Thus, Gowri or Devu is the Professor.

By 2), Devu is not the Professor, since her statement would then be a lie.
Thus, **Gowri is the Professor.**

By 1) Gowri told the truth. Thus, **Devu is the Advocate.**
Then, obviously, **Cima is the Doctor.**

ILLUSTRATION 5:

"All professors are researchers. Some scientists are professors."
Which of the given conclusions is logically valid and is inferred from the above arguments?

(a) All scientists are researchers
(b) All professors are scientists
(c) Some researchers are scientists
(d) No conclusion follows.

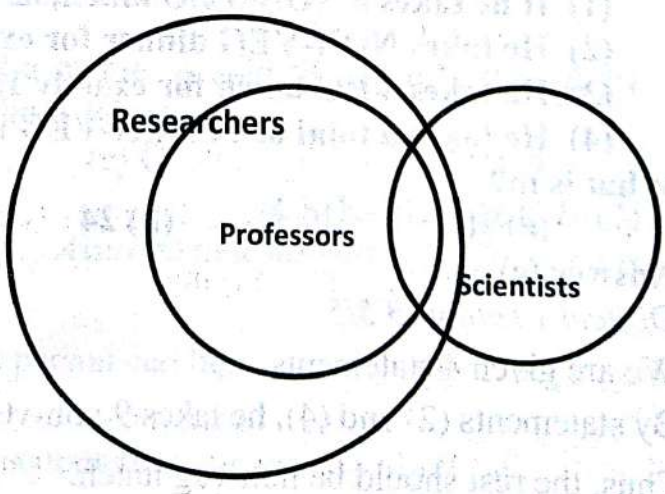
Answer (c)

The given data can be pictured as shown below.

Clearly, Professors' set is inside the Researchers' set.

We can check each of the option referring to the diagram.

(a) Wrong. Scientists' set is not inside the researcher set.
(b) Wrong. Professors' set is not inside the Scientists' set.
(c) Correct. There is intersection between researchers' and scientists' set.
(d) Wrong. Statement (c) is followed.



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15.2 CSIR-NET Exam Questions – Detailed Solution

1. Ajay, Bunty, Chinu and Deb were agent, baker, compounder and designer, but not necessarily in that order. Deb told baker that Chinu is on his way. Ajay is sitting across the designer and next to the compounder. The designer didn't say anything. What is each person's occupation?

(a) Ajay-Compounder, Bunty-Designer, Chinu-Baker, Deb-Agent
(b) Ajay-Compounder, Bunty-Baker, Chinu-Agent, Deb-Designer
(c) Ajay-Baker, Bunty-Agent, Chinu-Designer, Deb-Compounder
(d) Ajay-Baker, Bunty-Designer, Chinu-Agent, Deb-Compounder

Answer (d) **(CSIR-NET DEC 2014)**

Difficulty rating: 3.5/5 *Average time for solving: 120 seconds*

Observations:

(1) Given: Deb told baker that Chinu is on his way.
⇨ **Deb is not the baker, Chinu is not the baker.**

(2) Given: Ajay is sitting across the designer and next to the compounder.
⇨ **Ajay is not designer nor compounder.**

(3) Given: Chinu is on his way. So he is not sitting anywhere.
⇨ **Chinu is not designer nor compounder.**

(4) Given: The designer didn't say anything but Deb told something.
⇨ **Deb is not the designer.**

Now, let us deduce the correct option from our observations.

By observation 2), option (a) and (b) can be eliminated.

By observation 3), option (c) can be eliminated.

Therefore, our obvious choice is **option (d).**

2. Lunch-dinner pattern of a person of m days is given below. He has a choice of a VEG or a NON-VEG meal for his lunch/dinner

(1) If he takes a NON-VEG lunch, he will have only VEG for dinner
(2) He takes NON-VEG dinner for exactly 9 days
(3) He takes VEG lunch for exactly 15 days
(4) He takes a total of 14 NON-VEG meals.

What is m?

(a) 18 **(b) 24** **(c) 20** **(d) 38**

Answer (c) **(CSIR-NET DEC 2014)**

Difficulty rating: 3.5/5 *Average time for solving: 110 seconds*

We are given 4 statements.

By statements (2) and (4), he takes 9 non-veg dinners in a total of 14 non-veg meals.

Thus, the rest should be non-veg lunch.

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Hence, he takes non-veg lunch for (14 – 9) days = 5 days.

Then by statement (3), he takes VEG lunch for 15 days.

Total number of lunch he takes = 5 + 15 = 20.

Clearly, total number of lunch is the same as the total number of days, m.

Thus m = **20.**

3. If all horses are donkeys, some donkeys are monkeys, and some monkeys are men, then which statement must be true?

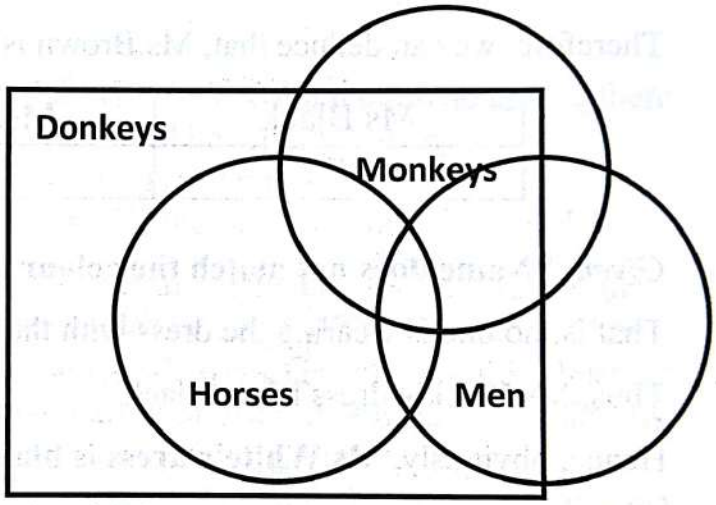
(a) All donkeys are men **(b) Some donkeys may be men**

(c) Some horses are men **(d) All horses are also monkeys**

Answer (b) **(CSIR-NET JUNE 2014)**

Difficulty rating: 3.5/5 *Average time for solving: 110 seconds*

The given data can be pictured using a Venn-diagram as shown. But note that this is not the only representation. For example, by the given conditions, we can also draw the Venn-diagram such a way that "Horses" set form no intersection with the "Men" set or the "Monkey" set.



However if we can eliminate three options for any one representation, we obtain the right answer.

Hence, checking the options, with respect to this representation.

(a) Wrong. Because there can be donkeys outside the Men set.

(b) Correct. Because of the use of the word "maybe". There is a possibility that some donkeys may be men.

(c) Wrong. Because the intersection of Horses and Men can be empty.

(d) Wrong. Because the Horse set is not inside the Monkey set.

4. One of the four - A, B, C and D - committed a crime. A said "I did it.", B said, "I didn't." C said, "B did it." D said, "A did it.". Who is lying?

(a) A **(b) B** **(c) C** **(d) D**

Answer (c) **(CSIR-NET DEC 2013)**

Difficulty rating: 3.5/5 *Average time for solving: 120 seconds*

Given that, only one person did the crime.

Since there can be only one correct option, only one person can lie.

A and D said, "A did it".

However, B and C are saying contradicting statements.

Therefore, either B or C is lying.

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B is not lying, because, if B is lying, then, "A and B both have done the crime". Only one person has done the crime.

Hence, **C is lying.**

5. During an evening party, when Ms.Black, Ms.Brown and Ms.White met, Ms.Brown remarked, "It is interesting that our dresses are white, black or brown, but for each of us the name does not match the colour of the dress!". Ms. White replied, "But your white dress does not suit you!". Pick the correct answer.

(a) Ms. White's dress was brown. **(b) Ms. Black's dress was white.**
(c) Ms. White's dress was black. **(d) Ms. Black's dress was black.**

Answer (c) **(CSIR-NET JUNE 2013)**

Difficulty rating: 3/5 *Average time for solving: 120 seconds*

Given: Ms. White replied (to Ms.Brown), "But your white dress does not suit you!"

Therefore, we can deduce that, Ms.Brown is wearing White dress.

Ms.Black	Ms.Brown	Ms.White
????	White	????

Given: **"Name does not match the colour of the dress."**

That is, no one is wearing the dress with the colours that matches their name.

Thus, Ms.Black's dress is not Black.

Hence, obviously, **Ms.White's dress is black** and Ms.Black's dress is Brown.

6. Amar, Akbar and Antony are 3 friends, one of whom is a doctor, another is an engineer and the third is a professor. Amar is not an engineer. Akbar is the shortest. The tallest person is a doctor. The engineer's height is the geometric mean of the heights of the other two. Then which of the following is true?

(a) Amar is a doctor and he is the tallest
(b) Akbar is a professor and he is the tallest
(c) Antony is a doctor and he is the shortest
(d) Antony is a doctor and he is the tallest

Answer (a) **(CSIR-NET DEC 2012)**

Difficulty rating: 3.5/5 *Average time for solving: 140 seconds*

Observations:

1) Given - Amar is not an engineer
   ⇨ Only Akbar or Antony can be the engineer.
2) Given - Akbar is the shortest.
3) Given - The tallest person is the doctor.
   ⇨ Akbar is not the doctor. (By observation 2)
   ⇨ Only Amar or Antony can be the doctor.

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4) Given - The engineer's height is the geometric mean of the heights of the other two.
Note: The value of geometric mean of two numbers will be always in between the numbers.

⇨ Height of engineer will be between the heights of the professor and the doctor.

5) Akbar is not the engineer. (By observation 2)

By observations 2), 3) and 4), Akbar is the professor and the shortest.

By observations 1), 5), and 3), Amar is the doctor and the tallest.

Hence **option (a) .**

15.3 PRACTICE QUESTIONS

1. Shafeeq is older than Salu and Sinu. Fawas is older than Emil. To find out who among them is the oldest, which of the following further information will be required, if any?
(a) Sinu is oldest than Salu (b) Emil is older than Salu
(c) Fawas is younger than Shafeeq (d) No further information is required

2. In a group of six women there are four dancers, four vocal musicians, one actress and three violinists. Nitha and Swathi are among the violinists while Gayatri and Thushara do not know how to play the violin. Thushara and Nayana are among the dancers. Gayatri, Swathi, Thushara and Nayana are all vocal musicians and two of them are violinists. If Aswathi is an actress, who among the following is certainly a dancer and a violinist?
(a) Gayatri (b) Aswathi (c) Thushara (d) Nayana

3. In a group, there are six persons A, B, C, D, E and F. The number of males equals that of females. A and E are sons of F. D is the mother of two, one boy and one girl. B is the son of A. There is only one married couple in the family at present.Which one of the following references can be drawn from the above?
(a) A, B and C are all females (b) A is the husband of D
(c) E and F are the children of D (d) D is the granddaughter of F

4. A, B, C, D, E and F not necessarily in that order are sitting in 6 chairs regularly placed around a round table. It is observed that, A is between D and F, C is opposite to D, D and E are not on neighbouring chairs. Which one of the following must be sitting on the neighbouring chairs?
(a) A and B (b) C and E (c) B and F (d) A and C

5. Five children were administered psychological tests to know their intellectual levels. In the report, psychologists pointed out that the child A is less intelligent than the child B. The child C is less intelligent than the child D. The child B is less intelligent than C and child A is more intelligent than the child E. Which child is the most intelligent?
(a) A (b) B (c) D (d) E

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6. P is taller than Q, R is shorter than P, S is taller than T but shorter than Q, then who among them is the tallest?

(a) P (b) Q (c) S (d) T

7. Elssa did an objective test with a particular marking scheme. Each correct answer earned 8 marks and each incorrect answer reduced the score by 4 marks. The test contained 30 questions. Elssa got a score of zero after the test. How many questions did she answer correctly?

(a) 8 (b) 10 (c) 16 (d) 20

8. In a group of 5 people, A, B and C are ambitious. C, D and E are honest. B, C and D are intelligent and A, D and E are industrious. Among these, neither industrious nor ambitious persons would include

(a) A alone (b) B and E (c) C and D (d) None

9. Six persons A, B, C, D, E and F are standing in a circle. B is between D and C. A is between E and C. F is to the right of D. Who is between A and F.

(a) B (b) C (c) D (d) E

10. A, B, C, D and E are five friends. A is shorter than B but taller than E. C is the tallest. D is shorter than B and taller than A. Who has two persons taller and two persons shorter than him or her?

(a) A (b) B (c) D (d) E

Answer Key

1. (c) 2. (d) 3. (b) 4. (b) 5. (c) 6. (a) 7. (b) 8. (d) 9. (d) 10. (c)

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MODEL QUESTION PAPERS – 7 SETS

(Use these question papers to take mock tests, strictly following the given instructions.)

To take real time mock tests, download the free android app
"CSIR-NET General Aptitude"

(More model question papers are updated time to time in the app.)

To download the app

Search "CSIR-NET General Aptitude" in play store.

OR

Scan the below QR Code with the QR code scanner in your android mobile phone.

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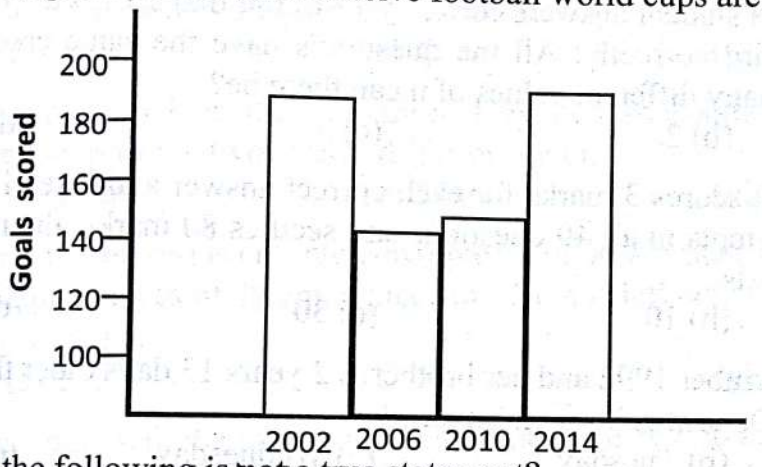
Model Question Paper – 1

Answer any 15 questions.
Correct Answer: 2 marks **Wrong Answer: –½ marks**
Maximum Marks: 30 **Maximum Time: 30 minutes**

1. Two trains 150 m and 170 m long run at the speed of 90 km/hr and 70 km/hr respectively in opposite directions on parallel tracks. The time (in seconds) which they take to cross each other, is:

(a) 6.2 (b) 6.6 (c) 7 (d) 7.2

2. The total goals scored in successive football world cups are given below.



Which of the following is not a true statement?

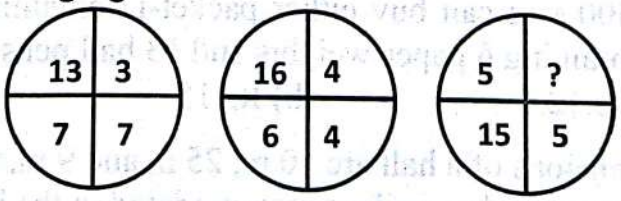
(a) In the year 2014 was scored the second largest number of goals among the given years.
(b) The difference in number of goals in 2002 and 2010 is more than that of 2006 and 2014.
(c) The total number of goals over the four world cups is more than 750.
(d) The total number of goals over the four world cups is less than 700.

3. A question paper must have a question on one of the eight poets: A, B, C, D, E, F, G, or H. The first four belong to the medieval period while the rest are considered modern poets. Generally, modern poets figure in the question paper in alternate years. Generally those who like H like G also; and those who like F like E also. The paper-setter does not like to ask about F as he has written a book on F, but he likes F. Last year, the paper contained a question on A. On the basis of the information given, this year's paper is most likely to contain a question on

(a) C (b) E (c) F (d) H

4. Find the missing number in the following figure.

(a) 1
(b) 3
(c) 5
(d) 7



5. Sreeram, Mahesh and Shameer start at the same time in the same direction to run around a circular stadium. Sreeram completes a round in 60 seconds, Mahesh in 75 seconds and

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Shameer in 100 seconds, all starting at the same point. After what time will they meet again at the starting point?

(a) 4 min (b) 5 min (c) 6 min (d) 8 min

6. If $A \times B = 16$, $B \times C = 22$, $C \times D = 55$, then $A \times D$ is

(a) 10 (b) 40 (c) 50 (d) 105

7. A total of 28 handshakes was exchanged at the conclusion of a party. Assuming that each participant was equally polite toward all the others the number of people present was:

(a) 14 (b) 28 (c) 56 (d) 8

8. On an exam of n questions, a student answers correctly 15 of the first 20. Of the remaining questions he answered one-third correctly. All the question/s have the same credit. If the student's mark is 50 %, how many different values of n can there be?

(a) 1 (b) 2 (c) 3 (d) 4

9. In an examination, a student scores 3 marks for each correct answer and loses 1 mark for every wrong answer. If he attempts in all 40 questions and secures 80 marks, the number of questions he attempts wrongly, is :

(a) 7 (b) 10 (c) 30 (d) 40

10. Dona was born on 13 September 1992 and her brother is 2 years 13 days older than she is. On which day was he born if Dona was born on Sunday?

(a) Monday (b) Tuesday (c) Wednesday (d) Sunday

11. Half of the villagers of a certain village have their own houses. One fifth of the villagers cultivate paddy. One third of the villagers are literate. Four fifth of the villagers are below twenty five. Then, which one of the following is certainly true?

(a) All the villagers who have their own houses are literate.
(b) Some villagers under twenty five are literate.
(c) A quarter of the villagers who have their own houses cultivate paddy.
(d) Half of the villagers who cultivate paddy are literate.

12. Find the next number in the series 7, 7, 8, 11, 9, 13, 10, 17, 11,...

(a) 12 (b) 18 (c) 19 (d) 21

13. On a Rs.10,000 order a merchant has a choice among three successive discounts of 20%, 20% and 10% and three successive discounts of 40%, 5% and 5%. By choosing the best offer, he can save:

(a) Rs.400 (b) Rs.330 (c) Rs.345 (d) Rs.360

14. For Rs.300 you can buy either packet-I containing 9 paper-weights and 7 ball-pens or packet-II containing 6 paper-weights and 13 ball pens. The price of each paper-weight is

(a) Rs.12 (b) Rs.15 (c) Rs.20 (d) Rs.24

15. The dimensions of a hall are 50 m, 25 m and 9 m. If each person requires 150 m^3, then the number of persons who can be accommodated in the hall are:

(a) 60 (b) 65 (c) 70 (d) 75

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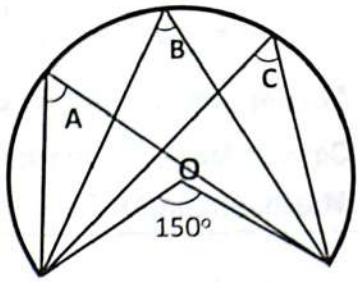
16. A segment of a circle (slightly greater than a semi-circle, whose center is O) is given. Identify the correct statement regarding the three angles A, B and C.



(a) A is equal to B but not equal to C.
(b) A, B and C are equal and have a value 75°.
(c) A, B and C are unequal.
(d) A, B and C are each equal to 85°.

17. The number of grams of water needed to reduce 9 grams shaving lotion containing 50% alcohol to a lotion containing 30% alcohol is:

(a) 3 (b) 4 (c) 5 (d) 6

18. A running track is the ring formed by two concentric circles. It is 10 m wide. The circumferences of the two circles differ by about:

(a) 15 m (b) 37 m (c) 63 m (d) 100 m

19. When an iron rod is cut into equal pieces of 30 cm each, a piece of 4cms is left out. When cut into equal pieces of 29 cm, a piece of 13 cm is left out. The minimum length of the rod (in cm) is

(a) 270 (b) 272 (c) 274 (d) 280

20. What angle will the two hands of a clock make at 4:50 am?

(a) 155° (b) 160° (c) 170° (d) 175°

Answer Key

1.(d)	2.(d)	3.(b)	4.(c)	5.(b)	6.(b)	7.(d)	8.(a)	9.(b)	10.(a)
11.(b)	12.(c)	13.(c)	14.(d)	15.(d)	16.(b)	17.(d)	18.(c)	19.(c)	20.(a)

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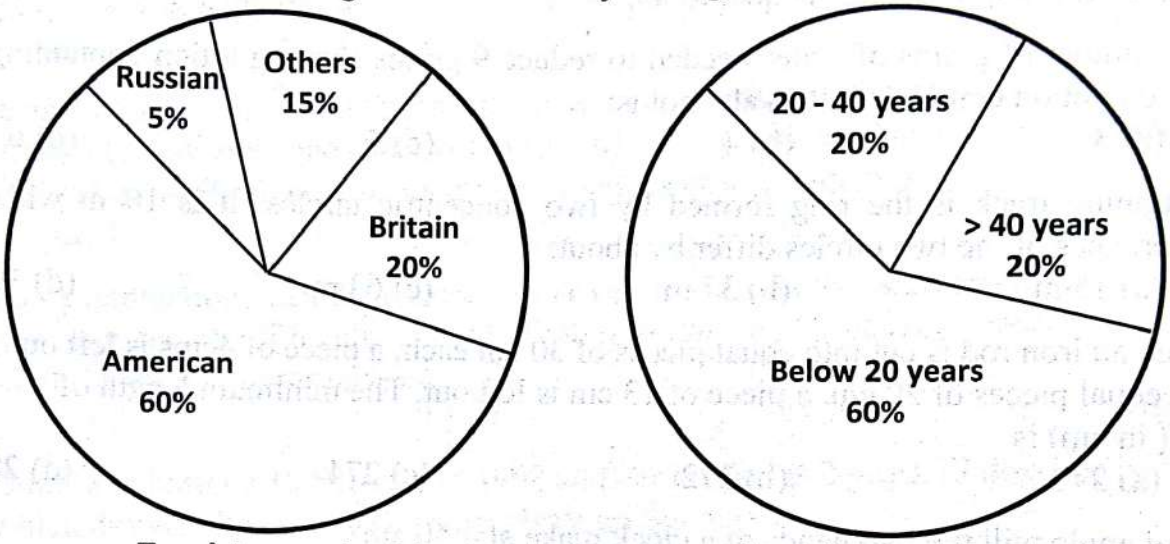
Model Question Paper – 2

Answer any 15 questions.

Correct Answer: 2 marks **Wrong Answer: –½ marks**

Maximum Marks: 30 **Maximum Time: 30 minutes**

1. The characteristics of foreign tourists in one year is given in the following pie chart.



Total number of tourists = 1,00,000

From the pie charts, find the number of American tourists in the age group 20-40 who visited during the year?

(a) 12000 (b) 20000 (c) 40000 (d) 60000

2. Two goods train each 500m long, are running in opposite directions on parallel tracks. Their speeds are 45 km/hr and 30 km/hr respectively. Find the time taken (in seconds) by the slower train to pass the driver of the faster train.

(a) 12 (b) 24 (c) 36 (d) 48

3. Six lectures A, B, C, D, E and F are to be organised in a span of seven days – from Sunday to Saturday, only one lecture on each day in accordance with the following:

(i) A should not be organised on Thursday.

(ii) C should be organised immediately after F.

(iii) There should be a gap of two days between E and D.

(iv) One day there will be no lecture (Friday is not that day), just before that day D will be organized.

(v) B should be organised on Tuesday and should not be followed by D.

On which day is there no lecture?

(a) Sunday (b) Friday (c) Monday (d) Tuesday

4. Find the next number in the series 1000, 1500, 2250,...

(a) 2500 (b) 2700 (c) 3375 (d) 3380

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5. Find the missing letter

(a) H
(b) J
(c) L
(d) N

A	E	C	G
B	J	F	?
C	O	I	U
D	T	L	B

6. In the numeration system with base 5, counting is as follows: 1, 2, 3, 4, 10, 11, 12, 13, 14, 20,... The number whose description in the decimal system is 69, when described in the base 5 system, is a number with:

(a) two consecutive digits
(b) two non-consecutive digits
(c) three consecutive digits
(d) three non-consecutive digits

7. Evaluate $3 - [3 - \{3 - 3(3 + 3)\}]$

(a) 15 (b) -12 (c) -15 (d) -18

8. Along a yard 300 metres long, 61 trees are planted at equal distances, one tree being at each end of the yard. What is the distance between two consecutive trees?

(a) 4.92 m (b) 5 m (c) 5.12 m (d) 6 m

9. In how many different ways can be the letters of the word 'HORROR' be arranged?

(a) 60 (b) 90 (c) 120 (d) 720

10. Today is Monday. After 100 days it will be?

(a) Wednesday (b) Thursday (c) Friday (d) Saturday

11. Jasir, Murali, Ligin, Sobin, Sreejesh and Toby live on different floors in a six-storeyed building (the ground floor is numbered 1, the floor above it 2, and so on). Jasir lives on an even-numbered floor. Murali does not live on an odd numbered floor. Ligin does not live on any of the floors below Toby's floor. Sobin does not live on floor number 2. Sreejesh does not live on a floor immediately above or immediately below Murali. Toby lives three floors above Sobin. Which of the following floor-person combination is correct?

	Jasir	Murali	Ligin	Sobin	Sreejesh	Toby
(a)	6	2	5	1	3	4
(b)	2	6	5	1	3	4
(c)	4	2	6	3	1	5
(d)	2	4	6	1	3	5

12. A father said to his son, "I was twice as old as you are at present at the time of your birth." If the father's age is 48 years now, the son's age (in years) five years back was :

(a) 14 (b) 11 (c) 15 (d) 13

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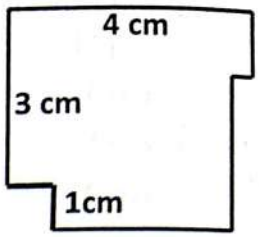
13. What is the perimeter of the given figure, where adjacent sides are right angles to each other?



(a) 16 cm
(b) 18 cm
(c) 20 cm
(d) Cannot be determined

14. Two high school classes took the same test. One class of 20 students made an average grade of 80%; the other class of 30 students made an average grade of 70%. The average grade for all students in both classes is:

(a) 75 % (b) 74 % (c) 72 % (d) 77 %

15. A horse and a cow were sold for Rs.15000 each. The horse was sold at loss of 20% and the cow at a gain of 20%. The entire transaction resulted in:

(a) No loss no gain
(b) Loss of Rs.1250
(c) Gain of Rs.2000
(d) Loss of Rs.2000

16. If the three points of contact of a circle inscribed in a triangle are joined, the angles of the resulting triangle:

(a) Are always equal to 60°
(b) Are always equal to each other.
(c) Are always one obtuse angle and two unequal acute angles.
(d) Are always acute angles.

17. For a journey, the cost of a child ticket is $\frac{1}{3}$ of the cost of an adult ticket. If the cost of tickets for 4 adults and 5 children is Rs.85, the cost of the child ticket is

(a) Rs.5 (b) Rs.6 (c) Rs.10 (d) Rs.15

18. X and Y are two cylinders of the same height. The base of X has diameter that is half the diameter of the base of Y. If the height of X is doubled, the volume of X becomes:

(a) Equal to the volume of Y
(b) Double the volume of Y
(c) Half the volume of Y
(d) Greater than the volume of Y

19. If the radius of the base and the height of a right circular cone are doubled, then its volume becomes:

(a) 2 times (b) 3 times (c) 4 times (d) 8 times

20. If a bacteria doubles itself in 4 minutes. What would be number of bacteria at the end of 20 minutes if it started with 3 bacteria

(a) 16 (b) 32 (c) 48 (d) 96

Answer Key

1.(a)	2.(d)	3.(c)	4.(c)	5.(d)	6.(c)	7.(c)	8.(b)	9.(a)	10.(a)
11.(b)	12.(b)	13.(a)	14.(b)	15.(b)	16.(d)	17.(a)	18.(c)	19.(d)	20.(d)

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Model Question Paper – 3

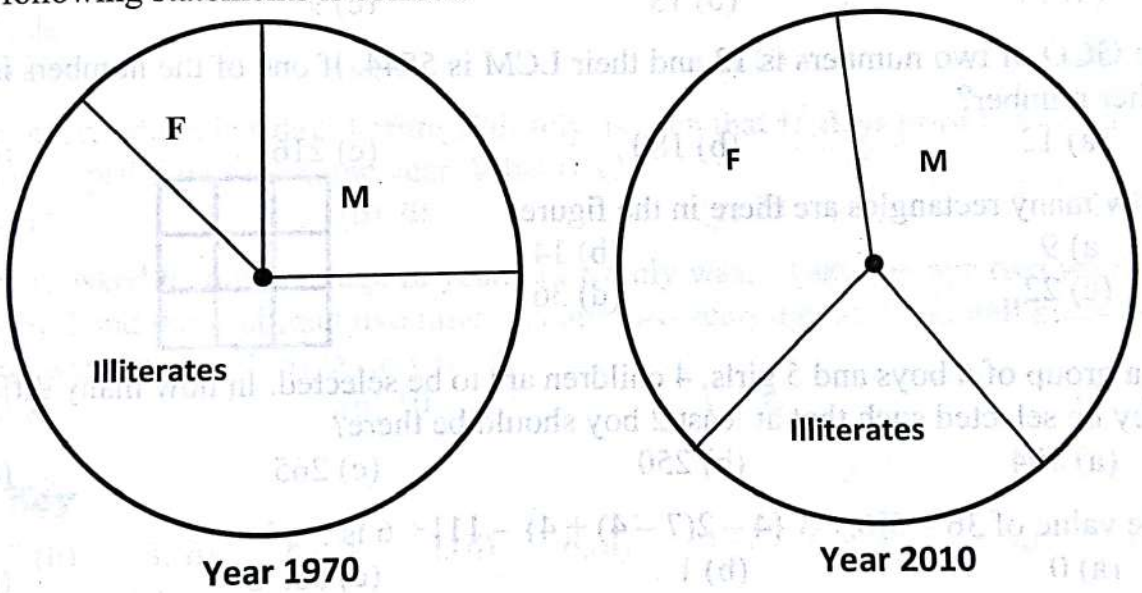
Answer any 15 questions.
Correct Answer: 2 marks
Maximum Marks: 30

Wrong Answer: –½ marks
Maximum Time: 30 minutes

1. If the length of a rectangle is decreased by 50% and its breadth is increased by 20%, then what is the increase or decrease per cent in the area of rectangle?
   (a) 30% increase
   (b) 40% decrease
   (c) 50% increase
   (d) 30% decrease

2. A train 125 m long passes a man, running at 5 km/hr in the same direction in which the train is going, in 10 seconds. The speed of the train is (in km/hr):
   (a) 45
   (b) 50
   (c) 54
   (d) 55

3. The given pie charts show the proportion of literates and illiterates in a country, in the years 1970 and 2010, and also the proportion of males(M) and females(F) among the literates. Which one of the following statements is correct?



   (a) In 1970, half of the illiterates were women.
   (b) The proportion of literate males to the total proportion of males remained the same over the years.
   (c) The male literacy did not improve over this period.
   (d) The ratio of female literates to male literates improved significantly over this period.

4. Six people – Akhil, Cheriyachen, Dileep, Murali, Sobin and Toby – were all born on the same day of the year, but each was born in a different year, during a single six year period.
   Akhil is older than Dileep.
   Sobin is older than both Toby and Cheriyachen.
   Murali is two years older than Toby.

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Akhil was born either in 1962 or in 1963.
The oldest member of the group was born in 1960.
If Murali is the oldest in the group, then which of the following must be true?

(a) Dileep was born in 1964 (b) Akhil was born in 1962
(c) Toby was born in 1961 (d) Sobin was born in 1961

5. What is the next number in the following sequence?
1, 1, 4, 4, 7, 9, 10, 16, 13,...
(a) 15 (b) 16 (c) 20 (d) 25

6. What is the angle between minute hand and hour hand of a clock, when the time is 9:54?
(a) 27° (b) 29° (c) 35° (d) 36°

7. What is the next number in the following sequence?
72, 46, 521, 612,...
(a) 343 (b) 678 (c) 428 (d) 476

8. A number system consists of digits 0,1,2,3,4 and 5. What is the decimal equivalent of 13 in this number system?
(a) 15 (b) 13 (c) 9 (d) 21

9. The GCD of two numbers is 12 and their LCM is 5544. If one of the numbers is 168, find the other number?
(a) 12 (b) 180 (c) 216 (d) 396

10. How many rectangles are there in the figure?
(a) 9 (b) 14
(c) 22 (d) 36

11. In a group of 6 boys and 5 girls, 4 children are to be selected. In how many different ways can they be selected such that at least 2 boy should be there?
(a) 194 (b) 250 (c) 265 (d) 300

12. The value of $36 - 6[6 + 3\{4 - 2(7 - 4) + 4\} - 11] \div 6$ is :
(a) 0 (b) 1 (c) 33.75 (d) 35

13. Find the missing number in the sequence?
3, 5, 9, 15, __, 33
(a) 21 (b) 23 (c) 24 (d) 30

14. Free notebooks were distributed equally among children of a class. The number of notebooks each child got was one-twelfth of the number of children. If the number of children had been half, each child would have got 4 notebooks. Total how many notebooks were distributed?
(a) 24 (b) 48 (c) 64 (d) 96

15. The average marks of 30 students in a Physics exam is 83. Later it is found that two marks 75 and 88 were wrongly entered as 81 and 79. The average after the mistakes were rectified is

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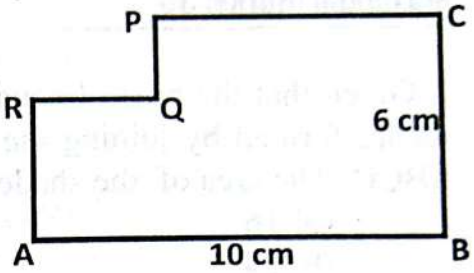
(a) 82.9 (b) 82 (c) 82.5 (d) 83.1

16. A reduction of 30% in the price of coffee enables a purchaser to obtain 3 kg more for Rs.70. The reduced price per kg of coffee is (in Rs.):

(a) 3 (b) 5 (c) 6 (d) 7

17. A rectangle of certain dimensions is chopped off from one corner of a larger rectangle as shown. AB = 10 cm and BC = 6 cm. The perimeter of the figure ABCPQRA (in cm) is:



(a) 24 cm
(b) 32 cm
(c) 36 cm
(d) 48 cm

18. A student on vacation for d days observed that

(A) It rained 7 times morning or afternoon.
(B) When it rained in the afternoon, it was clear in the morning.
(C) There were five clear afternoons.
(D) There were six clear mornings.

Then d equals:

(a) 7 (b) 9 (c) 10 (d) 11

19. A certain day which is x days before 29th July, is such that 15 days prior to that day, it was 3x days since April 5th of the same year. What is x?

(a) 15 (b) 20 (c) 25 (d) 30

20. Priya was asked to state her age in years. Her reply was, "Take my age two years hence, multiply it by 2 and then subtract two times my age two years ago and you will know how old I am." What was the age of Priya (in years)?

(a) 8 (b) 10 (c) 42 (d) 33

Answer Key

1.(b)	2.(b)	3.(d)	4.(d)	5.(d)	6.(a)	7.(a)	8.(c)	9.(d)	10.(d)
11.(c)	12.(d)	13.(b)	14.(b)	15.(a)	16.(d)	17.(b)	18.(b)	19.(c)	20.(a)

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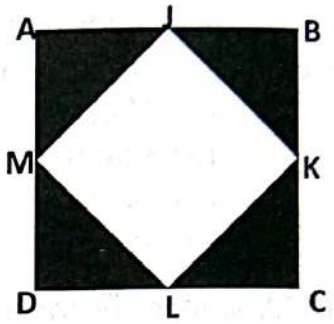
Model Question Paper – 4

Answer any 15 questions.

Correct Answer: 2 marks | **Wrong Answer: –½ marks**

Maximum Marks: 30 | **Maximum Time: 30 minutes**

1. Given that the area of square ABCD is 64 cm^2. Let JKLM be a square formed by joining the midpoints of the sides of the square ABCD. The area of the shaded region in the figure in cm^2 is



(a) 16
(b) 24
(c) 32
(d) 48

2. A 270 m long train running at the speed of 120 km/hr crosses another train running in opposite direction at the speed of 80 km/hr in 9 seconds. What is the length of the other train?

(a) 230 m (b) 240 m (c) 260 m (d) 320 m

3. Find the missing letter.

A	B	C	D
B	D	F	H
C	F	I	L
D	H	?	P

(a) L
(b) J
(c) K
(d) M

4. The angle between the hands of a clock is minimum at

(a) 12 o'clock (b) 4 o' clock (c) 5 o' clock (d) 6 o' clock

5. Find the missing number in the sequence?

2, 3, 6, 3, 4, __ , 4, 5, 20, …

(a) 3 (b) 5 (c) 7 (d) 12

6. The total number of digits used in numbering the pages of a book having 126 pages, is :

(a) 267 (b) 268 (c) 270 (d) 271

7. The number of girls is 6 times the number of boys. Which of the following cannot be the total number of children in the class?

(a) 14 (b) 25 (c) 35 (d) 84

8. Asif speaks truth in 70% cases and Toby in 60% of the cases. In what percentage of the cases are they likely to contradict each other, narrating the same incident?

(a) 12 % (b) 30 % (c) 46 % (d) 65 %

9. Timmy covered a distance of 50 km on his first trip. On a latter trip he travelled 300 km while going three times as fast. His new time compared with the old time was:

(a) Three times as much (b) Twice as much
(c) The same (d) Half as much

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10. The marks of 3 students in 4 different subjects are given in the following table.

Subject \ Student	S1	S2	S3
Chemistry	75	62	66
Physics	68	71	60
Mathematics	75	66	83
Zoology	82	81	71

Which of the following is correct?

(a) The average score in Mathematics of the 3 students is 83.
(b) The average score in Physics of the 3 students is 60.
(c) The average score for all subjects is the highest for S1.
(d) The average score for all subjects of S2 is higher than that of S3.

11. Anna's birthday was on a Tuesday and was exactly 43 days after Lakshmi's birthday. Sheela's birthday was 27 days before Anna's birthday. On what day of the week was Sheela's birthday?

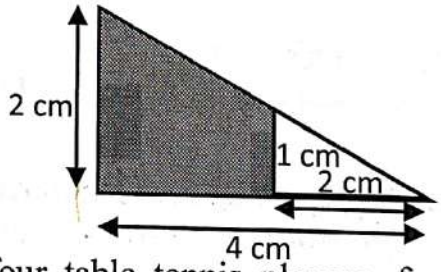
(a) Monday (b) Tuesday (c) Wednesday (d) Thursday

12. If the square of a number of two digits is decreased by the square of the number formed by reversing the digits, then the result is not always divisible by:

(a) 9
(b) The product of the digits
(c) the sum of the digits
(d) the difference of the digits

13. The area of the shaded region is

(a) 16 cm^2
(b) 8 cm^2
(c) 3 cm^2
(d) 12 cm^2



14. In a group of six women there are four table tennis players, four post graduates in mathematics, one postgraduate in physics and three bank employees. Nithina and Deeptha are among the bank employees while Silja and Muneera are unemployed. Muneera and Geethu are among the table tennis players. Silja, Deeptha, Muneera and Geethu are postgraduates in mathematics of whom two are bank employees. If Revathi is a postgraduate in physics who among the following is both a table tennis player and a bank employee?

(a) Muneera (b) Geethu (c) Nithina (d) Deeptha

15. The area (in m^2) of a triangular park of dimension 30 m, 40 m and 50 m is

(a) 600 (b) 1200 (c) 1250 (d) 1500

16. Elssa gets 3 marks for each correct sum and loses 2 marks for each wrong sum. She attempts 30 sums and obtains 40 marks. The number of sums solved correctly is:

(a) 10 (b) 15 (c) 20 (d) 25

17. There are five friends – Amjeesh, Dileep, Harris, Sabeer, and Vinod. Amjeesh is shorter than Dileep but taller than Vinod. Harris is the tallest. Sabeer is little shorter than Dileep and

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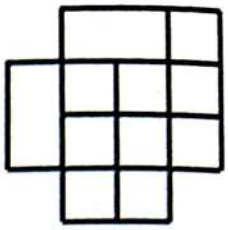
little taller than Amjeesh. Then who is the shortest?

(a) Amjeesh (b) Harris (c) Sabeer (d) Vinod

18. How many squares are there in this figure?



(a) 9
(b) 14
(c) 15
(d) 17

19. Three bells, toll at intervals of 36 sec, 40 sec and 48 sec respectively. They start ringing together at particular time. The next toll together after

(a) 6 minutes (b) 12 minutes (c) 18 minutes (d) 20 minutes

20. The difference between a two-digit number and the number obtained by interchanging the positions of its digits is 36. What is the difference between the two digits of that number?

(a) 1 (b) 2 (c) 3 (d) 4

Answer Key

1.(c)	2.(a)	3.(a)	4.(a)	5.(d)	6.(c)	7.(b)	8.(c)	9.(b)	10.(c)
11.(c)	12.(b)	13.(c)	14.(b)	15.(a)	16.(c)	17.(d)	18.(c)	19.(b)	20.(d)

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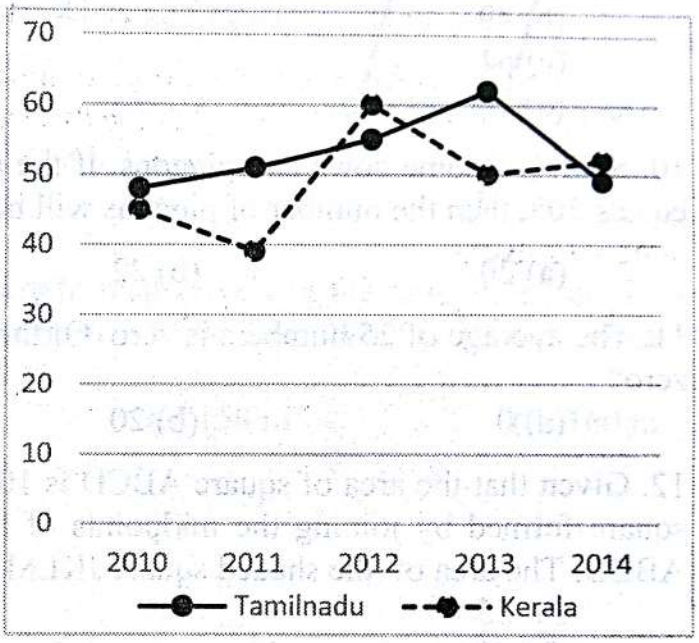
Model Question Paper – 5

Answer any 15 questions.
Correct Answer: 2 marks **Wrong Answer: -½ marks**
Maximum Marks: 30 **Maximum Time: 30 minutes**

1. On what dates of September 1999 will Sunday fall?
   (a) 3, 10, 17, 24, 31 (b) 4, 11, 18, 25
   (c) 5, 12, 19, 26 (d) 2, 9, 16, 23, 30

2. The given graph shows the distribution of annual rainfall (expressed in cm) in two states: Kerala and Tamil Nadu.



Study the graph and choose the correct statement among the following?
   (a) Kerala got more rain than Tamil Nadu in every years.
   (b) In 2015, Tamil Nadu is expected to get more rain than Kerala.
   (c) Rise in rainfall in Kerala during 2013-14 is less than the fall in rainfall in Tamil Nadu during the same period.
   (d) In 2013, Tamil Nadu got less rain than Kerala.

3. A toy factory manufactured a batch of electronic toys. If the toys were packed in boxes of 40 each, 17 boxes would not be enough to pack all of them. If the toys were packed in boxes of 50 each, 15 such boxes would not be filled completely. Coincidentally, in the end, the toys were packed in n boxes containing n toys each, without any remainder. The total number of toys was
   (a) 721 (b) 727 (c) 728 (d) 729

4. The gross income of a person is Rs.30000. 20% of his income is exempted from income tax and his net income is Rs.28560. The rate of income tax is:
   (a) 2 % (b) 3 % (c) 5 % (d) 6 %

5. A bell rings at intervals of 10 seconds. It was rung for the first time at 1 hr. 40 min 40 sec. p.m. and the last time at 3 hrs. 45mins. 50 sec. p.m. How many times was the bell rung?
   (a) 300 (b) 700 (c) 751 (d) 752

6. $(100 \div 5 + 6 - 4 \times 6) + (12 \times 5 - 15) =$
   (a) 32 (b) 47 (c) 79 (d) 85

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7. In how many different ways can the letters of the word 'PROTEIN' be arranged in such a way that the vowels occupy only the even positions?

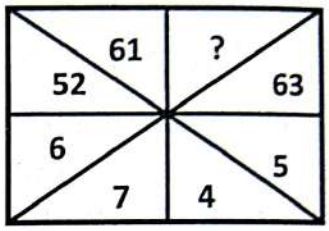
(a) 24 (b) 32 (c) 60 (d) 144

8. Electric poles occur at equal distances of 240 m along a road. Heaps of stones are put at equal distances of 320 m along the same road. If the first is at the foot of the first pole, then the next heap will lie at the foot of a pole after a distance of

(a) 480 m (b) 560 m (c) 960 m (d) 1200 m

9. Which one number can be placed at the sign of interrogation?

(a) 18
(b) 49
(c) 64
(d) 94



10. Sujin has some cows and pigeons. If the number of heads be 62 and the number of feet equals 208, then the number of pigeons will be:

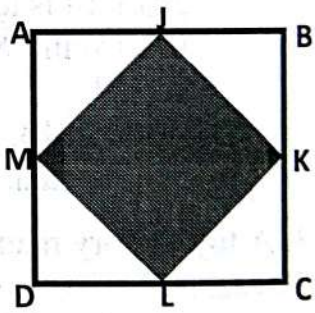
(a) 20 (b) 22 (c) 38 (d) 42

11. The average of 25 numbers is zero. Of them, at the most, how many may be greater than zero?

(a) 0 (b) 20 (c) 24 (d) 25

12. Given that the area of square ABCD is 100 cm^2. Let JKLM be a square formed by joining the midpoints of the sides of the square ABCD. The area of the shaded square JKLM in the figure in cm^2 is

(a) 25
(b) 40
(c) 50
(d) 75



13. A, B, C, D, E, F and G are members of a family consisting of 4 adults and 3 children, two of whom, F and G are girls. A and D are brothers and A is a doctor. E is an engineer married to one of the brothers and has two children. B is married to D and G is their child. Who is C?

(a) G's brother (b) F's father (c) E's daughter (d) A's son

14. A man walked diagonally across a square lot. Approximately what was the percent saved by not walking along the edges?

(a) 20 % (b) 24 % (c) 30 % (d) 33 %

15. A circular piece of metal of maximum size is cut out of a square piece and then a square piece of maximum size is cut out of the circular piece. The total amount of metal wasted is:

(a) quarter the area of the original square.
(b) half the area of the original square.
(c) the area of the circular piece.
(d) half the area of the circular piece

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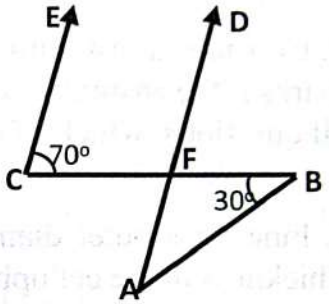
16. A, B, C, D, E and F not necessarily in that order arte sitting in 6 chairs regularly placed around a round table. It is observed that, A is between D and F,C is opposite to D,D and E are not on neighbouring chairs. Which one of the following must be true?

(a) A is the opposite to B
(b) D is the opposite to E.
(c) C and B are neighbours.
(d) B and E are neighbours

17. The angle between the hands of a clock is maximum at

(a) 12 o'clock (b) 3 o' clock (c) 5 o' clock (d) 6 o' clock

18. In the given diagram, CE is parallel to AD and the measures of two angles at B and C have been indicated. Then ∠DAB is equal to



(a) 30°
(b) 35°
(c) 40°
(d) 50°

19. Two trains are running at 84 km/hr and 48 km/hr respectively in the same direction. Fast train completely passes a man sitting in the slower train in 15 seconds. What is the approximate length of the fast train?

(a) 120 m (b) 125 m (c) 150 m (d) 160 m

20. The digit in the unit place of the number $(8^{71} - 2^{41})$

(a) 0 (b) 2 (c) 4 (d) 6

Answer Key

1.(c)	2.(c)	3.(d)	4.(d)	5.(d)	6.(b)	7.(d)	8.(c)	9.(d)	10.(a)
11.(c)	12.(c)	13.(d)	14.(c)	15.(b)	16.(d)	17.(d)	18.(c)	19.(c)	20.(a)

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Model Question Paper – 6

(Question paper of special CSIR-NET examination conducted in Guwahati Centre.)

Answer any 15 questions.

Correct Answer: 2 marks **Wrong Answer: –½ marks**

Maximum Marks: 30 **Maximum Time: 30 minutes**

1. Consider a test with 20 multiple-choice questions. For each question a student will get 2 marks if the answer is correct, -1 if it is incorrect, and 0 if it is unanswered. If a student answers all questions, which of the following scores is NOT possible?

(a) -11 (b) -8 (c) 1 (d) 8

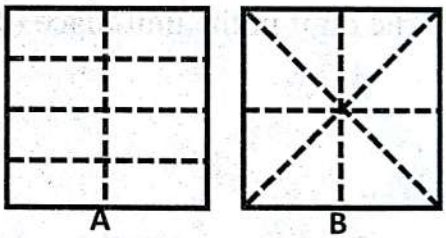
2. Inner and outer diameters of a roll of cellophane tape are 6 cm and 8 cm respectively. Thickness of the cellophane tape is 0.1 mm. What will be the approximate length of cellophane tape in the roll?

(a) 10 m (b) 11 m (c) 19 m (d) 22 m

3. If $1 + x + x^2 + \ldots = 2$, then $1 - x + x^2 - x^3 + \ldots =$

(a) 0 (b) 1 (c) 2/3 (d) 1/2

4. A cubic cake has a thin icing layer on the top and the sides. It is proposed to share it equally among 8 persons in one of the following ways (top view is shown). Which of the two is a fair division?



(a) Both A and B are fair divisions.
(b) A is a fair division, but B is not.
(c) B is a fair division, but B is not.
(d) Neither A, nor B is a fair division.

5. A man standing under a hemispherical dome at its periphery fires a gun. His friend is also standing under the dome but at diametrically opposite point. The radius of the dome is 165 m. What is the longest time up to which his friend can hear the echo of the sound reflected once from the dome? (speed of sound is 330 m/s)

(a) 1 s (b) 0.5 s (c) $0.5\sqrt{3}$ s (d) $\sqrt{2}$ s

6. Put inside a cube the largest sphere it can accommodate. Then put inside this sphere the largest cube it can accommodate. The ratio of the volume of the outer cube to the volume of the inner cube is

(a) $2\sqrt{2}$ (b) $\frac{1}{3\sqrt{3}}$ (c) $\frac{1}{2\sqrt{2}}$ (d) $3\sqrt{3}$

7. A chord of length 5 cm subtends an angle of 30° at the circumference of the circle. Another chord whose length is 10 cm subtends at the circumference an angle of

(a) 30° (b) 60° (c) 90° (d) 120°

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8. A series of squares whose sides are x, x^2, x^3,..., where $x < 1$, is available. It is found that the squares can be tiled to cover a rectangle of area x. Arrangement of the three largest tiles is as shown in the figure. Find x?

(a) $\frac{\sqrt{3}}{2}$ (b) $\frac{1}{\sqrt{2}}$ (c) $\frac{(\sqrt{5}-1)}{2}$ (d) $\frac{3}{4}$

9. Three circles of radius one unit each are touching one another externally. The area of the region bounded by them and not interior to any of them in square units is

(a) $\sqrt{3} - \frac{\pi}{2}$ (b) $2\sqrt{3} - \pi$ (c) $\pi - \sqrt{3}$ (d) $\sqrt{3} - \frac{\pi}{3}$

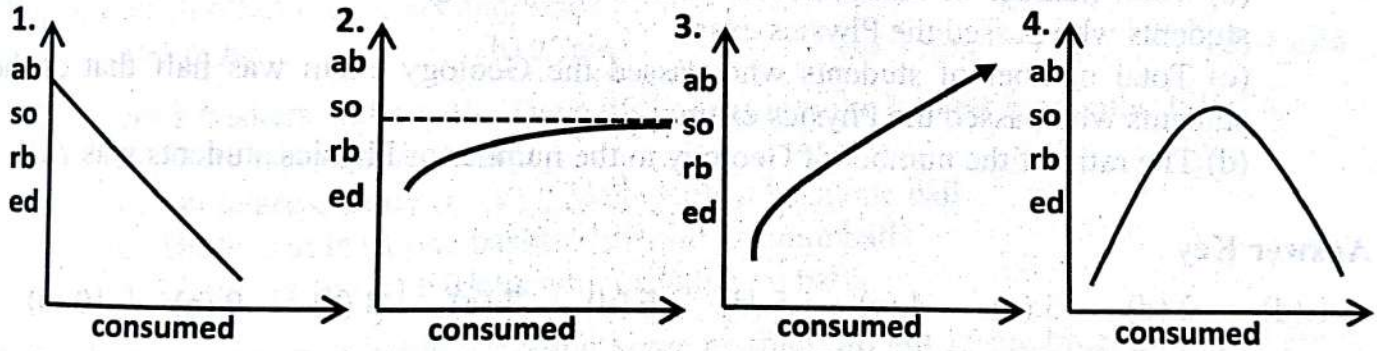
10. Given that $x+y+z = 1$ where x, y, z are real, which of the following is true?

(a) $x^2 + y^2 + z^2 < 0$
(b) $xy + yz + zx < 1$
(c) $xy + yz + zx > 1$
(d) $x > y > z$

11. An object kept at a point (x, y) is reflected in a mirror kept along x-axis. If the mirror is now rotated along y-axis, the image would move from

(a) (x, y) to (x, -y)
(b) (-x, y) to (-x, -y)
(c) (-x, -y) to (x, y)
(d) (x, -y) to (-x, y)

12. For a drug, the fraction absorbed can be given as a linear function of the amount consumed at the time with a negative slope. Which graph relates the amount consumed with the amount absorbed?



(a) 1 (b) 2 (c) 3 (d) 4

13. I walked some distance East and then turned and walked some distance North. I am now 10 km from my starting point. What could be the maximum distance walked by me, approximately?

(a) 15 km (b) 14 km (c) 10 km (d) 11 km

14. A and B are two hexadecimal digits and the numbers X, Y and Z are defined by the relations $X = \frac{BBBB}{AAAA}$, $Y = \frac{BBB}{AAA}$, $Z = \frac{BB}{AA}$. Then

(a) X = Y = Z (b) X > Y > Z (c) X < Y < Z (d) X > Y < Z

15. PA is tangent to a circle and a line PCB passes through its center. If $\angle P = 50^\circ$, $\angle B =$

(a) 45° (b) 50° (c) 40° (d) 20°

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16. A fixed square is divided into a 64 × 64 grid of tiny squares. A tiny square may be either filled or empty. Each pattern of filled and empty squares forms a code. The number of distinct codes that can be created is

(a) 2128 (b) 642 (c) 128 × 128 (d) 264 × 64

17. LPG cylinders having diameter of 40 cm are to be loaded in upright position in a single layer in a vehicle whose storage area is 160 cm × 110 cm. What is the maximum number of cylinders that can be loaded on the vehicle?

(a) 8 (b) 12 (c) 10 (d) 11

18. The three sides a, b, c of a triangle satisfy the equation $a^3 + b^3 + c^3 = 3abc$. Then the triangle

(a) must be equilateral
(b) is isosceles, but not equilateral
(c) must be right angled
(d) is neither equilateral nor isosceles

19. ABCD is a cyclic quadrilateral, right angled at B. AB = 8 cm, BC = 6 cm and D is at the distance of 3 cm from AC. The area of the quadrilateral (in sq. cm.) is

(a) 63 (b) 48 (c) 39 (d) 24

20. $\frac{3}{7}^{th}$ of the number of students who passed the Geology exam is the same as $\frac{6}{7}^{th}$ of the number of students who passed the Physics exam. Which of the following is necessarily true?

(a) The same number of students passed the Geology and Physics exams.
(b) Total number of students who passed the Geology exam was twice that of the students who passed the Physics exam.
(c) Total number of students who passed the Geology exam was half that of the students who passed the Physics exam.
(d) The ratio of the number of Geology to the number of Physics students was 7 : 3

Answer Key

1.(d)	2.(d)	3.(c)	4.(c)	5.(d)	6.(d)	7.(c)	8.(c)	9.(a)	10.(b)
11.(d)	12.(d)	13.(b)	14.(a)	15.(d)	16.(d)	17.(d)	18.(a)	19.(c)	20.(b)

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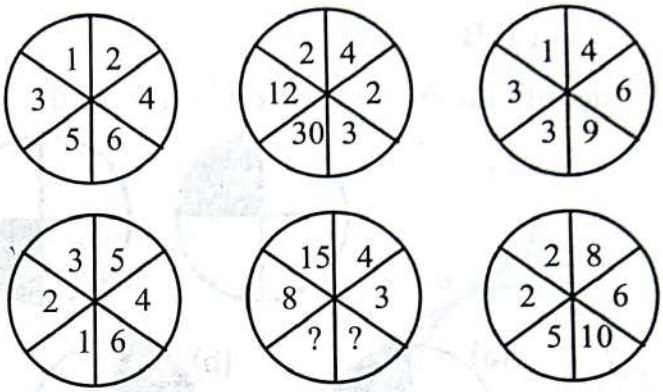
Model Question Paper – 7 (Issued by CSIR)

Answer any 15 questions.
Correct Answer: 2 marks **Wrong Answer: –½ marks**
Maximum Marks: 30 **Maximum Time: 30 minutes**

1. Find the missing numbers in the bottom middle circle.
(Clue: left halves of the central circles relate to the left circles and the right halves to the right circles)



(a) 10, 20
(b) 15, 15
(c) 21, 2
(d) 6, 2

2. A sphere of radius 4 cm is carved from a homogeneous sphere of radius 8 cm and mass 160 g. The mass of the smaller sphere is
(a) 80 g (b) 60 g (c) 40 g (d) 20 g

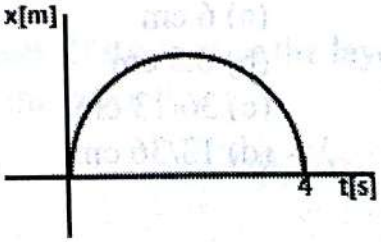
3. Two boys A and B are at two diametrically opposite points on a circle. At one instant the two start running on the circle; A anticlockwise with constant speed v and B clockwise with constant speed 2v. In 2 minutes, they pass each other for the first time. How much later will they pass each other for the second time?
(a) 1 min (b) 2 min (c) 3 min (d) 4 min

4. There are k baskets and n balls. The balls are put into the baskets randomly. If $k < n$;
(a) There is no empty basket
(b) There are exactly (n - k) baskets with at least one ball
(c) There is at least one basket with two or more balls
(d) There are (n - k) baskets with exactly two balls

5. In a bag containing only blue, red and green marbles, all but 15 are blue, all but 13 are red and all but 12 are green. How many are red?
(a) 13 (b) 7 (c) 25 (d) 20

6. A uniform cylindrical container is half filled with water. The height of the cylinder is twice its diameter. The cylinder is gradually tilted until the water touches the brim. At this instant, the container is inclined at
(a) 30° to vertical (b) 45° to vertical
(c) 60° to vertical (d) 75° to vertical

7. An ant is crawling along the x-axis such that the graph of its position on the x-axis versus time is a semi-circle (see figure). The total distance covered in the 4 s is



(a) 4 m (b) 2 m
(c) 2 cm (d) 4 cm

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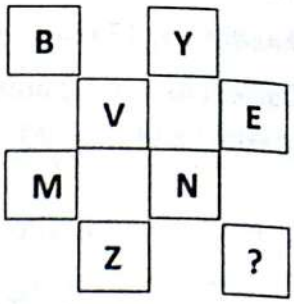
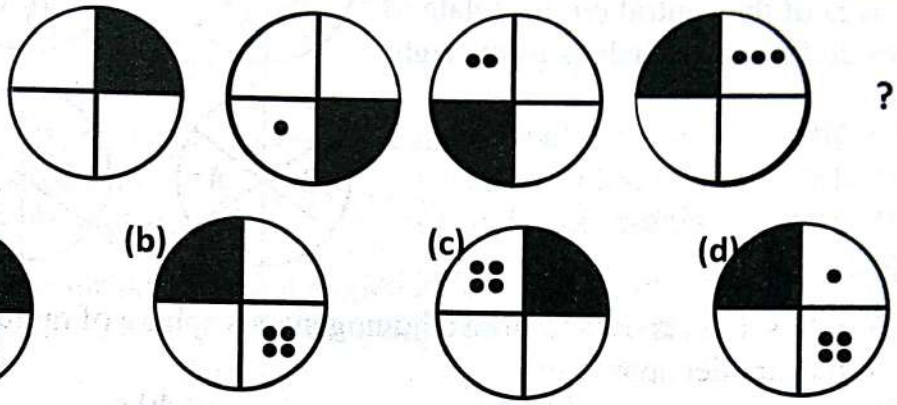
8. Identify the missing letter.



(a) W
(b) A
(c) X
(d) B

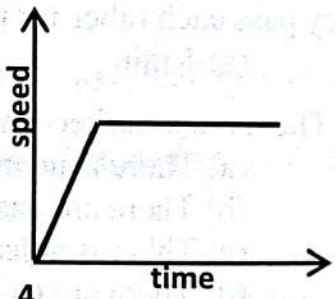
9. Identify the figure that comes next in the sequence.



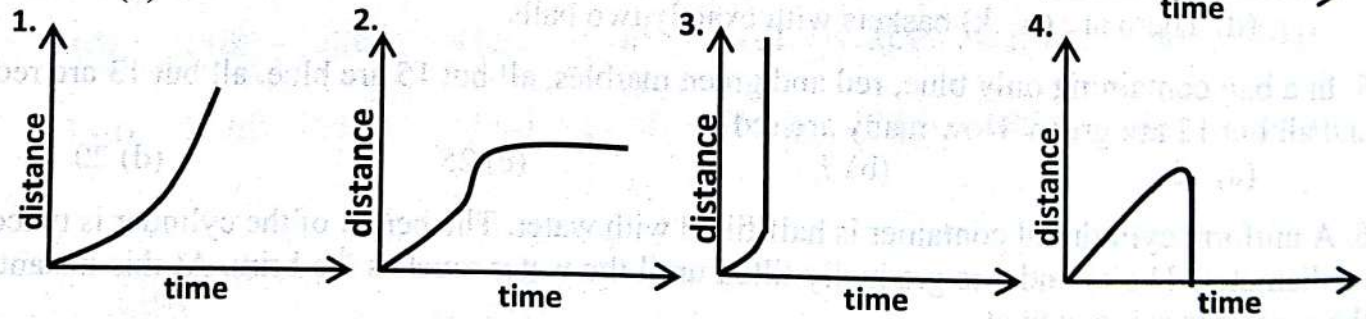
10. A person buys a shirt with marked price Rs.300 at 20% discount. In order to make a profit of 20% the person should sell the shirt for

(a) Rs. 288 (b) Rs. 300 (c) Rs. 240 (d) Rs. 360

11. A car is moving along a straight road. The graph shows how the speed varies with time. Which of the following graphs represents the distance covered by the car with time?

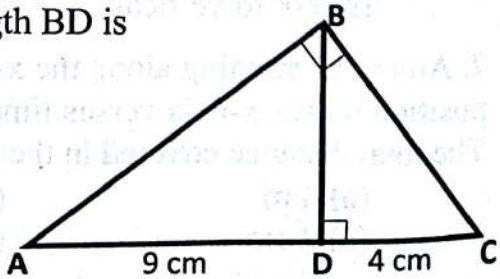


(a) 1
(b) 2
(c) 3
(d) 4



12. In triangle ABC, shown in the figure, AB is perpendicular to BC. Further, BD is perpendicular to AC. If AD = 9 cm and DC = 4 cm, the length BD is

(a) 6 cm
(b) 6.5 cm
(c) 36/13 cm
(d) 13/36 cm



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13. A box of sticks of equal lengths is provided. The minimum number sticks needed to build a frame to enclose a 3 dimensional volume is

(a) 6 (b) 12 (c) 3 (d) 8

14. 5 kg of adulterated rice has 2% stones in it and the rest is rice. Half of the stone content was removed. Now the percentage of stone content in it is

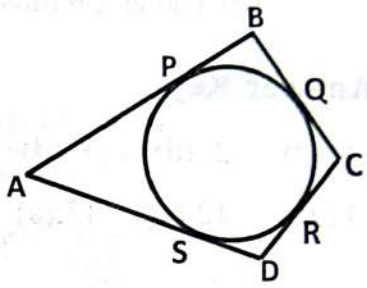
(a) 0.99 % (b) 1 % (c) 1.1 % (d) 1.01 %

15. A coin is tossed six times. The probability that heads will occur at least once is

(a) 63/64 (b) 1/3 (c) 1/64 (d) 3/2

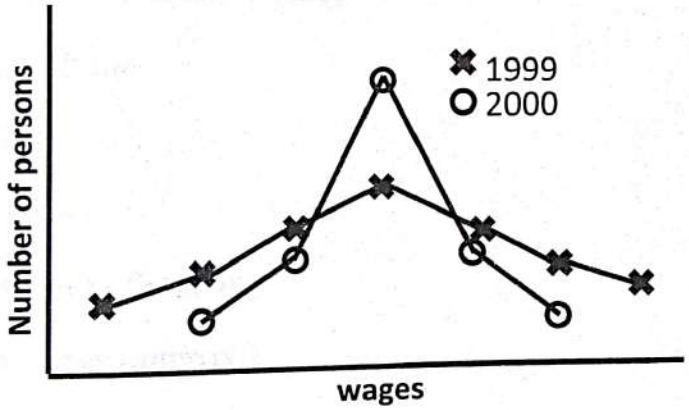
16. Circle PQR is inscribed in a quadrilateral ABCD. The circle touches side AD at point S. AP = 8 cm, QC = 3 cm and DC = 6 cm. The length of side AD is

(a) 9 cm
(b) 10 cm
(c) 11 cm
(d) 12 cm



17. The distribution of wages in a population is shown below for 2 years. The average wage(s)

(a) in 1999 is greater than that in 2000.
(b) in 1999 is less than that in 2000.
(c) in the two years are equal, but the variances are unequal.
(d) in the two years are unequal, but the variances are equal.



18. Two parameters TC and TF are related as shown in the table. Find the value of TF corresponding to a TC of 75.

(a) 167
(b) 162
(c) 150
(d) 200

TC	0	25	50	75	100
TF	32	77	122	?	212

19. Several identical cubes are arranged in a close-packed single layer. If the area of the layer is A and the volume of the layer is V , then the number of cubes in the collection is

(a) V/A (b) A^3/V^2 (c) V^2/A^3 (d) A/V

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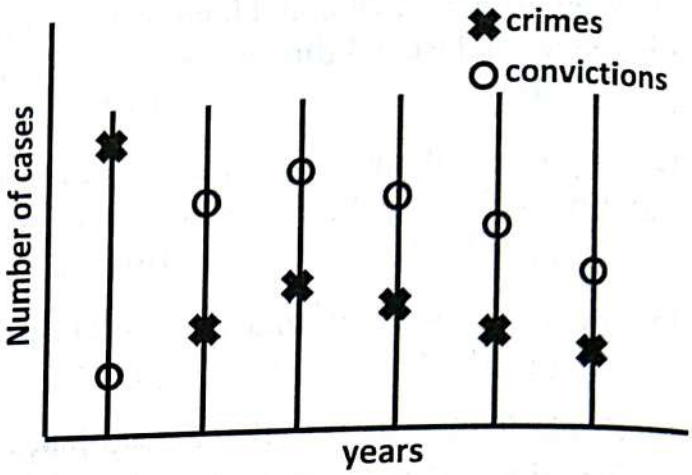
20. The graph shows crime rates and convictions for 5 years in a certain society. Which of the following is correct?

(a) The cause of convictions being on the rise is better law enforcement.
(b) Falling crime rates have slowly reduced the conviction rates also.
(c) There are fewer convictions because crime rate has fallen.
(d) The graph must be wrong.



Answer Key

1. (d)	2. (d)	3. (d)	4. (c)	5. (b)	6.(b)	7.(a)	8.(b)	9.(a)	10.(a)
11.(a)	12.(a)	13.(a)	14.(d)	15.(d)	16.(c)	17.(c)	18.(a)	19.(b)	20.(b)

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APPENDIX 1 - CONVERSION TABLE

Length

1 centimetre (cm) = 10 millimetre (mm)

1 metre (m) = 100 cm

1 kilometre (km) = 1000 m

1 inch = 2.54 cm

1 mile = 1.60934 (km)

Mass

1 gram (g) = 1000 milligram (mg)

1 kilogram (kg) = 1000g

1 kg = 2.20462 pound (lbs.)

1 litre (l) = 1000 millilitres (ml)

Time

1 minute (min) = 60 seconds (sec)

1 hour (hr.) = 60 min

1 day = 24 hrs.

1 week = 7 days

1 month = 4 weeks (approx.)

1 month = 30 days (approx.)

1 year = 12 months

1 year = 365 days (Ordinary year)

1 year = 366 days (Leap year)

1 decade = 10 years

1 century = 10 decades = 100 years

Digital storage

1 byte = 8 bits

1 kilobyte (KB) = 1024 bytes

1 megabyte (MB) = 1024 KB

1 gigabyte (GB) = 1024 MB

Temperature

$$x^o Celsius\ (C) = \frac{x^o Farhenheit\ (F) - 32}{1.8}$$

$$kelvin\ (K) = x^o C + 273$$

Constants

$$\pi = 3.14 = \frac{22}{7}\ (approx.)$$

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APPENDIX -2

- SYMBOLS AND ACRONYMS USED IN THIS TEXT BOOK

- LHS - Left Hand Side
- RHS - Right Hand Side
- $\approx$ - Approximately
- $\equiv$ - equivalent to
- $\Rightarrow$ - implies
- $\angle$ - angle
- Δ - Triangle
- $\neq$ - Not equal to
- ∞ - Infinity

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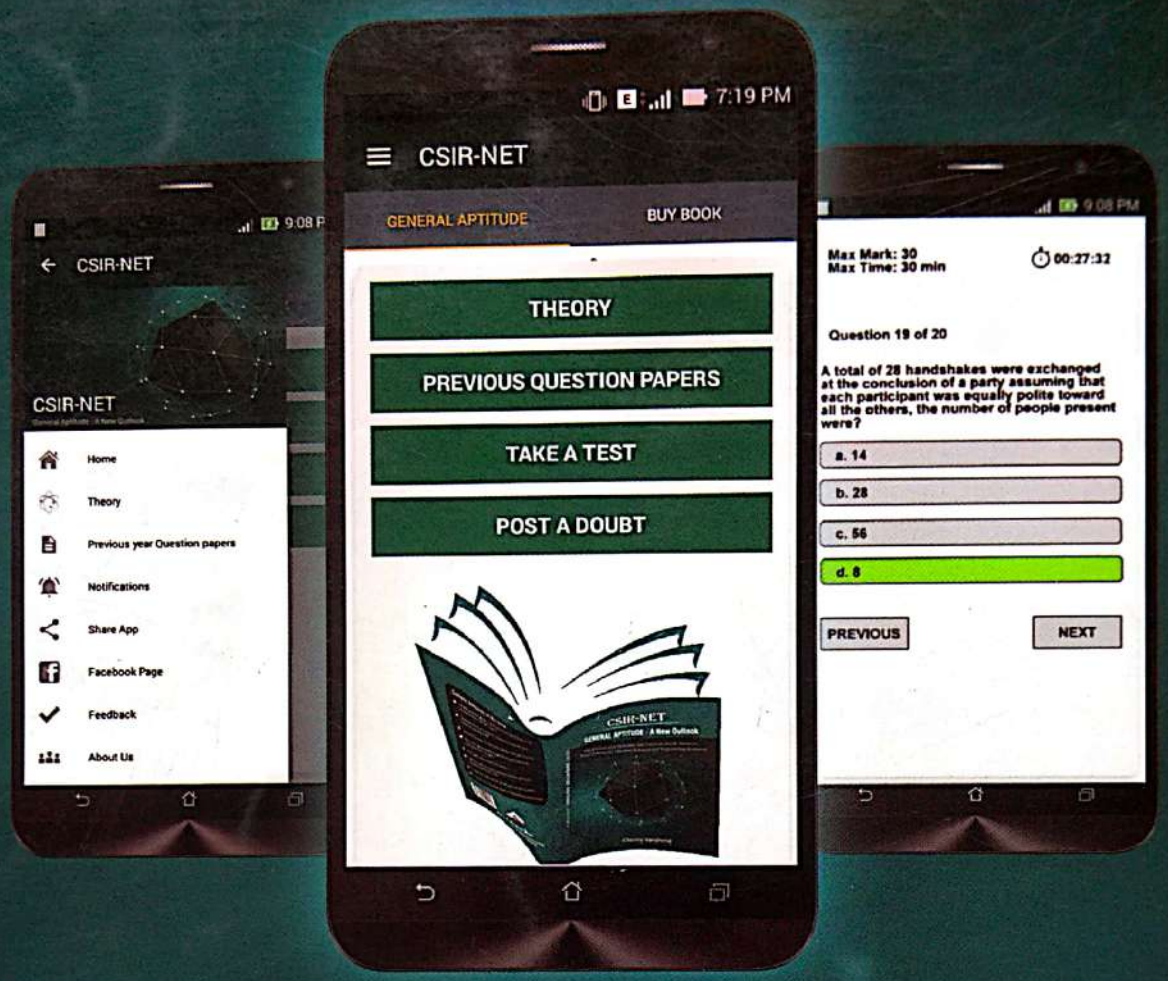

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